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MIL-HDBK-17-2D
Volume 2 of 3
23 JANUARY 1997

Superseding
MIL-HDBK-17-2C
25 FEBRUARY 1994

DEPARTMENT OF DEFENSE HANDBOOK

POLYMER MATRIX COMPOSITES

VOLUME 2. MATERIALS PROPERTIES



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FOREWORD

1. This handbook is approved for use by all Departments and Agencies of the Department of Defense.

2. This handbook is for guidance only. This handbook cannot be cited as a requirement. If it is, the contractor does not have to comply. This mandate is a DoD requirement only; it is not applicable to the Federal Aviation Administration (FAA) or other government agencies.

3. Every effort has been made to reflect the latest information on polymeric composites. The handbook is continually reviewed and revised to ensure its completeness and currentness. Documentation for the secretariat should be directed to: Materials Sciences Corporation, MIL-HDBK-17 Secretariat, 500 Office Center Drive, Suite 250, Fort Washington, PA 19034.

4. MIL-HDBK-17 provides guidelines and material properties for polymer (organic) matrix composite materials. The first three volumes of this handbook currently focus on, but are not limited to, polymeric composites intended for aircraft and aerospace vehicles. Metal matrix composites (MMC), ceramic matrix composites (CMC), and carbon/carbon composites (C/C) will be covered in separate volumes as developments occur.

5. This standardization handbook has been developed and is being maintained as a joint effort of the Department of Defense and the Federal Aviation Administration.

6. The information contained in this handbook was obtained from materials producers, industry, reports on Government sponsored research, the open literature, and by contact with research laboratories and those who participate in the MIL-HDBK-17 coordination activity.

7. All information and data contained in this handbook have been coordinated with industry and the US Army, Navy, Air Force, NASA, and Federal Aviation Administration prior to publication.

8. Copies of this document and revisions thereto may be obtained from the Standardization Document Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

9. Volume 2 has an annex. The annex contains technical data whose export is restricted. With the proper certification (DD Form 2345) the annex may be obtained from the Director, U.S. Army Research Laboratory, Weapons and Materials Directorate, ATTN: AMSRL-WM-M, Aberdeen Proving Ground, MD 21005-5069

10. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate, Attn: AMSRL-WM-M, Aberdeen Proving Ground, MD 21005-5069, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SUMMARY OF CHANGES IN REVISION MIL-HDBK-17-2D

Chapter 1

Chapter 1 has been reorganized. Section 1.4.2 and 1.4.3 have been moved from Volume 1, Sections 8.4.3 and 8.4.4.

Chapters 4 and 10

The format for the presentation of data in the handbook has been revised as documented in Section 1.4.2. This revision affected all data presented in Chapters 4 and 10.

New data sets were added in Sections 4.2.16 - 4.2.25. Additional data was added to Section 4.2.8. An annex for data with export control limitations has been established.

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CHAPTER 1 GENERAL INFORMATION

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1.1 INTRODUCTION

The standardization of a statistically-based mechanical property data base, procedures used, and overall material guidelines for characterization of composite material systems is recognized as being beneficial to both manufacturers and governmental agencies. It is also recognized that a complete characterization of the capabilities of any engineering material system is primarily dependent on the inherent material physical and chemical composition which precede, and are independent of, specific applications. Therefore, at the material system characterization level, the data and guidelines contained in this handbook are applicable to military and commercial products and provide the technical basis for establishing statistically valid design values acceptable to certificating or procuring agencies.

This standardization handbook has been developed and is maintained as a joint effort of the Department of Defense and the Federal Aviation Administration. It is oriented toward the standardization of methods used to develop and analyze mechanical property data on current and emerging composite materials.

1.2 PURPOSE AND SCOPE OF VOLUME 2

This handbook is for guidance only. This handbook cannot be cited as a requirement. If it is, the contractor does not have to comply. This mandate is a DoD requirement only; it is not applicable to the Federal Aviation Administration (FAA) or other government agencies.

The purpose of this handbook is to provide a standard source of statistically-based mechanical property data for current and emerging composite materials. Strength and strain-to-failure properties will be reported either in terms of B-values or S-values. A B-basis value is the mechanical property value above which at least 90 percent of the population is expected to fall, with a confidence of 95 percent. An S-basis value is the mechanical property value which is usually the specified minimum value of the appropriate government specification for the material. Physical, chemical, and mechanical values of the composite constituents - the fibers, matrix material, and prepreg - will be reported where applicable. Later chapters will include data summaries for the various composite systems. Individual chapters focus on particular reinforcement fibers.

Statistically-based strength properties will be defined for each composite material system over a range of potential usage conditions. The intent will be to provide data at the upper and lower limits of the potential environmental conditions for a particular material, so that application issues do not govern the mechanical property characterizations. If data are also available at intermediate environmental conditions, they will also be used to more exactly define the relationship between the material properties and the effect of the environment on those properties. The statistically-based strength data which are available are tabulated in Volume 2. These data are useful as a starting point for establishing structural design allowables when stress and strength analysis capabilities permit lamina level margin of safety checks. Depending on the application, some structural design allowables will have to be determined empirically at the laminate and composite level, since MIL-HDBK-17 does not provide these data. Additional information and properties will be added as they become available and are demonstrated to meet the guideline criteria.

All data included herein are based on test specimens only. Test specimen dimensions conform with those specified for the particular test method which is used. Standard test methods are recommended where possible (ASTM standards are the primary source). The designer and all users must be responsible for any translation of the data contained herein to other coupon dimensions, temperature, humidity, and other environmental conditions not covered in this document. Problems such as scale-up effects and the influence of the test method selected on properties are also not addressed in this document. The manner in which S-basis values are used is also up to the discretion of the designer. In general, decisions concerning which properties to use for a specific application or design are the responsibility of the designer and are outside the scope of this handbook.

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1.3 ORGANIZATION OF DATA IN HANDBOOK

The data in Volume 2 is divided into chapters of fiber properties, resin properties, and composite properties organized by fiber and then resin.

1.3.1 Fiber properties

Chapter 2 in Volume 2 will provide data for fiber properties. Sections are to be included for different types of fiber, e.g., glass fibers and carbon fibers. In each section, the general characteristics of the type of fiber will be given, as well as an index of suppliers, designations, and abbreviations. For each specific fiber, data will be organized in the following manner. The X's in the subsection number will be determined by the type of fiber and the specific fiber described.

2.X.X.1	Supplier and product data
2.X.X.2	Chemical and physical properties
2.X.X.2.1	Typical range of chemical constituents
2.X.X.2.2	Expected bound in physical properties
2.X.X.3	Electrical properties
2.X.X.4	Thermal-mechanical properties
2.X.X.4.1	Stress-strain curves
2.X.X.4.2	Environmental effects

1.3.2 Matrix properties

Matrix or resin properties will be included in Chapter 3 which will be divided into sections according to the type of resin. For example, Section 3.2 will give data for epoxies and Section 3.3 will provide data for polyester resins. The subsections for each resin will be the same as those in Chapter 2 given above.

1.3.3 Composite properties

The remaining chapters of Volume 2 will provide data for prepreg, lamina, laminate, and joint properties. There will be individual chapters for each family of composites based on fiber type. For example, Chapter 4 describes carbon fiber composites. Within each chapter, there is expected to be an index of suppliers, designations, and abbreviations. Sections will be included based on the resin type used with the fiber described in the chapter, e.g., Section 4.3 will provide properties for carbon/epoxy composites.

Properties will be organized in the following manner for each specific composite:

X.X.X.1	Supplier and product data
X.X.X.2	Prepreg chemical and physical properties
X.X.X.2.1	Physical description
X.X.X.2.2	Resin content
X.X.X.2.3	Fiber content
X.X.X.2.4	Volatiles content
X.X.X.2.5	Moisture content
X.X.X.2.6	Inorganic fillers and additives content
X.X.X.2.7	Areal weight
X.X.X.2.8	Tack and drape
X.X.X.2.9	Resin flow
X.X.X.2.10	Gel time
X.X.X.3	Lamina chemical properties
X.X.X.4	Lamina physical properties
X.X.X.5	Lamina mechanical properties

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X.X.X.5.1 Data Summaries
X.X.X.5.2 Typical stress-strain curves
X.X.X.6 Thermal properties
X.X.X.7 Electrical properties
X.X.X.8 Laminate thermal-mechanical properties
X.X.X.8.1 Index of properties by lay-up
X.X.X.8.2 Strength properties
a. Lay-up No. 1
b. Lay-up No.2
X.X.X.8.3 Thermal properties
X.X.X.8.4 Electrical properties
X.X.X.9 Joint thermal-mechanical properties
X.X.X.9.1 Index of properties by joint and composite system
X.X.X.9.2 Bearing strength
a. System No. 1
b. System No.2

1.4 PRESENTATION OF DATA

This section describes how the data are presented and organized in this volume (MIL-HDBK-17-2).

1.4.1 Properties and definitions

The properties and their definitions are found in the appropriate chapters of Volume 1. Fiber properties and methods for obtaining them are discussed in Chapter 3. Resin properties are presented in Chapter 4. Methods for characterizing prepreg materials are discussed in Chapter 5 and properties and definitions for laminae and laminates are presented in Chapter 6. Properties for structural elements are presented in Chapter 7. The statistical methods used in determining these properties are discussed in Chapter 8. Material system codes and laminate orientation codes are defined in this chapter.

1.4.1.1 Sign convention

All compressive values, represented by a superscript c, are reported as positive numbers. Thus, a positive compression strength indicates failure due to a load applied in the opposite direction of a positive tensile failure.

1.4.2 Table formats

The table formats for mechanical property data presentation are given in Tables 1.4.2(a), and 1.4.2(c) through (g). Table 1.4.2(a) shows the summary pages giving information about the material system and the properties for which data are available. The following notes apply to this table:

❶ Handbook section title and number. Sections are titles using the following information:

{Fiber} {Filament-Count}/{Matrix} {Tape/Weave Type} {Critical processing information}

The tape/weave type includes unidirectional tape (UT), plain weave (PW), or five-harness satin (5HS). Critical processing information, such as bleed or no-bleed, is included when it is necessary to discriminate between data sets. If a warning regarding data documentation is included for the data set, an asterisk follows the section title.

❷ The first set of information in a data section is a summary table containing information on the materials, processing, etc. The box with a heavy border in the upper right-hand corner identifies the first summary table.

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TABLE 1.4.2(a) *Summary table format,* continued on third page.

X.X.X {Fiber} {Filament-Count}/{Matrix} {Tape/Weave Type}* ❶

MATERIAL:	{Fiber} {Filament-Count}/{Matrix} {Tape/Weave Type} ❸			❷
PREPREG:	{Manufacturer} {Commercial Name} {Tape/weave type}			
FIBER:	{Manufacturer} {Commercial Name} {Filament-Count} {Sizing} {Twist}	MATRIX:	{Manufacturer} {Commercial Name}	
T_g(dry):	XXX°F T_g(wet): XXX°F	T_g METHOD:	{Method} ❹	
PROCESSING:	{Type of Process}: {Temperature}, {Duration}, {Pressure} ❺			

*{Warning} ❻

Date of fiber manufacture	MM/YY	Date of testing	MM/YY
Date of resin manufacture	MM/YY	Date of data submittal	MM/YY
Date of prepreg manufacture	MM/YY	Date of analysis	MM/YY
Date of composite manufacture	MM/YY		❼

LAMINA PROPERTY SUMMARY ❽

	{RTA}		{Ambient/dry, coldest to hottest}					{Wet, coldest to hottest}		
Tension, 1-axis										
Tension, 2-axis										
Tension, 3-axis										
Compression, 1-axis			Classes of approval are noted for each type test/direction/environmental condition combination							
Compression, 2-axis										
Compression, 3-axis										
Shear, 12-plane										
Shear, 23-plane										
Shear, 31-plane										
{Additional type test/direction}										
. . .										

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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{Fiber Class}/{Matrix Class} {Nominal FAW} - {Tape/Weave Type} {Fiber}/{Matrix} Summary

This box contains the fiber/matrix class of the material, such as carbon/epoxy, identified using the material system codes in Section 1.5.1 With the fiber and matrix class is the nominal fiber areal weight and the abbreviated tape/weave type. The material identification is summarized by the fiber and matrix names.

❸ Material information is presented for the composite, the prepreg, the fiber, and the matrix. Composite material identification is presented as:

{Fiber} {Filament-Count}/{Matrix} {Tape/Weave Type}

Prepreg identification is included as {Manufacturer} {Commercial Name} {Tape/weave type}. For fabric, information such as warp and fill fiber spacing is included when it is available. Fiber identification includes {Manufacturer} {Commercial Name} {Filament-Count} {Sizing} {Twist}. Resin identification is presented as {Manufacturer} {Commercial Name}.

❹ Glass transition temperature under dry and wet conditions is presented with the test method used to obtain these data (See Volume 1, Section 6.4.3).

❺ Basic processing information is presented. This includes the type of process, temperature, pressure, duration, and any other critical parameters for one or more processing steps.

❻ Any warning for limited data documentation is presented on each page of data presentation. On the first page of a data section, a warning is shown below the material identification block.

❼ The block below the material identification block presents various dates relevant to the fabrication and testing of the material. The date of data submittal determines the data documentation requirements that were used for the data set (Volume 1, Section 2.5.6) and the date of analysis determines the statistical analysis that was used (Volume 1, Section 8.3). Ranges of dates are presented where appropriate, such as a testing program which lasted several months.

❽ Lamina properties are summarized with the class of data provided for each property. The columns of the lamina property summary table define the environmental conditions. The first column contains room temperature ambient or dry data. Dry is used only if a drying procedure was used. Ambient refers to as-fabricated with subsequent storage in an ambient laboratory environment. The remaining columns are ordered from lowest to highest moisture content and within a given moisture content, from lowest to highest temperature. If there is enough space, a blank column separates the room temperature ambient/dry column from the other columns and each moisture condition from the others.

The rows of the lamina summary table identify the type test and direction. The basic mechanical properties are included in each summary table. If data are available, additional properties, such as the following, are included:

SB strength, 31-plane	G_{Ic}	CTE, 1-axis
SB strength, 23-plane	G_{IIc}	CTE, 2-axis
SB strength, 12-plane		CTE, 3-axis

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For each test type and direction, the class of data for the strength, modulus, Poisson's ratio, and strain-to-failure are provided, in that order. For example, if the entry under RTA and Tension, 1-axis is FI-S. There is room temperature ambient data for longitudinal tension strength, modulus, and strain-to-failure, but not Poisson's ratio. The strength data are fully approved, the modulus data are interim, and the strain-to-failure data are screening. The classes of data approval are defined in Volume 1, Section 2.5.1. Fully approved data requires a minimum of thirty data points from at least five batches with more complete documentation; screening data represent a small number of data points from one batch, and interim data represent a minimum of fifteen specimens from at least three batches. Certain test methods, for example, short beam shear, result only in screening data. (Note that SB strength is the result of a short beam shear test, based on current terminology in ASTM D 2344.)

Continuing on the second page of summary information (Table 1.4.2(a)):

① Any warning is placed at the top of this page.

② The box at the top of the second page of summary information presents basic physical parameters for the data set. The first data column contains nominal values, typically specification information.

③ The second data column presents the range of values for the data set submitted.

④ The last column presents the test method used to obtain these data. This information was not included in the early versions of data documentation requirements.

⑤ Laminate property data are summarized in the lower box in the same way as lamina property data are summarized on the previous page. Families of laminates are provided with properties listed below each laminate family. Specific lay-up information is provided in the detailed tables which follow. The type test and direction are included only if data are available and are based on Table 1.4.2(b).

TABLE 1.4.2(b) *Laminate type test and directions*

Type Test		Direction	
Tension	Compression After Impact	x-axis	xy-plane
Compression	Bearing	y-axis	yz-plane
Shear	CTE	z-axis	zx-plane
Open Hole Tension			
Open Hole Compression			

Unless otherwise noted, the x-axis corresponds to the 0 direction of the laminate lay-up. Data included for this material are indicated by the class of approval, identified in the footnote.

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TABLE 1.4.2(a) *Summary table format,* concluded.

Warning ①

		Nominal ②	As Submitted ③	Test Method ④
Fiber Density	(g/cm^3)	X.XX	{Minimum} - {Maximum}	{Method}
Resin Density	(g/cm^3)	X.XX	{Minimum} - {Maximum}	{Method}
Composite Density	(g/cm^3)	X.XX	{Minimum} - {Maximum}	{Method}
Fiber Areal Weight	(g/m^2)	XXX	{Minimum} - {Maximum}	{Method}
Ply Thickness	(in)	0.0XXX	{Minimum} - {Maximum}	{Method}

LAMINATE PROPERTY SUMMARY ⑤

	{RTA}		{Ambient/dry, coldest to hottest}		{Wet, coldest to hottest}		
{Laminate Family} {Type test/direction} . . .			Classes of approval are noted for each type/direction/environmental condition combination				

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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The format for a data table containing normalized material property information is shown in Table 1.4.2(b). Requirements and procedures for normalization are found in Volume 1, Section 2.5.7 and 2.4.3.

❶ Warnings are shown on each page for data sets which do not meet the data documentation requirements. Many of the data sets were submitted before the establishment of the data documentation requirements. Data sets which do not meet the first version of data documentation requirements or the data documentation requirements that were current when the data were submitted will not receive full approval.

❷ At the top right corner of each page is a box with a heavy border. This box contains information which identifies the data set, the type of test for which results are shown, specimen orientation, test conditions, and the classes of data.

{Table Number}
{Fiber Class}/{Matrix Class} {FAW}-{Form} - FAW, fiber areal weight
{Fiber Name}/{Matrix Name}
{Test Type}, {Direction}
{Lay-up}
{Test Temperature}/{Moisture Content} - repeated for each data column
{Classes of Data Approval}

❸ Material identification is provided for the composite material as

{Fiber} {Filament-Count}/{Matrix} {Tape/Weave Type}

The range of physical parameters, resin content, fiber volume, ply thickness, composite density, and void content are presented for the data on this particular page.

❹ The test method is identified with the organization, number, and date. For compression after impact, the nominal impact energy level used for the test is appended to the test method, since alternate levels are often used.

❺ The method of calculating the modulus is presented for mechanical property data. This includes the calculation method, and the location or range of measurements used for the calculation.

❻ The normalization method is presented for data that have been normalized (See Volume 1, Section 2.4.3). The fiber volume to which the data are normalized is also included. This value is typically 60% for unidirectional tape and 57% for weaves. Types of normalization as entered are:

Normalized by fiber volume to XX% (0.0XXX in. CPT)
Normalized by specimen thickness and batch fiber volume to XX% (0.0XXX in. CPT)
Normalized by specimen thickness and batch fiber areal weight to XX% fiber volume (0.0XXX in. CPT)

Corresponding cured ply thickness (CPT) values, based on a nominal fiber areal weight, are included for reference for each method.

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TABLE 1.4.2(c) *Table format for normalized data.*

{Warning} ❶

MATERIAL: {Fiber} {Filament count}/{Matrix} {Tape/weave type} ❸

RESIN CONTENT: XX - XX wt% COMP. DENSITY: X.XX-X.XX g/cm^3
FIBER VOLUME: XX - XX vol% VOID CONTENT: 0.X to X.X %
PLY THICKNESS: 0.0XXX - 0.0XXX in.

❷

TEST METHOD: ❹
{Organization} {Number} {Date}

MODULUS CALCULATION: ❺
{Method}, XXXX - XXXX με

NORMALIZED BY: {Method} to XX % (0.0XXX in. CPT) ❻

Temperature(°F) Moisture Content(%) Equilibrium at T, RH Source Code		❼					
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{tu} (ksi)	Mean Minimum Maximum C.V. (%) B-value Distribution C_1 C_2 No. Specimens No. Batches Approval Class		❽				
E_1^t (Msi)	Mean Minimum Maximum C.V. (%) No. Specimens No. Batches Approval Class		❾				
ν_{12}^t	Mean No. Specimens No. Batches Approval Class		❾				
ϵ_1^{tu} (με)	Mean Minimum Maximum C.V. (%) B-value Distribution C_1 C_2 No. Specimens No. Batches Approval Class		❽				

❿

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❼ At the top of each data column are the test conditions. Nominally dry conditions, for materials that are fabricated and stored under controlled conditions are noted. Wet conditions which are not conditioned to equilibrium are also noted. The source code provides a means for identifying data sets from the same source. No other source identification is provided.

❽ Strength data and strain-to-failure data are presented in the handbook with a full set of statistical parameters. The class of data approval is indicated for each property/condition combination. B-values are presented only for fully approved data. A-basis values are presented for fully approved data which meet the batch and specimen number requirements for A-basis values. The distribution of method of analysis is presented. The constants, C_1 and C_2, correspond to the distribution. These are as follows:

	C_1	C_2
Weibull	scale parameter	shape parameter
Normal	mean	standard deviation
Lognormal	mean of the natural log of the data	standard deviation of the natural log of the data
Nonparametric	rank	Hanson-Koopmans coefficient
ANOVA	sample between-batch variance	sample within-batch variance

All statistical parameters are presented for normalized and as-measured strength data. All statistical parameters are presented for as-measured strain-to-failure data.

❾ Modulus data are presented with only mean, minimum, maximum, coefficient of variation, batch size, and sample size. Values are presented for both normalized and as-measured data. Where available, Poisson's ratio data are presented with batch size and sample size information.

❿ Footnotes are presented wherever additional information is pertinent. Information frequently presented in footnotes include conditioning parameters, reasons for not presenting B-values, and deviations from standard test methods.

Symbols for properties are presented with property directions as subscripts and property type, for example, tension (t), as superscripts. The example table shows symbols for lamina tension in the fiber direction.

Table 1.4.2(d) shows an example table for material properties that are not normalized. The basic table format and information is identical to the table format for normalized data. Only as-measured data are presented in each column of information. The statistical parameters are the same provided for normalized data. Table 1.4.2(e) presents the format for bearing data, Table 1.4.2(f) the format for notched laminate data, and Table 1.4.2(g) the format for bearing/bypass data.

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TABLE 1.4.2(d) *Table format for as-measured data.*

{Warning} ❶

MATERIAL: {Fiber} {Filament count}/{Matrix} {Tape/weave type} ❸					❷
RESIN CONTENT: XX - XX wt%		COMP. DENSITY: X.XX-X.XX g/cm³			
FIBER VOLUME: XX - XX vol%		VOID CONTENT: 0.X to X.X %			
PLY THICKNESS: 0.0XXX - 0.0XXX in.					
TEST METHOD: ❹ {Organization} {Number} {Date}		MODULUS CALCULATION: ❺ {Method}, XXXX - XXXX με			
NORMALIZED BY: Not normalized ❻					
Temperature(°F) Moisture Content(%) Equilibrium at T, RH Source Code	❼				
F(1) (ksi) Mean Minimum Maximum C.V. (%) B-value Distribution C_1 C_2 No. Specimens No. Batches Approval Class	❽				
(Msi) Mean Minimum Maximum C.V. (%) No. Specimens No. Batches Approval Class	❾				
Mean No. Specimens No. Batches Approval Class	❾				
(με) Mean Minimum Maximum C.V. (%) B-value Distribution C_1 C_2 No. Specimens No. Batches Approval Class	❽				

❿

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TABLE 1.4.2(e) *Format for bearing strength property table.*

MATERIAL:	{Fiber} {Fil. Count} / {Matrix} {tape/weave type}		**{Table Number} {Material Class}/{Form} {Material Designation} {Property and Direction} {Laminate Family} {Test Temp/Moisture Conditions} {Data Approval Status}**
RESIN CONTENT:	XX-XX wt%	COMP. DENSITY:	0.0XX lb/in^3
FIBER VOLUME:	XX-XX %	VOID CONTENT:	X.X - X.X %
PLY THICKNESS:	0.00XX - 0.00XX in.		
TEST METHOD:		{Org. Method - Date}	
TYPE OF BEARING TEST:		{material/joint}	
JOINT CONFIGURATION			
Member 1 (t,w,layup):		{thickness, width, layup}	
Member 2 (t,w,layup):		{thickness, width, layup}	
FASTENER TYPE:		{ }	
TORQUE:		{ }	
NORMALIZED BY:		Not normalized	
FAILURE DESCRIPTION (3):			

Temperature (°F) Moisture Content (%) Equilibrium at T, RH (°F, %) Source Code		75 (1)		ET (2)	
Diameter		0.25	D2	0.25	D2
F^{bru}	Mean Minimum Maximum C.V. (%)				
(ksi)	No. Specimens No. Batches				
F^{br}_{offset}	Mean Minimum Maximum C.V. (%)				
(ksi)	No. Specimens No. Batches				
F^{br}_{pl}	Mean Minimum Maximum C.V. (%)				
(ksi)	No. Specimens No. Batches				

(1) As fabricated.
(2) Conditioned at XXX°F, XX% RH for XX days.
(3) See Table 8.3.2.1 for failure mode description.

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TABLE 1.4.2(f) *Format for notched laminate strength property table.*

MATERIAL:	{Fiber} {Fil. Count} / {Matrix} {tape/weave type}		**{Table Number} {Material Class}/{Form} {Material Designation} {Property and Direction} {Laminate Family} {Test Temp/Moisture Conditions} {Data Approval Status}**
RESIN CONTENT:	XX-XX wt%	COMP. DENSITY: 0.0XX lb/in^3	
FIBER VOLUME:	XX-XX %	VOID CONTENT: X.X - X.X %	
PLY THICKNESS:	0.00XX - 0.00XX in.		
TEST METHOD:		{Org. Method - Date}	
SPECIMEN GEOMETRY:		{thickness}	
FASTENER TYPE:		{ }	
TORQUE:		{ }	
NORMALIZED BY:		Not normalized	

Temperature (°F)		75			ET		
Moisture Content (%)		(1)			(2)		
Equilibrium at T, RH (°F, %)							
Source Code							
Width to Diameter Ratio		W1/D	W2/D	W3/D	W1/D	W2/D	W3/D
Diameter = 0.25	Mean						
	Minimum						
F_x^{cu}	Maximum						
	C.V. (%)						
(ksi)	No. Specimens						
	No. Batches						
Diameter = D2	Mean						
	Minimum						
F_x^{cu}	Maximum						
	C.V. (%)						
(ksi)	No. Specimens						
	No. Batches						
Diameter = D3	Mean						
	Minimum						
F_x^{cu}	Maximum						
	C.V. (%)						
(ksi)	No. Specimens						
	No. Batches						

(1) As fabricated.
(2) Conditioned at XXX°F, XX% RH for XX days.

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TABLE 1.4.2(g) *Format for bearing/bypass property table.*

MATERIAL:	{Fiber} {Fil. Count} / {Matrix} {tape/weave type}		**{Table Number} {Material Class}/{Form} {Material Designation} {Property and Direction} {Laminate Family} {Test Temp/Moisture Conditions} {Data Approval Status}**
RESIN CONTENT:	XX-XX wt%	COMP. DENSITY:	0.0XX lb/in^3
FIBER VOLUME:	XX-XX %	VOID CONTENT:	X.X - X.X %
PLY THICKNESS:	0.00XX - 0.00XX in.		
TEST METHOD:		{Org. Method - Date}	
SPECIMEN GEOMETRY:		{thickness}: { }	
FASTENER TYPE:		{ }	
TORQUE:		{ }	
NORMALIZED BY:		Not normalized	

Temperature (°F)		75	
Moisture Content (%)		(1)	
Equilibrium at T, RH (°F, %)			
Source Code			
Bearing/Bypass Ratio		0.50	0.75
F_{byp}^{cu}	Mean		
	Minimum		
(ksi)	Maximum		
	C.V. (%)		
F^{br}	Mean		
	Minimum		
(ksi)	Maximum		
	C.V. (%)		
	No. Specimens		
	No. Batches		
F_{byp}^{tu}	Mean		
	Minimum		
(ksi)	Maximum		
	C.V. (%)		
F^{br}	Mean		
	Minimum		
(ksi)	Maximum		
	C.V. (%)		
	No. Specimens		
	No. Batches		

(1) As fabricated.

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1.4.3 Sample graphs

Sample graphs for the stress-strain curves are given in Volume 1, Section 8.4.4. Figure 1.4.3(a) shows a graph of the tangent modulus and the secant modulus as a function of strain for the data of Volume 1, Figure 8.4.4(a).

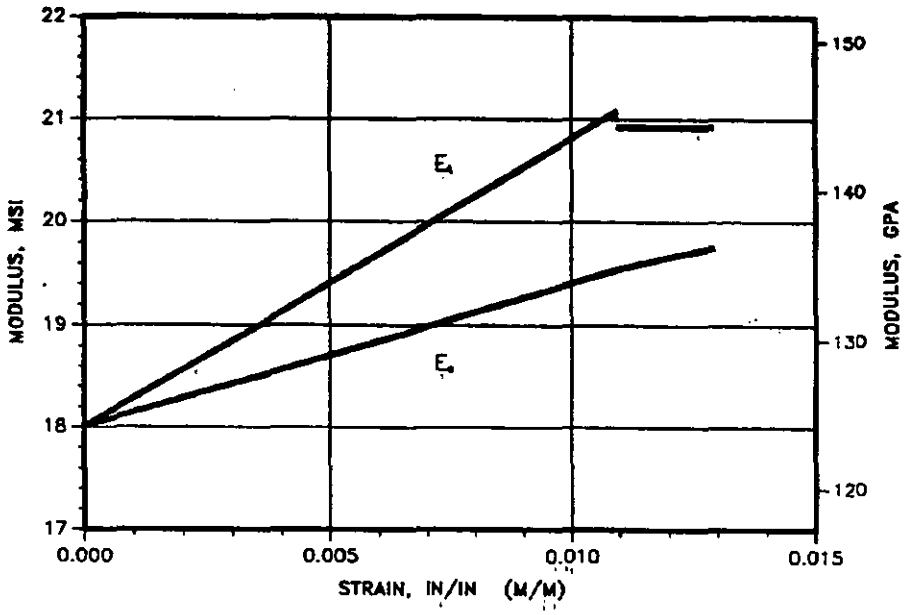


Figure 1.4.3(a) *Modulus Curves for AS4/3501-6, see Figure 8.3.7(a).*

For thermal properties, room temperature values will be listed in the data summary and additional values of the thermal expansion coefficient will be presented graphically as a function of temperature. The specific heat and the thermal conductivity will be presented in a similar fashion. When data are adequate to present curves showing specific heat, thermal conductivity, and mean coefficient of thermal expansion over a range of temperatures, graphical presentation is used in addition to tabular presentation. A smooth curve is drawn through the plotted points to depict the overall trend of the data. The smooth curve formats for specific heat, thermal conductivity, and thermal expansion are presented along with representative data in Figures 1.4.3(b) - (d), respectively. The reference temperature for thermal expansion should be shown on the figure. In Figure 1.4.3(d), the reference temperature of 70°F (21°C, 294 K) indicates that the mean coefficient of thermal expansion between 70°F (21°C, 294 K) and the indicated temperature is plotted.

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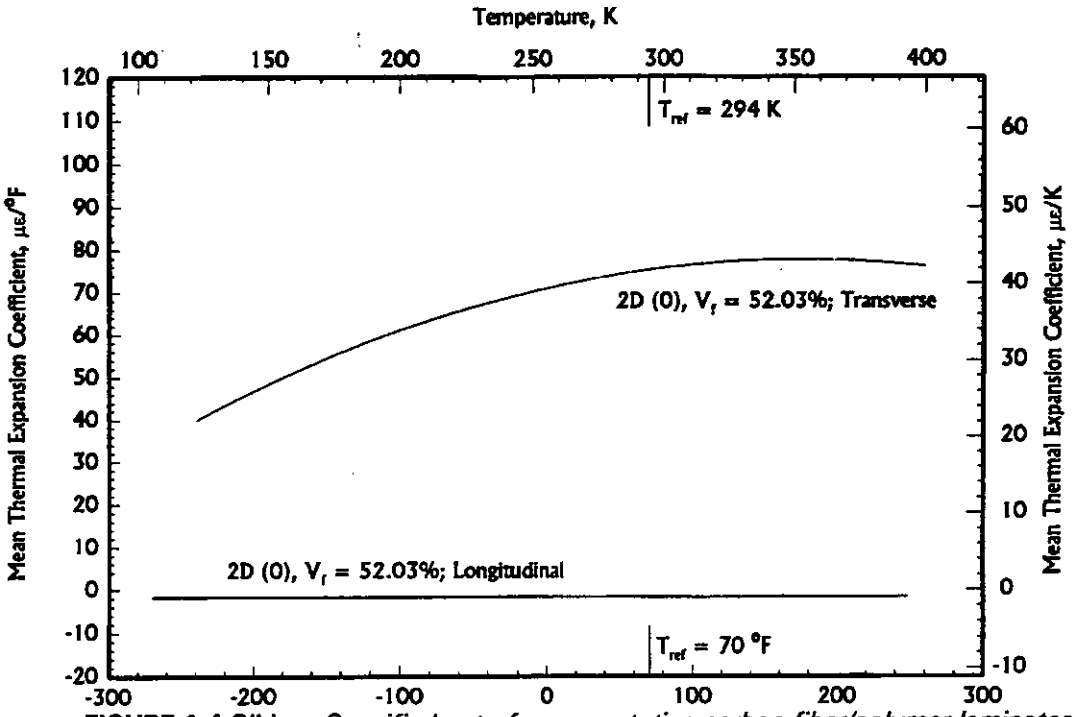


FIGURE 1.4.3(b) *Specific heat of representative carbon fiber/polymer laminates as a function of temperature.*

FIGURE 1.4.3(d) *Mean coefficient of thermal expansion of representative carbon fiber/epoxy laminate (P75 carbon/1962 epoxy) as a function of temperature.*

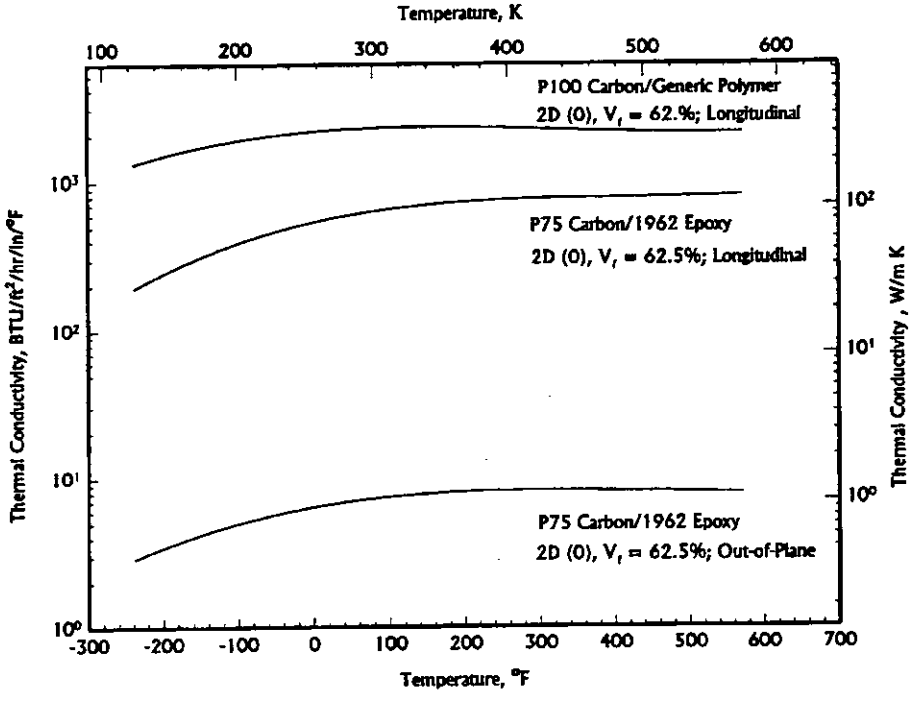


FIGURE 1.4.3(c) *Thermal conductivity of representative carbon fiber/polymer laminates as a function of temperature.*

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1.5 MATERIALS SYSTEMS

1.5.1 Materials system codes

The materials systems codes which are used in the handbook consist of a fiber system code and a matrix material code separated by a virgule (/). The codes for the fiber and matrix materials appear in Tables 1.5.1(a) and (b).

TABLE 1.5.1(a) *Fiber system codes.*

AlO	Alumina
Ar	Aramid
B	Boron
C	Carbon
DGl	D-Glass
EGl	E-Glass
Gl	Glass
Gr	Graphite
Li	Lithium
PAN	Polyacrylonitrile
PBT	Polybenzothiazole
Q	Quartz
Si	Silicon
SiC	Silicon carbide
SGl	S-Glass
Ti	Titanium

TABLE 1.5.1(b) *Matrix material codes.*

BMI	Bismaleimide
EP	Epoxy
PAI	Polyamide-imide
PEEK	Polyetheretherketone
PEI	Polyetherimide
PES	Polyethersulfone
PI	Polyimide
PPS	Polyphenylene sulfide
PSU	Polysulfone
SI	Silicone
TP	Thermoplastic
TPES	Thermoplastic polyester
	Fluorocarbon
	Phenolic
	Polybenzimidazole

1.5.2 Index of materials

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1.6 MATERIAL ORIENTATION CODES

1.6.1 Laminate orientation codes

The purpose of a laminate orientation code is to provide a simple, easily understood method of describing the lay-up of a laminate. The laminate orientation code is based largely on the code used in the Advanced Composites Design Guide (Reference 1.6.1(a)). The following information and the examples in Figure 1.6.1 describe the laminate orientation code used in MIL-HDBK-17.

1. The orientation of each lamina with respect to the x-axis is indicated by the angle between the fiber direction and the x-axis. Positive angles are measured counter-clockwise from the x-axis when looking toward the lay-up surface (right-hand rule).

2. When indicating the lay-up of a weave, the angle is measured between the warp direction and the x-axis.

3. Orientations of successive laminae with different absolute values are separated by a virgule (/).

4. Two or more adjacent laminae with the same orientation are indicated by adding a subscript, to the angle of the first such lamina, equal to the number of repetitions of laminae with that orientation.

5. Laminae are listed in order from the first laid up to the last. Brackets are used to indicate the beginning and the end of the code.

6. A subscript of 's' is used if the first half of the lay-up is indicated and the second half is symmetric with the first. When a symmetric lay-up with an odd number of laminae is shown, the layer which is not repeated is indicated by overlining the angle of that lamina.

7. A repeated set of laminae are enclosed in parentheses and the number of repetitions of the set indicated by a subscript.

8. The convention used for indicating materials is no subscript for a tape ply and a subscript "f" for a weave.

9. The laminate code for a hybrid has the different materials contained in the laminate indicated by subscripts on the laminae.

10. Since the majority of computer programs do not permit the use of subscripts and superscripts, the following modifications are recommended based on ASTM Committee E-49 guidelines (Reference 1.6.1(b)).

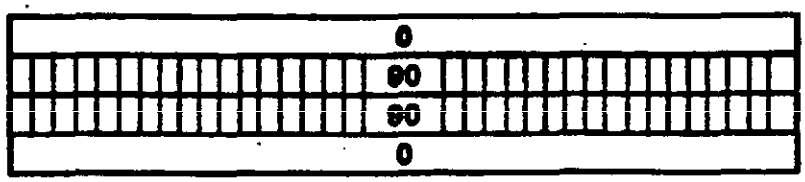
    a. Subscript information will be preceded by a colon (:), e.g., [90/0:2/45]:s.

    b. A bar over a ply (designating a non-repeated ply in a symmetric laminate) should be indicated by a backslash (\) after the ply, e.g., [0/45/90\]:s.

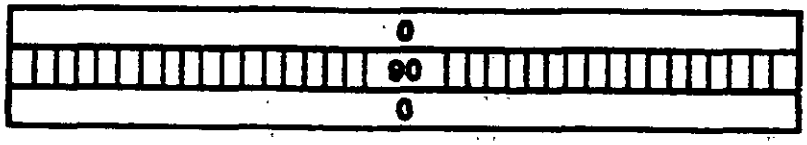
1.6.2 Braiding, orientation codes

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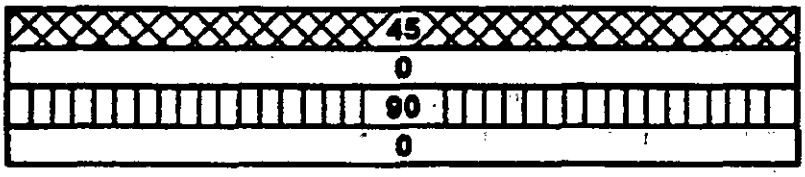
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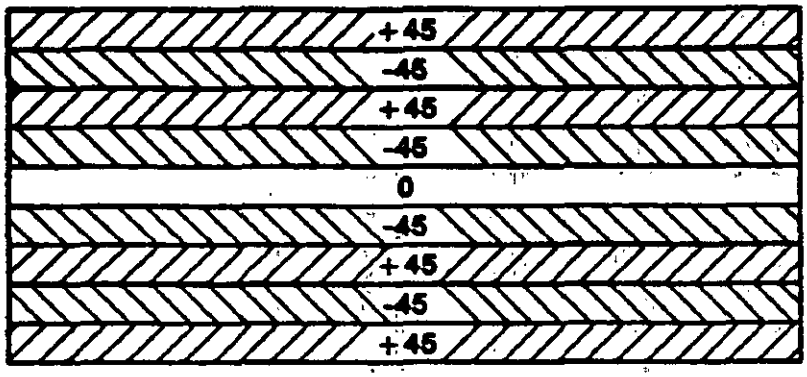
$[0/90]_s$



$[0/\overline{90}]_s$



$[45_f/0/90/0]$



$[(\pm 45)_2/\overline{0}]_s$

FIGURE 1.6.1 *Example laminate orientation codes.*

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1.7 SYMBOLS, ABBREVIATIONS, AND SYSTEMS OF UNITS

This section defines the symbols and abbreviations which are used within MIL-HDBK-17 and describes the system of units which is maintained. Common usage is maintained where possible. References 1.7(a) - 1.7(c) served as primary sources for this information.

1.7.1 Symbols and abbreviations

The symbols and abbreviations used in this document are defined in this section with the exception of statistical symbols. These latter symbols are defined in Volume 1, Chapter 8. The lamina/laminate coordinate axes used for all properties and a summary of the mechanical property notation are shown in Figure 1.7.1.

- The symbols f and m, when used as either subscripts or superscripts, always denote fiber and matrix, respectively.
- The type of stress (for example, cy - compression yield) is always used in the superscript position.
- Direction indicators (for example, x, y, z, 1, 2, 3, etc.) are always used in the subscript position.
- Ordinal indicators of laminae sequence (e.g., 1, 2, 3, etc.) are used in the superscript position and must be parenthesized to distinguish them from mathematical exponents.
- Other indicators may be used in either subscript or superscript position, as appropriate for clarity.
- Compound symbols (such as, basic symbols plus indicators) which deviate from these rules are shown in their specific form in the following list.

The following general symbols and abbreviations are considered standard for use in MIL-HDBK-17. Where exceptions are made, they are noted in the text and tables.

A - (1) area (m^2,in^2)
- (2) ratio of alternating stress to mean stress
- (3) A-basis for mechanical property values

a - (1) length dimension (mm,in)
- (2) acceleration (m/sec^2,ft/sec^2)
- (3) amplitude
- (4) crack or flaw dimension (mm,in)

B - (1) B-basis for mechanical property values
- (2) biaxial ratio

Btu - British thermal unit(s)

b - width dimension (mm,in), e.g., the width of a bearing or compression panel normal to load, or breadth of beam cross-section

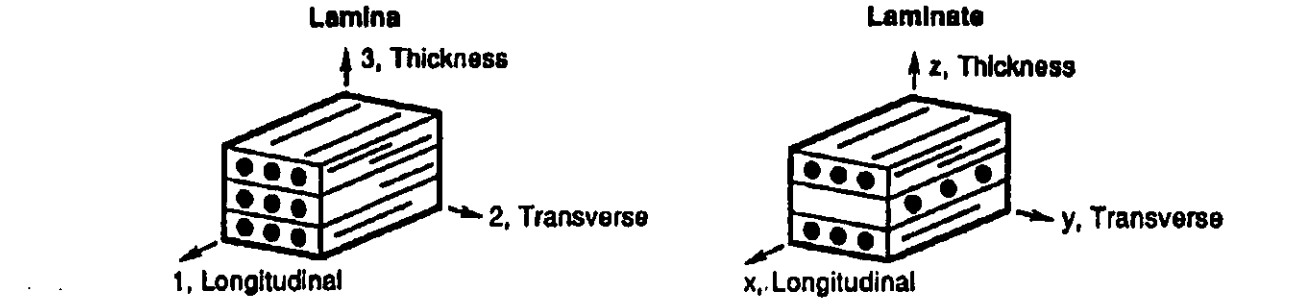
C - (1) specific heat (kJ/kg °C,Btu/lb °F)
- (2) Celsius

CF - centrifugal force (N,lbf)

CPF - crossply factor

CPT - cured ply thickness (mm, in.)

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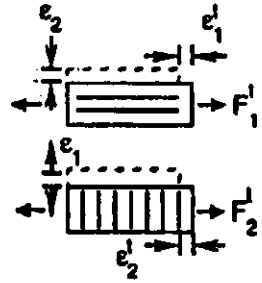
Notation = H_i^{jk}

Where,

$H = \begin{cases} \sigma, \tau; \text{Applied Normal, Shear Stress} \\ F; \text{Allowable Stress} \\ \varepsilon, \gamma; \text{Extensional, Shear Strain} \\ E, G; \text{Young's, Shear Modulus} \\ \nu; \text{Poisson's Ratio} \end{cases}$

Note: ν_{12}^l = Major Poisson's Ratio = $-\frac{\varepsilon_2^l}{\varepsilon_1^l}$

ν_{21}^l = Minor Poisson's Ratio = $-\frac{\varepsilon_1^l}{\varepsilon_2^l}$



$i = \begin{cases} \text{1; Longitudinal} \\ \text{2; Transverse} \\ \text{3; Thickness} \\ \text{12, 13, 32; Shear, Poisson's} \end{cases}$ Lamina

$\begin{cases} \text{x; Longitudinal} \\ \text{y; Transverse} \\ \text{z; Thickness} \\ \text{xy, xz, zy; Shear, Poisson's} \end{cases}$ Laminate

$j = \begin{cases} \text{c; Compression} \\ \text{t; Tension} \\ \text{s; Shear} \end{cases}$

$k = \begin{cases} \text{y; Yield} \\ \text{u; Ultimate, Not Used for Stiffness} \end{cases}$

Examples, F_2^{tu} = Lamina Ultimate Transverse Tensile Allowable Stress

E_z^c = Laminate Compressive Young's Modulus, Thickness Direction

FIGURE 1.7.1 *Mechanical property notation.*

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Check the source to verify that this is the current version before use.

CG - (1) center of mass, "center of gravity"
- (2) area or volume centroid

℄ - centerline

c - column buckling end-fixity coefficient

$\bar{c}$ - honeycomb sandwich core depth (mm,in)

cpm - cycles per minute

D - (1) diameter (mm,in)
- (2) hole or fastener diameter (mm,in)
- (3) plate stiffness (N-m,lbf-in)

d - mathematical operator denoting differential

E - modulus of elasticity in tension, average ratio of stress to strain for stress below proportional limit (GPa,Msi)

E' - storage modulus (GPa,Msi)

E" - loss modulus (GPa,Msi)

E_c - modulus of elasticity in compression, average ratio of stress to strain for stress below proportional limit (GPa,Msi)

E_c' - modulus of elasticity of honeycomb core normal to sandwich plane (GPa,Msi)

E^{scc} - secant modulus (GPa,Msi)

E^{tan} - tangent modulus (GPa,Msi)

e - minimum distance from a hole center to the edge of the sheet (mm,in)

e/D - ratio of edge distance to hole diameter (bearing strength)

F - (1) stress (MPa,ksi)
- (2) Fahrenheit

F^b - bending stress (MPa,ksi)

F^{ccr} - crushing or crippling stress (upper limit of column stress for failure) (MPa,ksi)

F^{su} - ultimate stress in pure shear (this value represents the average shear stress over the cross-section) (MPa,ksi)

FAW - fiber areal weight (g/m^2, lb/in^2)

FV - fiber volume (%)

f - (1) internal (or calculated) stress (MPa,ksi)
- (2) stress applied to the gross flawed section (MPa,ksi)
- (3) creep stress (MPa,ksi)

f^c - internal (or calculated) compressive stress (MPa,ksi)

f_c - (1) maximum stress at fracture (MPa,ksi)
- (2) gross stress limit (for screening elastic fracture data (MPa,ksi)

ft - foot, feet

G - modulus of rigidity (shear modulus) (GPa,Msi)

GPa - gigapascal(s)

g - (1) gram(s)
- (2) acceleration due to gravity (m/s^2,ft/s^2)

H/C - honeycomb (sandwich)

h - height dimension (mm,in) e.g. the height of a beam cross-section

hr - hour(s)

I - area moment of inertia (mm^4,in^4)

i - slope (due to bending) of neutral plane in a beam, in radians

in. - inch(es)

J - (1) torsion constant (= I_p for round tubes) (m^4,in^4)
- (2) Joule

K - (1) Kelvin
- (2) stress intensity factor ($MPa\sqrt{m}$,$ksi\sqrt{in}$)
- (3) coefficient of thermal conductivity (W/m °C, Btu/ft^2/hr/in/°F)
- (4) correction factor
- (5) dielectric constant

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K_{app} - apparent plane strain fracture toughness or residual strength (MPa$\sqrt{m}$,ksi$\sqrt{in}$)
K_c - critical plane strain fracture toughness, a measure of fracture toughness at point of crack growth instability (MPa$\sqrt{m}$,ksi$\sqrt{in}$)
K_{Ic} - plane strain fracture toughness (MPa$\sqrt{m}$,ksi$\sqrt{in}$)
K_N - empirically calculated fatigue notch factor
K_s - plate or cylinder shear buckling coefficient
K_t - (1) theoretical elastic stress concentration factor
- (2) t_w/c ratio in H/C sandwich
Kv - dielectric strength (KV/mm, V/mil)
K_x,K_y - plate or cylinder compression buckling coefficient
k - strain at unit stress (m/m,in/in)
L - cylinder, beam, or column length (mm,in)
L' - effective column length (mm,in)
lb - pound
M - applied moment or couple (N-m,in-lbf)
Mg - megagram(s)
MPa - megapascal(s)
MS - military standard
M.S. - margin of safety
MW - molecular weight
MWD - molecular weight distribution
m - (1) mass (kg,lb)
- (2) number of half wave lengths
- (3) metre
- (4) slope
N - (1) number of fatigue cycles to failure
- (2) number of laminae in a laminate
- (3) distributed in-plane forces on a panel (lbf/in)
- (4) Newton
- (5) normalized
NA - neutral axis
n - (1) number of times in a set
- (2) number of half or total wavelengths
- (3) number of fatigue cycles endured
P - (1) applied load (N,lbf)
- (2) exposure parameter
- (3) probability
- (4) specific resistance (Ω)
P^u - test ultimate load, (N,lb per fastener)
P^y - test yield load, (N,lb per fastener)
p - normal pressure (Pa,psi)
psi - pounds per square inch
Q - area static moment of a cross-section (mm^3,in^3)
q - shear flow (N/m,lbf/in)
R - (1) algebraic ratio of minimum load to maximum load in cyclic loading
- (2) reduced ratio
RA - reduction of area
R.H. - relative humidity
RMS - root-mean-square
RT - room temperature
r - (1) radius (mm,in)
- (2) root radius (mm,in)
- (3) reduced ratio (regression analysis)

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S - (1) shear force (N,lbf)
- (2) nominal stress in fatigue (MPa,ksi)
- (3) S-basis for mechanical property values

S_a - stress amplitude in fatigue (MPa,ksi)
S_e - fatigue limit (MPa,ksi)
S_m - mean stress in fatigue (MPa,ksi)
S_{max} - highest algebraic value of stress in the stress cycle (MPa,ksi)
S_{min} - lowest algebraic value of stress in the stress cycle (MPa,ksi)
S_R - algebraic difference between the minimum and maximum stresses in one cycle (MPa,ksi)
S.F. - safety factor
s - (1) arc length (mm,in)
- (2) H/C sandwich cell size (mm,in)

T - (1) temperature (°C,°F)
- (2) applied torsional moment (N-m,in-lbf)

T_d - thermal decomposition temperature (°C,°F)
T_F - exposure temperature (°C,°F)
T_g - glass transition temperature (°C,°F)
T_m - melting temperature (°C,°F)
t - (1) thickness (mm,in)
- (2) exposure time (s)
- (3) elapsed time (s)

V - (1) volume (mm^3,in^3)
- (2) shear force (N,lbf)

W - (1) weight (N,lbf)
- (2) width (mm,in)
- (3) Watt

x - distance along a coordinate axis
Y - nondimensional factor relating component geometry and flaw size
y - (1) deflection (due to bending) of elastic curve of a beam (mm,in)
- (2) distance from neutral axis to given point
- (3) distance along a coordinate axis

Z - section modulus, I/y (mm^3,in^3)
α - coefficient of thermal expansion (m/m/°C,in/in/°F)
γ - shear strain (m/m,in/in)
Δ - difference (used as prefix to quantitative symbols)
δ - elongation or deflection (mm,in)
ϵ - strain (m/m,in/in)
ϵ - elastic strain (m/m,in/in)
ϵ - plastic strain (m/m,in/in)
μ - permeability
η - plasticity reduction factor
$[\eta]$ - intrinsic viscosity
η^* - dynamic complex viscosity
ν - Poisson's ratio
ρ - (1) density (kg/m3,lb/in3)
- (2) radius of gyration (mm,in)

ρ_c - H/C sandwich core density (kg/m^3,lb/in^3)
Σ - total, summation
σ - standard deviation
σ_{ij}, τ_{ij} - stress in j direction on surface whose outer normal is in i direction ($i, j = 1, 2, 3$ or x, y, z) (MPa,ksi)
T - applied shear stress (MPa,ksi)
ω - angular velocity (radians/s)

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∞ - infinity

1.7.1.1 Constituent properties

The following symbols apply specifically to the constituent properties of a typical composite material.

E^f - Young's modulus of filament material (MPa,ksi)
E^m - Young's modulus of matrix material (MPa,ksi)
E_x^g - Young's modulus of impregnated glass scrim cloth in the filament direction or in the warp direction of a fabric (MPa,ksi)
E_y^g - Young's modulus of impregnated glass scrim cloth transverse to the filament direction or to the warp direction in a fabric (MPa,ksi)
G^f - shear modulus of filament material (MPa,ksi)
G^m - shear modulus of matrix (MPa,ksi)
G_{xy}^g - shear modulus of impregnated glass scrim cloth (MPa,ksi)
G_{cx}' - shear modulus of sandwich core along X-axis (MPa,ksi)
G_{cy}' - shear modulus of sandwich core along Y-axis (MPa,ksi)
ℓ - filament length (mm,in)
α^f - coefficient of thermal expansion for filament material (m/m/°C,in/in/°F)
α^m - coefficient of thermal expansion for matrix material (m/m/°C,in/in/°F)
α_x^g - coefficient of thermal expansion of impregnated glass scrim cloth in the filament direction or in the warp direction of a fabric (m/m/°C,in/in/°F)
α_y^g - coefficient of thermal expansion of impregnated glass scrim cloth transverse to the filament direction or to the warp direction in a fabric (m/m/°C,in/in/°F)
ν^f - Poisson's ratio of filament material
ν^m - Poisson's ratio of matrix material
ν_{xy}^g - glass scrim cloth Poisson's ratio relating to contraction in the transverse (or fill) direction as a result of extension in the longitudinal (or warp) direction
ν_{yx}^g - glass scrim cloth Poisson's ratio relating to contraction in the longitudinal (or warp) direction as a result of extension in the transverse (or fill) direction
σ - applied axial stress at a point, as used in micromechanics analysis (MPa,ksi)
τ - applied shear stress at a point, as used in micromechanics analysis (MPa,ksi)

1.7.1.2 Laminae and laminates

The following symbols, abbreviations, and notations apply to composite laminae and laminates. At the present time the focus in MIL-HDBK-17 is on laminae properties. However, commonly used nomenclature for both laminae and laminates are included here to avoid potential confusion.

A_{ij} (i,j = 1,2,6) - extensional rigidities (N/m,lbf/in)
B_{ij} (i,j = 1,2,6) - coupling matrix (N,lbf)
C_{ij} (i,j = 1,2,6) - elements of stiffness matrix (Pa,psi)
D_x, D_y - flexural rigidities (N-m,lbf-in)
D_{xy} - twisting rigidity (N-m,lbf-in)
D_{ij} (i,j = 1,2,6) - flexural rigidities (N-m,lbf-in)
E_1 - Young's modulus of lamina parallel to filament or warp direction (GPa,Msi)
E_2 - Young's modulus of lamina transverse to filament or warp direction (GPa,Msi)
E_x - Young's modulus of laminate along x reference axis (GPa,Msi)
E_y - Young's modulus of laminate along y reference axis (GPa,Msi)

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G_{12} - shear modulus of lamina in 12 plane (GPa,Msi)
G_{xy} - shear modulus of laminate in xy reference plane (GPa,Msi)
h_i - thickness of ith ply or lamina (mm,in)
M_x, M_y, M_{xy} - *bending and twisting moment components (N-m/m, in-lbf/in in plate and shell analysis)*
n_f - number of filaments per unit length per lamina
Q_x, Q_y - shear force parallel to z axis of sections of a plate perpendicular to x and y axes, respectively (N/m,lbf/in)
Q_{ij} (i,j = 1,2,6) - reduced stiffness matrix (Pa,psi)
u_x, u_y, u_z - components of the displacement vector (mm,in)
$u_x^\circ, u_y^\circ, u_z^\circ$ - components of the displacement vector at the laminate's midsurface (mm,in)
V_v - *void content (% by volume)*
V_f - filament content or fiber volume (% by volume)
V_g - glass scrim cloth content (% by volume)
V_m - matrix content (% by volume)
V_x, V_y - edge or support shear force (N/m,lbf/in)
W_f - filament content (% by weight)
W_g - glass scrim cloth content (% by weight)
W_m - matrix content (% by weight)
W_s - weight of laminate per unit surface area (N/m^2,lbf/in^2)
α_1 - lamina coefficient of thermal expansion along 1 axis (m/m/°C,in/in/°F)
α_2 - lamina coefficient of thermal expansion along 2 axis (m/m/°C,in/in/°F)
α_x - laminate coefficient of thermal expansion along general reference x axis (m/m/°C,in/in/°F)
α_y - laminate coefficient of thermal expansion along general reference y axis (m/m/°C,in/in/°F)
α_{xy} - laminate shear distortion coefficient of thermal expansion (m/m/°C,in/in/°F)
θ - angular orientation of a lamina in a laminate, i.e., angle between 1 and x axes (°)
λ_{xy} - product of ν_{xy} and ν_{yx}
ν_{12} - Poisson's ratio relating contraction in the 2 direction as a result of extension in the 1 direction[1]
ν_{21} - Poisson's ratio relating contraction in the 1 direction as a result of extension in the 2 direction[1]
ν_{xy} - Poisson's ratio relating contraction in the y direction as a result of extension in the x direction[1]
ν_{yx} - Poisson's ratio relating contraction in the x direction as a result of extension in the y direction[1]
ρ_c - density of a single lamina (kg/m^3,lb/in^3)
$\bar{\rho}_c$ - density of a laminate (kg/m^3,lb/in^3)
ϕ - (1) general angular coordinate, (°)
- (2) angle between x and load axes in off-axis loading (°)

1.7.1.3 Subscripts

The following subscript notations are considered standard in MIL-HDBK-17.

1, 2, 3 - laminae natural orthogonal coordinates (1 is filament or warp direction)
A - axial

[1]The convention for Poisson's ratio should be checked before comparing different sources as different conventions are used.

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a	- (1) adhesive
	- (2) alternating
app	- apparent
byp	- bypass
c	- composite system, specific filament/matrix composition. Composite as a whole, contrasted to individual constituents. Also, sandwich core when used in conjunction with prime (')
	- (4) critical
cf	- centrifugal force
e	- fatigue or endurance
eff	- effective
eq	- equivalent
f	- filament
g	- glass scrim cloth
H	- hoop
i	- i^{th} position in a sequence
L	- lateral
m	- (1) matrix
	- (2) mean
max	- maximum
min	- minimum
n	- (1) n^{th} (last) position in a sequence
	- (2) normal
p	- polar
s	- symmetric
st	- stiffener
T	- transverse
t	- value of parameter at time t
x, y, z	- general coordinate system
Σ	- total, or summation
o	- initial or reference datum
( )	- format for indicating specific, temperature associated with term in parentheses. RT - room temperature (21°C,70°F); all other temperatures in °F unless specified.

1.7.1.4 Superscripts

The following superscript notations are considered standard in MIL-HDBK-17.

b	- bending
br	- bearing
c	- (1) compression
	- (2) creep
cc	- compression crippling
cr	- compression buckling
e	- elastic
f	- filament
g	- glass scrim cloth
is	- interlaminar shear
(i)	- i^{th} ply or lamina
lim	- limit, used to indicate limit loading
m	- matrix
ohc	- open hole compression
oht	- open hole tension
p	- plastic
pl	- proportional limit

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rup	- rupture
s	- shear
scr	- shear buckling
sec	- secant (modulus)
so	- offset shear
T	- temperature or thermal
t	- tension
tan	- tangent (modulus)
u	- ultimate
y	- yield
'	- secondary (modulus), or denotes properties of H/C core when used with subscript c
CAI	- compression after impact

1.7.1.5 Acronyms

The following acronyms are used in MIL-HDBK-17.

AA	- atomic absorption
AES	- Auger electron spectroscopy
AIA	- Aerospace Industries Association
ANOVA	- analysis of variance
ARL	- US Army Research Laboratory - Materials Directorate
ASTM	- American Society for Testing and Materials
BMI	- bismaleimide
BVID	- barely visible impact damage
CAI	- compression after impact
CCA	- composite cylinder assemblage
CLS	- crack lap shear
CMCS	- Composite Motorcase Subcommittee (JANNAF)
CPT	- cured ply thickness
CTA	- cold temperature ambient
CTD	- cold temperature dry
CTE	- coefficient of thermal expansion
CV	- coefficient of variation
CVD	- chemical vapor deposition!
DCB	- double cantilever beam
DDA	- dynamic dielectric analysis
DLL	- design limit load
DMA	- dynamic mechanical analysis
DOD	- Department of Defense
DSC	- differential scanning calorimetry
DTA	- differential thermal analysis
DTRC	- David Taylor Research Center
ENF	- end notched flexure
EOL	- end-of-life
ESCA	- electron spectroscopy for chemical analysis
ESR	- electron spin resonance
ETW	- elevated temperature wet
FAA	- Federal Aviation Administration
FFF	- field flow fractionation
FMECA	- Failure Modes Effects Criticality Analysis
FOD	- foreign object damage
FTIR	- Fourier transform infrared spectroscopy
FWC	- finite width correction factor

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GC - gas chromatography
GSCS - Generalized Self Consistent Scheme
HDT - heat distortion temperature
HPLC - high performance liquid chromatography
ICAP - inductively coupled plasma emission
IITRI - Illinois Institute of Technology Research Institute
IR - infrared spectroscopy
ISS - ion scattering spectroscopy
JANNAF - Joint Army, Navy, NASA, and Air Force
LC - liquid chromatography
LPT - laminate plate theory
LSS - laminate stacking sequence
MMB - mixed mode bending
MOL - material operational limit
MS - mass spectroscopy
MSDS - material safety data sheet
MTBF - Mean Time Between Failure
NAS - National Aerospace Standard
NASA - National Aeronautics and Space Administration
NDI - nondestructive inspection
NMR - nuclear magnetic resonance
PEEK - polyether ether ketone
RDS - rheological dynamic spectroscopy
RH - relative humidity
RT - room temperature
RTA - room temperature ambient
RTD - room temperature dry
RTM - resin transfer molding
SACMA - Suppliers of Advanced Composite Materials Association
SAE - Society of Automotive Engineers
SANS - small-angle neutron scattering spectroscopy
SEC - size-exclusion chromatography
SEM - scanning electron microscopy
SFC - supercritical fluid chromatography
SI - International System of Units (Le Système Interational d'Unités)
SIMS - secondary ion mass spectroscopy
TBA - torsional braid analysis
TEM - transmission electron microscopy
TGA - thermogravimetric analysis
TLC - thin-layer chromatography
TMA - thermal mechanical analysis
TOS - thermal oxidative stability
TVM - transverse microcrack
UDC - unidirectional fiber composite
VNB - V-notched beam
XPS - X-ray photoelectron spectroscopy

1.7.2 System of units

To comply with Department of Defense Instructive 5000.2, Part 6, Section M, "Use of the Metric System," dated February 23, 1991, the data in MIL-HDBK-17 are generally presented in both the International System of Units (SI units) and the U. S. Customary (English) system of units. ASTM E-380, Standard for Metric Practice, provides guidance for the application for SI units which are intended as a basis for worldwide standardization of measurement units (Reference 1.5.2(a)). Further guidelines on the

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use of the SI system of units and conversion factors are contained in the following publications (References 1.7.2(b) - (e)):

(1) DARCOM P 706-470, *Engineering Design Handbook: Metric Conversion Guide*, July 1976.

(2) NBS Special Publication 330, "The International System of Units (SI)," National Bureau of Standards, 1986 edition.

(3) NBS Letter Circular LC 1035, "Units and Systems of Weights and Measures, Their Origin, Development, and Present Status," National Bureau of Standards, November 1985.

(4) NASA Special Publication 7012, "The International System of Units Physical Constants and Conversion Factors", 1964.

English to SI conversion factors pertinent to MIL-HDBK-17 data are contained in Table 1.7.2.

TABLE 1.7.2 *English to SI conversion factors.*

To convert from	to	Multiply by
Btu (thermochemical)/in^2-s	watt/meter^2 (W/m^2)	1.634 246 E+06
Btu-in/(s-ft^2-°F)	W/(m K)	5.192 204 E+02
degree Fahrenheit	degree Celsius (°C)	T = (T - 32)/1.8
degree Fahrenheit	kelvin (K)	T = (T + 459.67)/1.8
foot	meter (m)	3.048 000 E-01
ft^2	m^2	9.290 304 E-02
foot/second	meter/second (m/s)	3.048 000 E-01
ft/s^2	m/s^2	3.048 000 E-01
inch	meter (m)	2.540 000 E-02
in.^{2}	meter^2 (m^2)	6.451 600 E-04
in.^{3}	m^3	1.638 706 E-05
kilogram-force (kgf)	newton (N)	9.806 650 E+00
kgf/m^2	pascal (Pa)	9.806 650 E+00
kip (1000 lbf)	newton (N)	4.448 222 E+03
ksi (kip/in^2)	MPa	6.894 757 E+00
lbf-in	N-m	1.129 848 E-01
lbf-ft	N-m	1.355 818 E+00
lbf/in^2 (psi)	pascal (Pa)	6.894 757 E+03
lb/in^2	gm/m^2	7.030 696 E+05
lb/in^3	kg/m^3	2.767 990 E+04
Msi (10^6 psi)	GPa	6.894 757 E+00
pound-force (lbf)	newton (N)	4.488 222 E+00
pound-mass (lb avoirdupois)	kilogram (kg)	4.535 924 E-01
torr	pascal (Pa)	1.333 22 E+02

*The letter "E" following the conversion factor stands for exponent and the two digits after the letter "E" indicate the power of 10 by which the number is to be multiplied.

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1.8 DEFINITIONS

The following definitions are used within MIL-HDBK-17. This glossary of terms is not totally comprehensive but it does represent nearly all commonly used terms. Where exceptions are made, they are noted in the text and tables. For ease of identification the definitions have been organized alphabetically.

A-Basis (or A-Value) -- A statistically-based material property; a 95% lower confidence bound on the first percentile of a specified population of measurements. Also a 95% lower tolerance bound for the upper 99% of a specified population.

A-Stage -- An early stage in the reaction of thermosetting resins in which the material is still soluble in certain liquids and may be liquid or capable of becoming liquid upon heating. (Sometimes referred to as resol.)

Absorption -- A process in which one material (the absorbent) takes in or absorbs another (the absorbate).

Accelerator – A material which, when mixed with a catalyzed resin, will speed up the chemical reaction between the catalyst and the resin.

Accuracy – The degree of conformity of a measured or calculated value to some recognized standard or specified value. Accuracy involves the systematic error of an operation.

Addition Polymerization -- Polymerization by a repeated addition process in which monomers are linked together to form a polymer without splitting off of water or other simple molecules.

Adhesion – The state in which two surfaces are held together at an interface by forces or interlocking action or both.

Adhesive -- A substance capable of holding two materials together by surface attachment. In the handbook, the term is used specifically to designate structural adhesives, those which produce attachments capable of transmitting significant structural loads.

ADK – Notation used for the k-sample Anderson-Darling statistic, which is used to test the hypothesis that k batches have the same distribution.

Aliquot – A small, representative portion of a larger sample.

Aging -- The effect, on materials, of exposure to an environment for a period of time; the process of exposing materials to an environment for an interval of time.

Ambient -- The surrounding environmental conditions such as pressure or temperature.

Anelasticity – A characteristic exhibited by certain materials in which strain is a function of both stress and time, such that, while no permanent deformations are involved, a finite time is required to establish equilibrium between stress and strain in both the loading and unloading directions.

Angleply -- Same as **Crossply**.

Anisotropic -- Not isotropic; having mechanical and/or physical properties which vary with direction relative to natural reference axes inherent in the material.

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Aramid -- A manufactured fiber in which the fiber-forming substance consisting of a long-chain synthetic aromatic polyamide in which at least 85% of the amide (-CONH-) linkages are attached directly to two aromatic rings.

Areal Weight of Fiber -- The weight of fiber per unit area of prepreg. This is often expressed as grams per square meter. See Table 1.6.2 for conversion factors.

Artificial Weathering -- Exposure to laboratory conditions which may be cyclic, involving changes in temperature, relative humidity, radiant energy and any other elements found in the atmosphere in various geographical areas.

Aspect Ratio -- In an essentially two-dimensional rectangular structure (e.g., a panel), the ratio of the long dimension to the short dimension. However, in compression loading, it is sometimes considered to be the ratio of the load direction dimension to the transverse dimension. Also, in fiber micro-mechanics, it is referred to as the ratio of length to diameter.

Autoclave -- A closed vessel for producing an environment of fluid pressure, with or without heat, to an enclosed object which is undergoing a chemical reaction or other operation.

Autoclave Molding -- A process similar to the pressure bag technique. The lay-up is covered by a pressure bag, and the entire assembly is placed in an autoclave capable of providing heat and pressure for curing the part. The pressure bag is normally vented to the outside.

Axis of Braiding -- The direction in which the braided form progresses.

B-Basis (or B-Value) -- A statistically-based material property; a 95% lower confidence bound on the tenth percentile of a specified population of measurements. Also a 95% lower tolerance bound for the upper 90% of a specified population. (See Volume 1, Section 8.1.4)

B-Stage -- An intermediate stage in the reaction of a thermosetting resin in which the material softens when heated and swells when in contact with certain liquids but does not entirely fuse or dissolve. Materials are usually precured to this stage to facilitate handling and processing prior to final cure. (Sometimes referred to as resitol.)

Bag Molding -- A method of molding or laminating which involves the application of fluid pressure to a flexible material which transmits the pressure to the material being molded or bonded. Fluid pressure usually is applied by means of air, steam, water or vacuum.

Balanced Laminate -- A composite laminate in which all laminae at angles other than 0 degrees and 90 degrees occur only in $\pm$ pairs (not necessarily adjacent).

Batch (or Lot) -- For fibers and resins, a quantity of material formed during the same process and having identical characteristics throughout. For prepregs, laminae, and laminates, material made from one batch of fiber and one batch of resin.

Bearing Area -- The product of the pin diameter and the specimen thickness.

Bearing Load -- A compressive load on an interface.

Bearing Yield Strength -- The bearing stress at which a material exhibits a specified limiting deviation from the proportionality of bearing stress to bearing strain.

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Bend Test -- A test of ductility by bending or folding, usually with steadily applied forces. In some instances the test may involve blows to a specimen having a cross section that is essentially uniform over a length several times as great as the largest dimension of the cross section.

Binder -- A bonding resin used to hold strands together in a mat or preform during manufacture of a molded object.

Binomial Random Variable -- The number of successes in independent trials where the probability of success is the same for each trial.

Birefringence -- The difference between the two principal refractive indices (of a fiber) or the ratio between the retardation and thickness of a material at a given point.

Bleeder Cloth -- A nonstructural layer of material used in the manufacture of composite parts to allow the escape of excess gas and resin during cure. The bleeder cloth is removed after the curing process and is not part of the final composite.

Bobbin -- A cylinder or slightly tapered barrel, with or without flanges, for holding tows, rovings, or yarns.

Bond -- The adhesion of one surface to another, with or without the use of an adhesive as a bonding agent.

Braid -- A system of three or more yarns which are interwoven in such a way that no two yarns are twisted around each other.

Braid Angle -- The acute angle measured from the axis of braiding.

Braid, Biaxial -- Braided fabric with two-yarn systems, one running in the $+\theta$ direction, the other in the $-\theta$ direction as measured from the axis of braiding.

Braid Count -- The number of braiding yarn crossings per inch measured along the axis of a braided fabric.

Braid, Diamond -- Braided fabric with an over one, under one weave pattern, (1 x 1).

Braid, Flat -- A narrow bias woven tape wherein each yarn is continuous and is intertwined with every other yarn in the system without being intertwined with itself.

Braid, Hercules -- A braided fabric with an over three, under three weave pattern, (3 x 3).

Braid, Jacquard -- A braided design made with the aid of a jacquard machine, which is a shedding mechanism by means of which a large number of ends may be controlled independently and complicated patterns produced.

Braid, Regular -- A braided fabric with an over two, under two weave pattern (2 x 2).

Braid, Square -- A braided pattern in which the yarns are formed into a square pattern.

Braid, Two-Dimensional -- Braided fabric with no braiding yarns in the through thickness direction.

Braid, Three-Dimensional -- Braided fabric with one or more braiding yarns in the through thickness direction.

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Braid, Triaxial -- A biaxial braided fabric with laid in yarns running in the axis of braiding.

Braiding -- A textile process where two or more strands, yarns or tapes are intertwined in the bias direction to form an integrated structure.

Broadgoods -- A term loosely applied to prepreg material greater than about 12 inches in width, usually furnished by suppliers in continuous rolls. The term is currently used to designate both collimated uniaxial tape and woven fabric prepregs.

Buckling (Composite) -- A mode of structural response characterized by an out-of-plane material deflection due to compressive action on the structural element involved. In advanced composites, buckling may take the form not only of conventional general instability and local instability but also a micro-instability of individual fibers.

Bundle -- A general term for a collection of essentially parallel filaments or fibers.

C-Stage -- The final stage of the curing reaction of a thermosetting resin in which the material has become practically infusable and insoluble. (Normally considered fully cured and sometimes referred to as resite.)

Capstan -- A friction type take-up device which moves braided fabric away from the fell. The speed of which determines the braid angle.

Carbon Fibers -- Fibers produced by the pyrolysis of organic precursor fibers such as rayon, polyacrylonitrile (PAN), and pitch in an inert atmosphere. The term is often used interchangeably with "graphite"; however, carbon fibers and graphite fibers differ in the temperature at which the fibers are made and heat-treated, and the amount of carbon produced. Carbon fibers typically are carbonized at about 2400°F (1300°C) and assay at 93 to 95% carbon, while graphite fibers are graphitized at 3450 to 5450°F (1900 to 3000°C) and assay at more than 99% elemental carbon.

Carrier -- A mechanism for carrying a package of yarn through the braid weaving motion. A typical carrier consists of a bobbin spindle, a track follower, and a tensioning device.

Caul Plates -- Smooth metal plates, free of surface defects, the same size and shape as a composite lay-up, used immediately in contact with the lay-up during the curing process to transmit normal pressure and to provide a smooth surface on the finished laminate.

Censoring -- Data is right (left) censored at M, if, whenever an observation is less than or equal to M (greater than or equal to M), the actual value of the observation is recorded. If the observation exceeds (is less than) M, the observation is recorded as M.

Chain-Growth Polymerization -- One of the two principal polymerization mechanisms. In chain-growth polymerization, the reactive groups are continuously regenerated during the growth process. Once started, the polymer molecule grows rapidly by a chain of reactions emanating from a particular reactive initiator which may be a free radical, cation or anion.

Chromatogram -- A plot of detector response against peak volume of solution (eluate) emerging from the system for each of the constituents which have been separated.

Circuit -- One complete traverse of the fiber feed mechanism of a winding machine; one complete traverse of a winding band from one arbitrary point along the winding path to another point on a plane through the starting point and perpendicular to the axis.

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Cocuring -- The act of curing a composite laminate and simultaneously bonding it to some other prepared surface during the same cure cycle (see **Secondary Bonding**).

Coefficient of Linear Thermal Expansion -- The change in length per unit length resulting from a one-degree rise in temperature.

Coefficient of Variation -- The ratio of the population (or sample) standard deviation to the population (or sample) mean.

Collimated -- Rendered parallel.

Compatible -- The ability of different resin systems to be processed in contact with each other without degradation of end product properties. (See **Compatible**, Volume 1, Section 8.1.4)

Composite Class -- As used in the handbook, a major subdivision of composite construction in which the class is defined by the fiber system and the matrix class, e.g., organic-matrix filamentary laminate.

Composite Material -- Composites are considered to be combinations of materials differing in composition or form on a macroscale. The constituents retain their identities in the composite; that is, they do not dissolve or otherwise merge completely into each other although they act in concert. Normally, the components can be physically identified and exhibit an interface between one another.

Compound -- An intimate mixture of polymer or polymers with all the materials necessary for the finished product.

Condensation Polymerization -- This is a special type of step-growth polymerization characterized by the formation of water or other simple molecules during the stepwise addition of reactive groups.

Confidence Coefficient -- See **Confidence Interval**.

Confidence Interval -- A confidence interval is defined by a statement of one of the following forms:

$$(1)\ P\{a<\theta\} \le 1-\alpha$$
$$(2)\ P\{\theta<b\} \le 1-\alpha$$
$$(3)\ P\{a<\theta<b\} \le 1-\alpha$$

where $1-\alpha$ is called the confidence coefficient. A statement of type (1) or (2) is called a one-sided confidence interval and a statement of type (3) is called a two-sided confidence interval. In (1) a is a lower confidence limit and in (2) b is an upper confidence limit. With probability at least $1-\alpha$, the confidence interval will contain the parameter θ.

Constituent -- In general, an element of a larger grouping. In advanced composites, the principal constituents are the fibers and the matrix.

Continuous Filament -- A yarn or strand in which the individual filaments are substantially the same length as the strand.

Coupling Agent -- Any chemical substance designed to react with both the reinforcement and matrix phases of a composite material to form or promote a stronger bond at the interface. Coupling agents are applied to the reinforcement phase from an aqueous or organic solution or from a gas phase, or added to the matrix as an integral blend.

Coverage -- The measure of the fraction of surface area covered by the braid.

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Crazing -- Apparent fine cracks at or under the surface of an organic matrix.

Creel -- A framework arranged to hold tows, rovings, or yarns so that many ends can be withdrawn smoothly and evenly without tangling.

Creep -- The time dependent part of strain resulting from an applied stress.

Creep, Rate Of -- The slope of the creep-time curve at a given time.

Crimp -- The undulations induced into a braided fabric via the braiding process.

Crimp Angle -- The maximum acute angle of a single braided yarn's direction measured from the average axis of tow.

Crimp Exchange -- The process by which a system of braided yarns reaches equilibrium when put under tension or compression.

Critical Value(s) -- When testing a one-sided statistical hypothesis, a critical value is the value such that, if the test statistic is greater than (less than) the critical value, the hypothesis is rejected. When testing a two-sided statistical hypothesis, two critical values are determined. If the test statistic is either less than the smaller critical value or greater than the larger critical value, then the hypothesis is rejected. In both cases, the critical value chosen depends on the desired risk (often 0.05) of rejecting the hypothesis when it is true.

Crossply -- Any filamentary laminate which is not uniaxial. Same as Angleply. In some references, the term crossply is used to designate only those laminates in which the laminae are at right angles to one another, while the term angleply is used for all others. In the handbook, the two terms are used synonymously. The reservation of a separate terminology for only one of several basic orientations is unwarranted because a laminate orientation code is used.

Cumulative Distribution Function -- See Volume 1, Section 8.1.4.

Cure -- To change the properties of a thermosetting resin irreversibly by chemical reaction, i.e., condensation, ring closure, or addition. Cure may be accomplished by addition of curing (cross-linking) agents, with or without catalyst, and with or without heat. Cure may occur also by addition, such as occurs with anhydride cures for epoxy resin systems.

Cure Cycle -- The schedule of time periods at specified conditions to which a reacting thermosetting material is subjected in order to reach a specified property level.

Cure Stress -- A residual internal stress produced during the curing cycle of composite structures. Normally, these stresses originate when different components of a lay-up have different thermal coefficients of expansion.

Debond -- A deliberate separation of a bonded joint or interface, usually for repair or rework purposes. (See **Disbond, Unbond**).

Deformation -- The change in shape of a specimen caused by the application of a load or force.

Degradation -- A deleterious change in chemical structure, physical properties or appearance.

Delamination -- The separation of the layers of material in a laminate. This may be local or may cover a large area of the laminate. It may occur at any time in the cure or subsequent life of the laminate and may arise from a wide variety of causes.

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Denier – A direct numbering system for expressing linear density, equal to the mass in grams per 9000 meters of yarn, filament, fiber, or other textile strand.

Density – The mass per unit volume.

Desorption – A process in which an absorbed or adsorbed material is released from another material. Desorption is the reverse of absorption, adsorption, or both.

Deviation – Variation from a specified dimension or requirement, usually defining the upper and lower limits.

Dielectric Constant – The ratio of the capacity of a condenser having a dielectric constant between the plates to that of the same condenser when the dielectric is replaced by a vacuum; a measure of the electrical charge stored per unit volume at unit potential.

Dielectric Strength – The average potential per unit thickness at which failure of the dielectric material occurs.

Disbond – An area within a bonded interface between two adherends in which an adhesion failure or separation has occurred. It may occur at any time during the life of the structure and may arise from a wide variety of causes. Also, colloquially, an area of separation between two laminae in the finished laminate (in this case the term "delamination" is normally preferred.) (See **Debond, Unbond, Delamination.**)

Distribution – A formula which gives the probability that a value will fall within prescribed limits. (See **Normal, Weibull, and Lognormal Distributions**, also Volume 1, Section 8.1.4).

Dry -- a material condition of moisture equilibrium with a surrounding environment at 5% or lower relative humidity.

Dry Fiber Area – Area of fiber not totally encapsulated by resin.

Ductility – The ability of a material to deform plastically before fracturing.

Elasticity – The property of a material which allows it to recover its original size and shape immediately after removal of the force causing deformation.

Elongation – The increase in gage length or extension of a specimen during a tension test, usually expressed as a percentage of the original gage length.

Eluate – The liquid emerging from a column (in liquid chromatography).

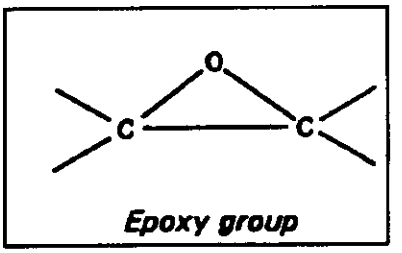
Eluent – The mobile phase used to sweep or elute the sample (solute) components into, through, and out of the column.

End – A single fiber, strand, roving or yarn being or already incorporated into a product. An end may be an individual warp yarn or cord in a woven fabric. In referring to aramid and glass fibers, an end is usually an untwisted bundle of continuous filaments.

Epoxy Equivalent Weight – The number of grams of resin which contain one chemical equivalent of the epoxy group.

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Epoxy Resin – Resins which may be of widely different structures but are characterized by the presence of the epoxy group. (The epoxy or epoxide group is usually present as a glycidyl ether, glycidyl amine, or as part of an aliphatic ring system. The aromatic type epoxy resins are normally used in composites.)



Epoxy group

Extensometer -- A device for measuring linear strain.

F-Distribution -- See Volume 1, Section 8.1.4.

Fabric, Nonwoven -- A textile structure produced by bonding or interlocking of fibers, or both, accomplished by mechanical, chemical, thermal, or solvent means, and combinations thereof.

Fabric, Woven -- A generic material construction consisting of interlaced yarns or fibers, usually a planar structure. Specifically, as used in this handbook, a cloth woven in an established weave pattern from advanced fiber yarns and used as the fibrous constituent in an advanced composite lamina. In a fabric lamina, the warp direction is considered the longitudinal direction, analogous to the filament direction in a filamentary lamina.

Fell -- The point of braid formation, which is defined as the point at which the yarns in a braid system cease movement relative to each other.

Fiber -- A general term used to refer to filamentary materials. Often, fiber is used synonymously with filament. It is a general term for a filament of finite length. A unit of matter, either natural or manmade, which forms the basic element of fabrics and other textile structures.

Fiber Content – The amount of fiber present in a composite. This is usually expressed as a percentage volume fraction or weight fraction of the composite.

Fiber Count – The number of fibers per unit width of ply present in a specified section of a composite.

Fiber Direction -- The orientation or alignment of the longitudinal axis of the fiber with respect to a stated reference axis.

Fiber System – The type and arrangement of fibrous material which comprises the fiber constituent of an advanced composite. Examples of fiber systems are collimated filaments or filament yarns, woven fabric, randomly oriented short-fiber ribbons, random fiber mats, whiskers, etc.

Filament – The smallest unit of a fibrous material. The basic units formed during spinning and which are gathered into strands of fiber, (for use in composites). Filaments usually are of extreme length and of very small diameter. Filaments normally are not used individually. Some textile filaments can function as a yarn when they are of sufficient strength and flexibility.

Filamentary Composites – A major form of advanced composites in which the fiber constituent consists of continuous filaments. Specifically, a filamentary composite is a laminate comprised of a number of laminae, each of which consists of a nonwoven, parallel, uniaxial, planar array of filaments (or filament yarns) embedded in the selected matrix material. Individual laminae are directionally oriented and combined into specific multiaxial laminates for application to specific envelopes of strength and stiffness requirements.

Filament Winding – A reinforced-plastics process that employs a series of continuous, resin-impregnated fibers applied to a mandrel in a predetermined geometrical relationship under controlled tension.

Filament Wound -- Pertaining to an object created by the filament winding method of fabrication.

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Fill (Filling) – In a woven fabric, the yarn running from selvage to selvage at right angles to the warp.

Filler – A relatively inert substance added to a material to alter its physical, mechanical, thermal, electrical, and other properties or to lower cost. Sometimes the term is used specifically to mean particulate additives.

Finish (or Size System) – A material, with which filaments are treated, which contains a coupling agent to improve the bond between the filament surface and the resin matrix in a composite material. In addition, finishes often contain ingredients which provide lubricity to the filament surface, preventing abrasive damage during handling, and a binder which promotes strand integrity and facilitates packing of the filaments.

Fixed Effect – A systematic shift in a measured quantity due to a particular level change of a treatment or condition. (See Volume 1, Section 8.1.4.)

Flash – Excess material which forms at the parting line of a mold or die, or which is extruded from a closed mold.

Former Plate – A die attached to a braiding machine which helps to locate the fell.

Fracture Ductility – The true plastic strain at fracture.

Gage Length – the original length of that portion of the specimen over which strain or change of length is determined.

Gel – The initial jelly-like solid phase that develops during formation of a resin from a liquid. Also, a semi-solid system consisting of a network of solid aggregates in which liquid is held.

Gel Coat – A quick-setting resin used in molding processes to provide an improved surface for the composite; it is the first resin applied to the mold after the mold-release agent.

Gel Point – The stage at which a liquid begins to exhibit pseudo-elastic properties. (This can be seen from the inflection point on a viscosity-time plot.)

Gel Time – The period of time from a pre-determined starting point to the onset of gelation (gel point) as defined by a specific test method.

Glass – An inorganic product of fusion which has cooled to a rigid condition without crystallizing. In the handbook, all reference to glass will be to the fibrous form as used in filaments, woven fabric, yarns, mats, chopped fibers, etc.

Glass Cloth – Conventionally-woven glass fiber material (see **Scrim**).

Glass Fibers – A fiber spun from an inorganic product of fusion which has cooled to a rigid condition without crystallizing.

Glass Transition – The reversible change in an amorphous polymer or in amorphous regions of a partially crystalline polymer from (or to) a viscous or rubbery condition to (or from) a hard and relatively brittle one.

Glass Transition Temperature – The approximate midpoint of the temperature range over which the glass transition takes place.

Graphite Fibers – See **Carbon Fibers**.

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Greige – Fabric that has received no finish.

Hand Lay-up – A process in which components are applied either to a mold or a working surface, and the successive plies are built up and worked by hand.

Hardness – Resistance to deformation; usually measured by indention. Types of standard tests include Brinell, Rockwell, Knoop, and Vickers.

Heat Cleaned – Glass or other fibers which have been exposed to elevated temperatures to remove preliminary sizings or binders which are not compatible with the resin system to be applied.

Heterogeneous – Descriptive term for a material consisting of dissimilar constituents separately identifiable; a medium consisting of regions of unlike properties separated by internal boundaries. (Note that all nonhomogeneous materials are not necessarily heterogeneous).

Homogeneous – Descriptive term for a material of uniform composition throughout; a medium which has no internal physical boundaries; a material whose properties are constant at every point, in other words, constant with respect to spatial coordinates (but not necessarily with respect to directional coordinates).

Horizontal Shear – Sometimes used to indicate interlaminar shear. This is not an approved term for use in this handbook.

Humidity, Relative – The ratio of the pressure of water vapor present to the pressure of saturated water vapor at the same temperature.

Hybrid – A composite laminate comprised of laminae of two or more composite material systems. Or, a combination of two or more different fibers such as carbon and glass or carbon and aramid into a structure (tapes, fabrics and other forms may be combined).

Hygroscopic – Capable of absorbing and retaining atmospheric moisture.

Hysteresis – The energy absorbed in a complete cycle of loading and unloading.

Inclusion – A physical and mechanical discontinuity occurring within a material or part, usually consisting of solid, encapsulated foreign material. Inclusions are often capable of transmitting some structural stresses and energy fields, but in a noticeably different manner from the parent material.

Integral Composite Structure – Composite structure in which several structural elements, which would conventionally be assembled by bonding or with mechanical fasteners after separate fabrication, are instead laid up and cured as a single, complex, continuous structure; e.g., spars, ribs, and one stiffened cover of a wing box fabricated as a single integral part. The term is sometimes applied more loosely to any composite structure not assembled by mechanical fasteners.

Interface – The boundary between the individual, physically distinguishable constituents of a composite.

Interlaminar – Descriptive term pertaining to some object (e.g., voids), event (e.g., fracture), or potential field (e.g., shear stress) referenced as existing or occurring between two or more adjacent laminae.

Interlaminar Shear – Shearing force tending to produce a relative displacement between two laminae in a laminate along the plane of their interface.

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Intermediate Bearing Stress -- The bearing stress at the point on the bearing load-deformation curve where the tangent is equal to the bearing stress divided by a designated percentage (usually 4%) of the original hole diameter.

Intralaminar -- Descriptive term pertaining to some object (e.g., voids), event (e.g., fracture), or potential field (e.g., temperature gradient) existing entirely within a single lamina without reference to any adjacent laminae.

Isotropic -- Having uniform properties in all directions. The measured properties of an isotropic material are independent of the axis of testing.

Jammed State -- The state of a braided fabric under tension or compression where the deformation of the fabric is dominated by the deformation properties of the yarn.

Knitting -- A method of constructing fabric by interlocking series of loops of one or more yarns.

Knuckle Area -- The area of transition between sections of different geometry in a filament wound part.

k-Sample Data -- A collection of data consisting of values observed when sampling from k batches.

Laid-In Yarns -- A system of longitudinal yarns in a triaxial braid which are inserted between the bias yarns.

Lamina -- A single ply or layer in a laminate made up of a series of layers.

Laminae -- Plural of lamina.

Laminate -- A product made by bonding together two or more layers or laminae of material or materials.

Laminate Orientation -- The configuration of a crossplied composite laminate with regard to the angles of crossplying, the number of laminae at each angle, and the exact sequence of the lamina lay-up.

Lattice Pattern -- A pattern of filament winding with a fixed arrangement of open voids.

Lay-up -- A process of fabrication involving the assembly of successive layers of resin-impregnated material.

Lognormal Distribution -- A probability distribution for which the probability that an observation selected at random from this population falls between a and b ($0 < a < b < B$) is given by the area under the normal distribution between log a and log b. The common (base 10) or the natural (base e) logarithm may be used. (See Volume 1, Section 8.1.4.)

Lower Confidence Bound -- See **Confidence Interval**.

Macro -- In relation to composites, denotes the gross properties of a composite as a structural element but does not consider the individual properties or identity of the constituents.

Macrostrain -- The mean strain over any finite gage length of measurement which is large in comparison to the material's interatomic distance.

Mandrel -- A form fixture or male mold used for the base in the production of a part by lay-up, filament winding or braiding.

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Mat -- A fibrous material consisting of randomly oriented chopped or swirled filaments loosely held together with a binder.

Material Acceptance -- The testing of incoming material to ensure that it meets requirements.

Material Qualification -- The procedures used to accept a material by a company or organization for production use.

Material System -- A specific composite material made from specifically identified constituents in specific geometric proportions and arrangements and possessed of numerically defined properties.

Material System Class -- As used in this handbook, a group consisting of material systems categorized by the same generic constituent materials, but without defining the constituents uniquely; e.g., the carbon/epoxy class.

Material Variability -- A source of variability due to the spatial and consistency variations of the material itself and due to variation in its processing. (See Volume 1, Section 8.1.4.)

Matrix-- The essentially homogeneous material in which the fiber system of a composite is embedded.

Mean -- See Sample Mean and Population Mean.

Mechanical Properties -- The properties of a material that are associated with elastic and inelastic reaction when force is applied, or the properties involving the relationship between stress and strain.

Median -- See Sample Median and Population Median.

Micro -- In relation to composites, denotes the properties of the constituents, i.e., matrix and reinforcement and interface only, as well as their effects on the composite properties.

Microstrain -- The strain over a gage length comparable to the material's interatomic distance.

Modulus, Chord -- The slope of the chord drawn between any two specified points on the stress-strain curve.

Modulus, initial -- The slope of the initial straight portion of a stress-strain curve.

Modulus, Secant -- The slope of the secant drawn from the origin to any specified point on the stress-strain curve.

Modulus, Tangent -- The ratio of change in stress to change in strain derived from the tangent to any point on a stress-strain curve.

Modulus, Young's -- The ratio of change in stress to change in strain below the elastic limit of a material. (Applicable to tension and compression).

Modulus of Rigidity (also Shear Modulus or Torsional Modulus) -- The ratio of stress to strain below the proportional limit for shear or torsional stress.

Modulus of Rupture, in Bending -- The maximum tensile or compressive stress (whichever causes failure) value in the extreme fiber of a beam loaded to failure in bending. The value is computed from the flexure equation:

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$$F^b = \frac{Mc}{I} \qquad 1.4(a)$$

where M = maximum bending moment computed from the maximum load and the original moment arm,
c = initial distance from the neutral axis to the extreme fiber where failure occurs,
I = the initial moment of inertia of the cross section about its neutral axis.

Modulus of Rupture, in Torsion – The maximum shear stress in the extreme fiber of a member of circular cross section loaded to failure in torsion calculated from the equation:

$$F^s = \frac{Tr}{J} \qquad 1.4(b)$$

where T = maximum twisting moment,
r = original outer radius,
J = polar moment of inertia of the original cross section.

Moisture Content -- The amount of moisture in a material determined under prescribed condition and expressed as a percentage of the mass of the moist specimen, i.e., the mass of the dry substance plus the moisture present.

Moisture Equilibrium -- The condition reached by a sample when it no longer takes up moisture from, or gives up moisture to, the surrounding environment.

Mold Release Agent -- A lubricant applied to mold surfaces to facilitate release of the molded article.

Molded Edge -- An edge which is not physically altered after molding for use in final form and particularly one which does not have fiber ends along its length.

Molding – The forming of a polymer or composite into a solid mass of prescribed shape and size by the application of pressure and heat.

Monolayer -- The basic laminate unit from which crossplied or other laminates are constructed.

Monomer – A compound consisting of molecules each of which can provide one or more constitutional units.

NDE -- Nondestructive evaluation. Broadly considered synonymous with NDI.

NDI – Nondestructive inspection. A process or procedure for determining the quality or characteristics of a material, part, or assembly without permanently altering the subject or its properties.

NDT -- Nondestructive testing. Broadly considered synonymous with NDI.

Necking -- A localized reduction in cross-sectional area which may occur in a material under tensile stress.

Negatively Skewed – A distribution is said to be negatively skewed if the distribution is not symmetric and the longest tail is on the left.

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Nominal Specimen Thickness -- The nominal ply thickness multiplied by the number of plies.

Nominal Value -- A value assigned for the purpose of a convenient designation. A nominal value exists in name only.

Normal Distribution -- A two parameter (μ,σ) family of probability distributions for which the probability that an observation will fall between a and b is given by the area under the curve

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \qquad 1.7(c)$$

between a and b. (See Volume 1, Section 8.1.4.)

Normalization -- A mathematical procedure for adjusting raw test values for fiber-dominated properties to a single (specified) fiber volume content.

Normalized Stress -- Stress value adjusted to a specified fiber volume content by multiplying the measured stress value by the ratio of specimen fiber volume to the specified fiber volume. This ratio may be obtained directly by experimentally measuring fiber volume, or indirectly by calculation using specimen thickness and fiber areal weight.

Observed Significance Level (OSL) -- The probability of observing a more extreme value of the test statistic when the null hypotheses is true.

Offset Shear Strength — (from valid execution of a material property shear response test) the value of shear stress at the intersection between a line parallel to the shear chord modulus of elasticity and the shear stress/strain curve, where the line has been offset along the shear strain axis from the origin by a specified strain offset value.

Oligomer -- A polymer consisting of only a few monomer units such as a dimer, trimer, etc., or their mixtures.

One-Sided Tolerance Limit Factor -- See Tolerance Limit Factor.

Orthotropic -- Having three mutually perpendicular planes of elastic symmetry.

Oven Dry -- The condition of a material that has been heated under prescribed conditions of temperature and humidity until there is no further significant change in its mass.

PAN Fibers -- Reinforcement fiber derived from the controlled pyrolysis of poly(acrylonitrile) fiber.

Parallel Laminate -- A laminate of woven fabric in which the plies are aligned in the same position as originally aligned in the fabric roll.

Parallel Wound -- A term used to describe yarn or other material wound into a flanged spool.

Peel Ply -- A layer of resin free material used to protect a laminate for later secondary bonding.

pH -- A measure of acidity or alkalinity of a solution, with neutrality represented by a value of 7, with increasing acidity corresponding to progressively smaller values, and increasing alkalinity corresponding to progressively higher values.

Pick Count -- The number of filling yarns per inch or per centimeter of woven fabric.

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Pitch Fibers -- Reinforcement fiber derived from petroleum or coal tar pitch.

Plastic -- A material that contains one or more organic polymers of large molecular weight, is solid in its finished state, and, at some state in its manufacture or processing into finished articles, can be shaped by flow.

Plasticizer -- A material of lower molecular weight added to a polymer to separate the molecular chains. This results in a depression of the glass transition temperature, reduced stiffness and brittleness, and improved processability. (Note, many polymeric materials do not need a plasticizer.)

Plied Yarn -- A yarn formed by twisting together two or more single yarns in one operation.

Poisson's Ratio -- The absolute value of the ratio of transverse strain to the corresponding axial strain resulting from uniformly distributed axial stress below the proportional limit of the material.

Polymer -- An organic material composed of molecules characterized by the repetition of one or more types of monomeric units.

Polymerization -- A chemical reaction in which the molecules of monomers are linked together to form polymers via two principal reaction mechanisms. Addition polymerizations proceed by chain growth and most condensation polymerizations through step growth.

Population -- The set of measurements about which inferences are to be made or the totality of possible measurements which might be obtained in a given testing situation. For example, "all possible ultimate tensile strength measurements for carbon/epoxy system A, conditioned at 95% relative humidity and room temperature". In order to make inferences about a population, it is often necessary to make assumptions about its distributional form. The assumed distributional form may also be referred to as the population. (See Volume 1, Section 8.1.4.)

Population Mean -- The average of all potential measurements in a given population weighted by their relative frequencies in the population. (See Volume 1, Section 8.1.4.)

Population Median -- That value in the population such that the probability of exceeding it is 0.5 and the probability of being less than it is 0.5. (See Volume 1, Section 8.1.4.)

Population Variance -- A measure of dispersion in the population.

Porosity -- A condition of trapped pockets of air, gas, or vacuum within a solid material, usually expressed as a percentage of the total nonsolid volume to the total volume (solid plus nonsolid) of a unit quantity of material.

Positively Skewed -- A distribution is said to be positively skewed if the distribution is not symmetric and the longest tail is on the right.

Postcure -- Additional elevated temperature cure, usually without pressure, to increase the glass transition temperature, to improve final properties, or to complete the cure.

Pot Life -- The period of time during which a reacting thermosetting composition remains suitable for its intended processing after mixing with a reaction initiating agent.

Precision -- The degree of agreement within a set of observations or test results obtained. Precision involves repeatability and reproducibility.

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Precursor (for Carbon or Graphite Fiber) – Either the PAN or pitch fibers from which carbon and graphite fibers are derived.

Preform -- An assembly of dry fabric and fibers which has been prepared for one of several different wet resin injection processes. A preform may be stitched or stabilized in some other way to hold its A shape. A commingled preform may contain thermoplastic fibers and may be consolidated by elevated temperature and pressure without resin injection.

Preply -- Layers of prepreg material, which have been assembled according to a user specified stacking sequence.

Prepreg -- Ready to mold or cure material in sheet form which may be tow, tape, cloth, or mat impregnated with resin. It may be stored before use.

Pressure – The force or load per unit area.

Probability Density Function -- See Volume 1, Section 8.1.4.

Proportional Limit – The maximum stress that a material is capable of sustaining without any deviation from the proportionality of stress to strain (also known as Hooke's law).

Quasi-Isotropic Laminate – A laminate approximating isotropy by orientation of plies in several or more directions.

Random Effect – A shift in a measured quantity due to a particular level change of an external, usually uncontrollable, factor. (See Volume 1, Section 8.1.4.)

Random Error – That part of the data variation that is due to unknown or uncontrolled factors and that affects each observation independently and unpredictably. (See Volume 1, Section 8.1.4.)

Reduction of Area -- The difference between the original cross sectional area of a tension test specimen and the area of its smallest cross section, usually expressed as a percentage of the original area.

Refractive Index - The ratio of the velocity of light (of specified wavelength) in air to its velocity in the substance under examination. Also defined as the sine of the angle of incidence divided by the sine of the angle of refraction as light passes from air into the substance.

Reinforced Plastic – A plastic with relatively high stiffness or very high strength fibers embedded in the composition. This improves some mechanical properties over that of the base resin.

Release Agent – See **Mold Release Agent**.

Resilience -- A property of a material which is able to do work against restraining forces during return from a deformed condition.

Resin -- An organic polymer or prepolymer used as a matrix to contain the fibrous reinforcement in a composite material or as an adhesive. This organic matrix may be a thermoset or a thermoplastic, and may contain a wide variety of components or additives to influence; handleability, processing behavior and ultimate properties.

Resin Content -- The amount of matrix present in a composite either by percent weight or percent volume.

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Resin Starved Area -- Area of composite part where the resin has a non-continuous smooth coverage of the fiber.

Resin System -- A mixture of resin, with ingredients such as catalyst, initiator, diluents, etc. required for the intended processing and final product.

Room Temperature Ambient (RTA) -- 1) an environmental condition of 73 ± 5°F (23 ± 3°C) at ambient laboratory relative humidity; 2) a material condition where, immediately following consolidation/cure, the material is stored at 73 ± 5°F (23 ± 3°C) and at a maximum relative humidity of 60%.

Roving -- A number of strands, tows, or ends collected into a parallel bundle with little or no twist. In spun yarn production, an intermediate state between sliver and yarn.

S-Basis (or S-Value) -- The mechanical property value which is usually the specified minimum value of the appropriate government specification or SAE Aerospace Material Specification for this material.

Sample -- A small portion of a material or product intended to be representative of the whole. Statistically, a sample is the collection of measurements taken from a specified population. (See Volume 1, Section 8.1.4.)

Sample Mean -- The arithmetic average of the measurements in a sample. The sample mean is an estimator of the population mean. (See Volume 1, Section 8.1.4.)

Sample Median -- Order the observation from smallest to largest. Then the sample median is the value of the middle observation if the sample size is odd; the average of the two central observations if n is even. If the population is symmetric about its mean, the sample median is also an estimator of the population mean. (See Volume 1, Section 8.1.4.)

Sample Standard Deviation -- The square root of the sample variance. (See Volume 1, Section 8.1.4.)

Sample Variance -- The sum of the squared deviations from the sample mean, divided by n-1. (See Volume 1, Section 8.1.4.)

Sandwich Construction -- A structural panel concept consisting in its simplest form of two relatively thin, parallel sheets of structural material bonded to, and separated by, a relatively thick, light-weight core.

Saturation -- An equilibrium condition in which the net rate of absorption under prescribed conditions falls essentially to zero.

Scrim (also called Glass Cloth, Carrier) -- A low cost fabric woven into an open mesh construction, used in the processing of tape or other B-stage material to facilitate handling.

Secondary Bonding -- The joining together, by the process of adhesive bonding, of two or more already-cured composite parts, during which the only chemical or thermal reaction occurring is the curing of the adhesive itself.

Selvage or Selvedge -- The woven edge portion of a fabric parallel to the warp.

Set -- The strain remaining after complete release of the force producing the deformation.

Shear Fracture (for crystalline type materials) -- A mode of fracture resulting from translation along slip planes which are preferentially oriented in the direction of the shearing stress.

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Shelf Life – The length of time a material, substance, product, or reagent can be stored under specified environmental conditions and continue to meet all applicable specification requirements and/or remain suitable for its intended function.

Short Beam Strength (SBS) – a test result from valid execution of ASTM test method D 2344.

Significant – Statistically, the value of a test statistic is significant if the probability of a value at least as extreme is less than or equal to a predetermined number called the significance level of the test.

Significant Digit – Any digit that is necessary to define a value or quantity.

Size System – See **Finish.**

Sizing – A generic term for compounds which are applied to yarns to bind the fiber together and stiffen the yarn to provide abrasion-resistance during weaving. Starch, gelatin, oil, wax, and man-made polymers such as polyvinyl alcohol, polystyrene, polyacrylic acid, and polyacetatates are employed.

Skewness – See **Positively Skewed, Negatively Skewed.**

Sleeving – A common name for tubular braided fabric.

Slenderness Ratio – The unsupported effective length of a uniform column divided by the least radius of gyration of the cross-sectional area.

Sliver – A continuous strand of loosely assembled fiber that is approximately uniform in cross-sectional area and has no twist.

Solute – The dissolved material.

Specific Gravity – The ratio of the weight of any volume of a substance to the weight of an equal volume of another substance taken as standard at a constant or stated temperature. Solids and liquids are usually compared with water at 39°F (4°C).

Specific Heat – The quantity of heat required to raise the temperature of a unit mass of a substance one degree under specified conditions.

Specimen – A piece or portion of a sample or other material taken to be tested. Specimens normally are prepared to conform with the applicable test method.

Spindle – A slender upright rotation rod on a spinning frame, roving frame, twister or similar machine.

Standard Deviation – See **Sample Standard Deviation.**

Staple – Either naturally occurring fibers or lengths cut from filaments.

Step-Growth Polymerization – One of the two principal polymerization mechanisms. In sep-growth polymerization, the reaction grows by combination of monomer, oligomer, or polymer molecules through the consumption of reactive groups. Since average molecular weight increases with monomer consumption, high molecular weight polymers are formed only at high degrees of conversion.

Strain – the per unit change, due to force, in the size or shape of a body referred to its original size or shape. Strain is a nondimensional quantity, but it is frequently expressed in inches per inch, meters per meter, or percent.

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Strand – Normally an untwisted bundle or assembly of continuous filaments used as a unit, including slivers, tow, ends, yarn, etc. Sometimes a single fiber or filament is called a strand.

Strength – the maximum stress which a material is capable of sustaining.

Stress – The intensity at a point in a body of the forces or components of forces that act on a given plane through the point. Stress is expressed in force per unit area (pounds-force per square inch, megapascals, etc.).

Stress Relaxation – The time dependent decrease in stress in a solid under given constraint conditions.

Stress-Strain Curve (Diagram) – A graphical representation showing the relationship between the change in dimension of the specimen in the direction of the externally applied stress and the magnitude of the applied stress. Values of stress usually are plotted as ordinates (vertically) and strain values as abscissa (horizontally).

Structural Element – a generic element of a more complex structural member (for example, skin, stringer, shear panels, sandwich panels, joints, or splices).

Structured Data – See Volume 1, Section 8.1.4.

Surfacing Mat – A thin mat of fine fibers used primarily to produce a smooth surface on an organic matrix composite.

Symmetrical Laminate – A composite laminate in which the sequence of plies below the laminate midplane is a mirror image of the stacking sequence above the midplane.

Tack – Stickiness of the prepreg.

Tape – Prepreg fabricated in widths up to 12 inches wide for carbon and 3 inches for boron. Cross stitched carbon tapes up to 60 inches wide are available commercially in some cases.

Tenacity – The tensile stress expressed as force per unit linear density of the unstrained specimen i.e., grams-force per denier or grams-force per tex.

Tex – A unit for expressing linear density equal to the mass or weight in grams of 1000 meters of filament, fiber, yarn or other textile strand.

Thermal Conductivity – Ability of a material to conduct heat. The physical constant for quantity of heat that passes through unit cube of a substance in unit time when the difference in temperature of two faces is one degree.

Thermoplastic – A plastic that repeatedly can be softened by heating and hardened by cooling through a temperature range characteristic of the plastic, and when in the softened stage, can be shaped by flow into articles by molding or extrusion.

Thermoset – A plastic that is substantially infusible and insoluble after having been cured by heat or other means.

Tolerance – The total amount by which a quantity is allowed to vary.

Tolerance Limit – A lower (upper) confidence limit on a specified percentile of a distribution. For example, the B-basis value is a 95% lower confidence limit on the tenth percentile of a distribution.

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Tolerance Limit Factor -- The factor which is multiplied by the estimate of variability in computing the tolerance limit.

Toughness -- A measure of a material's ability to absorb work, or the actual work per unit volume or unit mass of material that is required to rupture it. Toughness is proportional to the area under the load-elongation curve from the origin to the breaking point.

Tow -- An untwisted bundle of continuous filaments. Commonly used in referring to man-made fibers, particularly carbon and graphite fibers, in the composites industry.

Transformation -- A transformation of data values is a change in the units of measurement accomplished by applying a mathematical function to all data values. For example, if the data is given by x, then $y = x + 1$, x , $1/x$, $\log x$, and $\cos x$ are transformations.

Transition, First Order -- A change of state associated with crystallization or melting in a polymer.

Transversely Isotropic -- Descriptive term for a material exhibiting a special case of orthotropy in which properties are identical in two orthotropic dimensions, but not the third; having identical properties in both transverse directions but not the longitudinal direction.

Traveller -- A small piece of the same product (panel, tube, etc.) as the test specimen, used for example to measure moisture content as a result of conditioning.

Twist -- The number of turns about its axis per unit of length in a yarn or other textile strand. It may be expressed as turns per inch (tpi) or turns per centimeter (tpcm).

Twist, Direction of -- The direction of twist in yarns and other textile strands is indicated by the capital letters S and Z. Yarn has S twist if, when held in a vertical position, the visible spirals or helices around its central axis are in the direction of slope of the central portion of the letter S, and Z twist is in the other direction.

Twist multiplier -- The ratio of turns per inch to the square root of the cotton count.

Typical Basis -- A typical property value is a sample mean. Note that the typical value is defined as the simple arithmetic mean which has a statistical connotation of 50% reliability with a 50% confidence.

Unbond -- An area within a bonded interface between two adherends in which the intended bonding action failed to take place. Also used to denote specific areas deliberately prevented from bonding in order to simulate a defective bond, such as in the generation of quality standards specimens. (See **Disbond, Debond**).

Unidirectional Laminate -- A laminate with nonwoven reinforcements and all layers laid up in the same direction.

Unit Cell -- The term applied to the path of a yarn in a braided fabric representing a unit cell of a repeating geometric pattern. The smallest element representative of the braided structure.

Unstructured Data -- See Volume 1, Section 8.1.4.

Upper Confidence Limit -- See **Confidence Interval**.

Vacuum Bag Molding -- A process in which the lay-up is cured under pressure generated by drawing a vacuum in the space between the lay-up and a flexible sheet placed over it and sealed at the edges.

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Variance -- See Sample Variance.

Viscosity -- The property of resistance to flow exhibited within the body of a material.

Void -- A physical and mechanical discontinuity occurring within a material or part which may be two-dimensional (e.g., disbonds, delaminations) or three-dimensional (e.g., vacuum-, air-, or gas-filled pockets). Porosity is an aggregation of micro-voids. Voids are essentially incapable of transmitting structural stresses or nonradiative energy fields. (See Inclusion.)

Warp -- The longitudinally oriented yarn in a woven fabric (see Fill); a group of yarns in long lengths and approximately parallel.

Wet Lay-up -- A method of making a reinforced product by applying a liquid resin system while or after the reinforcement is put in place.

Weibull Distribution (Two - Parameter) -- A probability distribution for which the probability that a randomly selected observation from this population lies between a and b ($0 < a < b < \infty$) is given by Equation 1.7(d) where α is called the scale parameter and β is called the shape parameter. (See Volume 1, Section 8.1.4.)

$$\exp\left[-\left(\frac{a}{\alpha}\right)^{\beta}\right] - \exp\left[-\left(\frac{b}{\alpha}\right)^{\beta}\right] \qquad 1.7(d)$$

Wet Lay-up -- A method of making a reinforced product by applying a liquid resin system while the reinforcement is put in place.

Wet Strength -- The strength of an organic matrix composite when the matrix resin is saturated with absorbed moisture. (See Saturation).

Wet Winding -- A method of filament winding in which the fiber reinforcement is coated with the resin system as a liquid just prior to wrapping on a mandrel.

Whisker -- A short single crystal fiber or filament. Whisker diameters range from 1 to 25 microns, with aspect ratios between 100 and 15,000.

Work Life -- The period during which a compound, after mixing with a catalyst, solvent, or other compounding ingredient, remains suitable for its intended use.

Woven Fabric Composite -- A major form of advanced composites in which the fiber constituent consists of woven fabric. A woven fabric composite normally is a laminate comprised of a number of laminae, each of which consists of one layer of fabric embedded in the selected matrix material. Individual fabric laminae are directionally oriented and combined into specific multiaxial laminates for application to specific envelopes of strength and stiffness requirements.

Yarn -- A generic term for strands or bundles of continuous filaments or fibers, usually twisted and suitable for making textile fabric.

Yarn, Plied -- Yarns made by collecting two or more single yarns together. Normally, the yarns are twisted together though sometimes they are collected without twist.

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Yield Strength -- The stress at which a material exhibits a specified limiting deviation from the proportionality of stress to strain. (The deviation is expressed in terms of strain such as 0.2 percent for the Offset Method or 0.5 percent for the Total Extension Under Load Method.)

X-Axis -- In composite laminates, an axis in the plane of the laminate which is used as the 0 degree reference for designating the angle of a lamina.

X-Y Plane -- In composite laminates, the reference plane parallel to the plane of the laminate.

Y-Axis -- In composite laminates, the axis in the plane of the laminate which is perpendicular to the x-axis.

Z-Axis -- In composite laminates, the reference axis normal to the plane of the laminate.

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4.1 INTRODUCTION

4.2 CARBON - EPOXY COMPOSITES

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4.2.1 T-500 12k/976 unidirectional tape*

				C/Ep T-500/976 Summary
MATERIAL:	T-500 12k/976 unidirectional tape			
PREPREG:	Fiberite Hy-E 3076P unidirectional tape			
FIBER:	Union Carbide Thornel T-500 12k	MATRIX:	Fiberite 976	
T_g(ambient):	361°F	T_g(wet):	T_g METHOD:	
PROCESSING:	240°F, 1 hour, 85 psi; 350°F, 2 hours, 100 psi			

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	6/88
Date of prepreg manufacture	12/83	Date of analysis	1/93
Date of composite manufacture			

LAMINA PROPERTY SUMMARY

	75°F/A		-65°F/A		250°F/A			
Tension, 1-axis	II-I		II-I		II-I			
Tension, 2-axis	II-I		II-I		II-I			
Tension, 3-axis								
Compression, 1-axis								
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
Shear, 31-plane								

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.79		
Resin Density	(g/cm^3)	1.28		
Composite Density	(g/cm^3)	1.59	1.57 - 1.61	
Fiber Areal Weight	(g/m^2)	142	142 - 146	
Ply Thickness	(in)	0.0053	0.0050 - 0.0057	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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Table 4.2.1(a)
C/Ep 142-UT
T-500/976
Tension, 1-axis
$[90]_8$
75/A, -65/A, 200/A
Interim

MATERIAL: T-500 12k/976 unidirectional tape

RESIN CONTENT: 28 - 34 wt%
FIBER VOLUME: 59 - 64 %
PLY THICKNESS: 0.0050 - 0.0057 in.

COMP. DENSITY: 1.57 - 1.61 g/cm^3
VOID CONTENT: 0.3 - 1.7 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Chord, 20 - 40 % of ultimate load

NORMALIZED BY: Specimen thickness to 60% fiber volume

Temperature(°F)		75 ambient		-65 ambient		250 ambient	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		13		13		13	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	295	298	213	213	273	276
	Minimum	257	270	163	196	236	258
	Maximum	329	328	243	235	302	310
	C.V. (%)	6.41	5.74	9.78	5.02	7.39	6.05
F_1^{tu}	B-value	(1)		(1)		(1)	
	Distribution	ANOVA		Weibull		Weibull	
(ksi)	C_1	20.5		221		282	
	C_2	4.64		13.1		15.7	
	No. Specimens	15		15		15	
	No. Batches	3		3		3	
	Approval Class	Interim		Interim		Interim	
	Mean	21.9	22.0	19.0	19.1	22.2	22.4
	Minimum	20.9	20.5	15.9	17.7	18.6	21.0
	Maximum	24.7	24.0	21.5	21.5	25.1	23.8
E_1^t	C.V. (%)	4.42	4.15	8.11	5.76	6.91	4.17
(Msi)	No. Specimens	15		15		15	
	No. Batches	3		3		3	
	Approval Class	Interim		Interim		Interim	
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		13000		10700		11800
	Minimum		11700		9300		10800
	Maximum		13900		12000		12900
	C.V. (%)		4.98		5.98		5.32
ϵ_1^{tu}	B-value		(1)		(1)		(1)
	Distribution		ANOVA		Weibull		Weibull
($\mu\epsilon$)	C_1		706		11000		12100
	C_2		4.75		18.8		21.6
	No. Specimens	15		15		15	
	No. Batches	3		3		3	
	Approval Class	Interim		Interim		Interim	

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.1(b)
C/Ep 142-UT
T-500/976
Tension, 2-axis
$[90]_8$
75/A, -65/A, 200/A
Interim

MATERIAL: T-500 12k/976 unidirectional tape

RESIN CONTENT: 28 - 34 wt%
FIBER VOLUME: 59 - 64 %
PLY THICKNESS: 0.0050 - 0.0057 in.
COMP. DENSITY: 1.57 - 1.61 lb/in^3
VOID CONTENT: 0.3 - 1.7 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Chord, 20 - 40 % of ultimate load

NORMALIZED BY: Not normalized

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		75 ambient 13	-65 ambient 13	250 ambient 13			
	Mean	10.2	10.3	7.90			
	Minimum	9.40	9.40	7.00			
	Maximum	11.3	12.1	8.80			
	C.V. (%)	5.59	6.61	5.35			
F_2^{tu}	B-value	(1)	(1)	(1)			
	Distribution	ANOVA	Lognormal	Weibull			
(ksi)	C_1	0.594	2.33	8.09			
	C_2	3.48	0.0636	19.7			
	No. Specimens	15	15	15			
	No. Batches	3	3	3			
	Approval Class	Interim	Interim	Interim			
	Mean	1.3	1.5[1]	1.2			
	Minimum	1.3	1.4	1.1			
	Maximum	1.7	1.6	1.3			
E_2^t	C.V. (%)	7.8	4.8	7.0			
(Msi)	No. Specimens	15	15	15			
	No. Batches	3	3	3			
	Approval Class	Interim	Interim	Interim			
	Mean						
ν_{21}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean	7750	7110	6930			
	Minimum	5800	6200	5900			
	Maximum	8900	8600	8000			
	C.V. (%)	10.3	8.28	8.32			
ϵ_2^{tu}	B-value	(1)	(1)	(1)			
	Distribution	Weibull	Weibull	Weibull			
($\mu\epsilon$)	C_1	8080	7390	7180			
	C_2	12.4	11.5	13.7			
	No. Specimens	15	15	15			
	No. Batches	3	3	3			
	Approval Class	Interim	Interim	Interim			

(1) B-values are presented only for fully approved data.

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4.2.2 HITEX 33 6k/E7K8 unidirectional tape*

MATERIAL:	HITEX 33 6k/E7K8 unidirectional tape	C/Ep HITEX 33/E7K8 Summary
PREPREG:	U.S. Polymeric HITEX 33 6k/E7K8 unidirectional tape, grade 145	
FIBER:	Hitco HITEX 33 6k, no twist MATRIX: U.S. Polymeric E7K8	
T_g(ambient):	T_g(wet): T_g METHOD:	
PROCESSING:	Autoclave cure: 300 - 310°F, 120 - 130 min., 55 psi	

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	1/88
Date of prepreg manufacture	Date of analysis	1/93
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	75°F/A		-65°F/A	180°F/A		75°F/W	180°F/W	
Tension, 1-axis	SSSS		SS-S			SSS-	SSS-	
Tension, 2-axis	SS--							
Tension, 3-axis								
Compression, 1-axis	SS-S		SS-S			SS--	SS--	
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane	S---			S---		S---	S---	
Shear, 23-plane								
Shear, 31-plane								

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.80		
Resin Density	(g/cm^3)	1.27		
Composite Density	(g/cm^3)	1.59	1.56 - 1.61	
Fiber Areal Weight	(g/m^2)	145		
Ply Thickness	(in)	0.0057	0.0053 - 0.0058	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.2(a)
C/Ep 145-UT
HITEX 33/E7K8
Tension, 1-axis
$[0]_{10}$
75/A, -65/A, 75/1.5%
Screening

MATERIAL: HITEX 33 6k/E7K8 unidirectional tape

RESIN CONTENT: 34 wt% COMP. DENSITY: 1.58 g/cm^3
FIBER VOLUME: 58 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0057 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Fiber volume fraction to 60%

Temperature(°F)		75 ambient		-65 ambient		75 1.5 (1)	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	313	304	296	288	318	310
	Minimum	292	283	267	259	280	272
	Maximum	339	330	327	319	345	335
	C.V. (%)	4.80	4.84	9.19	9.20	7.63	7.65
F_1^{tu}	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	Weibull	Weibull	Normal	Normal	Normal	Normal
(ksi)	C_1	320	311	296	288	318	310
	C_2	22.2	21.9	27.2	26.5	24.3	23.7
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	18.2	17.7	18.5	18.0	18.5	18.0
	Minimum	17.5	17.0	18.1	17.7	18.3	17.8
	Maximum	19.0	18.5	18.6	18.1	18.7	18.2
E_1^t	C.V. (%)	2.58	2.60	1.06	1.07	0.79	0.79
(Msi)	No. Specimens	18		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	0.310				0.310	
ν_{12}^t	No. Specimens	5				5	
	No. Batches	1				1	
	Approval Class	Screening				Screening	
	Mean	15900		16100			
	Minimum	15200		15500			
	Maximum	17100		17000			
	C.V. (%)	4.81		3.61			
ϵ_1^{tu}	B-value	(2)		(2)			
	Distribution	Normal		Normal			
($\mu\epsilon$)	C_1	15900		16200			
	C_2	765		582			
	No. Specimens	5		5			
	No. Batches	1		1			
	Approval Class	Screening		Screening			

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.2(b)
C/Ep 145-UT
HITEX 33/E7K8
Tension, 1-axis
$[0]_{10}$
180/1.5%
Screening

MATERIAL: HITEX 33 6k/E7K8 unidirectional tape

RESIN CONTENT: 34 wt% COMP. DENSITY: 1.58 g/cm^3
FIBER VOLUME: 58 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0057 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Fiber volume fraction to 60%

Temperature(°F)		180			
Moisture Content(%)		1.5			
Equilibrium at T, RH		(1)			
Source Code		20			
		Normalized	Measured		
	Mean	308	300		
	Minimum	296	288		
	Maximum	318	309		
	C.V. (%)	2.65	2.65		
F_1^{tu}	B-value	(2)	(2)		
	Distribution	Normal	Normal		
(ksi)	C_1	308	300		
	C_2	8.17	7.95		
	No. Specimens	5			
	No. Batches	1			
	Approval Class	Screening			
	Mean	18.7	18.2		
	Minimum	17.8	17.3		
	Maximum	19.5	19.0		
E_1^t	C.V. (%)	3.64	3.65		
(Msi)	No. Specimens	5			
	No. Batches	1			
	Approval Class	Screening			
	Mean	0.300			
ν_{12}^t	No. Specimens	5			
	No. Batches	1			
	Approval Class	Screening			
	Mean				
	Minimum				
	Maximum				
	C.V. (%)				
ϵ_1^{tu}	B-value				
	Distribution				
($\mu\epsilon$)	C_1				
	C_2				
	No. Specimens				
	No. Batches				
	Approval Class				

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are presented only for fully approved data.

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Table 4.2.2(c)
C/Ep 145-UT
HITEX 33/E7K8
Tension, 2-axis
$[90]_{20}$
75/A
Screening

MATERIAL: HITEX 33 6k/E7K8 unidirectional tape

RESIN CONTENT: 34 wt% COMP. DENSITY: 1.58 g/cm^3
FIBER VOLUME: 58 % VOID CONTENT: 0.39 %
PLY THICKNESS: 0.0058 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Not normalized

Temperature(°F)		75					
Moisture Content(%)		ambient					
Equilibrium at T, RH							
Source Code		20					
	Mean	6.90					
	Minimum	5.58					
	Maximum	8.07					
	C.V. (%)	11.2					
F_2^{tu}	B-value	(1)					
	Distribution	Weibull					
(ksi)	C_1	7.23					
	C_2	10.9					
	No. Specimens	20					
	No. Batches	1					
	Approval Class	Screening					
	Mean	1.25					
	Minimum	1.23					
	Maximum	1.27					
E_2^t	C.V. (%)	0.977					
(Msi)	No. Specimens	20					
	No. Batches	1					
	Approval Class	Screening					
	Mean						
ν_{21}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_2^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.2(d)
C/Ep 145-UT
HITEX 33/E7K8
Compression, 1-axis
$[0]_{10}$
75/A, -65/A, 75/1.5%
Screening

MATERIAL: HITEX 33 6k/E7K8 unidirectional tape

RESIN CONTENT: 34 - 35 wt% COMP. DENSITY: 1.57 - 1.58 g/cm³
FIBER VOLUME: 57 - 58 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0057 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Fiber volume fraction to 60%

Temperature(°F)		75 ambient		-65 ambient		75 1.5 (1)	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	209	204	230	224	198	193
	Minimum	168	164	209	204	178	174
	Maximum	234	228	254	248	217	211
	C.V. (%)	9.41	9.41	7.98	8.04	8.13	8.03
F_1^{cu}	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	Weibull	Weibull	Normal	Normal	Normal	Normal
(ksi)	C_1	218	212	230	224	198	193
	C_2	13.7	13.7	18.3	17.9	16.1	15.7
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	17.1	16.2	17.9	16.9	18.0	17.0
	Minimum	16.1	15.2	17.5	16.5	17.5	16.6
	Maximum	17.8	16.8	18.1	17.1	18.8	17.8
E_1^c	C.V. (%)	2.89	2.94	1.23	1.35	3.04	5.59
(Msi)	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		12600		13600		
	Minimum		12000		13600		
	Maximum		13400		13700		
	C.V. (%)		2.92		0.48		
ϵ_1^{cu}	B-value		(2)		(2)		
	Distribution		Weibull		Normal		
($\mu\epsilon$)	C_1		12800		13600		
	C_2		35.7		65.7		
	No. Specimens		20		5		
	No. Batches		1		1		
	Approval Class		Screening		Screening		

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are presented only for fully approved data.

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Table 4.2.2(e)
C/Ep 145-UT
HITEX 33/E7K8
Compression, 1-axis
$[0]_{10}$
180/1.5%
Screening

MATERIAL: HITEX 33 6k/E7K8 unidirectional tape

RESIN CONTENT: 34 - 35 wt% COMP. DENSITY: 1.57 - 1.58 g/cm³
FIBER VOLUME: 57 - 58 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0057 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Fiber volume fraction to 60%

Temperature(°F)		180					
Moisture Content(%)		1.5					
Equilibrium at T, RH		(1)					
Source Code		20					
		Normalized	Measured				
	Mean	136	132				
	Minimum	111	108				
	Maximum	161	157				
	C.V. (%)	13.4	13.6				
F_1^{cu}	B-value	(2)	(2)				
	Distribution	Normal	Normal				
(ksi)	C_1	136	132				
	C_2	18.3	17.8				
	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					
	Mean	17.6	16.6				
	Minimum	17.0	16.1				
	Maximum	18.0	17.0				
E_1^c	C.V. (%)	2.47	2.47				
(Msi)	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{cu}	B-value						
	Distribution						
$(\mu\epsilon)$	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.2(f)
C/Ep 145-UT
HITEX 33/E7K8
Shear, 12-plane
$[(\pm 45)_2/45]_s$
75/A, 180/A, 75/1.5%, 180/1.5%
Screening

MATERIAL: HITEX 33 6k/E7K8 unidirectional tape

RESIN CONTENT: 29 - 30 wt% COMP. DENSITY: 1.59 - 1.61 g/cm^3
FIBER VOLUME: 62 - 64 % VOID CONTENT: 0.05 - 0.91 %
PLY THICKNESS: 0.0053 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3518-76

NORMALIZED BY: Not normalized

Temperature(°F)		75	180	75	180		
Moisture Content(%)		ambient	ambient	1.5	1.5		
Equilibrium at T, RH				(1)	(1)		
Source Code		20	20	20	20		
	Mean	15.0	13.2	16.3	11.7		
	Minimum	13.5	13.1	15.8	11.5		
	Maximum	15.8	13.3	16.7	11.9		
	C.V. (%)	3.52	0.655	2.20	1.27		
F_{12}^{su}	B-value	(2)	(2)	(2)	(2)		
	Distribution	Weibull	Normal	Normal	Normal		
(ksi)	C_1	15.2	13.2	16.3	11.7		
	C_2	34.8	0.0865	0.357	0.148		
	No. Specimens	20	5	5	5		
	No. Batches	1	1	1	1		
	Approval Class	Screening	Screening	Screening	Screening		
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
γ_{12}^{su}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
G_{12}^{s}	C.V. (%)						
(Msi)	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are presented only for fully approved data.

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Check the source to verify that this is the current version before use.

4.2.3 AS4 12k/E7K8 unidirectional tape*

MATERIAL:	AS4 12k/E7K8 unidirectional tape		C/Ep AS4/E7K8 Summary
PREPREG:	U.S. Polymeric AS4 12k/E7K8 unidirectional tape		
FIBER:	Hercules AS4 12k	MATRIX:	U.S. Polymeric E7K8
T_g(ambient):	T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure: 300 - 310°F, 120 - 130 min., 55 psi		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	1/88
Date of prepreg manufacture	Date of analysis	1/93
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	75°F/A		-65°F/A	180°F/A		75°F/W	180°F/W	
Tension, 1-axis	IISS		SS-S			SSSS	SSSS	
Tension, 2-axis	SS--							
Tension, 3-axis								
Compression, 1-axis	SS-S		SS-S			SS--	SS--	
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane	I---			S---		S---	S---	
Shear, 23-plane								
Shear, 31-plane								

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.80		
Resin Density	(g/cm^3)	1.28		
Composite Density	(g/cm^3)	1.59	1.52 - 1.59	
Fiber Areal Weight	(g/m^2)	145		
Ply Thickness	(in)	0.0054	0.0054 - 0.0057	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.3(a)
C/Ep 145-UT
AS4/E7K8
Tension, 1-axis
$[0]_{10}$
75/A, -65/A, 75/0.77%
Interim, Screening

MATERIAL: AS4 12k/E7K8 unidirectional tape

RESIN CONTENT: 32 - 37 wt%
FIBER VOLUME: 53 - 60 %
PLY THICKNESS: 0.0054 in.
COMP. DENSITY: 1.53 - 1.59 g/cm³
VOID CONTENT: 0.64 - 2.2 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Slope of initial linear portion of load-displacement curve

NORMALIZED BY: Fiber volume to 60%

Temperature(°F)		75 ambient		-65 ambient		75 0.77 (1)	
Moisture Content(%) Equilibrium at T, RH							
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	303	293	291	273	304	294
	Minimum	253	252	255	239	286	276
	Maximum	345	347	327	306	317	306
	C.V. (%)	8.26	8.94	8.93	8.90	4.16	4.22
F_1^{tu}	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	ANOVA	ANOVA	Normal	Normal	Normal	Normal
(ksi)	C_1	26.7	32.4	291	273	304	294
	C_2	4.40	7.49	26.0	24.4	12.7	12.2
	No. Specimens	20		5		5	
	No. Batches	2		1		1	
	Approval Class	Interim		Screening		Screening	
	Mean	19.3	18.7	20.1	18.8	19.6	18.9
	Minimum	18.5	17.4	19.7	18.4	19.0	18.4
	Maximum	21.3	21.4	20.6	19.3	20.1	19.4
E_1^t	C.V. (%)	3.79	6.10	1.67	1.79	2.04	1.96
(Msi)	No. Specimens	20		5		5	
	No. Batches	2		1		1	
	Approval Class	Interim		Screening		Screening	
	Mean	0.320				0.288	
ν_{12}^t	No. Specimens	5				5	
	No. Batches	1				1	
	Approval Class	Screening				Screening	
	Mean	13900		13500		14600	
	Minimum	12500		12000		13700	
	Maximum	16000		14800		15000	
	C.V. (%)	11.0		8.24		3.83	
ϵ_1^{tu}	B-value	(2)		(2)		(2)	
	Distribution	Normal		Normal		Normal	
$(\mu\epsilon)$	C_1	13900		13500		14600	
	C_2	1530		1110		561	
	No. Specimens	5		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are presented only for fully approved data.

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Table 4.2.3(b)
C/Ep 145-UT
AS4/E7K8
Tension, 1-axis
$[0]_{10}$
180/0.77%
Screening

MATERIAL: AS4 12k/E7K8 unidirectional tape

RESIN CONTENT: 32 - 37 wt%
FIBER VOLUME: 53 - 60 %
PLY THICKNESS: 0.0054 in.

COMP. DENSITY: 1.53 - 1.59 g/cm³
VOID CONTENT: 0.64 - 2.2 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Slope of initial linear portion of load-displacement curve

NORMALIZED BY: Fiber volume to 60%

Temperature(°F)		180			
Moisture Content(%)		0.77			
Equilibrium at T, RH		(1)			
Source Code		20			
		Normalized	Measured		
	Mean	310	296		
	Minimum	284	274		
	Maximum	326	306		
	C.V. (%)	5.87	4.76		
F_1^{tu}	B-value	(2)	(2)		
	Distribution	Normal	Normal		
(ksi)	C_1	310	296		
	C_2	18.2	13.9		
	No. Specimens	5			
	No. Batches	1			
	Approval Class	Screening			
	Mean	20.1	19.2		
	Minimum	19.1	18.5		
	Maximum	21.8	20.4		
E_1^t	C.V. (%)	5.65	4.01		
(Msi)	No. Specimens	5			
	No. Batches	1			
	Approval Class	Screening			
	Mean		0.288		
ν_{12}^t	No. Specimens	5			
	No. Batches	1			
	Approval Class	Screening			
	Mean		14600		
	Minimum		13900		
	Maximum		15400		
	C.V. (%)		4.21		
ϵ_1^{tu}	B-value		(2)		
	Distribution		Normal		
($\mu\epsilon$)	C_1		14600		
	C_2		616		
	No. Specimens	5			
	No. Batches	1			
	Approval Class	Screening			

(1) Conditioned for 14 days at 160°F, 85% RH.
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Table 4.2.3(c)
C/Ep 145-UT
AS4/E7K8
Tension, 2-axis
$[90]_{20}$
75/A
Screening

MATERIAL: AS4 12k/E7K8 unidirectional tape

RESIN CONTENT: 32 - 38 wt%
FIBER VOLUME: 53 - 60 %
PLY THICKNESS: 0.0057 in.

COMP. DENSITY: 1.54 - 1.59 g/cm^3
VOID CONTENT: 0.64 - 0.75 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Slope of initial linear portion of load-displacement curve

NORMALIZED BY: Not normalized

Temperature(°F)		75					
Moisture Content(%)		ambient					
Equilibrium at T, RH							
Source Code		20					
F_2^{tu} (ksi)	Mean	5.47					
	Minimum	4.10					
	Maximum	7.01					
	C.V. (%)	13.2					
	B-value	(1)					
	Distribution	Weibull					
	C_1	5.79					
	C_2	8.04					
	No. Specimens	20					
	No. Batches	1					
	Approval Class	Screening					
E_2^t (Msi)	Mean	1.23					
	Minimum	1.16					
	Maximum	1.32					
	C.V. (%)	3.76					
	No. Specimens	20					
	No. Batches	1					
	Approval Class	Screening					
ν_{21}^t	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_2^{tu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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Table 4.2.3(d)
C/Ep 145-UT
AS4/E7K8
Compression, 1-axis
$[0]_{10}$
75/A, -65/A, 75/0.77%
Screening

MATERIAL: AS4 12k/E7K8 unidirectional tape

RESIN CONTENT: 35 - 40 wt%
FIBER VOLUME: 51 - 57 %
PLY THICKNESS: 0.0054 in.

COMP. DENSITY: 1.52 - 1.58 g/cm³
VOID CONTENT: 1.4 - 2.3 %

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Slope of initial linear portion of load-displacement curve

NORMALIZED BY: Fiber volume to 60%

Temperature(°F)		75 ambient		-65 ambient		75 0.77 (1)	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	245	209	276	235	215	182
	Minimum	207	176	251	213	196	166
	Maximum	269	229	299	254	238	202
	C.V. (%)	8.00	7.80	6.57	6.60	7.78	7.75
F_1^{cu}	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	Weibull	Weibull	Normal	Normal	Normal	Normal
(ksi)	C_1	254	216	276	235	215	183
	C_2	16.3	16.3	18.1	15.4	16.7	14.2
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	19.0	17.9	17.6	16.5	18.5	17.4
	Minimum	17.3	16.3	16.6	15.7	17.7	16.7
	Maximum	20.4	19.2	18.0	17.0	19.0	17.9
E_1^c	C.V. (%)	4.58	4.54	3.16	3.14	2.95	2.86
(Msi)	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		11700		14400		
	Minimum		10800		13900		
	Maximum		13100		15100		
	C.V. (%)		4.81		3.89		
ϵ_1^{cu}	B-value		(2)		(2)		
	Distribution		Normal		Normal		
($\mu\epsilon$)	C_1		11700		14400		
	C_2		564		559		
	No. Specimens	20		5			
	No. Batches	1		1			
	Approval Class	Screening		Screening			

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are presented only for fully approved data.

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Table 4.2.3(e)
C/Ep 145-UT
AS4/E7K8
Compression, 1-axis
$[0]_{10}$
180/0.77%
Screening

MATERIAL: AS4 12k/E7K8 unidirectional tape

RESIN CONTENT: 35 - 40 wt%
FIBER VOLUME: 51 - 57 %
PLY THICKNESS: 0.0054 in.
COMP. DENSITY: 1.52 - 1.58 g/cm^3
VOID CONTENT: 1.4 - 2.3 %

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Slope of initial linear portion of load-displacement curve

NORMALIZED BY: Fiber volume to 60%

Temperature(°F)		180					
Moisture Content(%)		0.77					
Equilibrium at T, RH		(1)					
Source Code		20					
		Normalized	Measured				
	Mean	150	127				
	Minimum	125	106				
	Maximum	176	150				
	C.V. (%)	14.8	15.0				
F_1^{cu} (ksi)	B-value	(2)	(2)				
	Distribution	Normal	Normal				
	C_1	150	127				
	C_2	22.2	18.9				
	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					
E_1^c (Msi)	Mean	18.0	17.0				
	Minimum	17.4	16.4				
	Maximum	18.4	17.3				
	C.V. (%)	2.46	2.41				
	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					
ν_{12}^c	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_1^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are presented only for fully approved data.

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Table 4.2.3(f)
C/Ep 145-UT
AS4/E7K8
Shear, 12-plane
$[(\pm 45)_2/45]_s$
75/A, 180/A,
75/0.77%, 180/0.77%
Interim, Screening

MATERIAL: AS4 12k/E7K8 unidirectional tape

RESIN CONTENT: 33 - 36 wt%
FIBER VOLUME: 55 - 57 %
PLY THICKNESS: 0.0055 in.

COMP. DENSITY: 1.54 - 1.55 g/cm^3
VOID CONTENT: 1.9 - 2.3 %

TEST METHOD: ASTM D 3518-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75	180	75	180		
Moisture Content(%)		ambient	ambient	0.77	0.77		
Equilibrium at T, RH				(1)	(1)		
Source Code		20	20	20	20		
	Mean	16.5	14.6	15.1	13.4		
	Minimum	13.8	14.2	13.5	13.0		
	Maximum	17.0	14.9	15.8	13.8		
	C.V. (%)	6.41	1.90	6.04	2.44		
F_{12}^{su}	B-value	(2)	(2)	(2)	(2)		
	Distribution	ANOVA	Normal	Normal	Normal		
(ksi)	C_1	2.46	14.6	15.1	13.4		
	C_2	7.58	0.277	0.905	0.328		
	No. Specimens	20	5	5	5		
	No. Batches	2	1	1	1		
	Approval Class	Interim	Screening	Screening	Screening		
	Mean						
	Minimum						
	Maximum						
G_{12}^{s}	C.V. (%)						
(Msi)	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
γ_{12}^{su}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are presented only for fully approved data.

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4.2.4 Celion 12k/E7K8 unidirectional tape*

MATERIAL:	Celion 12k/E7K8 unidirectional tape	C/Ep Celion 12k/E7K8 Summary
PREPREG:	U.S. Polymeric Celion 12k/E7K8 unidirectional tape, grade 280	
FIBER:	Celanese Celion 12k, no twist MATRIX: U.S. Polymeric E7K8	
T_g(ambient):	T_g(wet): T_g METHOD:	
PROCESSING:	Autoclave cure: 300 - 310°F, 120 - 130 min., 55 psi	

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	1/88
Date of prepreg manufacture	Date of analysis	1/93
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	75°F/A		-65°F/A	180°F/A		75°F/W	180°F/W	
Tension, 1-axis	SSSS		SS-S			SSS-	SSSS	
Tension, 2-axis	SS--							
Tension, 3-axis								
Compression, 1-axis	SS-S		SS-S			SS--	SS--	
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane	S---			S---		S---	S---	
Shear, 23-plane								
Shear, 31-plane								

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.8		
Resin Density	(g/cm^3)	1.28		
Composite Density	(g/cm^3)	1.59	1.59 - 1.61	
Fiber Areal Weight	(g/m^2)	280		
Ply Thickness	(in)	0.011	0.010 - 0.011	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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Table 4.2.4(a)
C/Ep 280-UT
Celion 12k/E7K8
Tension, 1-axis
$[0]_6$
75/A, -65/A, 75/0.77%
Screening

MATERIAL: Celion 12k/E7K8 unidirectional tape

RESIN CONTENT: 29 wt% COMP. DENSITY: 1.61 g/cm^3
FIBER VOLUME: 63 - 64 % VOID CONTENT: 0.53 - 1.0 %
PLY THICKNESS: 0.011 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Fiber volume to 60%

Temperature(°F)		75 ambient		-65 ambient		75 0.77 (1)	
Moisture Content(%) Equilibrium at T, RH							
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	293	309	281	302	300	314
	Minimum	265	285	268	287	292	306
	Maximum	317	332	307	330	315	330
	C.V. (%)	4.52	4.52	5.44	5,44	3.22	3.60
F_1^{tu}	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	Weibull	Weibull	Normal	Normal	Normal	Normal
(ksi)	C_1	299	316	281	302	300	314
	C_2	25.6	25.9	15.3	16.4	9.67	10.1
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	20.0	21.1	19.2	20.6	19.0	19.9
	Minimum	18.7	20.1	18.6	20.0	18.5	19.4
	Maximum	21.9	23.0	20.3	21.8	20.0	21.0
E_1^t	C.V. (%)	4.48	4.25	3.40	3.80	3.22	3.60
(Msi)	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	0.286				0.292	
ν_{12}^t	No. Specimens	5				5	
	No. Batches	1				1	
	Approval Class	Screening				Screening	
	Mean	14300		14800			
	Minimum	13500		14200			
	Maximum	14700		15800			
	C.V. (%)	3.34		3.87			
ϵ_1^{tu}	B-value	(2)		(2)			
	Distribution	Normal		Normal			
($\mu\epsilon$)	C_1	14300		14800			
	C_2	478		573			
	No. Specimens	5		5			
	No. Batches	1		1			
	Approval Class	Screening		Screening			

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are presented only for fully approved data.

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Table 4.2.4(b)
C/Ep 280-UT
Celion 12k/E7K8
Tension, 1-axis
$[0]_5$
180/0.77%
Screening

MATERIAL: Celion 12k/E7K8 unidirectional tape

RESIN CONTENT: 29 wt% COMP. DENSITY: 1.61 g/cm^3
FIBER VOLUME: 63 - 64 % VOID CONTENT: 0.53 - 1.0 %
PLY THICKNESS: 0.011 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Fiber volume to 60%

Temperature(°F)		180					
Moisture Content(%)		0.77					
Equilibrium at T, RH		(1)					
Source Code		20					
		Normalized	Measured				
	Mean	293	311				
	Minimum	269	286				
	Maximum	316	335				
	C.V. (%)	6.43	7.19				
F_1^{tu}	B-value	(2)	(2)				
	Distribution	Normal	Normal				
(ksi)	C_1	293	311				
	C_2	18.9	20.0				
	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					
	Mean	19.8	21.0				
	Minimum	19.4	20.6				
	Maximum	20.1	21.4				
E_1^t	C.V. (%)	1.61	1.81				
(Msi)	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					
	Mean		0.322				
ν_{12}^t	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					
	Mean		13800				
	Minimum		12300				
	Maximum		15400				
	C.V. (%)		10.4				
ϵ_1^{tu}	B-value		(2)				
	Distribution		Normal				
($\mu\epsilon$)	C_1		13800				
	C_2		1440				
	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are presented only for fully approved data.

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Table 4.2.4(c)
C/Ep 280-UT
Celion 12k/E7K8
Tension, 2-axis
$[90]_{12}$
75/A
Screening

MATERIAL: Celion 12k/E7K8 unidirectional tape

RESIN CONTENT: 31 - 33 wt%
FIBER VOLUME: 59 - 61 %
PLY THICKNESS: 0.011 in.
COMP. DENSITY: 1.59 - 1.60 g/cm^3
VOID CONTENT: 0.68 - 0.74 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75					
Moisture Content(%)		ambient					
Equilibrium at T, RH							
Source Code		20					
	Mean	6.00					
	Minimum	5.21					
	Maximum	6.89					
	C.V. (%)	8.79					
F_2^{tu}	B-value	(1)					
	Distribution	Weibull					
(ksi)	C_1	6.24					
	C_2	12.6					
	No. Specimens	20					
	No. Batches	1					
	Approval Class	Screening					
	Mean	1.28					
	Minimum	1.19					
	Maximum	1.36					
E_2^t	C.V. (%)	4.52					
(Msi)	No. Specimens	20					
	No. Batches	1					
	Approval Class	Screening					
	Mean						
ν_{21}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_2^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.4(d)
C/Ep 280-UT
Celion 12k/E7K8
Compression, 1-axis
$[0]_8$
75/A, -65/A, 75/0.77%
Screening

MATERIAL: Celion 12k/E7K8 unidirectional tape

RESIN CONTENT: 29 - 30 wt%
FIBER VOLUME: 62 - 64 %
PLY THICKNESS: 0.010 in.

COMP. DENSITY: 1.60 - 1.61 g/cm^3
VOID CONTENT: 0.78 - 0.79 %

TEST METHOD: SACMA 1-88

MODULUS CALCULATION:

NORMALIZED BY: Fiber volume to 60%

Temperature(°F)		75 ambient		-65 ambient		75 0.77 (1)	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{cu} (ksi)	Mean	206	213	221	229	207	214
	Minimum	171	177	198	205	198	205
	Maximum	247	255	267	276	219	227
	C.V. (%)	8.62	8.62	12.2	12.2	5.06	5.06
	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	Weibull	Weibull	Normal	Normal	Normal	Normal
	C_1	214	221	221	228	207	214
	C_2	12.1	12.1	27.0	28.0	10.5	10.8
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
E_1^c (Msi)	Mean	19.9	21.1	22.9	24.3	21.6	22.3
	Minimum	18.1	19.2	20.8	22.0	20.2	21.0
	Maximum	21.7	22.3	23.8	25.1	22.8	23.6
	C.V. (%)	4.95	5.08	5.28	5.90	5.25	5.86
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
ν_{12}^c	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_1^{cu} ($\mu\epsilon$)	Mean		11200		9870		
	Minimum		10800		9210		
	Maximum		11800		10600		
	C.V. (%)		3.59		5.32		
	B-value		(2)		(2)		
	Distribution		Normal		Normal		
	C_1		11200		9870		
	C_2		401		526		
	No. Specimens	5		5			
	No. Batches	1		1			
	Approval Class	Screening		Screening			

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.4(e)
C/Ep 280-UT
Celion 12k/E7K8
Compression, 1-axis
$[0]_5$
180/0.77%
Screening

MATERIAL: Celion 12k/E7K8 unidirectional tape

RESIN CONTENT: 29 - 30 wt%
FIBER VOLUME: 62 - 64 %
PLY THICKNESS: 0.010 in.
COMP. DENSITY: 1.60 - 1.61 g/cm^3
VOID CONTENT: 0.78 - 0.79 %

TEST METHOD: SACMA 1-88

MODULUS CALCULATION:

NORMALIZED BY: Fiber volume to 60%

Temperature(°F)		180					
Moisture Content(%)		0.77					
Equilibrium at T, RH		(1)					
Source Code		20					
		Normalized	Measured				
	Mean	185	192				
	Minimum	158	164				
	Maximum	220	228				
	C.V. (%)	12.9	12.9				
F_1^{cu}	B-value	(2)	(2)				
	Distribution	Normal	Normal				
(ksi)	C_1	185	192				
	C_2	24.0	24.8				
	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					
	Mean	21.1	22.3				
	Minimum	19.5	20.6				
	Maximum	23.1	24.5				
E_1^c	C.V. (%)	6.80	7.63				
(Msi)	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned at 160°F, 85% RH.
(2) B-values are presented only for fully approved data.

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Table 4.2.4(f)
C/Ep 280-UT
Celion 12k/E7K8
Shear, 12-plane
[±45/45]$_s$
75/A, 180/A,
75/0.77%, 180/0.77%
Screening

MATERIAL: Celion 12k/E7K8 unidirectional tape

RESIN CONTENT: 30 - 31 wt%
FIBER VOLUME: 61 - 62 %
PLY THICKNESS: 0.011 in.

COMP. DENSITY: 1.60 g/cm^3
VOID CONTENT: 0.41 - 0.61 %

TEST METHOD: ASTM D 3518-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75	180	75	180		
Moisture Content(%)		ambient	ambient	0.77	0.77		
Equilibrium at T, RH				(1)	(1)		
Source Code		20	20	20	20		
	Mean	9.9	10.0	12.0	10.0		
	Minimum	9.3	8.1	11.3	8.2		
	Maximum	11.1	11.1	12.3	11.4		
	C.V. (%)	4.16	11.7	3.41	11.7		
F_{12}^{su}	B-value	(2)	(2)	(2)	(2)		
	Distribution	Nonpara.	Normal	Normal	Normal		
(ksi)	C_1	10	10.0	12.0	10.0		
	C_2	1.25	1.17	0.407	1.17		
	No. Specimens	20	5	5	5		
	No. Batches	1	1	1	1		
	Approval Class	Screening	Screening	Screening	Screening		
	Mean						
	Minimum						
	Maximum						
G_{12}^{s}	C.V. (%)						
(Msi)	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
γ_{12}^{su}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned at 160°F, 85% RH.
(2) B-values are presented only for fully approved data.

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4.2.5 AS4 12k/938 unidirectional tape*

				C/Ep AS4/938 Summary
MATERIAL:	AS4 12k/938 unidirectional tape			
PREPREG:	Fiberite Hy-E 1338H unidirectional tape, grade 145			
FIBER:	Hercules AS4 12k, unsized, no twist	MATRIX:	Fiberite 938	
T_g(ambient):	°F	T_g(wet):	260 °F	T_g METHOD:
PROCESSING:	Autoclave cure: 350 ± 10°F, 120 - 135 min., 100 ± 15 psi			

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	8/85
Date of resin manufacture		Date of data submittal	4/89
Date of prepreg manufacture	7/85	Date of analysis	1/93
Date of composite manufacture			

LAMINA PROPERTY SUMMARY

	75°F/A		-65°F/A	200°F/A		200°F/W		
Tension, 1-axis	II--		II--	II--				
Tension, 2-axis	II--			II--				
Tension, 3-axis								
Compression, 1-axis	II--					II--		
Compression, 2-axis	S---							
Compression, 3-axis								
Shear, 12-plane	S---			I---				
Shear, 23-plane								
Shear, 31-plane								

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.80	1.77 - 1.79	
Resin Density	(g/cm^3)	1.30	1.30	
Composite Density	(g/cm^3)	1.60	1.55 - 1.58	
Fiber Areal Weight	(g/m^2)	145	144 - 146	
Ply Thickness	(in)	0.0055	0.0048 - 0.0065	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.5(a)
C/Ep 145-UT
AS4/938
Tension, 1-axis
$[0]_8$
75/A, -65/A, 200/A
Interim

MATERIAL: AS4 12k/938 unidirectional tape

RESIN CONTENT: 35 - 41 wt% COMP. DENSITY: 1.55 - 1.57 g/cm³
FIBER VOLUME: 52 - 57 % VOID CONTENT: 0.0 - <1.0 %
PLY THICKNESS: 0.0042 - 0.0052 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76 (1)

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		75 ambient		-65 ambient		200 ambient	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		12		12		12	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	314	272	296	238	321	274
	Minimum	270	230	198	174	263	229
	Maximum	351	330	363	287	356	322
	C.V.(%)	7.45	8.79	14.4	11.0	7.79	8.10
F_1^{tu}	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	Weibull	ANOVA	ANOVA	ANOVA	ANOVA	Weibull
(ksi)	C_1	324	26.3	49.1	249	26.9	284
	C_2	16.5	4.12	4.64	11.1	3.78	13.3
	No. Specimens	22		22		20	
	No. Batches	3		3		3	
	Approval Class	Interim		Interim		Interim	
	Mean	22.4	19.4	19.5	19.0	20.4	20.8
	Minimum	18.8	17.1	18.5	16.9	18.4	18.4
	Maximum	26.9	21.0	21.5	22.0	24.0	22.4
E_1^t	C.V.(%)	9.88	4.66	4.07	5.13	7.23	6.06
(Msi)	No. Specimens	22		22		20	
	No. Batches	3		3		3	
	Approval Class	Interim		Interim		Interim	
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
ϵ_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Gage length 2.0 inches
(2) B-values are presented only for fully approved data.

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Table 4.2.5(b)
C/Ep 145-UT
AS4/938
Tension, 2-axis
$[90]_{16}$
75/A, 200/A
Interim

MATERIAL: AS4 12k/938 unidirectional tape

RESIN CONTENT: 35 - 40 wt%
FIBER VOLUME: 52 - 58 %
PLY THICKNESS: 0.0053 - 0.0063 in.

COMP. DENSITY: 1.56 - 1.58 g/cm^3
VOID CONTENT: 0.0 - <1.0 %

TEST METHOD: ASTM D 303 (1)

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75.0 ambient	200 ambient			
Moisture Content(%)						
Equilibrium at T, RH						
Source Code		12	12			
F_2^{tu} (ksi)	Mean	8.96	8.84			
	Minimum	6.50	6.85			
	Maximum	12.0	10.3			
	C.V.(%)	15.2	12.2			
	B-value	(2)	(2)			
	Distribution	Weibull	ANOVA			
	C_1	9.54	1.18			
	C_2	7.10	3.96			
	No. Specimens	19	17			
	No. Batches	3	3			
	Approval Class	Interim	Interim			
E_2^t (Msi)	Mean	1.29	1.23			
	Minimum	0.970	1.05			
	Maximum	1.72	1.40			
	C.V.(%)	7.89	7.81			
	No. Specimens	19	17			
	No. Batches	3	3			
	Approval Class	Interim	Interim			
ν_{21}^t	Mean					
	No. Specimens					
	No. Batches					
	Approval Class					
ϵ_2^{tu} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) Gage length 2.0 inches.
(2) B-values are presented only for fully approved data.

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Table 4.2.5(c)
C/Ep 145-UT
AS4/938
Compression, 1-axis
$[0]_8$
75/A, 200/W
Interim, Screening

MATERIAL: AS4 12k/938 unidirectional tape

RESIN CONTENT: 33 - 38 wt%
FIBER VOLUME: 54 - 60 %
PLY THICKNESS: 0.0048 - 0.0060 in.
COMP. DENSITY: 1.55 - 1.58 g/cm^3
VOID CONTENT: 0.0 - <1.0 %

TEST METHOD: SACMA 1-88

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

		Temperature(°F)	75		200		
		Moisture Content(%)	ambient		(1)		
		Equilibrium at T, RH					
		Source Code	12		12		
			Normalized	Measured	Normalized	Measured	
F_1^{cu} (ksi)		Mean	228	211	190	168	
		Minimum	186	172	158	138	
		Maximum	265	251	223	194	
		C.V.(%)	9.31	10.2	8.96	9.29	
		B-value	(2)	(2)	(2)	(2)	
		Distribution	Weibull	ANOVA	ANOVA	ANOVA	
		C_1	224	22.4	19.0	17.6	
		C_2	12.5	3.31	4.40	4.57	
		No. Specimens	25		24		
		No. Batches	3		3		
		Approval Class	Interim		Interim		
E_1^c (Msi)		Mean	18.2	18.4	19.1	18.4	
		Minimum	15.7	15.9	16.9	16.6	
		Maximum	21.0	22.5	24.0	21.0	
		C.V.(%)	9.13	12.4	12.8	9.10	
		No. Specimens	15		13		
		No. Batches	2		2		
		Approval Class	Interim		Screening		
ν_{12}^c		Mean					
		No. Specimens					
		No. Batches					
		Approval Class					
ϵ_1^{cu} ($\mu\epsilon$)		Mean					
		Minimum					
		Maximum					
		C.V.(%)					
		B-value					
		Distribution					
		C_1					
		C_2					
		No. Specimens					
		No. Batches					
		Approval Class					

(1) Specimens conditioned at 140°F and 95% relative humidity for one month.
(2) B-values are presented only for fully approved data.

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Table 4.2.5(d)
C/Ep 145-UT
AS4/938
Compression, 2-axis
$[90]_8$
75/A
Screening

MATERIAL: AS4 12k/938 unidirectional tape

RESIN CONTENT: 36 wt%
FIBER VOLUME: 56 %
PLY THICKNESS: 0.0058 in.

COMP. DENSITY: 1.56 g/cm^3
VOID CONTENT: 0.0 %

TEST METHOD:
SACMA 1-88

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75.0					
Moisture Content(%)		ambient					
Equilibrium at T, RH							
Source Code		12					
	Mean	30.4					
	Minimum	26.2					
	Maximum	39.7					
	C.V.(%)	16.4					
F_2^{cu} (ksi)	B-value	(1)					
	Distribution	Nonpara.					
	C_1	6					
	C_2	2.14					
	No. Specimens	10					
	No. Batches	1					
	Approval Class	Screening					
	Mean						
	Minimum						
	Maximum						
E_2^c	C.V.(%)						
(Msi)	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
ν_{21}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
ϵ_2^{cu} ($\mu\epsilon$)	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.5(e)
C/Ep 145-UT
AS4/938
Shear, 12-plane
$[\pm 45]_{2s}$
200/A
Screening, Interim

MATERIAL: AS4 12k/938 unidirectional tape

RESIN CONTENT: 35 - 37 wt% COMP. DENSITY: 1.56 - 1.58 g/cm^3
FIBER VOLUME: 54 - 57 % VOID CONTENT: 0.0 - <1.0 %
PLY THICKNESS: 0.0051 - 0.0063 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3518-76

NORMALIZED BY: Not normalized

Temperature(°F)		75.0 ambient	200 ambient			
Moisture Content(%)						
Equilibrium at T, RH						
Source Code		12	12			
	Mean	13.0	13.9			
	Minimum	10.8	11.9			
	Maximum	13.9	16.0			
	C.V.(%)	6.36	7.63			
F_{12}^{su}	B-value	(1)	(1)			
	Distribution	Weibull	ANOVA			
(ksi)	C_1	13.4	1.26			
	C_2	25.4	4.96			
	No. Specimens	13	18			
	No. Batches	3	3			
	Approval Class	Screening	Interim			
	Mean					
	Minimum					
	Maximum					
G_{12}^{s}	C.V.(%)					
	No. Specimens					
(Msi)	No. Batches					
	Approval Class					
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{12}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) B-values are presented only for fully approved data.

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4.2.6 T-300 3k/934 plain weave*

				C/Ep T-300/934 Summary
MATERIAL:	T-300 3k/934 plain weave			
PREPREG:	Fiberite HMF-322/34 plain weave			
FIBER:	Toray T-300 3k	MATRIX:	Fiberite 934	
T_g(ambient):	410°F T_g(wet):	T_g METHOD:	DSC	
PROCESSING:	Autoclave cure: 355 ± 10°F, 120 - 130 min., 85-100 psig			

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	6/88
Date of prepreg manufacture	2/84	Date of analysis	1/93
Date of composite manufacture			

LAMINA PROPERTY SUMMARY

	75°F/A		-65°F/A	250°F/A		160°F/W	250°F/W	
Tension, 1-axis	IS-I		IS-I	SS-S		II--	II--	
Tension, 2-axis	II-I		II-I	SS-S		II--	II--	
Tension, 3-axis								
Compression, 1-axis	II--		II--	SI--		I---	I---	
Compression, 2-axis	II--		II--	SI--		I---	I---	
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
SB Strength, 31-plane	S---		S---	S---				

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)		1.73 - 1.74	
Resin Density	(g/cm^3)	1.30		
Composite Density	(g/cm^3)	1.55	1.54 - 1.57	
Fiber Areal Weight	(g/m^2)	194	1.92 - 2.00	
Ply Thickness	(in)		0.0073 - 0.0084	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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Table 4.2.6(a)
C/Ep 194-PW
T-300/934
Tension, 1-axis
$[0_f]_{12}$
75/A, -65/A, 250/A
Interim, Screening

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.54 - 1.57 g/cm^3
FIBER VOLUME: 58 - 60 % VOID CONTENT: <0.5 - 1.2 %
PLY THICKNESS: 0.0074 - 0.0082 in.

TEST METHOD: ASTM D 3039-76 (2)

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		75 ambient 12		-65 ambient 12		250 ambient 12	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{tu} (ksi)	Mean	91	94	83	85	109	113
	Minimum	82	85	78	79	104	109
	Maximum	99	100	87	90	114	118
	C.V.(%)	4.1	4.0	3.2	3.3	3.54	3.42
	B-value	(1)	(1)	(1)	(1)	(1)	(1)
	Distribution	Weibull	Weibull	Weibull	Weibull	Normal	Normal
	C_1	93.0	96	83.7	86	86.0	113
	C_2	28.2	31	35.8	36	2.86	3.87
	No. Specimens	20		20		5	
	No. Batches	4		4		1	
	Approval Class	Interim		Interim		Screening	
E_1^t (Msi)	Mean	9.1	9.4	10.	10.	9.3	9.7
	Minimum	8.4	8.7	8.6	9.0	9.1	9.4
	Maximum	9.5	9.9	12	12	10.0	10.7
	C.V.(%)	3.3	3.6	11	10.	4.6	5.6
	No. Specimens	20		20		5	
	No. Batches	4		4		1	
	Approval Class	Screening		Screening		Screening	
ν_{12}^t	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_1^{tu} ($\mu\epsilon$)	Mean		9780		8990		11300
	Minimum		8880		7990		10900
	Maximum		11200		9800		11800
	C.V.(%)		5.61		6.07		3.11
	B-value		(1)		(1)		(1)
	Distribution		ANOVA		ANOVA		Normal
	C_1		577		592		11300
	C_2		3.12		3.61		351
	No. Specimens	20		20		5	
	No. Batches	4		4		1	
	Approval Class	Interim		Interim		Screening	

(1) B-values are presented only for fully approved data.
(2) Width 0.5 inch, speed of testing 0.05 in./in./min, gage length below recommendation.

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Table 4.2.6(b)
C/Ep 194-PW
T-300/934
Tension, 1-axis
$[0_t]_{12}$
160/W, 250/W
Interim

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt%
FIBER VOLUME: 58 - 60 %
PLY THICKNESS: 0.0074 - 0.0082 in.
COMP. DENSITY: 1.54 - 1.57 g/cm³
VOID CONTENT: <0.5 - 1.2 %

TEST METHOD: ASTM D 3039-76 (3)

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

		160 (1)		250 (1)			
Temperature(°F)							
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		12		12			
		Normalized	Measured	Normalized	Measured		
F_1^{tu} (ksi)	Mean	96	98	79	82		
	Minimum	84	88	61	66		
	Maximum	104	106	95	97		
	C.V.(%)	5.7	5.11	14	11		
	B-value	(2)	(2)	(2)	(2)		
	Distribution	ANOVA	Weibull	ANOVA	Weibull		
	C_1	6.0	101	12	86		
	C_2	4.8	24	5.3	11		
	No. Specimens	15		15			
	No. Batches	3		3			
	Approval Class	Interim		Interim			
E_1^t (Msi)	Mean	9.8	10.0	9.4	9.7		
	Minimum	8.1	8.6	6.8	7.1		
	Maximum	11.0	11.7	12.0	13.0		
	C.V.(%)	8.7	8.7	17.	18		
	No. Specimens	15		15			
	No. Batches	3		3			
	Approval Class	Interim		Interim			
ν_{12}^t	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_1^{tu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Immersed in water at 160°F for 14 days.
(2) B-values are presented only for fully approved data.
(3) Width 0.5 inch, speed of testing 0.05 in./in./min, gage length below recommendation.

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Table 4.2.6(c)
C/Ep 194-PW
T-300/934
Tension, 2-axis
$[90_t]_{12}$
75/A, -65/A, 250/A
Interim, Screening

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.54 - 1.57 g/cm³
FIBER VOLUME: 58 - 60 % VOID CONTENT: <0.5 - 1.2 %
PLY THICKNESS: 0.0074 - 0.0082 in.

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75 ambient		-65 ambient		250 ambient	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		12		12		12	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_2^{tu} (ksi)	Mean	88	91	80.	82	94	98
	Minimum	80.	82	70.	72	90.	94
	Maximum	97	99	91	95	97	101
	C.V. (%)	5.7	5.5	6.2	6.5	2.6	2.7
	B-value	(1)	(1)	(1)	(1)	(1)	(1)
	Distribution	ANOVA	ANOVA	ANOVA	ANOVA	Normal	Normal
	C_1	5.4	5.4	5.2	5.7	93.7	97.8
	C_2	3.5	3.4	3.3	3.4	2.47	2.59
	No. Specimens	20		20		5	
	No. Batches	4		4		1	
	Approval Class	Interim		Interim		Screening	
E_2^t (Msi)	Mean	9.0	9.3	9.1	9.5	8.1	8.5
	Minimum	8.3	8.7	8.1	8.3	8.0	8.3
	Maximum	9.9	10.3	10.8	11.1	8.2	8.6
	C.V. (%)	5.0	4.8	9.3	9.2	1.1	1.5
	No. Specimens	20		20		5	
	No. Batches	4		4		1	
	Approval Class	Interim		Interim		Screening	
ν_{21}^t	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_2^{tu} ($\mu\epsilon$)	Mean		9630		9100		11400
	Minimum		8680		7750		10400
	Maximum		11100		10700		12400
	C.V. (%)		6.18		7.44		8.59
	B-value		(1)		(1)		(1)
	Distribution		ANOVA		ANOVA		Normal
	C_1		616		710		11400
	C_2		2.82		3.08		981
	No. Specimens	20		20		5	
	No. Batches	4		4		1	
	Approval Class	Interim		Interim		Screening	

(1) B-values are presented only for fully approved data.

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Table 4.2.6(d)
C/Ep 194-PW
T-300/934
Tension, 2-axis
$[90_t]_{12}$
160/W, 250/W
Interim

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.54 - 1.57 g/cm^3
FIBER VOLUME: 58 - 60 % VOID CONTENT: <0.5 - 1.2 %
PLY THICKNESS: 0.0074 - 0.0082 in.

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

		160 (1)		250 (1)	
Temperature(°F)					
Moisture Content(%)					
Equilibrium at T, RH					
Source Code		12		12	
		Normalized	Measured	Normalized	Measured
	Mean	97	100.	81	83
	Minimum	90.	92	73	75
	Maximum	111	113	89	91
	C.V. (%)	6.8	6.3	5.1	4.8
F_2^{tu}	B-value	(2)	(2)	(2)	(2)
	Distribution	ANOVA	ANOVA	ANOVA	ANOVA
(ksi)	C_1	7.3	6.8	4.4	4.2
	C_2	4.8	4.5	4.5	4.2
	No. Specimens	15		15	
	No. Batches	3		3	
	Approval Class	Interim		Interim	
	Mean	10.	10.	9.9	10.
	Minimum	8.0	8.2	8.2	8.5
	Maximum	11.8	12.1	11.9	12.1
E_2^t	C.V. (%)	11	11	11	11
(Msi)	No. Specimens	15		15	
	No. Batches	3		3	
	Approval Class	Interim		Interim	
	Mean				
ν_{21}^t	No. Specimens				
	No. Batches				
	Approval Class				
	Mean				
	Minimum				
	Maximum				
	C.V. (%)				
ϵ_2^{tu}	B-value				
	Distribution				
($\mu\epsilon$)	C_1				
	C_2				
	No. Specimens				
	No. Batches				
	Approval Class				

(1) Immersed in water at 160°F for 14 days.
(2) B-values are presented only for fully approved data.

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Table 4.2.6(e)
C/Ep 194-PW
T-300/934
Compression, 1-axis
$[0_f]_{12}$
75/A, -65/A, 250/A
Interim, Screening

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt%
FIBER VOLUME: 58 - 60 %
PLY THICKNESS: 0.0074 - 0.0082 in.

COMP. DENSITY: 1.54 - 1.57 g/cm³
VOID CONTENT: <0.5 - 1.2 %

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75 ambient		-65 ambient		250 ambient	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		12		12		12	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{cu} (2) (ksi)	Mean	95	98	104	108	100.	105
	Minimum	83	87	87	90.	94	98
	Maximum	120	125	133	139	107	111
	C.V.(%)	10.	10.	13	14	5.6	5.1
	B-value	(1)	(1)	(1)	(1)	(1)	(1)
	Distribution	ANOVA	ANOVA	ANOVA	ANOVA	Normal	Normal
	C_1	10.	11	15	16	100.	105
	C_2	3.9	3.9	3.7	3.8	5.64	5.4
	No. Specimens	20		20		5	
	No. Batches	4		4		1	
	Approval Class	Interim		Interim		Screening	
E_1^c (3) (Msi)	Mean	8.4	8.8	8.2	8.6	8.4	8.9
	Minimum	7.7	8.0	7.4	7.8	7.9	8.1
	Maximum	9.0	9.4	8.9	9.7	10.0	10.1
	C.V.(%)	5.1	5.3	5.1	5.7	6.3	6.4
	No. Specimens	20		20		19	
	No. Batches	4		4		4	
	Approval Class	Interim		Interim		Interim	
ν_{12}^c	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_1^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.
(2) Tab thickness of 0.112 - 0.120 inch is larger than 0.070 inch nominal thickness per method.
(3) Specimen thickness of 0.09 - 0.10 inch is less than nominal 0.12 inch thickness per method.

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Table 4.2.6(f)
C/Ep 194-PW
T-300/934
Compression, 1-axis
$[0_f]_{12}$
160/W, 250/W
Interim

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt%
FIBER VOLUME: 58 - 60 %
PLY THICKNESS: 0.0074 - 0.0082 in.
COMP. DENSITY: 1.54 - 1.57 g/cm³
VOID CONTENT: <0.5 - 1.2 %

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		160 (1)		250 (1)			
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		12		12			
		Normalized	Measured	Normalized	Measured		
F_1^{cu} (3) (ksi)	Mean	74	76	44	46		
	Minimum	67	68	40	41		
	Maximum	81	84	49	51		
	C.V.(%)	6.9	5.6	6.2	6.2		
	B-value	(2)	(2)	(2)	(2)		
	Distribution	ANOVA	ANOVA	Weibull	Weibull		
	C_1	5.6	6.2	45.4	46.8		
	C_2	4.9	5.0	17.4	16.9		
	No. Specimens	15		15			
	No. Batches	3		3			
	Approval Class	Interim		Interim			
E_1^c (Msi)	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
	No. Specimens						
	No. Batches						
	Approval Class						
ν_{12}^c	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_1^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Immersed in water at 160°F for 14 days.
(2) B-values are presented only for fully approved data.
(3) Tab thickness of 0.112 - 0.120 inch is larger than 0.070 inch nominal thickness per method.

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Table 4.2.6(g)
C/Ep 194-PW
T-300/934
Compression, 2-axis
$[90_t]_{12}$
75/A, -65/A, 250/A
Interim, Screening

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.54 - 1.57 g/cm³
FIBER VOLUME: 58 - 60 % VOID CONTENT: <0.5 - 1.2 %
PLY THICKNESS: 0.0074 - 0.0082 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75 ambient		-65 ambient		250 ambient	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		12		12		12	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_2^{cu} (2) (ksi)	Mean	90.	93	103	106	82	85
	Minimum	81	85	94	98	77	81
	Maximum	100.	104	116	121	84	88
	C.V. (%)	5.9	6.0	6.2	6.1	3.4	3.4
	B-value	(1)	(1)	(1)	(1)	(1)	(1)
	Distribution	ANOVA	ANOVA	Normal	Normal	Normal	Normal
	C_1	5.6	5.9	103	106	81.7	85.3
	C_2	3.2	3.2	6.18	6.4	2.74	2.86
	No. Specimens	20		20		5	
	No. Batches	4		4		1	
	Approval Class	Interim		Interim		Screening	
E_2^c (3) (Msi)	Mean	8.3	8.6	8.4	8.8	8.8	9.0
	Minimum	7.4	7.7	7.5	7.7	7.9	8.1
	Maximum	9.3	9.5	9.0	9.4	10.2	10.6
	C.V. (%)	7.0	6.6	5.1	5.5	8.4	8.9
	No. Specimens	20		20		20	
	No. Batches	4		4		4	
	Approval Class	Interim		Interim		Interim	
ν_{21}^c	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_2^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.
(2) Tab thickness of 0.112 - 0.120 inch is larger than 0.070 inch nominal thickness per method.
(3) Specimen thickness of 0.09 - 0.10 inch is less than nominal 0.120 inch thickness per method.

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Table 4.2.6(h)
C/Ep 194-PW
T-300/934
Compression, 2-axis
$[90_t]_{12}$
160/W, 250/W
Interim

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt%
FIBER VOLUME: 58 - 60 %
PLY THICKNESS: 0.0074 - 0.0082 in.

COMP. DENSITY: 1.54 - 1.57 g/cm³
VOID CONTENT: <0.5 - 1.2 %

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		160 wet		250 wet		
Moisture Content(%)		(1)		(1)		
Equilibrium at T, RH		12		12		
Source Code						
		Normalized	Measured	Normalized	Measured	
F_2^{cu} (3) (ksi)	Mean	75	77	46	47	
	Minimum	63	66	38	39	
	Maximum	81	83	59	60	
	C.V. (%)	7.2	6.5	11	11	
	B-value	(2)	(2)	(2)	(2)	
	Distribution	ANOVA	ANOVA	ANOVA	ANOVA	
	C_1	6.0	5.4	5.9	5.8	
	C_2	5.0	4.7	5.1	5.0	
	No. Specimens	15		15		
	No. Batches	3		3		
	Approval Class	Interim		Interim		
E_2^c (Msi)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	No. Specimens					
	No. Batches					
	Approval Class					
ν_{21}^c	Mean					
	No. Specimens					
	No. Batches					
	Approval Class					
ϵ_2^{cu} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) Immersed in water at 160°F for 14 days.
(2) B-values are presented only for fully approved data.
(3) Tab thickness of 0.112 - 0.120 inch is larger then 0.070 nominal thickness per method.

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Table 4.2.6(i)
C/Ep 194-PW
T-300/934
SBS, 31-plane
$[0_t]_{12}$
75/A, -65/A, 250/A
Screening

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT:	33 - 35 wt%	COMP. DENSITY:	1.54 - 1.57 g/cm^3
FIBER VOLUME:	58 - 60 %	VOID CONTENT:	<0.5 - 1.2 %
PLY THICKNESS:	0.0074 - 0.0082 in.		

TEST METHOD: ASTM D 2344-68 (1)

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Not normalized

Temperature(°F)		75 ambient	-65 ambient	250 ambient		
Moisture Content(%)						
Equilibrium at T, RH						
Source Code		12	12	12		
	Mean	12.0	11.9	9.2		
	Minimum	10.5	10.0	9.1		
	Maximum	13.4	13.9	9.5		
	C.V.(%)	6.89	8.38	2.1		
F_{31}^{sbs}	B-value	(2)	(2)	(2)		
	Distribution	ANOVA	ANOVA	Normal		
(ksi)	C_1	1.07	0.901	9.2		
	C_2	3.41	3.71	0.20		
	No. Specimens	20	20	5		
	No. Batches	4	4	1		
	Approval Class	Screening	Screening	Screening		

(1) Length-to-thickness ratio is approximately 11.
(2) B-values are presented only for fully approved data.

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4.2.7 Celion 12k/938 unidirectional tape*

MATERIAL:	Celion 12k/938 unidirectional tape	C/Ep Celion 12k/938 Summary
PREPREG:	Fiberite HY-E 1638N	
FIBER:	Celanese Celion 12k, EP06, no twist	MATRIX: Fiberite 938
T_g(ambient):	T_g(wet):	T_g METHOD:
PROCESSING:	Autoclave cure: 355 ± 10°F, 120 - 130 min., 85 - 100 psig	

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	5/85	Date of testing	7/85
Date of resin manufacture		Date of data submittal	6/88
Date of prepreg manufacture		Date of analysis	1/93
Date of composite manufacture			

LAMINA PROPERTY SUMMARY

	75°F/A		-67°F/A	250°F/A		180°F/W		
Tension, 1-axis	IIII		SSSS	IISI		IISI		
Tension, 2-axis	II-I		II-I	SS-S		II-I		
Tension, 3-axis								
Compression, 1-axis	II--		II--	II--		II--		
Compression, 2-axis	II--		II--	SI--		I---		
Compression, 3-axis								
Shear, 12-plane	I---		S---	S---		I---		
Shear, 23-plane								
SB Strength, 31-plane	I---							

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.78		
Resin Density	(g/cm^3)	1.30		
Composite Density	(g/cm^3)		1.54 - 1.61	
Fiber Areal Weight	(g/m^2)	145	144 - 147	
Ply Thickness	(in)		0.0040 - 0.0073	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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Table 4.2.7(a)
C/Ep 145-UT
Celion 12k/938
Tension, 1-axis
$[0]_7$
75/A, -67/A, 250/A
Interim, Screening

MATERIAL: Celion 12k/938 unidirectional tape

RESIN CONTENT: 28 - 36 wt%
FIBER VOLUME: 56 - 65 %
PLY THICKNESS: 0.0040 - 0.0063 in.

COMP. DENSITY: 1.55 - 1.61 g/cm^3
VOID CONTENT: <1.1 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Secant at 25% of typical ultimate load

NORMALIZED BY: Fiber volume 60%

Temperature(°F)		75 ambient		-67 ambient		250 ambient	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		12		12		12	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	273	271	262	278	309	319
	Minimum	223	207	235	254	295	306
	Maximum	324	319	290	303	328	337
	C.V.(%)	7.56	9.76	7.67	6.25	3.00	2.82
F_1^{tu}	B-value	(1)	(1)	(1)	(1)	(1)	(1)
	Distribution	ANOVA	ANOVA	ANOVA	ANOVA	Weibull	Weibull
(ksi)	C_1	21.0	29.3	25.1	20.9	314	323
	C_2	2.42	4.36	18.0	16.2	34.5	36.1
	No. Specimens	102		10		15	
	No. Batches	3		2		3	
	Approval Class	Interim		Screening		Interim	
	Mean	19.7	19.5	19.0	20.2	20.1	20.7
	Minimum	16.9	16.5	17.3	18.1	16.9	17.9
	Maximum	23.1	21.8	20.3	22.0	23.4	23.4
E_1^t	C.V.(%)	5.22	5.59	4.94	5.94	9.12	7.49
(Msi)	No. Specimens	102		10		15	
	No. Batches	3		2		3	
	Approval Class	Interim		Screening		Interim	
	Mean	0.317		0.279		0.280	
ν_{12}^t (2)	No. Specimens	102		10		10	
	No. Batches	3		2		2	
	Approval Class	Inteirm		Screening		Interim	
	Mean	13100		12800		14800	
	Minimum	10600		11500		12900	
	Maximum	14800		14000		16100	
	C.V.(%)	6.95		6.72		5.81	
ϵ_1^{tu}	B-value	(1)		(1)		(1)	
	Distribution	ANOVA		ANOVA		Weibull	
($\mu\epsilon$)	C_1	946		1060		15100	
	C_2	3.14		17.2		21.4	
	No. Specimens	102		10		15	
	No. Batches	3		2		3	
	Approval Class	Interim		Screening		Interim	

(1) B-values are presented only for fully approved data.
(2) Poisson's ratio measured at 25% of typical ultimate load.

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Table 4.2.7(b)
C/Ep 145-UT
Celion 12k/938
Tension, 1-axis
[0]$_7$
180/1.1%
Interim, Screening

MATERIAL: Celion 12k/938 unidirectional tape

RESIN CONTENT: 28 - 36 wt%
FIBER VOLUME: 56 - 64 %
PLY THICKNESS: 0.0044 - 0.0063 in.
COMP. DENSITY: 1.55 - 1.59 g/cm^3
VOID CONTENT: <1.4 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Secant at 25% of typical ultimate load

NORMALIZED BY: Fiber volume fraction to 60%

Temperature(°F)		180					
Moisture Content(%)		1.1					
Equilibrium at T, RH		(1)					
Source Code		12					
		Normalized	Measured				
	Mean	277	282				
	Minimum	236	219				
	Maximum	307	328				
	C.V.(%)	8.89	14.3				
F_1^{tu}	B-value	(3)	(3)				
	Distribution	ANOVA	ANOVA				
(ksi)	C_1	27.7	46.7				
	C_2	5.36	5.89				
	No. Specimens	15					
	No. Batches	3					
	Approval Class	Interim					
	Mean	18.9	19.2				
	Minimum	17.7	16.4				
	Maximum	20.5	21.9				
E_1^t	C.V.(%)	4.81	9.74				
(Msi)	No. Specimens	15					
	No. Batches	3					
	Approval Class	Interim					
	Mean	0.345					
ν_{12}^t (2)	No. Specimens	14					
	No. Batches	3					
	Approval Class	Screening					
	Mean	14000					
	Minimum	11800					
	Maximum	15700					
	C.V.(%)	8.13					
ϵ_1^{tu}	B-value	(3)					
	Distribution	ANOVA					
($\mu\epsilon$)	C_1	1180					
	C_2	3.36					
	No. Specimens	15					
	No. Batches	3					
	Approval Class	Interim					

(1) Conditioned at 160°F, 88% R.H. until weight gain was between 1.0 and 1.2%.
(2) Poisson's ratio measured at 25% of typical ultimate load.
(3) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.7(c)
C/Ep 145-UT
Celion 12k/938
Tension, 2-axis
$[90]_{20}$
75/A, -67/A, 250/A, 180/1.1%
Interim, Screening

MATERIAL: Celion 12k/938 unidirectional tape

RESIN CONTENT: 32 - 37 wt%
FIBER VOLUME: 55 - 60 %
PLY THICKNESS: 0.0053 - 0.0064 in.
COMP. DENSITY: 1.55 - 1.58 g/cm³
VOID CONTENT: <1.3 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Secant at 25% of typical ultimate load

NORMALIZED BY: Not normalized

Temperature(°F)		75	-67	250	180		
Moisture Content(%)		ambient	ambient	ambient	1.1		
Equilibrium at T, RH					(1)		
Source Code		12	12	12	12		
	Mean	9.6	9.5	8.8	5.8		
	Minimum	7.5	8.5	7.1	5.0		
	Maximum	13.9	10.4	10.7	6.6		
	C.V.(%)	13	6.6	11	8.4		
F_2^{tu}	B-value	(2)	(2)	(2)	(2)		
	Distribution	ANOVA	Weibull	Weibull	ANOVA		
(ksi)	C_1	1.3	9.8	9.2	0.54		
	C_2	2.7	18	10.	5.1		
	No. Specimens	101	15	10	15		
	No. Batches	3	3	2	3		
	Approval Class	Interim	Interim	Screening	Interim		
	Mean	1.35	1.35	1.22	1.19		
	Minimum	1.14	1.25	0.94	1.03		
	Maximum	1.82	1.51	1.52	1.36		
E_2^t	C.V.(%)	9.29	4.96	12.5	8.65		
(Msi)	No. Specimens	101	15	10	15		
	No. Batches	3	3	2	3		
	Approval Class	Interim	Interim	Screening	Interim		
	Mean						
ν_{21}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean	7200	6700	7600	4900		
	Minimum	1300	5500	6900	4200		
	Maximum	9500	7900	9300	5800		
	C.V.(%)	15	9.2	9.5	8.6		
ϵ_2^{tu}	B-value	(2)	(2)	(2)	(2)		
	Distribution	Nonpara.	Weibull	Normal	Weibull		
$(\mu\epsilon)$	C_1	5	7000	7600	5100		
	C_2		12	720	12		
	No. Specimens	97	15	10	15		
	No. Batches	3	3	2	3		
	Approval Class	Interim	Interim	Screening	Interim		

(1) Conditioned at 160°F, 88% R.H. until weight gain was between 1.0 and 1.2%.
(2) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.7(d)
C/Ep 145-UT
Celion 12k/938
Compression, 1-axis
$[0]_7$
75/A, -67/A, 250/A
Interim

MATERIAL: Celion 12k/938 unidirectional tape

RESIN CONTENT: 26 - 35 wt% COMP. DENSITY: 1.56 - 1.61 g/cm³
FIBER VOLUME: 57 - 67 % VOID CONTENT: <1.5 %
PLY THICKNESS: 0.0046 - 0.0073 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Chord modulus between 20% and 40% of typical ultimate load

NORMALIZED BY: Fiber volume fraction to 60%

Temperature(°F)		75 ambient		-67 ambient		250 ambient	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		12		12		12	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	201	198	240	240	195	201
	Minimum	166	172	204	216	180	179
	Maximum	255	246	286	276	214	229
	C.V.(%)	9.88	8.99	11.3	8.25	5.48	7.26
F_1^{cu}	B-value	(1)	(1)	(1)	(1)	(1)	(1)
	Distribution	ANOVA	ANOVA	ANOVA	ANOVA	ANOVA	ANOVA
(ksi)	C_1	21.4	18.7	31.1	21.9	11.9	16.7
	C_2	3.93	3.35	5.59	4.97	5.07	5.59
	No. Specimens	102		15		15	
	No. Batches	3		3		3	
	Approval Class	Interim		Interim		Interim	
	Mean	17.2	18.2	18.8	19.1	18.1	18.1
	Minimum	14.7	15.0	16.6	16.6	17.1	16.3
	Maximum	21.0	21.5	21.7	22.5	19.1	20.3
E_1^c	C.V.(%)	6.87	7.64	7.14	9.74	3.73	7.07
(Msi)	No. Specimens	97		15		15	
	No. Batches	3		3		3	
	Approval Class	Interim		Interim		Interim	
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
ϵ_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.7(e)
C/Ep 145-UT
Celion 12k/938
Compression, 1-axis
$[0]_7$
180/1.1%
Interim

MATERIAL: Celion 12k/938 unidirectional tape

RESIN CONTENT: 28 - 34 wt% COMP. DENSITY: 1.58 - 1.60 g/cm³
FIBER VOLUME: 58 - 65 % VOID CONTENT: <1.0 %
PLY THICKNESS: 0.0044 - 0.0073 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Chord modulus between 20% and 40% of typical ultimate load

NORMALIZED BY: Fiber volume fraction to 60%

Temperature(°F)		180					
Moisture Content(%)		1.1					
Equilibrium at T, RH		(1)					
Source Code		12					
		Normalized	Measured				
	Mean	185	188				
	Minimum	157	160				
	Maximum	206	217				
	C.V.(%)	7.40	7.55				
F_1^{cu} (ksi)	B-value	(2)	(2)				
	Distribution	Weibull	Weibull				
	C_1	191	194				
	C_2	16.3	14.4				
	No. Specimens	15					
	No. Batches	3					
	Approval Class	Interim					
E_1^c (Msi)	Mean	18.2	19.2				
	Minimum	15.7	15.8				
	Maximum	22.3	23.7				
	C.V.(%)	8.88	10.5				
	No. Specimens	15					
	No. Batches	3					
	Approval Class	Interim					
ν_{12}^c	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_1^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned at 160°F, 88% R.H. until weight gain was between 1.0 and 1.2%.
(2) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.7(f)
C/Ep 145-UT
Celion 12k/938
Shear, 12-plane
$[\pm 45]_{2s}$
75/A, -65/A, 250/A, 180/1.1%
Interim, Screening

MATERIAL: Celion 12k/938 unidirectional tape

RESIN CONTENT: 28 - 34 wt%
FIBER VOLUME: 58 - 65 %
PLY THICKNESS: 0.0044 - 0.0064 in.
COMP. DENSITY: 1.57 - 1.61 g/cm³
VOID CONTENT: <1.4 %

TEST METHOD: ASTM D 3518-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75	-67	250	180	
Moisture Content(%)		ambient	ambient	ambient	1.1	
Equilibrium at T, RH					(1)	
Source Code		12	12	12	12	
F_{12}^{su} (ksi)	Mean	14	16	14	14	
	Minimum	11	14	13	13	
	Maximum	16	18	15	14	
	C.V.(%)	7.3	10.	6.1	3.6	
	B-value	(2)	(2)	(2)	(2)	
	Distribution	ANOVA	ANOVA	Weibull	ANOVA	
	C_1	1.1	1.8	14	0.53	
	C_2	4.4	5.8	19	4.6	
	No. Specimens	102	14	14	15	
	No. Batches	3	3	3	3	
	Approval Class	Interim	Screening	Screening	Interim	
G_{12}^{s} (Msi)	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
	No. Specimens					
	No. Batches					
	Approval Class					
γ_{12}^{su} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) Conditioned at 160°F, 88% R.H. until weight gain was between 1.0 and 1.2%.
(2) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.7(g)
C/Ep 145-UT
Celion 12k/938
SBS, 31-plane
$[0]_{14}$
75/A
Screening

MATERIAL: Celion 12k/938 unidirectional tape

RESIN CONTENT: 31 - 40 wt%
FIBER VOLUME: 52 - 62 %
PLY THICKNESS: 0.0051 - 0.0064 in.

COMP. DENSITY: 1.54 - 1.59 g/cm^3
VOID CONTENT: <1.0 %

TEST METHOD: ASTM D 2344-68

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75				
Moisture Content(%)		ambient				
Equilibrium at T, RH						
Source Code		12				
	Mean	18.3				
	Minimum	16.6				
	Maximum	19.7				
	C.V.(%)	3.29				
F_{31}^{sbs}	B-value	(1)				
(ksi)	Distribution	ANOVA				
	C_1	0.619				
	C_2	2.76				
	No. Specimens	102				
	No. Batches	3				
	Approval Class	Screening				

(1) B-values are presented only for fully approved data.

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4.2.8 AS4 12k/3502 unidirectional tape*

MATERIAL:	AS4 12k/3502 unidirectional tape		C/Ep AS4/3502 Summary
PREPREG:	Hercules AS4/3502 unidirectional tape		
FIBER:	Hercules AS4 12k, surface-treated, 0 twist	MATRIX: Hercules 3502	
T_g(ambient):	407°F T_g(wet): °F	T_g METHOD: TMA	
PROCESSING:	Autoclave cure: 280 ± 5°F, 90 min, 85 + 15-0 psi; 350°F, 120 min.		

* Additional data set found on p. 70.

Date of fiber manufacture	4/83 - 6/83	Date of testing	11/83-7/84
Date of resin manufacture	6/83	Date of data submittal	12/93, 5/94
Date of prepreg manufacture	6/83 - 7/83	Date of analysis	8/94
Date of composite manufacture	8/83 - 5/84		

LAMINA PROPERTY SUMMARY

	75°F/A		-65°F/A		180°F/W	250°F/W		
Tension, 1-axis	FF--		FF--		FF--	FF--		
Tension, 2-axis	FF--		FF--		FF--	FF--		
Tension, 3-axis								
Compression, 1-axis	FF--		II--		FF--	FF--		
Compression, 2-axis	FF--		II--		FF--	FF--		
Compression, 3-axis								
Shear, 12-plane	FF--		II--		FF--	II--		
Shear, 23-plane								
Shear, 31-plane								

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.79	1.77 - 1.80	
Resin Density	(g/cm^3)	1.26	1.24 - 1.29	
Composite Density	(g/cm^3)	1.57	1.56 - 1.59	
Fiber Areal Weight	(g/m^2)	147	146 - 150	
Ply Thickness	(in)	0.0055	0.0049 - 0.0061	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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Table 4.2.8(a)
C/Ep 147 - UT
AS4/3502
Tension, 1-axis
$[0]_8$
75/A, -65/A, 180/W
Fully Approved

MATERIAL: AS4 12k/3502 unidirectional tape

RESIN CONTENT: 30 - 33 wt% COMP. DENSITY: 1.56-1.59 g/cm³
FIBER VOLUME: 59 - 61 % VOID CONTENT: 0.0 - 1.0 %
PLY THICKNESS: 0.0049 - 0.0061 in.

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Ply thickness to 0.0055 in. (59%)

Temperature(°F)		75 ambient		-65 ambient		180 1.1 - 1.3 (1)	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		49		49		49	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	258	253	231	227	261	255
	Minimum	191	186	162	151	140	135
	Maximum	317	302	285	280	317	315
	C.V. (%)	9.83	9.13	13.4	13.6	14.8	15.2
F_1^{tu}	B-value	205		173		200	
	Distribution	Weibull		Weibull		Weibull	
(ksi)	C_1	269		244		276	
	C_2	11.2		8.82		9.39	
	No. Specimens	36		38		40	
	No. Batches	5		5		5	
	Approval Class	Fully Approved		Fully Approved		Fully Approved	
	Mean	19.3		19.2		19.7	
	Minimum	15.6		16.8		15.1	
	Maximum	21.0		23.2		23.3	
E_1^t	C.V. (%)	5.74		6.31		6.87	
(Msi)	No. Specimens	36		38		40	
	No. Batches	5		5		5	
	Approval Class	Fully Approved		Fully Approved		Fully Approved	
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned at 160°F, 95-100% relative humidity until the moisture content was between 1.1 and 1.3%.

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Table 4.2.8(b)
C/Ep 147 - UT
AS4/3502
Tension, 1-axis
$[0]_8$
250/W
Fully Approved

MATERIAL: AS4 12k/3502 unidirectional tape

RESIN CONTENT: 30 - 33 wt%
FIBER VOLUME: 59 - 61 %
PLY THICKNESS: 0.0055 - 0.0059 in.

COMP. DENSITY: 1.56-1.59 g/cm³
VOID CONTENT: 0.0 - 1.0 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Ply thickness to 0.0055 in. (59%)

Temperature(°F)		250					
Moisture Content(%)		1.1 - 1.3					
Equilibrium at T, RH		(1)					
Source Code		49					
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	256	250				
	Minimum	200	195				
	Maximum	301	305				
	C.V. (%)	9.39	10.2				
F_1^{tu}	B-value	191					
	Distribution	ANOVA					
(ksi)	C_1	25.0					
	C_2	2.61					
	No. Specimens	30					
	No. Batches	5					
	Approval Class	Fully Approved					
	Mean	20.1					
	Minimum	17.8					
	Maximum	23.9					
E_1^t	C.V. (%)	7.32					
(Msi)	No. Specimens	30					
	No. Batches	5					
	Approval Class	Fully Approved					
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{tu}	B-value						
	Distribution						
$(\mu\epsilon)$	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned at 160°F, 95-100% relative humidity until the moisture content was between 1.1 and 1.3%.

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Table 4.2.8(c)
C/Ep 147 - UT
AS4/3502
Tension, 2-axis
$[90]_{24}$
75/A, -65/A, 180/W, 250/W
Fully Approved

MATERIAL: AS4 12k/3502 unidirectional tape

RESIN CONTENT: 31 - 33 wt%
FIBER VOLUME: 59 - 60 %
PLY THICKNESS: 0.0052 - 0.0059 in.
COMP. DENSITY: 1.56-1.59 g/cm^3
VOID CONTENT: 0.0 - 1.0 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Not normalized

Temperature(°F)		75	-65	180	250	
Moisture Content(%)		ambient	ambient	1.1 - 1.3	1.1 - 1.3	
Equilibrium at T, RH				(1)	(1)	
Source Code		49	49	49	49	
	Mean	7.76	6.65	4.39	2.68	
	Minimum	6.26	2.48	3.52	2.13	
	Maximum	10.2	8.93	5.20	3.40	
	C.V. (%)	10.7	18.0	8.44	12.3	
F_2^t	B-value	6.28	4.57	3.46	1.65	
	Distribution	Normal	Weibull	ANOVA	ANOVA	
(ksi)	C_1	7.76	7.09	0.380	0.348	
	C_2	0.832	7.20	2.43	2.94	
	No. Specimens	30	30	30	30	
	No. Batches	5	5	5	5	
	Approval Class	Fully Approved	Fully Approved	Fully Approved	Fully Approved	
	Mean	1.35	1.44	1.21	0.958	
	Minimum	1.28	1.32	1.14	0.912	
	Maximum	1.49	1.58	1.35	1.06	
E_2^t	C.V. (%)	4.26	4.16	4.02	3.61	
(Msi)	No. Specimens	30	30	30	30	
	No. Batches	5	5	5	5	
	Approval Class	Fully Approved	Fully Approved	Fully Approved	Fully Approved	
	Mean					
ν_{21}^t	No. Specimens					
	No. Batches					
	Approval Class					
	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
ϵ_2^t	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) Conditioned at 160°F, 95-100% relative humidity until the moisture content was between 1.1 and 1.3%.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:29Z
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MATERIAL: AS4 12k/3502 unidirectional tape

RESIN CONTENT: 33 - 37 wt% COMP. DENSITY: 1.56-1.57 g/cm^3
FIBER VOLUME: 55 - 59 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0054 - 0.0060 in.

TEST METHOD: ASTM D 3410A-75

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Ply thickness to 0.0055 in. (59%)

Table 4.2.8(d)
C/Ep 147 - UT
AS4/3502
Compression, 1-axis
$[0]_{19}$
75/A, -65/A, 180/W
Fully Approved, Interim

		75 ambient 49		-65 ambient 49		180 1.1 - 1.3 (1) 49	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	204	198	233	225	176	171
	Minimum	168	160	207	201	146	137
	Maximum	226	221	252	251	200	197
	C.V. (%)	6.45	6.62	5.63	6.23	6.31	8.09
F_1^{cu} (ksi)	B-value	171		(2)		145	
	Distribution	ANOVA		Weibull		ANOVA	
	C_1	13.5		238		11.5	
	C_2	2.44		23.0		2.65	
	No. Specimens	30		15		30	
	No. Batches	5		5		5	
	Approval Class	Fully Approved		Interim		Fully Approved	
	Mean	18.0		18.8		18.6	
	Minimum	16.9		17.1		17.5	
	Maximum	19.4		20.5		20.0	
E_1^c	C.V. (%)	3.19		5.43		3.36	
(Msi)	No. Specimens	30		16		30	
	No. Batches	5		5		5	
	Approval Class	Fully Approved		Interim		Fully Approved	
	Mean						
ν_{21}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned at 160°F, 95-100% relative humidity until the moisture content was between 1.1 and 1.3%.
(2) B-values are presented only for fully approved data.

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Table 4.2.8(e)
C/Ep 147 - UT
AS4/3502
Compression, 1-axis
$[0]_{19}$
250/W
Fully Approved

MATERIAL: AS4 12k/3502 unidirectional tape

RESIN CONTENT: 33 - 37 wt% COMP. DENSITY: 1.56-1.57 g/cm^3
FIBER VOLUME: 55 - 59 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0054 - 0.0060 in.

TEST METHOD: ASTM D 3410A-75

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Ply thickness to 0.0055 in. (59%)

Temperature(°F)		250					
Moisture Content(%)		1.1 - 1.3					
Equilibrium at T, RH		(1)					
Source Code		49					
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{cu} (ksi)	Mean	147	143				
	Minimum	118	110				
	Maximum	170	163				
	C.V. (%)	9.42	9.40				
	B-value	119					
	Distribution	Weibull					
	C_1	153					
	C_2	12.5					
	No. Specimens	30					
	No. Batches	5					
	Approval Class	Fully Approved					
E_1^c (Msi)	Mean	18.7					
	Minimum	17.3					
	Maximum	20.6					
	C.V. (%)	3.99					
	No. Specimens	30					
	No. Batches	5					
	Approval Class	Fully Approved					
ν_{12}^c	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_1^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned at 160°F, 95-100% relative humidity until the moisture content was between 1.1 and 1.3%.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:29Z
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MATERIAL: AS4 12k/3502 unidirectional tape

RESIN CONTENT: 31 - 33 wt% COMP. DENSITY: 1.56-1.59 g/cm³
FIBER VOLUME: 59 - 60 % VOID CONTENT: 0.0 - 1.0 %
PLY THICKNESS: 0.0054 - 0.0058 in.

TEST METHOD: ASTM D695M (1)

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Not normalized

Table 4.2.8(f)
C/Ep 147 - UT
AS4/3502
Compression, 2-axis
$[90]_{24}$
75/A, -65/A, 180/W, 250/W
Fully Approved, Interim

Temperature(°F)		75	-65	180	250	
Moisture Content(%)		ambient	ambient	1.1 - 1.3	1.1 - 1.3	
Equilibrium at T, RH				(2)	(2)	
Source Code		49	49	49	49	
F_2^{cu} (ksi)	Mean	34.6	49.8	24.7	18.4	
	Minimum	27.5	42.5	23.0	17.0	
	Maximum	40.4	57.2	26.7	19.9	
	C.V. (%)	9.53	10.4	3.23	4.99	
	B-value	26.6	37.3	22.3	15.3	
	Distribution	ANOVA	Weibull	ANOVA	ANOVA	
	C_1	3.37	52.1	0.836	0.990	
	C_2	2.38	11.3	2.80	3.18	
	No. Specimens	30	15	30	30	
	No. Batches	5	5	5	5	
	Approval Class	Fully Approved	Interim	Fully Approved	Fully Approved	
E_2^c (Msi)	Mean	1.41	1.68	1.24	1.09	
	Minimum	1.29	1.57	1.14	0.973	
	Maximum	1.60	1.95	1.41	1.41	
	C.V. (%)	4.86	6.07	4.90	9.44	
	No. Specimens	30	15	30	30	
	No. Batches	5	5	5	5	
	Approval Class	Fully Approved	Interim	Fully Approved	Interim	
ν_{21}^c	Mean					
	No. Specimens					
	No. Batches					
	Approval Class					
ϵ_2^{cu} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) Tabbed specimen - length 3.12 inch, width 0.50 inch, gage length 0.50 inch.
(2) Conditioned at 160°F, 95-100% relative humidity until the moisture content was between 1.1 and 1.3%.

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Table 4.2.8(g)
C/Ep 147 - UT
AS4/3502
Shear, 12-plane
$[\pm 45]_{4s}$
75/A, -65/A, 180/W, 250/W
Fully Approved, Interim

MATERIAL: AS4 12k/3502 unidirectional tape

RESIN CONTENT: 31 - 33 wt% COMP. DENSITY: 1.56-1.59 g/cm^3
FIBER VOLUME: 59 - 60 % VOID CONTENT: 0.0 - 1.0 %
PLY THICKNESS: 0.0053 - 0.0059 in.

TEST METHOD: ASTM D 3518-76

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Not normalized

Temperature(°F)		75	-65	180	250	
Moisture Content(%)		ambient	ambient	1.1 - 1.3	1.1 - 1.3	
Equilibrium at T, RH				(1)	(1)	
Source Code		49	49	49	49	
F_{12}^{su} (ksi)	Mean	14.8	15.3	13.5	11.5	
	Minimum	13.7	13.3	12.5	10.5	
	Maximum	15.8	16.2	14.1	12.4	
	C.V. (%)	3.18	4.58	3.39	4.27	
	B-value	13.4	(2)	11.8	10.3	
	Distribution	ANOVA	ANOVA	ANOVA	ANOVA	
	C_1	0.503	0.706	0.502	0.503	
	C_2	2.91	2.04	3.24	2.32	
	No. Specimens	36	23	37	42	
	No. Batches	5	5	5	5	
	Approval Class	Fully Approved	Interim	Fully Approved	Fully Approved	
G_{12}^{s} (Msi)	Mean	0.543	0.769	0.217	0.141	
	Minimum	0.496	0.738	0.169	0.103	
	Maximum	0.593	0.863	0.260	0.205	
	C.V. (%)	5.16	3.69	9.25	17.9	
	No. Specimens	33	23	33	41	
	No. Batches	5	5	5	5	
	Approval Class	Fully Approved	Interim	Fully Approved	Fully Approved	
γ_{12}^{su} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) Conditioned at 160°F, 95-100% relative humidity until the moisture content was between 1.1 and 1.3%
(2) B-values are presented only for fully approved data.

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:29Z
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MATERIAL:	AS4 12k/3502 unidirectional tape			C/Ep AS4/3502 Summary
PREPREG:	Hercules AS4/3502 unidirectional tape			
FIBER:	Hercules AS4 12k, surface-treated	MATRIX:	Hercules 3502	
T_g(ambient):	460°F T_g(wet):	T_g METHOD:	TMA	
PROCESSING:	Autoclave cure: 275°F, 45 min., 350°F, 2 hours, 85 psig; Postcure: 400°F, 4 hours			

* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	12/80 - 2/82	Date of testing	
Date of resin manufacture		Date of data submittal	6/90
Date of prepreg manufacture	12/80 - 2/82	Date of analysis	1/93
Date of composite manufacture			

LAMINA PROPERTY SUMMARY

	75°F/A		-65°F/A	265°F/A		75°F/W	265°F/W	
Tension, 1-axis	IIII			IIII			IIII	
Tension, 2-axis	II-I					II-I	II-I	
Tension, 3-axis								
Compression, 1-axis			II-I	II-I			II-I	
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
Shear, 31-plane								

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.79	1.78 - 1.81	
Resin Density	(g/cm^3)	1.26		
Composite Density	(g/cm^3)	1.58		
Fiber Areal Weight	(g/m^2)			
Ply Thickness	(in)		0.0047 - 0.0062	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.8(h)
C/Ep 147 -UT
AS4/3502
Tension, 1-axis
$[0]_6$
75/A, 265/A, 265/W
Interim

MATERIAL: AS4 12k/3502 unidirectional tape

RESIN CONTENT: 25 - 29 wt% COMP. DENSITY: 1.59 - 1.62 g/cm³
FIBER VOLUME: 63 - 68 % VOID CONTENT:
PLY THICKNESS: 0.0055 - 0.0058 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		75 ambient		265 ambient		265 wet	
Moisture Content(%)		(1)		(1)		(2)	
Equilibrium at T, RH							
Source Code		26		26		26	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{tu} (ksi)	Mean	253	275	269	292	251	273
	Minimum	212	226	148	165	183	196
	Maximum	294	323	314	358	287	315
	C.V.(%)	8.35	9.49	15.2	16.5	9.09	10.4
	B-value	(3)	(3)	(3)	(3)	(3)	(3)
	Distribution	ANOVA	ANOVA	ANOVA	ANOVA	ANOVA	ANOVA
	C_1	21.5	27.2	24.0	30.2	24.0	30.2
	C_2	2.20	2.60	2.83	3.01	2.83	3.01
	No. Specimens	30		20		25	
	No. Batches	5		4		5	
	Approval Class	Interim		Interim		Interim	
E_1^t (Msi)	Mean	18.7	20.4	18.4	20.0	19.0	20.6
	Minimum	17.3	18.9	17.4	19.1	18.0	19.2
	Maximum	20.2	22.2	19.7	20.8	19.7	22.1
	C.V.(%)	3.88	3.37	3.52	2.59	3.53	3.22
	No. Specimens	29		20		25	
	No. Batches	5		4		5	
	Approval Class	Interim		Interim		Interim	
ν_{12}^t	Mean	0.340		0.356		0.280	
	No. Specimens	30		20		25	
	No. Batches	5		4		5	
	Approval Class	Interim		Interim		Interim	
ϵ_1^{tu} ($\mu\epsilon$)	Mean	12400		13900		12400	
	Minimum	10200		10400		9220	
	Maximum	14400		15700		13900	
	C.V.(%)	8.65		12.0		8.95	
	B-value	(3)		(3)		(3)	
	Distribution	ANOVA		ANOVA		ANOVA	
	C_1	1120		1850		1170	
	C_2	2.62		3.92		2.87	
	No. Specimens	30		20		25	
	No. Batches	5		4		5	
	Approval Class	Interim		Interim		Interim	

(1) Conditioned at 180°F, ambient relative humidity for 2 days
(2) Conditioned at 180°F, 75% relative humidity for 10 days
(3) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.8(i)
C/Ep 147 -UT
AS4/3502
Tension, 2-axis
$[90]_{16}$
75/A, 75/W, 265/W
Interim

MATERIAL: AS4 12k/3502 unidirectional tape

RESIN CONTENT: 25 - 29 wt% COMP. DENSITY: 1.59 - 1.62 g/cm³
FIBER VOLUME: 63 - 68 % VOID CONTENT:
PLY THICKNESS: 0.0055 - 0.0059 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Not normalized

Temperature(°F)		75	75	265		
Moisture Content(%)		ambient	wet	wet		
Equilibrium at T, RH		(1)	(2)	(2)		
Source Code		26	26	26		
	Mean	8.04	3.27	3.29		
	Minimum	5.93	2.54	2.62		
	Maximum	10.6	4.15	4.15		
	C.V.(%)	13.5	16.3	13.0		
F_2^{tu}	B-value	(3)	(3)	(3)		
	Distribution	ANOVA	ANOVA	ANOVA		
(ksi)	C_1	1.11	0.560	0.452		
	C_2	2.36	3.79	3.16		
	No. Specimens	30	15	20		
	No. Batches	5	3	4		
	Approval Class	Interim	Interim	Interim		
	Mean	1.50	1.04	1.04		
	Minimum	1.43	0.95	0.95		
	Maximum	1.58	1.10	1.10		
E_2^t	C.V.(%)	2.76	5.1	4.3		
(Msi)	No. Specimens	30	15	20		
	No. Batches	5	3	4		
	Approval Class	Interim	Interim	Interim		
	Mean					
ν_{21}^t	No. Specimens					
	No. Batches					
	Approval Class					
	Mean	5500	3320	3440		
	Minimum	4000	2750	2840		
	Maximum	7390	4200	4200		
	C.V.(%)	13.7	13.3	12.1		
ϵ_2^{tu}	B-value	(3)	(3)	(3)		
	Distribution	Weibull	ANOVA	ANOVA		
($\mu\epsilon$)	C_1	5820	506	456		
	C_2	7.67	5.66	3.79		
	No. Specimens	30	15	20		
	No. Batches	5	3	4		
	Approval Class	Interim	Interim	Interim		

(1) Conditioned at 180°F, ambient relative humidity for 2 days
(2) Conditioned at 180°F, 75% relative humidity for 63 days
(3) B-values are presented only for fully approved data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:29Z
Check the source to verify that this is the current version before use.

* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.8(j)
C/Ep 147 -UT
AS4/3502
Compression, 1-axis
$[0]_6$
-65/A, 265/A, 265/W
Interim

MATERIAL: AS4 12k/3502 unidirectional tape

RESIN CONTENT: 25 - 29 wt%
FIBER VOLUME: 63 - 68 %
PLY THICKNESS: 0.0047 - 0.0062 in.

COMP. DENSITY: 1.59 - 1.62 g/cm³
VOID CONTENT:

TEST METHOD: ASTM D 3410C

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		-65		265		265	
Moisture Content(%)		ambient		ambient		wet	
Equilibrium at T, RH		(1)		(1)		(2)	
Source Code		26		26		26	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	226	253	228	249	176	192
	Minimum	173	206	142	150.	139	146
	Maximum	307	325	275	292	208	228
	C.V.(%)	16.8	14.1	15.0	15.1	11.5	13.3
F_1^{cu}	B-value	(3)	(3)	(3)	(3)	(3)	(3)
	Distribution	Weibull	Weibull	Weibull	Weibull	Weibull	Weibull
(ksi)	C_1	242	269	241	264	184	203
	C_2	6.23	7.45	8.66	9.19	10.6	9.32
	No. Specimens	15		15		15	
	No. Batches	3		3		3	
	Approval Class	Interim		Interim		Interim	
	Mean	19.3	21.1	21.2	23.2	19.6	21.4
	Minimum	17.1	19.3	17.1	19.3	18.5	20.5
	Maximum	21.8	23.7	23.1	26.3	20.6	22.5
E_1^c	C.V.(%)	6.63	7.30	9.53	9.70	3.85	3.70
(Msi)	No. Specimens	15		15		15	
	No. Batches	3		3		3	
	Approval Class	Interim		Interim		Interim	
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		16200		13400		10500
	Minimum		11100		7370		7770
	Maximum		21200		16000		12800
	C.V.(%)		17.4		16.2		14.1
ϵ_1^{cu}	B-value		(3)		(3)		(3)
	Distribution		Weibull		Weibull		Weibull
($\mu\epsilon$)	C_1		17400		14200		11100
	C_2		6.39		8.53		8.71
	No. Specimens	15		15		15	
	No. Batches	3		3		3	
	Approval Class	Interim		Interim		Interim	

(1) Conditioned at 180°F, ambient relative humidity for 2 days
(2) Conditioned at 150°F, 98% relative humidity for 14 days
(3) B-values are presented only for fully approved data.

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4.2.9 Celion 3000/E7K8 plain weave*

MATERIAL:	Celion 3000/E7K8 plain weave	**C/Ep Celion 3000/E7K8 Summary**
PREPREG:	U.S. Polymeric Celion 3000/E7K8 Plain Weave, Grade 195	
FIBER:	Celanese Celion 3000 MATRIX: U.S. Polymeric E7K8	
T_g(ambient):	T_g(wet): T_g METHOD:	
PROCESSING:	Autoclave: 310°F, 2 hours, 85 psig	

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	1/88
Date of prepreg manufacture	2/86 - 3/86	Date of analysis	1/93
Date of composite manufacture			

LAMINA PROPERTY SUMMARY

	75°F/A		-65°F/A	180°F/A		75°F/W	180°F/W	
Tension, 1-axis	SS-S		SS--			SSSS	SSS-	
Tension, 2-axis	SS-S		SS-S			SS-S	SS-S	
Tension, 3-axis								
Compression, 1-axis	SS-S		SS-S	SS-S		SS-S	SS-S	
Compression, 2-axis	SS-S		SS--	SS--		SS-S	SS--	
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
SB Strength, 31-plane	S---		S---	S---		S---	S---	

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.8		
Resin Density	(g/cm^3)	1.28		
Composite Density	(g/cm^3)	1.54	1.37 - 1.55	
Fiber Areal Weight	(g/m^2)	195		
Ply Thickness	(in)	0.0075	0.0078 - 0.011	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(a)
C/Ep 195-PW
Celion 3000/E7K8
Tension, 1-axis
$[0_f]_{10}$
75/A, -65/A
Screening

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 37 - 38 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 55 - 56 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0078 - 0.0085 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75 ambient		-65 ambient		
Moisture Content(%)						
Equilibrium at T, RH		20		20		
Source Code						
		Normalized	Measured	Normalized	Measured	
	Mean	132	128	110	106	
	Minimum	120	115	101	98.4	
	Maximum	143	140	118	113	
	C.V.(%)	4.7	5.8	6.2	5.4	
F_1^{tu}	B-value	(1)	(1)	(1)	(1)	
	Distribution	Weibull	Weibull	Normal	Normal	
(ksi)	C_1	135	132	110	106	
	C_2	25.7	21.4	6.88	5.74	
	No. Specimens	20		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		
	Mean	9.67	9.38	9.98	9.66	
	Minimum	9.49	8.85	9.82	9.46	
	Maximum	9.98	9.74	10.0	9.90	
E_1^t	C.V.(%)	1.2	2.5	1.0	1.8	
(Msi)	No. Specimens	20		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		
	Mean	0.0580				
ν_{12}^t	No. Specimens	5				
	No. Batches	1				
	Approval Class	Screening		Screening		
	Mean	13700		11000		
	Minimum	12300		10200		
	Maximum	14800		11600		
	C.V.(%)	4.5		5.4		
ϵ_1^{tu}	B-value	(1)		(1)		
	Distribution	Weibull		Normal		
($\mu\epsilon$)	C_1	14000		11000		
	C_2	26.8		592		
	No. Specimens	20		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(b)
C/Ep 195-PW
Celion 3000/E7K8
Tension, 1-axis
$[0_f]_{10}$
75/W, 180/W
Screening

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 37 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 55 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0078 - 0.0081 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)			75		180		
Moisture Content(%)			wet		wet		
Equilibrium at T, RH			(1)		(1)		
Source Code			20		20		
			Normalized	Measured	Normalized	Measured	
		Mean	125	122	123	120	
		Minimum	111	105	114	112	
		Maximum	130	129	131	127	
		C.V.(%)	6.3	8.1	6.5	6.3	
F_1^{tu}		B-value	(2)	(2)	(2)	(2)	
		Distribution	Normal	Normal	Normal	Normal	
(ksi)		C_1	125	122	123	120	
		C_2	7.93	9.93	7.99	7.52	
		No. Specimens	5		5		
		No. Batches	1		1		
		Approval Class	Screening		Screening		
		Mean	9.23	9.01	9.55	9.33	
		Minimum	8.93	8.81	9.37	9.15	
		Maximum	9.53	9.20	9.84	9.63	
E_1^t		C.V.(%)	2.5	1.7	1.9	2.0	
(Msi)		No. Specimens	5		5		
		No. Batches	1		1		
		Approval Class	Screening		Screening		
		Mean		0.0620		0.0560	
ν_{12}^t		No. Specimens	5		5		
		No. Batches	1		1		
		Approval Class	Screening		Screening		
		Mean		13700		12800	
		Minimum		12100		11200	
		Maximum		14300		14100	
		C.V.(%)		6.9		9.6	
ϵ_1^{tu}		B-value		(2)		(2)	
		Distribution		Normal		Normal	
($\mu\epsilon$)		C_1		13700		12800	
		C_2		939		1230	
		No. Specimens	5		5		
		No. Batches	1		1		
		Approval Class	Screening		Screening		

(1) Conditioned at 160°F, 85% relative humidity for 7 days
(2) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(c)
C/Ep 195-PW
Celion 3000/E7K8
Tension, 1-axis
$[0_t]_{12}$
75/A, -65/A
Screening

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 39 - 44 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 51 - 54 % VOID CONTENT: 0.04 - 0.5 %
PLY THICKNESS: 0.0079 - 0.0084 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)			75		-65		
Moisture Content(%)			ambient		ambient		
Equilibrium at T, RH							
Source Code			20		20		
			Normalized	Measured	Normalized	Measured	
		Mean	132	122	122	115	
		Minimum	106	100	117	111	
		Maximum	147	136	126	123	
		C.V.(%)	7.5	7.5	2.8	4.3	
F_1^{tu}		B-value	(1)	(1)	(1)	(1)	
		Distribution	Weibull	Weibull	Normal	Normal	
(ksi)		C_1	136	126	122	116	
		C_2	16.4	17.3	3.44	4.97	
		No. Specimens	20		5		
		No. Batches	1		1		
		Approval Class	Screening		Screening		
		Mean	9.96	9.21	9.29	8.82	
		Minimum	9.30	8.74	8.95	8.51	
		Maximum	9.98	9.78	9.66	9.41	
E_1^t		C.V.(%)	1.2	2.5	2.8	4.0	
(Msi)		No. Specimens	20		5		
		No. Batches	1		1		
		Approval Class	Screening		Screening		
		Mean					
ν_{12}^t		No. Specimens					
		No. Batches					
		Approval Class					
		Mean		14100			
		Minimum		13600			
		Maximum		14600			
		C.V.(%)		2.6			
ϵ_1^{tu}		B-value		(1)			
		Distribution		Normal			
($\mu\epsilon$)		C_1		14100			
		C_2		371			
		No. Specimens		5			
		No. Batches		1			
		Approval Class		Screening			

(1) B-values are presented only for fully approved data.

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Table 4.2.9(d)
C/Ep 195-PW
Celion 3000/E7K8
Tension, 1-axis
$[0_f]_{12}$
75/W, 180/W
Screening

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 42 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 51 % VOID CONTENT: 0.48 %
PLY THICKNESS: 0.0081 - 0.0083 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		180		
Moisture Content(%)		wet		wet		
Equilibrium at T, RH		(1)		(1)		
Source Code		20		20		
		Normalized	Measured	Normalized	Measured	
	Mean	145	129	148	133	
	Minimum	143	125	139	124	
	Maximum	148	131	154	142	
	C.V.(%)	1.6	1.8	4.0	5.6	
F_1^{tu}	B-value	(2)	(2)	(2)	(2)	
	Distribution	Normal	Normal	Normal	Normal	
(ksi)	C_1	145	129	148	133	
	C_2	2.23	2.37	5.94	7.50	
	No. Specimens	5		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		
	Mean	10.6	9.42	10.3	9.21	
	Minimum	10.1	8.79	10.1	8.91	
	Maximum	11.4	10.0	10.5	9.53	
E_1^t	C.V.(%)	4.9	5.0	1.3	2.7	
(Msi)	No. Specimens	5		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		
	Mean		0.0560		0.0560	
ν_{12}^t	No. Specimens	5		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		
	Mean		13400			
	Minimum		12300			
	Maximum		14300			
	C.V.(%)		5.30			
ϵ_1^{tu}	B-value		(2)			
	Distribution		Normal			
($\mu\epsilon$)	C_1		13400			
	C_2		713			
	No. Specimens	5				
	No. Batches	1				
	Approval Class	Screening				

(1) Conditioned at 160°F, 85% relative humidity for 7 days
(2) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(e)
C/Ep 195-PW
Celion 3000/E7K8
Tension, 2-axis
$[90_t]_{10}$
75/A, -65/A
Screening

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 36 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 56 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0078 - 0.0084 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

			75 ambient 20		-65 ambient 20			
			Normalized	Measured	Normalized	Measured		
F_2^{tu} (ksi)		Mean	128	127	113	111		
		Minimum	120	115	101	100		
		Maximum	137	134	125	122		
		C.V.(%)	3.6	3.7	9.1	8.9		
		B-value	(1)	(1)	(1)	(1)		
		Distribution	Normal	Normal	Normal	Normal		
		C_1	128	127	113	111		
		C_2	4.64	4.69	10.3	9.89		
		No. Specimens	20		5			
		No. Batches	1		1			
		Approval Class	Screening		Screening			
E_2^t (Msi)		Mean	9.50	9.37	9.51	9.34		
		Minimum	9.36	9.04	9.29	9.20		
		Maximum	9.69	9.71	9.65	9.68		
		C.V.(%)	0.98	1.8	1.6	2.1		
		No. Specimens	20		5			
		No. Batches	1		1			
		Approval Class	Screening		Screening			
ν_{21}^t		Mean						
		No. Specimens						
		No. Batches						
		Approval Class						
ϵ_2^{tu} ($\mu\epsilon$)		Mean		13400		11700		
		Minimum		12600		10700		
		Maximum		14200		12700		
		C.V.(%)		3.5		7.7		
		B-value		(1)		(1)		
		Distribution		Weibull		Normal		
		C_1		13600		11700		
		C_2		32.5		902		
		No. Specimens	20		5			
		No. Batches	1		1			
		Approval Class	Screening		Screening			

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(f)
C/Ep 195-PW
Celion 3000/E7K8
Tension, 2-axis
$[90_f]_{10}$
75/W, 180/W
Screening

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 36 wt% COMP. DENSITY: 1.55 g/cm³
FIBER VOLUME: 56 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0078 - 0.0084 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		180			
Moisture Content(%)		wet		wet			
Equilibrium at T, RH		(1)		(1)			
Source Code		20		20			
		Normalized	Measured	Normalized	Measured		
	Mean	119	117	130	128		
	Minimum	105	104	129	125		
	Maximum	130	126	132	131		
	C.V.(%)	7.8	7.3	0.89	1.8		
F_2^{tu}	B-value	(2)	(2)	(2)	(2)		
	Distribution	Normal	Normal	Normal	Normal		
(ksi)	C_1	119	117	130	128		
	C_2	9.35	8.51	1.16	2.35		
	No. Specimens	5		5			
	No. Batches	1		1			
	Approval Class	Screening		Screening			
	Mean	9.08	8.92	9.35	9.18		
	Minimum	8.98	8.73	9.26	8.96		
	Maximum	9.21	9.14	9.48	9.38		
E_2^t	C.V.(%)	1.2	1.6	1.2	1.8		
(Msi)	No. Specimens	5		5			
	No. Batches	1		1			
	Approval Class	Screening		Screening			
	Mean						
ν_{21}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		13100		14200		
	Minimum		11400		13700		
	Maximum		14400		14800		
	C.V.(%)		8.7		3.5		
ϵ_2^{tu}	B-value		(2)		(2)		
	Distribution		Normal		Normal		
($\mu\epsilon$)	C_1		13100		14200		
	C_2		1135		490		
	No. Specimens	5		5			
	No. Batches	1		1			
	Approval Class	Screening		Screening			

(1) Conditioned at 160°F, 85% relative humidity for 7 days
(2) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(g)
C/Ep 195-PW
Celion 3000/E7K8
Compression, 1-axis
$[0_f]_{10}$
75/A, -65/A, 180/A
Screening

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 36 - 40 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 53 - 55 % VOID CONTENT: 0.0 - 0.75 %
PLY THICKNESS: 0.0079 - 0.0084 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75 ambient		-65 ambient		180 ambient	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	104	101	121	118	97.4	94.5
	Minimum	90.5	87.7	113	111	87.5	85.1
	Maximum	122	120	132	126	105	100
	C.V.(%)	8.3	8.7	5.9	4.7	7.2	7.1
F_1^{cu}	B-value	(1)	(1)	(1)	(1)	(1)	(1)
	Distribution	Weibull	Weibull	Normal	Normal	Normal	Normal
(ksi)	C_1	108	105	121	118	97.4	94.5
	C_2	13.0	12.1	7.19	5.58	7.00	6.72
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	9.88	9.02	9.83	9.33	9.45	9.16
	Minimum	9.56	8.65	9.75	9.20	9.14	8.89
	Maximum	10.3	9.29	9.95	9.48	9.66	9.37
E_1^c	C.V.(%)	2.3	2.0	1.0	1.1	2.3	2.0
(Msi)	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		10900		12200		10400
	Minimum		10500		12000		10200
	Maximum		11200		12300		10800
	C.V.(%)		2.2		1.0		2.3
ϵ_1^{cu}	B-value		(1)		(1)		(1)
	Distribution		Weibull		Normal		Normal
($\mu\epsilon$)	C_1		11000		12200		10400
	C_2		54.2		122		239
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(h)
C/Ep 195-PW
Celion 3000/E7K8
Compression, 1-axis
$[0_f]_{10}$
75/W, 180/W
Screening

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 36 - 37 wt% COMP. DENSITY: 1.55 g/cm³
FIBER VOLUME: 54 - 56 % VOID CONTENT: 0.0 - 0.70 %
PLY THICKNESS: 0.0073 - 0.0086 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		180		
Moisture Content(%)		wet		wet		
Equilibrium at T, RH		(1)		(1)		
Source Code		20		20		
		Normalized	Measured	Normalized	Measured	
	Mean	94.9	92.6	78.9	77.6	
	Minimum	89.7	88.2	72.7	70.5	
	Maximum	102	98.8	83.2	82.3	
	C.V.(%)	5.5	4.9	5.7	6.0	
F_1^{cu}	B-value	(2)	(2)	(2)	(2)	
	Distribution	Normal	Normal	Normal	Normal	
(ksi)	C_1	94.9	92.6	78.9	77.6	
	C_2	5.47	4.57	4.53	4.65	
	No. Specimens	5		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		
	Mean	9.39	8.92	8.97	8.52	
	Minimum	8.80	8.12	8.45	8.18	
	Maximum	10.2	9.79	9.54	8.80	
E_1^c	C.V.(%)	6.3	6.8	4.4	3.5	
(Msi)	No. Specimens	5		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		
	Mean					
ν_{12}^c	No. Specimens					
	No. Batches					
	Approval Class					
	Mean		9800		8130	
	Minimum		8970		7620	
	Maximum		10400		8600	
	C.V.(%)		6.0		4.4	
ϵ_1^{cu}	B-value		(2)		(2)	
	Distribution		Normal		Normal	
($\mu\epsilon$)	C_1		9800		8130	
	C_2		590		356	
	No. Specimens	5		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		

(1) Conditioned at 160°F, 85% relative humidity for 7 days
(2) B-values are presented only for fully approved data.

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Table 4.2.9(i)
C/Ep 195-PW
Celion 3000/E7K8
Compression, 1-axis
$[0_f]_{12}$
75/A, -65/A, 180/A
Screening

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 38 - 40 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 52 - 54 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0078 - 0.0084 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75 ambient		-65 ambient		180 ambient	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	114	107	133	122	103	97.6
	Minimum	86.4	84.4	127	116	96.0	89.2
	Maximum	128	121	139	129	114	107
	C.V.(%)	9.5	9.1	3.9	4.6	6.8	7.2
F_1^{cu}	B-value	(1)	(1)	(1)	(1)	(1)	(1)
	Distribution	Weibull	Weibull	Normal	Normal	Normal	Normal
(ksi)	C_1	118	111	133	122	103	97.6
	C_2	13.8	14.0	5.22	5.60	6.99	7.04
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	8.22	7.80	8.45	7.71	8.40	7.67
	Minimum	8.07	7.51	8.27	7.43	8.20	7.58
	Maximum	8.50	8.05	8.73	8.09	8.54	7.84
E_1^c	C.V.(%)	1.6	2.2	2.3	3.4	1.5	1.4
(Msi)	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
ν_{12}^c	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		13500				
	Minimum		13000				
	Maximum		13700				
	C.V.(%)		1.6				
ϵ_1^{cu}	B-value		(1)				
	Distribution		Nonpara.				
($\mu\epsilon$)	C_1		10				
	C_2		1.25				
	No. Specimens	20					
	No. Batches	1					
	Approval Class	Screening					

(1) B-values are presented only for fully approved data.

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Table 4.2.9(j)
C/Ep 195-PW
Celion 3000/E7K8
Compression, 1-axis
$[0_t]_{12}$
75/W, 180/W
Screening

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 38 - 40 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 52 - 54 % VOID CONTENT: 0.0 - 0.04 %
PLY THICKNESS: 0.0080 - 0.0084 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		180		
Moisture Content(%)		wet		wet		
Equilibrium at T, RH		(1)		(1)		
Source Code		20		20		
		Normalized	Measured	Normalized	Measured	
	Mean	96.1	90.7	80.2	75.7	
	Minimum	83.9	78.4	74.4	72.2	
	Maximum	107	101	83.3	79.9	
	C.V.(%)	9.3	9.4	4.7	4.4	
F_1^{cu}	B-value	(2)	(2)	(2)	(2)	
	Distribution	Normal	Normal	Normal	Normal	
(ksi)	C_1	96.1	90.7	80.2	75.7	
	C_2	8.91	8.55	3.73	3.31	
	No. Specimens	5		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		
	Mean	9.08	8.30	9.36	8.54	
	Minimum	8.84	7.91	9.14	8.20	
	Maximum	9.17	8.62	9.57	8.84	
E_1^c	C.V.(%)	1.5	3.2	2.0	2.9	
(Msi)	No. Specimens	5		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		
	Mean					
ν_{12}^c	No. Specimens					
	No. Batches					
	Approval Class					
	Mean		10700			
	Minimum		10600			
	Maximum		11000			
	C.V.(%)		1.5			
ϵ_1^{cu}	B-value		(2)			
	Distribution		Normal			
($\mu\epsilon$)	C_1		10700			
	C_2		164			
	No. Specimens	5				
	No. Batches	1				
	Approval Class	Screening				

(1) Conditioned at 160°F, 85% relative humidity for 7 days
(2) B-values are presented only for fully approved data.

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Table 4.2.9(k)
C/Ep 195-PW
Celion 3000/E7K8
SBS, 31-plane
$[0_f]_{14}$
75/A, -65/A, 180/A, 75/W, 180/W
Screening

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 36 - 39 wt%
FIBER VOLUME: 54 - 56 %
PLY THICKNESS: 0.0079 - 0.0081 in.

COMP. DENSITY: 1.55 g/cm^3
VOID CONTENT: 0.0 - 0.75 %

TEST METHOD: ASTM D 2344-68

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75	-65	180	75	180
Moisture Content(%)		ambient	ambient	ambient	wet (1)	wet (1)
Equilibrium at T, RH						
Source Code		20	20	20	20	20
	Mean	10.3	11.6	9.70	9.81	6.92
	Minimum	9.43	10.7	9.34	9.24	6.60
	Maximum	11.4	13.6	9.94	10.4	7.22
	C.V.(%)	5.7	10.8	3.0	7.0	3.4
F_{31}^{sbs} (ksi)	B-value	(2)	(2)	(2)	(2)	(2)
	Distribution	Normal	Normal	Normal	Normal	Normal
	C_1	10.3	11.6	9.70	9.81	6.92
	C_2	0.446	1.25	0.293	0.505	0.237
	No. Specimens	20	5	5	5	5
	No. Batches	1	1	1	1	1
	Approval Class	Screening	Screening	Screening	Screening	Screening

(1) Conditioned at 160°F, 85% relative humidity for 7 days
(2) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(l)
C/Ep 195-PW
Celion 3000/E7K8
SBS, 31-plane
$[0_f]_{12}$
75/A, -65/A, 180/A, 75/W, 180/W
Screening

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 39 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 54 % VOID CONTENT: 0.29 %
PLY THICKNESS: 0.0080 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 2344-68

NORMALIZED BY: Not normalized

Temperature(°F)		75	-65	180	75	180
Moisture Content(%)		ambient	ambient	ambient	wet	wet
Equilibrium at T, RH					(1)	(1)
Source Code		20	20	20	20	20
	Mean	9.76	10.2	9.72	9.72	8.72
	Minimum	9.00	9.54	8.76	8.76	8.35
	Maximum	10.7	10.5	10.3	10.3	9.00
	C.V.(%)	4.8	3.9	6.1	6.1	2.8
F_{31}^{sbs} (ksi)	B-value	(2)	(2)	(2)	(2)	(2)
	Distribution	Normal	Normal	Normal	Normal	Normal
	C_1	9.76	10.2	9.72	9.72	8.72
	C_2	0.470	0.395	0.591	0.591	0.247
	No. Specimens	20	5	5	5	5
	No. Batches	1	1	1	1	1
	Approval Class	Screening	Screening	Screening	Screening	Screening

(1) Conditioned at 160°F, 85% relative humidity for 7 days
(2) B-values are presented only for fully approved data.

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4.2.10 HITEX 33 6k/E7K8 plain weave *

MATERIAL:	HITEX 33 6k/E7K8 plain weave		C/Ep HITEX 33/E7K8 Summary
PREPREG:	U.S. Polymeric Hitex 33 6k/E7K8 Plain weave		
FIBER:	Hitco HITEX 33 6k G'	MATRIX:	U.S. Polymeric E7K8
T_g(ambient):	T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave: 310°F, 2 hours, 85 psig		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	1/88
Date of prepreg manufacture	Date of analysis	1/93
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	75°F/A		-65°F/A	180°F/A		75°F/W	180°F/W	
Tension, 1-axis								
Tension, 2-axis	SSSS		SS-S			SSSS	SSSS	
Tension, 3-axis								
Compression, 1-axis	SS-S		SS--	SS--		SS-S	SS--	
Compression, 2-axis	SS-S		SS--	SS--		SS-S	SS--	
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
SB Strength, 31-plane	S---		S---			S---	S---	

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.77		
Resin Density	(g/cm^3)	1.27		
Composite Density	(g/cm^3)	1.56		
Fiber Areal Weight	(g/m^2)	195		
Ply Thickness	(in)	0.0085	0.0077 - 0.0099	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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Table 4.2.10(a)
C/Ep 195-PW
HITEX 33/E7K8
Tension, 2-axis
$[90_f]_{12}$
75/A, -65/A, 75/W
Screening

MATERIAL: HITEX 33 6k/E7K8 Plain weave

RESIN CONTENT: 37 - 41 wt% COMP. DENSITY: 1.53 - 1.55 g/cm^3
FIBER VOLUME: 51 - 55 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0087 - 0.0098 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75 ambient		-65 ambient		75 wet (1)	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	131	124	126	122	134	119
	Minimum	120	103	122	111	130	114
	Maximum	139	136	131	131	137	125
	C.V. (%)	4.3	6.8	3.1	6.7	2.8	3.8
F_2^{tu}	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	Weibull	Weibull	Normal	Normal	Normal	Normal
(ksi)	C_1	134	128	126	122	134	120
	C_2	28.2	17.8	3.88	8.16	3.69	4.55
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	8.65	8.14	8.10	7.82	9.61	8.55
	Minimum	8.01	7.52	7.73	7.54	9.26	8.20
	Maximum	9.65	8.62	8.29	8.26	9.94	9.13
E_2^t	C.V. (%)	6.2	3.1	2.7	3.4	2.8	4.1
(Msi)	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	0.0460				0.0540	
ν_{21}^t	No. Specimens	5				5	
	No. Batches	1				1	
	Approval Class	Screening				Screening	
	Mean	14300		15600		10500	
	Minimum	13700		14600		9930	
	Maximum	14900		16500		10800	
	C.V. (%)	3.8		4.4		3.2	
ϵ_2^{tu}	B-value	(2)		(2)		(2)	
	Distribution	Normal		Normal		Normal	
($\mu\epsilon$)	C_1	14300		15600		10500	
	C_2	541		687		335	
	No. Specimens	5		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	

(1) Conditioned at 160°F, 85% relative humidity for 14 days
(2) B-values are presented only for fully approved data.

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Table 4.2.10(b)
C/Ep 195-PW
HITEX 33/E7K8
Tension, 2-axis
$[90_t]_{12}$
180/W
Screening

MATERIAL: HITEX 33 6k/E7K8 Plain weave

RESIN CONTENT: 41 wt% COMP. DENSITY: 1.53 g/cm^3
FIBER VOLUME: 51 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0089 - 0.0094 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		180			
Moisture Content(%)		wet			
Equilibrium at T, RH		(1)			
Source Code		20			
		Normalized	Measured	Normalized	Measured
	Mean	138	122		
	Minimum	120	107		
	Maximum	155	135		
	C.V. (%)	10.2	9.1		
F_2^{tu}	B-value	(2)	(2)		
	Distribution	Normal	Normal		
(ksi)	C_1	138	123		
	C_2	14.1	11.1		
	No. Specimens	5			
	No. Batches	1			
	Approval Class	Screening			
	Mean	9.91	8.80		
	Minimum	9.11	8.23		
	Maximum	10.7	9.23		
E_2^t	C.V. (%)	7.2	5.3		
(Msi)	No. Specimens	5			
	No. Batches	1			
	Approval Class	Screening			
	Mean	0.0700			
ν_{21}^t	No. Specimens	5			
	No. Batches	1			
	Approval Class	Screening			
	Mean	10400			
	Minimum	9840			
	Maximum	10800			
	C.V. (%)	3.6			
ϵ_2^{tu}	B-value	(2)			
	Distribution	Normal			
($\mu\epsilon$)	C_1	10400			
	C_2	372			
	No. Specimens	5			
	No. Batches	1			
	Approval Class	Screening			

(1) Conditioned at 160°F, 85% relative humidity for 14 days
(2) B-values are presented only for fully approved data.

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Table 4.2.10(c)
C/Ep 195-PW
HITEX 33/E7K8
Compression, 1-axis
$[0_f]_{12}$
75/A, -65/A, 180/A
Screening

MATERIAL: HITEX 33 6k/E7K8 Plain weave

RESIN CONTENT: 45 wt% COMP. DENSITY: 1.51 g/cm^3
FIBER VOLUME: 47 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0079 - 0.0099 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		75 ambient 20		-65 ambient 20		180 ambient 20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	136	112	155	128	130	107
	Minimum	111	98.4	147	118	118	94.9
	Maximum	158	128	164	139	139	117
	C.V.(%)	8.4	7.5	5.5	7.5	6.3	7.8
F_1^{cu} (ksi)	B-value	(1)	(1)	(1)	(1)	(1)	(1)
	Distribution	Weibull	Weibull	Normal	Normal	Normal	Normal
	C_1	141	116	155	128	130	107
	C_2	13.3	14.5	8.51	9.57	8.21	8.22
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	9.11	7.53	10.1	8.30	9.37	7.75
	Minimum	8.64	6.83	9.72	7.74	9.15	7.38
	Maximum	9.63	8.17	10.8	8.76	9.66	8.66
E_1^c	C.V.(%)	3.0	5.2	4.0	5.1	2.4	7.1
(Msi)	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		14400				
	Minimum		13700				
	Maximum		15200				
	C.V.(%)		3.1				
ϵ_1^{cu}	B-value		(1)				
	Distribution		Weibull				
($\mu\epsilon$)	C_1		14600				
	C_2		34.7				
	No. Specimens	20					
	No. Batches	1					
	Approval Class	Screening					

(1) B-values are presented only for fully approved data.

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Table 4.2.10(d)
C/Ep 195-PW
HITEX 33/E7K8
Compression, 1-axis
$[0_f]_{12}$
75/W, 180/W
Screening

MATERIAL: HITEX 33 6k/E7K8 Plain weave

RESIN CONTENT: 45 wt% COMP. DENSITY: 1.51 g/cm³
FIBER VOLUME: 47 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0081 - 0.0098 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		180		
Moisture Content(%)		wet		wet		
Equilibrium at T, RH		(1)		(1)		
Source Code		20		20		
		Normalized	Measured	Normalized	Measured	
	Mean	133	110	68.5	56.4	
	Minimum	130	100	54.2	46.7	
	Maximum	139	116	75.8	62.2	
	C.V.(%)	2.8	5.8	13.6	12.0	
F_1^{cu}	B-value	(2)	(2)	(2)	(2)	
	Distribution	Normal	Normal	Normal	Normal	
(ksi)	C_1	133	110	68.5	56.4	
	C_2	3.71	6.36	9.31	6.79	
	No. Specimens	5		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		
	Mean	8.78	7.24	9.43	7.78	
	Minimum	8.41	7.04	9.32	7.69	
	Maximum	9.07	7.51	9.64	7.89	
E_1^c	C.V.(%)	3.2	2.5	1.4	9.5	
(Msi)	No. Specimens	5		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		
	Mean					
ν_{12}^c	No. Specimens					
	No. Batches					
	Approval Class					
	Mean		14600			
	Minimum		14000			
	Maximum		15400			
	C.V.(%)		3.6			
ϵ_1^{cu}	B-value		(2)			
	Distribution		Normal			
($\mu\epsilon$)	C_1		14600			
	C_2		525			
	No. Specimens	5				
	No. Batches	1				
	Approval Class	Screening				

(1) Conditioned at 160°F, 85% relative humidity for 14 days
(2) B-values are presented only for fully approved data.

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Table 4.2.10(e)
C/Ep 195-PW
HITEX 33/E7K8
Compression, 2-axis
$[90_f]_6$
75/A, -65/A, 180/A
Screening

MATERIAL: HITEX 33 6k/E7K8 Plain weave

RESIN CONTENT: 39 - 41 wt% COMP. DENSITY: 1.53 g/cm^3
FIBER VOLUME: 51 - 52 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0083 - 0.0087 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		75 ambient 20		-65 ambient 20		180 ambient 20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	104	92.4	128	114	99.4	88.6
	Minimum	77.9	70.4	111	98.8	86.4	77.0
	Maximum	125	109	138	123	113	101
	C.V. (%)	13.1	12.6	8.0	8.1	12.0	12.0
F_2^{cu} (ksi)	B-value	(1)	(1)	(1)	(1)	(1)	(1)
	Distribution	Weibull	Weibull	Normal	Normal	Normal	Normal
	C_1	110	97.4	128	114	99.4	88.6
	C_2	9.70	10.5	10.3	9.18	11.9	10.6
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	8.92	8.21	9.49	8.74	9.07	8.35
	Minimum	8.50	7.78	9.36	8.65	8.95	8.20
	Maximum	9.40	8.77	9.58	8.93	9.18	8.52
E_2^c	C.V. (%)	2.5	3.4	0.9	1.3	1.3	1.7
(Msi)	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean						
ν_{21}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		10900				
	Minimum		10400				
	Maximum		11400				
	C.V. (%)		2.4				
ϵ_2^{cu}	B-value		(1)				
	Distribution		Weibull				
($\mu\epsilon$)	C_1		11100				
	C_2		46.5				
	No. Specimens	20					
	No. Batches	1					
	Approval Class	Screening					

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.10(f)
C/Ep 195-PW
HITEX 33/E7K8
Compression, 2-axis
$[90_t]_6$
75/W, 180/W
Screening

MATERIAL: HITEX 33:6k/E7K8 Plain weave

RESIN CONTENT: 39 - 41 wt% COMP. DENSITY: 1.53 g/cm³
FIBER VOLUME: 51 - 52 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0080 - 0.0083 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		180			
Moisture Content(%)		wet		wet			
Equilibrium at T, RH		(1)		(1)			
Source Code		20		20			
		Normalized	Measured	Normalized	Measured		
	Mean	99.2	88.5	84.0	74.9		
	Minimum	80.9	72.2	74.2	66.1		
	Maximum	112	100	88.8	79.2		
	C.V. (%)	12.1	12.1	7.0	6.9		
F_2^{cu}	B-value	(2)	(2)	(2)	(2)		
	Distribution	Normal	Normal	Normal	Normal		
(ksi)	C_1	99.2	88.5	84.0	74.9		
	C_2	12.0	10.7	5.8	5.20		
	No. Specimens	5		5			
	No. Batches	1		1			
	Approval Class	Screening		Screening			
	Mean	9.30	8.56	8.96	8.25		
	Minimum	8.74	7.98	8.69	8.03		
	Maximum	9.56	8.78	9.31	8.43		
E_2^c	C.V. (%)	3.5	3.9	2.9	2.0		
(Msi)	No. Specimens	5		5			
	No. Batches	1		1			
	Approval Class	Screening		Screening			
	Mean						
ν_{21}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		10200				
	Minimum		9910				
	Maximum		10900				
	C.V. (%)		3.7				
ϵ_2^{cu}	B-value		(2)				
	Distribution		Normal				
($\mu\epsilon$)	C_1		10200				
	C_2		381				
	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					

(1) Conditioned at 160°F, 85% relative humidity for 14 days
(2) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.10(g)
C/Ep 195-PW
HITEX 33/E7K8
Compression, 2-axis
$[90_t]_{12}$
75/A, -65/A, 180/A
Screening

MATERIAL: HITEX 33 6k/E7K8 Plain weave

RESIN CONTENT: 45 wt% COMP. DENSITY: 1.51 g/cm^3
FIBER VOLUME: 47 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0080 - 0.0097 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75 ambient		-65 ambient		180 ambient	
Moisture Content(%)		20		20		20	
Equilibrium at T, RH							
Source Code							
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	132	110	147	122	132	110
	Minimum	114	97.9	138	115	128	106
	Maximum	145	118	161	127	146	117
	C.V. (%)	5.7	5.3	6.0	4.1	5.9	4.7
F_2^{cu}	B-value	(1)	(1)	(1)	(1)	(1)	(1)
	Distribution	Weibull	Weibull	Normal	Normal	Normal	Normal
(ksi)	C_1	136	113	147	122	132	110
	C_2	21.6	23.4	8.78	5.02	7.73	5.12
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	8.74	7.27	9.09	7.54	9.11	7.57
	Minimum	8.41	6.70	8.12	7.07	8.61	7.41
	Maximum	9.20	8.06	10.1	7.90	9.49	7.71
E_2^c	C.V. (%)	2.6	4.7	9.1	5.6	3.8	1.5
(Msi)	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean						
ν_{21}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		14100				
	Minimum		13400				
	Maximum		14700				
	C.V. (%)		2.6				
ϵ_2^{cu}	B-value		(1)				
	Distribution		Weibull				
($\mu\epsilon$)	C_1		14300				
	C_2		46.4				
	No. Specimens	20					
	No. Batches	1					
	Approval Class	Screening					

(1) B-values are presented only for fully approved data.

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Table 4.2.10(h)
C/Ep 195-PW
HITEX 33/E7K8
Compression, 2-axis
$[90_f]_{12}$
75/W, 180/W
Screening

MATERIAL: HITEX 33 6k/E7K8 Plain weave

RESIN CONTENT: 45 wt% COMP. DENSITY: 1.51 g/cm^3
FIBER VOLUME: 47 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0080 - 0.0097 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		180		
Moisture Content(%)		wet		wet		
Equilibrium at T, RH		(1)		(1)		
Source Code		20		20		
		Normalized	Measured	Normalized	Measured	
	Mean	117	97.4	61.1	50.8	
	Minimum	107	88.4	52.2	44.1	
	Maximum	132	105	66.4	57.2	
	C.V. (%)	9.1	6.9	9.9	9.9	
F_2^{cu}	B-value	(2)	(2)	(2)	(2)	
	Distribution	Normal	Normal	Normal	Normal	
(ksi)	C_1	117	97.4	61.1	50.8	
	C_2	10.6	6.74	6.04	5.01	
	No. Specimens	5		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		
	Mean	8.99	7.48	9.26	7.71	
	Minimum	8.48	7.08	8.76	7.32	
	Maximum	9.54	7.8	9.69	8.39	
E_2^c	C.V. (%)	4.5	4.0	4.0	6.2	
(Msi)	No. Specimens	5		5		
	No. Batches	1		1		
	Approval Class	Screening		Screening		
	Mean					
ν_{21}^c	No. Specimens					
	No. Batches					
	Approval Class					
	Mean		13500			
	Minimum		12700			
	Maximum		14200			
	C.V. (%)		4.2			
ϵ_2^{cu}	B-value		(2)			
	Distribution		Normal			
($\mu\epsilon$)	C_1		13500			
	C_2		564			
	No. Specimens		5			
	No. Batches		1			
	Approval Class		Screening			

(1) Conditioned at 160°F, 85% relative humidity for 14 days
(2) B-values are presented only for fully approved data.

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Table 4.2.10(i)
C/Ep 195-PW
HITEX 33/E7K8
SBS, 31-plane
$[90_t]_6$
75/A, -65/A, 180/A
Screening

MATERIAL: HITEX 33 6k/E7K8 Plain weave

RESIN CONTENT: 44 wt% COMP. DENSITY: 1.51 g/cm^3
FIBER VOLUME: 48 % VOID CONTENT: 0.18 %
PLY THICKNESS: 0.0077 - 0.0093 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 2344-76

NORMALIZED BY: Not normalized

Temperature(°F)		75.0	-65.0	75.0	180.0	
Moisture Content(%)		ambient	ambient	wet	wet	
Equilibrium at T, RH				(1)	(1)	
Source Code		20	20	20	20	
	Mean	8.67	8.83	9.40	8.35	
	Minimum	7.77	8.14	9.20	7.83	
	Maximum	9.40	9.37	9.73	8.80	
	C.V.(%)	5.0	6.3	2.1	4.5	
F_{31}^{sbs}	B-value	(2)	(2)	(2)	(2)	
	Distribution	Weibull	Normal	Normal	Normal	
(ksi)	C_1	8.86	8.83	9.40	8.35	
	C_2	23.6	0.554	0.202	0.379	
	No. Specimens	20	5	5	5	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	

(1) Conditioned at 160°F, 85% relative humidity for 14 days
(2) B-values are presented only for fully approved data.

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4.2.11 AS4 3k/E7K8 plain weave *

MATERIAL:	AS4 3k/E7K8 plain weave		C/Ep AS4/E7K8 Summary
PREPREG:	U.S. Polymeric AS4/E7K8 Plain weave		
FIBER:	Hercules AS4 3k	MATRIX:	U.S. Polymeric E7K8
T_g(ambient):	T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave: 290°F, 2 hours, 85 psig		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	1/88, 6/90
Date of prepreg manufacture	2/86 - 7/89	Date of analysis	1/93
Date of composite manufacture			

LAMINA PROPERTY SUMMARY

	75°F/A							
Tension, 1-axis								
Tension, 2-axis								
Tension, 3-axis								
Compression, 1-axis	II-I							
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
SB Strength, 31-plane	S---							

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.77		
Resin Density	(g/cm^3)	1.28		
Composite Density	(g/cm^3)	1.56		
Fiber Areal Weight	(g/m^2)	195		
Ply Thickness	(in)	0.0087	0.0074 - 0.0088	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.11(a)
C/Ep 195-PW
AS4/E7K8
Compression, 1-axis
$[0_f]_{12}$
75/A
Interim

MATERIAL: AS4 3k/E7K8 Plain weave

RESIN CONTENT: 37 - 48 wt% COMP. DENSITY: 1.52 - 1.54 g/cm^3
FIBER VOLUME: 48 - 55 % VOID CONTENT: 0.0 - 1.9 %
PLY THICKNESS: 0.0074 - 0.0085 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75					
Moisture Content(%)		ambient					
Equilibrium at T, RH							
Source Code		20,27					
		Normalized	Measured				
	Mean	111	988				
	Minimum	64.4	58.0				
	Maximum	138	122				
	C.V. (%)	11.7	11.3				
F_1^{cu}	B-value	(1)	(1)				
	Distribution	ANOVA	ANOVA				
(ksi)	C_1	13.3	11.3				
	C_2	1.81	1.80				
	No. Specimens	206					
	No. Batches	18					
	Approval Class	Interim					
	Mean	9.02	8.07				
	Minimum	7.87	7.07				
	Maximum	10.5	9.04				
E_1^c	C.V. (%)	5.24	4.28				
(Msi)	No. Specimens	210					
	No. Batches	18					
	Approval Class	Interim					
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		11600				
	Minimum		8820				
	Maximum		15000				
	C.V. (%)		14.5				
ϵ_1^{cu}	B-value		(1)				
	Distribution		ANOVA				
($\mu\epsilon$)	C_1		1730				
	C_2		1.97				
	No. Specimens	190					
	No. Batches	17					
	Approval Class	Interim					

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.11(b)
C/EP 195-PW
AS4/E7K8
SBS, 31-plane
$[0_t]_{12}$
75/A
Screening

MATERIAL: AS4/E7K8 Plain weave

RESIN CONTENT: 38 - 48 wt%
FIBER VOLUME: 48 - 55 %
PLY THICKNESS: 0.0074 - 0.0085 in.

COMP. DENSITY: 1.52 - 1.54 g/cm^3
VOID CONTENT: 0.0 - 1.9 %

TEST METHOD: ASTM D 2344 - 84

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75				
Moisture Content(%)		ambient				
Equilibrium at T, RH						
Source Code		20,27				
	Mean	9.68				
	Minimum	7.53				
	Maximum	14.2				
	C.V. (%)	12.0				
F_{31}^{sbs} (ksi)	B-value	(1)				
	Distribution	ANOVA				
	C_1	1.20				
	C_2	1.95				
	No. Specimens	170				
	No. Batches	16				
	Approval Class	Screening				

(1) B-values are presented only for fully approved data.

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4.2.12 AS4/3501-6 (bleed) unidirectional tape*

MATERIAL:	AS4/3501-6 unidirectional tape		C/Ep AS4/3501-6 Summary
PREPREG:	Hercules AS4/3501-6 unidirectional tape		
FIBER:	Hercules AS4	MATRIX:	Hercules 3501-6 TMA
T_g(ambient):	390°F T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure: 240 ± 10°F, 60 min., 85 psig; 350 ± 10°F, 120 ± 10 min., 100 ± 10 psig, bleed		

* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	6/90
Date of prepreg manufacture	Date of analysis	1/93
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	75°F/A		200°F/A		75°F/W	200°F/W		
Tension, 1-axis	II--							
Tension, 2-axis	SS--							
Tension, 3-axis								
Compression, 1-axis	IS--		II--		SS--	SS--		
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
SB Strength, 31-plane	S---		S---		S---	S---		

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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Check the source to verify that this is the current version before use.

* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.8		
Resin Density	(g/cm^3)	1.27		
Composite Density	(g/cm^3)	1.59		
Fiber Areal Weight	(g/m^2)	145		
Ply Thickness	(in)		0.0041 - 0.0059	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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Check the source to verify that this is the current version before use.

* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.12(a)
C/Ep 145-UT
AS4/3501-6
Tension, 1-axis
$[0]_8$
75/A
Interim

MATERIAL: AS4/3501-6 (bleed) unidirectional tape

RESIN CONTENT: 38 - 34 wt% COMP. DENSITY: 1.56 g/cm^3
FIBER VOLUME: 58 - 65 % VOID CONTENT:
PLY THICKNESS: 0.0048 - 0.0057 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		75					
Moisture Content(%)		ambient					
Equilibrium at T, RH							
Source Code		26					
		Normalized	Measured				
	Mean	291	295				
	Minimum	263	271				
	Maximum	326	326				
	C.V. (%)	6.09	5.05				
F_1^{tu}	B-value	(1)	(1)				
	Distribution	Weibull	Weibull				
(ksi)	C_1	300	302				
	C_2	18.4	20.3				
	No. Specimens	21					
	No. Batches	7					
	Approval Class	Interim					
	Mean	19.6	19.9				
	Minimum	18.0	18.3				
	Maximum	21.1	22.6				
E_1^t	C.V. (%)	3.73	6.48				
(Msi)	No. Specimens	21					
	No. Batches	7					
	Approval Class	Interim					
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.12(b)
C/Ep 145-UT
AS4/3501-6
Tension, 2-axis
$[90]_8$
75/A
Screening

MATERIAL: AS4/3501-6 (bleed) unidirectional tape

RESIN CONTENT: 28 - 29 wt%
FIBER VOLUME: 63 - 64 %
PLY THICKNESS: 0.0048 - 0.0057 in.

COMP. DENSITY: 1.60 - 1.61 g/cm^3
VOID CONTENT:

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75				
Moisture Content(%)		ambient				
Equilibrium at T, RH						
Source Code		26				
	Mean	7.78				
	Minimum	7.00				
	Maximum	9.50				
	C.V. (%)	12.1				
F_2^{tu}	B-value	(1)				
	Distribution	Normal				
(ksi)	C_1	7.78				
	C_2	0.941				
	No. Specimens	6				
	No. Batches	2				
	Approval Class	Screening				
	Mean	1.48				
	Minimum	1.40				
	Maximum	1.50				
E_2^t	C.V. (%)	2.75				
(Msi)	No. Specimens	6				
	No. Batches	2				
	Approval Class	Screening				
	Mean					
ν_{12}^t	No. Specimens					
	No. Batches					
	Approval Class					
	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
ϵ_2^{tu}	B-value					
	Distribution					
$(\mu\epsilon)$	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.12(c)
C/Ep 145-UT
AS4/3501-6
Compression, 1-axis
$[0]_8$
75/A, 200/A, 75/W
Interim, Screening

MATERIAL: AS4/3501-6 (bleed) unidirectional tape

RESIN CONTENT: 28 - 34 wt% COMP. DENSITY: 1.58 - 1.61 g/cm³
FIBER VOLUME: 58 - 65 % VOID CONTENT:
PLY THICKNESS: 0.0041 - 0.0055 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		75 ambient		200 ambient		75 wet (1)	
Moisture Content(%)		26		26		26	
Equilibrium at T, RH							
Source Code							
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	210	214	196	201	202	213
	Minimum	144	161	148	165	165	179
	Maximum	269	260	242	237	274	266
	C.V. (%)	16.0	13.5	13.6	10.7	18.0	14.1
F_1^{cu}	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	ANOVA	ANOVA	ANOVA	ANOVA	Weibull	Weibull
(ksi)	C_1	34.7	27.7	27.7	22.3	217	226
	C_2	2.39	2.52	2.52	2.35	5.89	7.82
	No. Specimens	26		27		10	
	No. Batches	7		7		2	
	Approval Class	Interim		Interim		Screening	
	Mean	17.8	18.8	16.3	17.4	17.4	18.5
	Minimum	15.1	16.4	13.0	14.3	15.6	17.1
	Maximum	20.3	20.0	18.7	19.6	20.3	20.6
E_1^c	C.V. (%)	7.50	7.18	10.7	10.1	9.14	5.84
(Msi)	No. Specimens	14		15		10	
	No. Batches	3		3		2	
	Approval Class	Screening		Interim		Screening	
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned at 140°F, 95% relative humidity for 30 days.
(2) B-values are presented only for fully approved data.

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Check the source to verify that this is the current version before use.

* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.12(d)
C/Ep 145-UT
AS4/3501-6
Compression, 1-axis
$[0]_8$
200/W
Screening

MATERIAL: AS4/3501-6 (bleed) unidirectional tape

RESIN CONTENT: 28 - 34 wt% COMP. DENSITY: 1.58 - 1.61 g/cm³
FIBER VOLUME: 58 - 65 % VOID CONTENT:
PLY THICKNESS: 0.0041 - 0.0055 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		200			
Moisture Content(%)		wet			
Equilibrium at T, RH					
Source Code		26			
		Normalized	Measured		
	Mean	169	179		
	Minimum	100	107		
	Maximum	212	226		
	C.V. (%)	22.2	22.9		
F_1^{cu}	B-value	(1)	(1)		
	Distribution	ANOVA	ANOVA		
(ksi)	C_1	41.7	46.6		
	C_2	5.28	5.72		
	No. Specimens	10			
	No. Batches	3			
	Approval Class	Screening			
	Mean	17.7	18.7		
	Minimum	12.1	13.4		
	Maximum	27.2	25.5		
E_1^c	C.V. (%)	21.6	15.8		
(Msi)	No. Specimens	10			
	No. Batches	3			
	Approval Class	Screening			
	Mean				
ν_{12}^c	No. Specimens				
	No. Batches				
	Approval Class				
	Mean				
	Minimum				
	Maximum				
	C.V. (%)				
ϵ_1^{cu}	B-value				
	Distribution				
$(\mu\epsilon)$	C_1				
	C_2				
	No. Specimens				
	No. Batches				
	Approval Class				

(1) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.12(e)
C/Ep 145-UT
AS4/3501-6
SBS, 31-plane
$[0]_8$
75/A, 200/A, 75/W, 200/W
Screening

MATERIAL: AS4/3501-6 (bleed) unidirectional tape

RESIN CONTENT: 30 - 34 wt%
FIBER VOLUME: 58 - 62 %
PLY THICKNESS: 0.0047 - 0.0055 in.

COMP. DENSITY: 1.58 - 1.60 g/cm³
VOID CONTENT:

TEST METHOD: ASTM D 2344

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75	200	75	200	
Moisture Content(%)		ambient	ambient	wet (1)	wet (1)	
Equilibrium at T, RH						
Source Code		26	26	26	26	
	Mean	17.3	13.0	13.9	9.0	
	Minimum	14.1	11.1	13.1	8.3	
	Maximum	19.4	14.9	15.5	10.1	
	C.V. (%)	7.63	11.6	6.13	6.4	
F_{31}^{sbs} (ksi)	B-value	(2)	(2)	(2)	(2)	
	Distribution	ANOVA	ANOVA	Normal	Normal	
	C_1	1.38	1.59	13.9	9.0	
	C_2	2.62	2.77	0.852	0.58	
	No. Specimens	21	21	6	9	
	No. Batches	7	7	2	3	
	Approval Class	Screening	Screening	Screening	Screening	

(1) Conditioned at 140°F, 95% relative humidity for 30 days.
(2) B-values are presented only for fully approved data.

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Table 4.2.12(f)
C/Ep 145-UT
AS4/3501-6
Tension, x-axis
$[0/45/90/-45]_s$
75/A
Screening

MATERIAL: AS4/3501-6 (bleed) unidirectional tape

RESIN CONTENT: 29 - 32 wt%
FIBER VOLUME: 60 - 63 %
PLY THICKNESS: 0.0055 - 0.0062 in.

COMP. DENSITY: 1.59 - 1.60 g/cm³
VOID CONTENT:

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: NA

Temperature(°F)		75				
Moisture Content(%)		ambient				
Equilibrium at T, RH						
Source Code		26				
	Mean	95.8				
	Minimum	90.6				
	Maximum	106				
	C.V. (%)	5.95				
F_x^{tu}	B-value					
	Distribution	ANOVA				
(ksi)	C_1	29.9				
	C_2	14.5				
	No. Specimens	6				
	No. Batches	2				
	Approval Class	Screening				
	Mean	7.22				
	Minimum	6.60				
	Maximum	8.40				
E_x^t	C.V. (%)	9.74				
(Msi)	No. Specimens	6				
	No. Batches	2				
	Approval Class	Screening				
	Mean					
ν_{xy}^t	No. Specimens					
	No. Batches					
	Approval Class					
	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
ϵ_x^{tu}	B-value					
	Distribution					
$(\mu\epsilon)$	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

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Table 4.2.12(g)
C/Ep 145-UT
AS4/3501-6
Open Hole Tension, x-axis
[0/45/90/-45]$_s$
75/A
Screening

MATERIAL: AS4/3501-6 (bleed) unidirectional tape

RESIN CONTENT: 29 - 32 wt%
FIBER VOLUME: 60 - 63 %
PLY THICKNESS: 0.0055 - 0.0057 in.

COMP. DENSITY: 1.59 - 1.60 g/cm^3
VOID CONTENT:

TEST METHOD: SACMA 5-88 (1)

MODULUS CALCULATION:

NORMALIZED BY: NA

Temperature(°F)		75				
Moisture Content(%)		ambient				
Equilibrium at T, RH						
Source Code		26				
F_x^{oht} (ksi)	Mean	62.0				
	Minimum	59.2				
	Maximum	65.1				
	C.V. (%)	3.13				
	B-value					
	Distribution	Normal				
	C_1	62.0				
	C_2	1.94				
	No. Specimens	6				
	No. Batches	2				
	Approval Class	Screening				
E_x^{oht} (Msi)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	No. Specimens					
	No. Batches					
	Approval Class					
ϵ_x^{oht} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) Note SACMA 5-88 uses a $[+45/0/-45/90]_{2s}$ lay-up.

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4.2.13 AS4/3501-6 (no bleed) unidirectional tape *

C/Ep AS4/3501-6 Summary

MATERIAL:	AS4/3501-6 unidirectional tape		
PREPREG:	Hercules AS4/3501-6 unidirectional tape		
FIBER:	Hercules AS4, unsized	MATRIX:	Hercules 3501-6 TMA
T_g(ambient):	390°F T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure: 240 ± 10°F, 60 min., 85 psig; 350 ± 10°F, 120 ± 10 min., 100 ± 10 psig, no bleed		

* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	~12/82 - 8/89	Date of testing	~6/83 - ~4/91
Date of resin manufacture		Date of data submittal	6/90
Date of prepreg manufacture	1/83 - 11/89	Date of analysis	1/93
Date of composite manufacture			

LAMINA PROPERTY SUMMARY

	75°F/A		-65°F/A	200°F/A		200°F/W		
Tension, 1-axis	II--		SS--	SS--				
Tension, 2-axis	SS--							
Tension, 3-axis								
Compression, 1-axis	II--			I---		II--		
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
SB Strength, 31-plane	S---			S---				

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.8		
Resin Density	(g/cm^3)	1.27		
Composite Density	(g/cm^3)	1.59		
Fiber Areal Weight	(g/m^2)	145	142 - 149	
Ply Thickness	(in)		0.0055 - 0.0063	

LAMINATE PROPERTY SUMMARY

	75°F/A							
[10/45/90/-45] family								
Tension, x-axis	S---							
OHT, x-axis	S---							

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.13(a)
C/Ep 145-UT
AS4/3501-6
Tension, 1-axis
$[0]_8$
75/A, -65/A, 200/A
Interim, Screening

MATERIAL: AS4/3501-6 (no bleed)

RESIN CONTENT: 36 - 39 wt% COMP. DENSITY: 1.55 - 1.57 g/cm^3
FIBER VOLUME: 52 - 56 % VOID CONTENT:
PLY THICKNESS: 0.0055 - 0.0060 in.

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Initial tangent

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		75 ambient		-65 ambient		200 ambient	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		26		26		26	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	290	262	261	237	315	286
	Minimum	262	235	207	187	278	247
	Maximum	322	286	300	274	330	297
	C.V. (%)	5.62	5.38	12.4	12.8	4.89	5.59
F_1^{tu}	B-value	(1)	(1)	(1)	(1)	(1)	(1)
	Distribution	ANOVA	ANOVA	ANOVA	ANOVA	Nonpara.	Nonpara.
(ksi)	C_1	16.5	14.3	34.9	33.1	6	6
	C_2	2.05	2.01	4.69	5.05	2.25	2.25
	No. Specimens	30		9		9	
	No. Batches	10		3		3	
	Approval Class	Interim		Screening		Screening	
	Mean	18.9	17.1	21.1	19.2	20.8	18.9
	Minimum	17.0	15.5	19.7	17.7	19.4	17.4
	Maximum	20.3	17.9	22.3	21.4	22.0	20.2
E_1^t	C.V. (%)	4.00	3.20	4.60	5.78	4.72	4.70
(Msi)	No. Specimens	30		9		9	
	No. Batches	10		3		3	
	Approval Class	Interim		Screening		Screening	
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.13(b)
C/Ep 145-UT
AS4/3501-6
Tension, 2-axis
$[90]_s$
75/A
Screening

MATERIAL: AS4/3501-6 (no bleed)

RESIN CONTENT: 37 wt%
FIBER VOLUME: 54 - 55 %
PLY THICKNESS: 0.0060 - 0.0062 in.

COMP. DENSITY: 1.56 g/cm^3
VOID CONTENT:

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Initial tangent

NORMALIZED BY: Not normalized

Temperature(°F)		75				
Moisture Content(%)		ambient				
Equilibrium at T, RH						
Source Code		26				
	Mean	8.0				
	Minimum	6.8				
	Maximum	9.3				
	C.V. (%)	10.				
F_2^{tu}	B-value	(1)				
	Distribution	Normal				
(ksi)	C_1	8.0				
	C_2	0.81				
	No. Specimens	9				
	No. Batches	3				
	Approval Class	Screening				
	Mean	1.2				
	Minimum	1.1				
	Maximum	1.4				
E_2^t	C.V. (%)	8.9				
(Msi)	No. Specimens	9				
	No. Batches	3				
	Approval Class	Screening				
	Mean					
ν_{21}^t	No. Specimens					
	No. Batches					
	Approval Class					
	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
ϵ_2^{tu}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.13(c)
C/Ep 145-UT
AS4/3501-6
Compression, 1-axis
$[0]_8$
75/A, 200/A, 200/W
Interim

MATERIAL: AS4/3501-6 (no bleed)

RESIN CONTENT: 36 - 39 wt%
FIBER VOLUME: 52 - 56 %
PLY THICKNESS: 0.0056 - 0.0060 in.

COMP. DENSITY: 1.55 - 1.57 g/cm^3
VOID CONTENT:

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Initial tangent

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		75 ambient		200 ambient		200 wet (1)	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		26		26		26	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	233	211	213	193	191	173
	Minimum	200	186	174	157	142	128
	Maximum	260	234	267	243	220	201
	C.V. (%)	6.39	6.16	9.74	10.0	11.0	11.4
F_1^{cu}	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	ANOVA	ANOVA	ANOVA	ANOVA	ANOVA	ANOVA
(ksi)	C_1	15.2	13.4	21.0	19.6	22.4	21.1
	C_2	2.21	2.23	2.00	2.03	4.17	4.25
	No. Specimens	30		30		15	
	No. Batches	8		10		3	
	Approval Class	Interim		Interim		Interim	
	Mean	18.8	17.0			18.3	16.6
	Minimum	17.9	16.2			17.5	15.7
	Maximum	19.7	17.8			19.1	17.3
E_1^c	C.V. (%)	3.21	3.53			2.62	3.16
(Msi)	No. Specimens	15				15	
	No. Batches	3				3	
	Approval Class	Interim				Interim	
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned at 140°F, 95% relative humidity for 30 days.
(2) B-values are presented only for fully approved data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:29Z
Check the source to verify that this is the current version before use.

* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.13(d)
C/Ep 145-UT
AS4/3501-6
SBS, 31-plane
$[0]_8$
75/A, 200/A
Screening

MATERIAL: AS4/3501-6 (no bleed)

RESIN CONTENT: 36 - 39 wt%
FIBER VOLUME: 52 - 56 %
PLY THICKNESS: 0.0057 - 0.0063 in.

COMP. DENSITY: 1.55 - 1.57 g/cm^3
VOID CONTENT:

TEST METHOD: ASTM D 2344-76

MODULUS CALCULATION: Initial tangent

NORMALIZED BY: Not normalized

Temperature(°F)		75	200			
Moisture Content(%)		ambient	ambient			
Equilibrium at T, RH						
Source Code		26	26			
	Mean	17.9	14.0			
	Minimum	16.5	12.9			
	Maximum	19.0	15.4			
	C.V. (%)	4.46	4.73			
F_{31}^{sbs} (ksi)	B-value	(1)	(1)			
	Distribution	ANOVA	ANOVA			
	C_1	0.824	0.683			
	C_2	2.36	2.34			
	No. Specimens	30	30			
	No. Batches	8	10			
	Approval Class	Screening	Screening			

(1) B-values are presented only for fully approved data.

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Table 4.2.13(e)
C/Ep 145 - UT
AS4/3501-6
Tension, x-axis
$[0/45/90/-45]_s$
75/A
Screening

MATERIAL: AS4/3501-6 (no bleed)

RESIN CONTENT: 36 - 37 wt% COMP. DENSITY: 1.56-1.57 g/cm³
FIBER VOLUME: 54 - 56 % VOID CONTENT:
PLY THICKNESS: 0.0057 - 0.0062 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: NA

Temperature(°F)		75				
Moisture Content(%)		ambient				
Equilibrium at T, RH						
Source Code		26				
	Mean	87.4				
	Minimum	83.2				
	Maximum	92.8				
	C.V. (%)	3.43				
F_x^{tu}	B-value					
	Distribution	Normal				
(ksi)	C_1	87.4				
	C_2	3.00				
	No. Specimens	9				
	No. Batches	3				
	Approval Class	Screening				
	Mean					
	Minimum					
	Maximum					
E_x^t	C.V. (%)					
(Msi)	No. Specimens					
	No. Batches					
	Approval Class					
	Mean					
ν_{xy}^t	No. Specimens					
	No. Batches					
	Approval Class					
	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
ϵ_x^{tu}	B-value					
	Distribution					
$(\mu\epsilon)$	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:29Z
Check the source to verify that this is the current version before use.

Table 4.2.13(f)
C/Ep 145 - UT
AS4/3501-6
Open Hole Tension, x-axis
$[0/45/90/-45]_s$
75/A
Screening

MATERIAL: AS4/3501-6 (no bleed)
RESIN CONTENT: 36 - 37 wt%
COMP. DENSITY: 1.56-1.57 g/cm³
FIBER VOLUME: 54 - 56 %
VOID CONTENT:
PLY THICKNESS: 0.0060 - 0.0064 in.
TEST METHOD: SACMA 5-88 (1)
MODULUS CALCULATION:
NORMALIZED BY: NA

Temperature(°F)		75				
Moisture Content(%)		ambient				
Equilibrium at T, RH						
Source Code		26				
	Mean	56.8				
	Minimum	54.4				
	Maximum	60.8				
	C.V. (%)	3.75				
F_x^{oht}	B-value					
(ksi)	Distribution	Normal				
	C_1	56.8				
	C_2	2.13				
	No. Specimens	9				
	No. Batches	3				
	Approval Class	Screening				
	Mean					
	Minimum					
	Maximum					
E_x^{oht}	C.V. (%)					
(Msi)	No. Specimens					
	No. Batches					
	Approval Class					
	Mean					
ν_{xy}^{t}	No. Specimens					
	No. Batches					
	Approval Class					
	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
ϵ_x^{oht}	B-value					
($\mu\epsilon$)	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) Note SACMA 5-88 uses a $[45/0/-45/90]_{2s}$ lay-up.

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4.2.14 AS4 3k/3501-6 Plain weave *

MATERIAL:	AS4 3k/3501-6 Plain weave		C/Ep AS4/3501-6 Summary
PREPREG:	Hercules AW193P		
FIBER:	Hercules AS4 3k W	MATRIX:	Hercules 3501-6
T_g(ambient):	T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure: 240 ± 10°F, 60 min., 85 psig; 350 ± 10°F, 120 ± 10 min., 100 ± 10 psig, no bleed		

* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	6/88
Date of prepreg manufacture	Date of analysis	1/93
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	75°F/A		-65°A/F	200°F/A		75°F/W	200°F/W	
Tension, 1-axis	SS--		SS--	SS--				
Tension, 2-axis								
Tension, 3-axis								
Compression, 1-axis	II--			II--		II--	II--	
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
SB Strength, 31-plane	S---			S---		S---	S---	

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.80		
Resin Density	(g/cm^3)	1.28		
Composite Density	(g/cm^3)	1.58	1.54 - 1.56	
Fiber Areal Weight	(g/m^2)	193	193	
Ply Thickness	(in)	0.0070	0.0074 - 0.0086	

LAMINATE PROPERTY SUMMARY

	75°F/A							
$[0_t/90_t/\pm 45_t]$ Family								
Tension, x-axis	SS--							
$[\pm 45_t/0_t/90_t]$ Family								
OHT, x-axis	S---							

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.14(a)
C/Ep 193-PW
AS4/3501-6
Tension, 1-axis
$[0_f]_8$
75/A, -65/A, 200/A
Screening

MATERIAL: AS4 3k/3501-6 Plain weave

RESIN CONTENT: 38 wt% COMP. DENSITY: 1.56 g/cm^3
FIBER VOLUME: 53 - 54 % VOID CONTENT:
PLY THICKNESS: 0.0074 - 0.0080 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		75 ambient 26		-65 ambient 26		200 ambient 26	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	124	117	112	105	126	119
	Minimum	117	111	103	98.1	116	108
	Maximum	133	124	120	112	133	126
	C.V. (%)	4.18	3.56	4.63	4.00	4.79	5.88
F_1^{tu}	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	Normal	Normal	Normal	Normal	Normal	Normal
(ksi)	C_1	124	117	112	105	126	119
	C_2	5.17	4.15	5.17	4.21	6.05	7.00
	No. Specimens	9		9		9	
	No. Batches	3		3		3	
	Approval Class	Screening		Screening		Screening	
	Mean	9.8	9.2	10.5	9.9	10.1	9.5
	Minimum	9.4	8.8	9.7	9.1	7.1	6.7
	Maximum	10.2	9.5	11.1	10.4	10.7	10.1
E_1^t	C.V. (%)	3,0	2.5	4.6	4.2	11	11
(Msi)	No. Specimens	9		9		9	
	No. Batches	3		3		3	
	Approval Class	Screening		Screening		Screening	
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.14(b)
C/Ep 193-PW
AS4/3501-6
Compression, 1-axis
$[0_f]_{14}$
75/A, 200/A, 75/W
Interim

MATERIAL: AS4 3k/3501-6 Plain weave

RESIN CONTENT: 39 - 41 wt% COMP. DENSITY: 1.54 - 1.55 g/cm^3
FIBER VOLUME: 51 - 52 % VOID CONTENT:
PLY THICKNESS: 0.0081 - 0.0086 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		200		75 (1)	
Moisture Content(%)		ambient		ambient		wet	
Equilibrium at T, RH		26		26		26	
Source Code							
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{cu} (ksi)	Mean	130	117	108	97.3	112	101
	Minimum	115	104	92.8	83.0	99.6	88.0
	Maximum	140	127	121	109	122	109
	C.V. (%)	6.45	6.49	7.44	7.71	5.56	5.65
	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	Nonpara.	Nonpara.	Weibull	Normal	ANOVA	ANOVA
	C_1	8	8	112	97.3	6.83	6.32
	C_2	1.54	1.54	15.1	7.51	4.85	5.09
	No. Specimens	15		15		15	
	No. Batches	3		3		3	
	Approval Class	Interim		Interim		Interim	
E_1^c (Msi)	Mean	9.2	8.3	9.8	8.8	9.4	8.4
	Minimum	8.5	7.7	9.2	8.4	8.8	8.1
	Maximum	9.8	8.8	10.2	9.1	9.9	8.8
	C.V. (%)	3.4	4.3	3.5	2.5	3.0	2.4
	No. Specimens	15		15		15	
	No. Batches	3		3		3	
	Approval Class	Interim		Interim		Interim	
ν_{12}^c	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_1^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned at 140°F, 95% relative humidity for 30 days.
(2) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.14(c)
C/Ep 193-PW
AS4/3501-6
Compression, 1-axis
$[0_f]_{14}$
200/W
Interim

MATERIAL: AS4 3k/3501-6 Plain weave

RESIN CONTENT: 39 - 41 wt% COMP. DENSITY: 1.54 - 1.55 g/cm³
FIBER VOLUME: 51 - 52 % VOID CONTENT:
PLY THICKNESS: 0.0081 - 0.0086 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		200			
Moisture Content(%)		(1)			
Equilibrium at T, RH		wet			
Source Code		26			
		Normalized	Measured		
	Mean	58.7	52.7		
	Minimum	51.7	46.2		
	Maximum	65.4	59.7		
	C.V. (%)	7.27	7.58		
F_1^{cu}	B-value	(2)	(2)		
	Distribution	Weibull	Weibull		
(ksi)	C_1	60.6	54.5		
	C_2	15.6	15.2		
	No. Specimens	15			
	No. Batches	3			
	Approval Class	Interim			
	Mean	9.1	8.1		
	Minimum	8.7	7.8		
	Maximum	9.4	8.5		
E_1^c	C.V. (%)	2.4	2.9		
(Msi)	No. Specimens	15			
	No. Batches	3			
	Approval Class	Interim			
	Mean				
ν_{12}^c	No. Specimens				
	No. Batches				
	Approval Class				
	Mean				
	Minimum				
	Maximum				
	C.V. (%)				
ϵ_1^{cu}	B-value				
	Distribution				
($\mu\epsilon$)	C_1				
	C_2				
	No. Specimens				
	No. Batches				
	Approval Class				

(1) Conditioned at 140°F, 95% relative humidity for 30 days.
(2) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

MATERIAL:	AS4 3k/3501-6 Plain weave		
RESIN CONTENT:	39 - 41 wt%	COMP. DENSITY:	1.54 - 1.55 g/cm^3
FIBER VOLUME:	51 - 52 %	VOID CONTENT:	
PLY THICKNESS:	0.0077 - 0.0082 in.		
TEST METHOD:	ASTM D 2344	MODULUS CALCULATION:	
NORMALIZED BY:	Not normalized		

Table 4.2.14(d)
C/Ep 193-PW
AS4/3501-6
SBS, 31-plane
$[0_f]_{14}$
75/A, 200/A, 75/W, 200/W
Screening

Temperature(°F)		75 ambient	200 ambient	75 wet (1)	200 wet (1)	
Moisture Content(%)						
Equilibrium at T, RH						
Source Code		26	26	26	26	
	Mean	10.9	8.4	10.9	5.3	
	Minimum	9.7	8.1	10.0	5.2	
	Maximum	11.9	8.8	11.4	5.5	
	C.V. (%)	6.09	2.5	3.47	2.3	
F_{31}^{sbs} (ksi)	B-value	(2)	(2)	(2)	(2)	
	Distribution	Weibull	Normal	Weibull	Nonpara.	
	C_1	11.2	8.4	11.0	7	
	C_2	20.1	0.21	35.4	1.81	
	No. Specimens	15	9	15	12	
	No. Batches	3	3	3	3	
	Approval Class	Screening	Screening	Screening	Screening	

(1) Conditioned at 140°F, 95% relative humidity for 30 days.
(2) B-values are presented only for fully approved data.

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MATERIAL: AS4 3k/3501-6 Plain weave

RESIN CONTENT: 37 - 38 wt% COMP. DENSITY: 0.056 lb/in^3
FIBER VOLUME: 53 - 54 % VOID CONTENT:
PLY THICKNESS: 0.0080 - 0.0085 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: NA

Table 4.2.14(e)
C/Ep 193-PW
AS4/3501-6
Tension, x-axis
$[0_f/90_f/\pm 45_f]_{2s}$
75/A
Screening

Temperature(°F)		75				
Moisture Content(%)		ambient				
Equilibrium at T, RH						
Source Code		26				
	Mean	68.5				
	Minimum	62.0				
	Maximum	75.1				
	C.V. (%)	7.60				
F_x^{tu}	B-value	(1)				
	Distribution	Normal				
(ksi)	C_1	68.5				
	C_2	5.21				
	No. Specimens	9				
	No. Batches	3				
	Approval Class	Screening				
	Mean	6.0				
	Minimum	5.6				
	Maximum	6.3				
E_x^t	C.V. (%)	3.6				
(Msi)	No. Specimens	9				
	No. Batches	3				
	Approval Class	Screening				
	Mean					
ν_{xy}^t	No. Specimens					
	No. Batches					
	Approval Class					
	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
ϵ_x^{tu}	B-value					
	Distribution					
$(\mu\epsilon)$	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) B-values are presented only for fully approved data.

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MATERIAL: AS4 3k/3501-6 Plain weave						**Table 4.2.14(f)** **C/Ep 193-PW** **AS4/3501-6** **Open Hole Tension,** **x-axis** **$[\pm 45_t/0_t/90_t]_{2s}$** **75/A** **Screening**
RESIN CONTENT: 37 - 38 wt%		COMP. DENSITY: 0.056 lb/in^3				
FIBER VOLUME: 53 - 54 %		VOID CONTENT:				
PLY THICKNESS: 0.0080 - 0.0085 in.						
TEST METHOD:		MODULUS CALCULATION:				
SACMA 5-88 (1)						
NORMALIZED BY: NA						
Temperature(°F)		75				
Moisture Content(%)		ambient				
Equilibrium at T, RH						
Source Code		26				
	Mean	51.4				
	Minimum	48.6				
	Maximum	53.8				
	C.V. (%)	3.40				
F_x^{oht}	B-value	(2)				
	Distribution	ANOVA				
(ksi)	C_1	2.46				
	C_2	1.20				
	No. Specimens	9				
	No. Batches	3				
	Approval Class	Screening				
	Mean					
	Minimum					
	Maximum					
E_x^t	C.V. (%)					
(Msi)	No. Specimens					
	No. Batches					
	Approval Class					
	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
ϵ_x^{tu}	B-value					
	Distribution					
$(\mu\epsilon)$	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) SACMA 5-88 uses a [+45/0/-45/90]$_s$ lay-up.
(2) B-values are presented only for fully approved data.

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4.2.15 AS4 3k/3501-6S 5-harness satin weave *

MATERIAL:	AS4 3k/3501-6S 5-harness satin weave		**C/Ep AS4/3501-6S Summary**
PREPREG:	Hercules AW280-5HS		
FIBER:	Hercules AS4 3k W	MATRIX:	Hercules 3501-6S
T_g(ambient):	T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure: 240 ± 10°F, 60 min., 85 psig; 350 ± 10°F, 120 ± 10 min., 100 ± 10 psig, no bleed		

* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	6/88
Date of prepreg manufacture	Date of analysis	1/93
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	75°F/A		200°F/A					
Tension, 1-axis	II--							
Tension, 2-axis								
Tension, 3-axis								
Compression, 1-axis	I---		I---					
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
SB Strength, 31-plane	S---		S---					

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.80		
Resin Density	(g/cm^3)	1.28		
Composite Density	(g/cm^3)	1.58	1.58 - 1.59	
Fiber Areal Weight	(g/m^2)	280	279 - 284	
Ply Thickness	(in)		0.0106 - 0.0107	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.15(a)
C/Ep 280-5HS
AS4/3501-6S
Tension, 1-axis
$[0_f]_8$
75/A
Interim

MATERIAL: AS4 3k/3501-6S 5-harness satin weave

RESIN CONTENT: 33 - 35 wt%
FIBER VOLUME: 57 - 60 %
PLY THICKNESS: 0.0106 - 0.0107 in.

COMP. DENSITY: 1.58 - 1.59 g/cm^3
VOID CONTENT:

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75					
Moisture Content(%)		ambient					
Equilibrium at T, RH							
Source Code		26					
		Normalized	Measured				
	Mean	112	115				
	Minimum	97.6	100				
	Maximum	123	126				
	C.V. (%)	5.78	5.55				
F_1^{tu}	B-value	(1)	(1)				
	Distribution	ANOVA	ANOVA				
(ksi)	C_1	6.63	6.55				
	C_2	2.26	2.25				
	No. Specimens	30					
	No. Batches	10					
	Approval Class	Interim					
	Mean	9.73	10.0				
	Minimum	8.93	9.20				
	Maximum	10.1	10.3				
E_1^t	C.V. (%)	2.48	2.31				
(Msi)	No. Specimens	30					
	No. Batches	10					
	Approval Class	Interim					
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.15(b)
C/Ep 280-5HS
AS4/3501-6S
Compression, 1-axis
$[0_f]_6$
75/A, 200/A
Interim

MATERIAL: AS4 3k/3501-6S 5-harness satin weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.58 - 1.59 g/cm³
FIBER VOLUME: 57 - 60 % VOID CONTENT:
PLY THICKNESS: 0.0106 - 0.0107 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75 ambient		200 ambient			
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		26		26			
		Normalized	Measured	Normalized	Measured		
	Mean	124	128	110	113		
	Minimum	108	111	96.1	99.0		
	Maximum	144	148	122	125		
	C.V. (%)	6.73	6.74	6.31	6.24		
F_1^{cu}	B-value	(1)	(1)	(1)	(1)		
	Distribution	Weibull	Weibull	ANOVA	ANOVA		
(ksi)	C_1	128	132	7.04	7.15		
	C_2	15.4	15.3	2.10	2.09		
	No. Specimens	30		30			
	No. Batches	10		10			
	Approval Class	Interim		Interim			
	Mean						
	Minimum						
	Maximum						
E_1^c	C.V. (%)						
(Msi)	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.15(c)
C/Ep 280-5HS
AS4/3501-6S
SBS, 31-plane
$[0_f]_8$
75/A, 200/A
Screening

MATERIAL: AS4 3k/3501-6S 5-harness satin weave

RESIN CONTENT: 33 - 35 wt%
FIBER VOLUME: 57 - 60 %
PLY THICKNESS: 0.0106 - 0.0107 in.

COMP. DENSITY: 1.58 - 1.59 g/cm^3
VOID CONTENT:

TEST METHOD: ASTM D 2344

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75	200			
Moisture Content(%)		ambient	ambient			
Equilibrium at T, RH						
Source Code		26	26			
	Mean	11.0	9.53			
	Minimum	9.00	8.40			
	Maximum	13.2	10.8			
	C.V. (%)	10.8	6.70			
F_{31}^{sbs}	B-value	(1)	(1)			
	Distribution	ANOVA	ANOVA			
(ksi)	C_1	1.22	0.66			
	C_2	2.18	2.32			
	No. Specimens	30	30			
	No. Batches	10	10			
	Approval Class	Screening	Screening			

(1) B-values are presented only for fully approved data.

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4.2.16 AS4 6k/3502-6S 5-harness satin weave*

MATERIAL:	AS4 6k/3502 5-harness satin weave	C/Ep AS4/3502 Summary
PREPREG:	Hercules A370-5H/3502, 5-harness satin weave, 11 × 11 tow/in.	
FIBER:	Hercules AS4 6k, surface-treated "W"*, 0 twist	MATRIX: Hercules 3502
T_g(ambient):	404°F T_g(wet): 313°F	T_g METHOD: TMA
PROCESSING:	Autoclave cure: 280 ± 5°F, 90 min, 85 + 15-0 psi; 350°F, 120 min.	

* now "G"

Date of fiber manufacture	10/82-3/83	Date of testing	9/83-1/84
Date of resin manufacture	5/83	Date of data submittal	12/93, 5/94
Date of prepreg manufacture	5/83	Date of analysis	8/94
Date of composite manufacture	8/83 - 9/83		

LAMINA PROPERTY SUMMARY

	75°F/A		-65°F/A		180°F/W	250°F/W		
Tension, 1-axis	FF--		FF--		FF--	FF--		
Tension, 2-axis								
Tension, 3-axis								
Compression, 1-axis	FF--		IS--		FF--	FF--		
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane	FF--		FF--		FS--	FS--		
Shear, 23-plane								
Shear, 31-plane								

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.79		
Resin Density	(g/cm^3)	1.26		
Composite Density	(g/cm^3)	1.57	1.55 - 1.60	
Fiber Areal Weight	(g/m^2)	365	361 - 372	
Ply Thickness	(in)	0.0145	0.0142 - 0.0158	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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Table 4.2.16(a)
C/Ep - 365 - 5HS
AS4/3502
Tension, 1-axis
$[0/90/0/90/90/0]_t$
75/A, -65/A, 180/W
Fully Approved

MATERIAL: AS4 6k/3502 5-harness satin weave

RESIN CONTENT: 36 - 37 wt% COMP. DENSITY: 1.55-1.56 g/cm³
FIBER VOLUME: 56 - 57 % VOID CONTENT: 0.0 - 0.2 %
PLY THICKNESS: 0.0146 - 0.0157 in.

TEST METHOD: BMS 8-168D

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Ply thickness to 0.0145 in. (57%)

Temperature(°F)		75 ambient		-65 ambient		180 1.1 - 1.3 (1)	
Moisture Content(%) Equilibrium at T, RH							
Source Code		49		49		49	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	114	109	105	101	117	111
	Minimum	97.1	91.9	87.9	84.0	102	96.1
	Maximum	126	121	116	112	128	123
	C.V. (%)	6.87	7.01	5.33	5.53	5.29	5.36
F_1^{tu}	B-value	91.9		95.0		102	
	Distribution	ANOVA		Normal		ANOVA	
(ksi)	C_1	8.15		104.9		6.31	
	C_2	2.70		5.59		2.33	
	No. Specimens	30		30		30	
	No. Batches	5		5		5	
	Approval Class	Fully Approved		Fully Approved		Fully Approved	
	Mean	9.61		9.67		10.5	
	Minimum	9.29		9.09		9.74	
	Maximum	10.4		10.1		10.9	
E_1^t	C.V. (%)	3.08		2.35		2.75	
(Msi)	No. Specimens	30		30		30	
	No. Batches	5		5		5	
	Approval Class	Fully Approved		Fully Approved		Fully Approved	
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned at 160°F, 95-100% relative humidity until the moisture content was between 1.1 and 1.3%.

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Table 4.2.16(b)
C/Ep - 365 - 5HS
AS4/3502
Tension, 1-axis
$[0/90/0/90/90/0]_f$
250/W
Fully Approved

MATERIAL: AS4 6k/3502 5-harness satin weave

RESIN CONTENT: 36 - 37 wt% COMP. DENSITY: 1.55-1.56 g/cm³
FIBER VOLUME: 56 - 57 % VOID CONTENT: 0.0 - 0.2 %
PLY THICKNESS: 0.0150 - 0.0157 in.

TEST METHOD: BMS 8-168D

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Ply thickness to 0.0145 in. (57%)

Temperature(°F)		250					
Moisture Content(%)		1.1 - 1.3					
Equilibrium at T, RH							
Source Code		49					
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	108	102				
	Minimum	96.8	91.8				
	Maximum	119	113				
	C.V. (%)	4.62	4.86				
F_1^{tu}	B-value	96.6					
	Distribution	Weibull					
(ksi)	C_1	111					
	C_2	23.1					
	No. Specimens	30					
	No. Batches	5					
	Approval Class	Fully Approved					
	Mean	10.1					
	Minimum	9.29					
	Maximum	10.7					
E_1^t	C.V. (%)	3.65					
(Msi)	No. Specimens	30					
	No. Batches	5					
	Approval Class	Fully Approved					
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{tu}	B-value						
	Distribution						
$(\mu\epsilon)$	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned at 160°F, 95-100% relative humidity until the moisture content was between 1.1 and 1.3%.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:29Z
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Table 4.2.16(c)
C/Ep - 365 - 5HS
AS4/3502
Compression, 1-axis
$[0/90/0/90/90/0]_t$
75/A, -65/A, 180/W
Fully Approved, Interim

MATERIAL: AS4 6k/3502 5-harness satin weave

RESIN CONTENT: 36 - 37 wt% COMP. DENSITY: 1.55-1.56 g/cm³
FIBER VOLUME: 56 - 57 % VOID CONTENT: 0.0 - 0.2 %
PLY THICKNESS: 0.0142 - 0.0157 in.

TEST METHOD: ASTM D 695M (1)

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Ply thickness to 0.0145 in. (57%)

		75 ambient (2) 1		-65 ambient (2) 49		180 1.1 - 1.3 (2) 49	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	104	99.7	108	103	65.9	63.3
	Minimum	79.7	75.5	85.0	79.0	52.1	49.7
	Maximum	122	116	118	114	76.7	74.0
	C.V. (%)	10.1	10.7	8.62	9.56	9.81	10.0
F_1^{cu} (ksi)	B-value	83.7		(3)		52.4	
	Distribution	Weibull		Weibull		Weibull	
	C_1	109		111		68.7	
	C_2	12.1		16.4		11.7	
	No. Specimens	30		15		30	
	No. Batches	5		5		5	
	Approval Class	Fully Approved		Interim		Fully Approved	
E_1^c (Msi)	Mean	8.49		8.90		9.21	
	Minimum	8.15		7.70		6.25	
	Maximum	8.86		11.0		12.5	
	C.V. (%)	2.13		10.3		18.2	
	No. Specimens	30		14		30	
	No. Batches	5		5		5	
	Approval Class	Fully Approved		Interim		Fully Approved	
ν_{12}^t	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_1^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Tabbed specimen, length 3.12 inch, width 0.50 inch, gage length 0.50 inch.
(2) Conditioned at 160°F, 95-100% relative humidity until the moisture content was between 1.1 and 1.3%.
(3) B-values are presented only for fully approved data.

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Table 4.2.16(d)
C/Ep - 365 - 5HS
AS4/3502
Compression, 1-axis
$[0/90/0/90/90/0]_t$
250/W
Fully Approved

MATERIAL: AS4 6k/3502 5-harness satin weave

RESIN CONTENT: 36 - 37 wt% COMP. DENSITY: 1.55-1.56 g/cm³
FIBER VOLUME: 56 - 57 % VOID CONTENT: 0.0 - 0.2 %
PLY THICKNESS: 0.0142 - 0.0157 in.

TEST METHOD: ASTM D 695M (1)

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Ply thickness to 0.0145 in. (57%)

Temperature(°F)		250					
Moisture Content(%)		1.1 - 1.3					
Equilibrium at T, RH							
Source Code		49					
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	56.3	53.2				
	Minimum	45.5	42.6				
	Maximum	75.2	70.8				
	C.V. (%)	16.0	16.2				
F_1^{cu}	B-value	30.5					
	Distribution	ANOVA					
(ksi)	C_1	9.41					
	C_2	2.75					
	No. Specimens	30					
	No. Batches	5					
	Approval Class	Fully Approved					
	Mean	10.3					
	Minimum	8.88					
	Maximum	12.4					
E_1^c	C.V. (%)	6.60					
(Msi)	No. Specimens	30					
	No. Batches	5					
	Approval Class	Fully Approved					
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{cu}	B-value						
	Distribution						
$(\mu\epsilon)$	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Tabbed specimen, length 3.12 inch, width 0.50 inch, gage length 0.50 inch.
(2) Conditioned at 160°F, 95-100% relative humidity until the moisture content was between 1.1 and 1.3%.

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Table 4.2.16(e)
C/Ep - 365 - 5HS
AS4/3502
Shear, 12-plane
[±45/±45/∓45]$_r$,
75/A, -65/A, 180/W, 250/W
Fully Approved, Screening

MATERIAL: AS4 6k/3502 5-harness satin weave

RESIN CONTENT: 36 - 37 wt% COMP. DENSITY: 1.55-1.56 g/cm^3
FIBER VOLUME: 56 - 57 % VOID CONTENT: 0.0 - 0.2 %
PLY THICKNESS: 0.0145 - 0.0158 in.

TEST METHOD: ASTM D 3518-76

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Not normalized

Temperature(°F)		75	-65	180	250	
Moisture Content(%)		ambient	ambient	1.1 - 1.3	1.1 - 1.3	
Equilibrium at T, RH				(1)	(1)	
Source Code		49	49	49	49	
F_{12}^{su} (ksi)	Mean	12.6	14.0	11.7	9.30	
	Minimum	11.4	12.1	10.7	8.27	
	Maximum	13.7	15.4	12.9	10.5	
	C.V. (%)	5.61	7.47	5.24	6.76	
	B-value	10.1	10.1	9.53	6.95	
	Distribution	ANOVA	ANOVA	ANOVA	ANOVA	
	C_1	0.775	1.16	0.669	0.698	
	C_2	3.21	3.36	3.20	3.37	
	No. Specimens	36	36	36	36	
	No. Batches	5	5	5	5	
	Approval Class	Fully Approved	Fully Approved	Fully Approved	Fully Approved	
G_{12}^{s} (Msi)	Mean	0.514	0.682	0.204	0.174	
	Minimum	0.485	0.638	0.196	0.147	
	Maximum	0.553	0.731	0.212	0.203	
	C.V. (%)	3.68	3.40	2.82	11.8	
	No. Specimens	36	36	6	5	
	No. Batches	5	5	1	1	
	Approval Class	Fully Approved	Fully Approved	Screening	Screening	
γ_{12}^{su} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) Conditioned at 160°F, 95-100% relative humidity until the moisture content was between 1.1 and 1.3%.

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4.2.17 T-300 15k/976 unidirectional tape*

MATERIAL:	T300 15k/976 unidirectional tape			C/Ep T300 15k/976 Summary
PREPREG:	Fiberite T300/976 unidirectional tape			
FIBER:	Union Carbide T300 15k	MATRIX:	Fiberite 976	
T_g(dry):	518°F T_g(wet): 493°F	T_g METHOD:	DMA	
PROCESSING:	*Autoclave cure: 250°F, 100 psi, 45 min; 350°F, 2 hours*			

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	2/82
Date of prepreg manufacture	7/80	Date of analysis	9/94
Date of composite manufacture			

LAMINA PROPERTY SUMMARY

	72°F/A		-67°F/A	260°F/A	350°F/A			
Tension, 1-axis	SSSS		SSSS	SSSS	SSSS			
Tension, 2-axis	SS-S		SS-S	SS-S	SS-S			
Tension, 3-axis								
Compression, 1-axis	SS-S		SS-S	SS-S	SS-S			
Compression, 2-axis	SS-S		SS-S	SS-S	SS-S			
Compression, 3-axis								
Shear, 12-plane	SS--		SS--	SS--	SS--			
Shear, 23-plane								
SB Strength, 31-plane	S---		S---	S---	S---			

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.78		
Resin Density	(g/cm^3)	1.28		
Composite Density	(g/cm^3)	1.62	1.58 - 1.65	
Fiber Areal Weight	(g/m^2)	152		
Ply Thickness	(in)		0.0049 - 0.0053	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.17(a)
C/Ep - UT
T300 15k/976
Tension, 1-axis
$[0]_8$
72/A, -67/A, 260/A
Screening

MATERIAL: T300 15k/976 unidirectional tape

RESIN CONTENT: 35 wt%
FIBER VOLUME: 59 %
PLY THICKNESS: 0.0053 in.

COMP. DENSITY: 1.60 g/cm^3
VOID CONTENT: approx. 0.0 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Fiber Volume to 60 %

Temperature(°F)		72 ambient		-67 ambient		260 ambient	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		48		48		48	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	211	207	199	197	236	232
	Minimum	185	191	187	173	205	212
	Maximum	235	219	220	214	256	255
	C.V. (%)	11.2	6.47	6.83	7.67	9.88	6.84
F_1^{tu}	B-value	(1)		(1)		(1)	
	Distribution	Normal		Normal		Normal	
(ksi)	C_1	211		199		236	
	C_2	23.6		13.6		23.3	
	No. Specimens	5		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	19.6	19.3	20.8	20.4	22.6	22.4
	Minimum	17.8	18.2	19.5	19.6	20.5	21.2
	Maximum	21.2	20.4	22.6	21.0	24.9	22.9
E_1^t	C.V. (%)	6.09	5.18	5.88	2.74	8.97	2.19
(Msi)	No. Specimens	5		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	0.318		0.318		0.312	
ν_{12}^t	No. Specimens	5		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	10400		8600		9900	
	Minimum	10000		8000		9500	
	Maximum	10800		9000		10500	
	C.V. (%)	3.42		5.29		4.46	
ϵ_1^{tu}	B-value	(1)		(1)		(1)	
	Distribution	Normal		Normal		Normal	
($\mu\epsilon$)	C_1	10400		8600		9900	
	C_2	356		454		442	
	No. Specimens	5		4		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.17(b)
C/Ep - UT
T300 15k/976
Tension, 1-axis
$[0]_6$
350/A
Screening

MATERIAL: T300 15k/976 unidirectional tape

RESIN CONTENT: 35 wt% COMP. DENSITY: 1.60 g/cm^3
FIBER VOLUME: 59 % VOID CONTENT: approx. 0.0 %
PLY THICKNESS: 0.0053 in.

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Fiber Volume to 60 %

Temperature(°F)		350					
Moisture Content(%)		ambient					
Equilibrium at T, RH							
Source Code		48					
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	232	228				
	Minimum	212	219				
	Maximum	248	242				
	C.V. (%)	7.11	3.77				
F_1^{tu}	B-value	(1)					
	Distribution	Normal					
(ksi)	C_1	232					
	C_2	16.5					
	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					
	Mean	22.4	22.1				
	Minimum	21.0	20.2				
	Maximum	24.2	23.9				
E_1^t	C.V. (%)	5.59	6.19				
(Msi)	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					
	Mean	0.348					
ν_{12}^t	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					
	Mean	9930					
	Minimum	9600					
	Maximum	10700					
	C.V. (%)	5.29					
ϵ_1^{tu}	B-value	(2)					
	Distribution	Normal					
($\mu\epsilon$)	C_1	9930					
	C_2	525					
	No. Specimens	4					
	No. Batches	1					
	Approval Class	Screening					

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.17(c)
C/Ep - UT
T300 15k/976
Tension, 2-axis
$[90]_{15}$
72/A, -67/A, 260/A, 350/A
Screening

MATERIAL: T300 15k/976 unidirectional tape

RESIN CONTENT: 25 wt%
FIBER VOLUME: 69 %
PLY THICKNESS: 0.0049 in.

COMP. DENSITY: 1.64 g/cm^3
VOID CONTENT: approx. 0.0 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Not normalized

Temperature(°F)		72 ambient	-67 ambient	260 ambient	350 ambient	
Moisture Content(%)						
Equilibrium at T, RH						
Source Code		48	48	48	48	
	Mean	5.66	4.73	3.81	3.47	
	Minimum	4.53	3.23	2.87	2.67	
	Maximum	6.52	6.29	4.68	3.83	
	C.V. (%)	15.4	25.1	17.4	13.2	
F_2^{tu}	B-value	(1)	(1)	(1)	(1)	
	Distribution	Normal	Normal	Normal	Normal	
(ksi)	C_1	5.66	4.73	3.812	3.47	
	C_2	0.870	1.19	0.664	0.458	
	No. Specimens	5	5	5	5	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	
	Mean	1.34	1.69	1.37	1.30	
	Minimum	1.28	1.49	1.16	1.25	
	Maximum	1.39	1.88	1.55	1.43	
E_2^t	C.V. (%)	3.13	9.01	10.1	5.83	
(Msi)	No. Specimens	5	5	5	5	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	
	Mean					
ν_{21}^t	No. Specimens					
	No. Batches					
	Approval Class					
	Mean	3900	2760	2640	2620	
	Minimum	3200	1900	2100	2200	
	Maximum	4600	3300	3400	3000	
	C.V. (%)	14.6	20.4	19.1	13.3	
ϵ_2^{tu}	B-value	(1)	(1)	(1)	(1)	
	Distribution	Normal	Normal	Normal	Normal	
($\mu\epsilon$)	C_1	3900	2760	2640	2620	
	C_2	570	564	503	349	
	No. Specimens	5	5	5	5	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	

(1) B-values are presented only for fully approved data.

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Table 4.2.17(d)
C/Ep - UT
T300 15k/976
Compression, 1-axis
$[0]_{20}$
72/A, -67/A, 260/A
Screening

MATERIAL: T300 15k/976 unidirectional tape

RESIN CONTENT: 24 wt%
FIBER VOLUME: 70 %
PLY THICKNESS: 0.0050 in.

COMP. DENSITY: 1.63 g/cm³
VOID CONTENT: approx. 0.0 %

TEST METHOD: ASTM D 3410A-75

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Fiber Volume to 60 %

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		72 ambient 48		-67 ambient 48		260 ambient 48	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	188	218	192	223	147	171
	Minimum	139	162	169	196	95.6	111
	Maximum	214	248	218	254	177	205
	C.V. (%)	15.9	15.9	9.76	9.76	21.7	21.7
F_1^{cu} (ksi)	B-value	(1)		(1)		(1)	
	Distribution	Normal		Normal		Normal	
	C_1	188		192		147	
	C_2	29.9		18.8		31.9	
	No. Specimens	5		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	18.7	21.8	18.8	21.9	18.4	21.4
	Minimum	14.9	17.3	16.2	18.8	10.8	12.6
	Maximum	21.9	25.5	25.5	29.6	22.6	26.2
E_1^c	C.V. (%)	13.4	13.4	20.1	20.1	26.5	26.5
(Msi)	No. Specimens	5		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		12500		14500		8860
	Minimum		9500		9900		6300
	Maximum		19600		20000		12600
	C.V. (%)		32.2		31.5		30.2
ϵ_1^{cu}	B-value		(1)		(1)		(1)
	Distribution		Normal		Normal		Normal
($\mu\epsilon$)	C_1		12500		14500		8860
	C_2		404		4560		2670
	No. Specimens	5		5		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	

(1) B-values are presented only for fully approved data.

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Table 4.2.17(e)
C/Ep - UT
T300 15k/976
Compression, 1-axis
$[0]_{20}$
350/A
Screening

MATERIAL: T300 15k/976 unidirectional tape

RESIN CONTENT: 24 wt%
FIBER VOLUME: 70 %
PLY THICKNESS: 0.0050 in.

COMP. DENSITY: 1.63 g/cm^3
VOID CONTENT: approx. 0.0 %

TEST METHOD: ASTM D 3410A-75

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Fiber Volume to 60 %

Temperature(°F)		350					
Moisture Content(%)		ambient					
Equilibrium at T, RH							
Source Code		48					
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	136	158				
	Minimum	107	124				
	Maximum	160	186				
	C.V. (%)	18.5	18.5				
F_1^{cu}	B-value	(1)					
	Distribution	Normal					
(ksi)	C_1	136					
	C_2	25.2					
	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					
	Mean	19.7	22.9				
	Minimum	16.5	19.1				
	Maximum	23.0	26.7				
E_1^c	C.V. (%)	13.2	13.2				
(Msi)	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		9400				
	Minimum		5000				
	Maximum		14000				
	C.V. (%)		39.7				
ϵ_1^{cu}	B-value		(2)				
	Distribution		Normal				
($\mu\epsilon$)	C_1		9400				
	C_2		3730				
	No. Specimens	5					
	No. Batches	1					
	Approval Class	Screening					

(1) B-values are presented only for fully approved data.

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Table 4.2.17(f)
C/Ep - UT
T300 15k/976
Compression, 2-axis
$[90]_{20}$
72/A, -67/A, 260/A, 350/A
Screening

MATERIAL: T300 15k/976 unidirectional tape

RESIN CONTENT: 24 wt%
FIBER VOLUME: 70 %
PLY THICKNESS: 0.0050 in.

COMP. DENSITY: 1.63 g/cm^3
VOID CONTENT: approx. 0.0 %

TEST METHOD: ASTM D 3410A-75

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Not normalized

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		72 ambient 48	-67 ambient 48	260 ambient 48	350 ambient 48	
	Mean	30.0	35.1	22.6	19.1	
	Minimum	26.7	26.7	19.4	17.3	
	Maximum	31.9	44.9	25.7	22.8	
	C.V. (%)	7.10	18.9	10.7	11.7	
F_2^{cu}	B-value	(1)	(1)	(1)	(1)	
	Distribution	Normal	Normal	Normal	Normal	
(ksi)	C_1	30.0	35.1	22.6	19.1	
	C_2	2.13	6.62	2.42	2.24	
	No. Specimens	5	5	5	5	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	
	Mean	1.46	1.84	1.84	1.64	
	Minimum	1.32	1.46	1.37	1.25	
	Maximum	1.73	2.18	3.03	2.02	
E_2^c	C.V. (%)	11.1	17.0	36.7	19.6	
(Msi)	No. Specimens	5	5	5	5	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	
	Mean					
ν_{21}^c	No. Specimens					
	No. Batches					
	Approval Class					
	Mean	32300	22100	14900	14200	
	Minimum	7900	13000	9600	6900	
	Maximum	46300	27700	21400	21300	
	C.V. (%)	44.7	31.1	40.1	47.2	
ϵ_2^{cu}	B-value	(1)	(1)	(2)	(1)	
	Distribution	Normal	Normal		Normal	
($\mu\epsilon$)	C_1	32300	22100		14200	
	C_2	14400	6880		6720	
	No. Specimens	5	5	3	5	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	

(1) B-values are presented only for fully approved data.
(2) The statistical analysis is not completed for less than four specimens.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.17(g)
C/Ep - UT
T300 15k/976
Shear, 12-plane
$[\pm 45]_{2s}$
72/A, -67/A, 260/A, 350/A
Screening

MATERIAL: T300 15k/976 unidirectional tape

RESIN CONTENT: 25 wt%
FIBER VOLUME: 69 %
PLY THICKNESS: 0.0052 in.

COMP. DENSITY: 1.63 g/cm^3
VOID CONTENT: approx. 0.1 %

TEST METHOD: ASTM D 3518-76

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Not normalized

Temperature(°F)		72 ambient	-67 ambient	260 ambient	350 ambient	
Moisture Content(%)						
Equilibrium at T, RH						
Source Code		48	48	48	48	
	Mean	11.1	13.7	8.25	8.30	
	Minimum	11.0	13.2	7.78	7.67	
	Maximum	11.4	15.5	8.72	9.36	
	C.V. (%)	1.23	6.99	4.78	7.80	
F_{12}^{su}	B-value	(1)	(1)	(1)	(1)	
	Distribution	Normal	Nonpara.	Normal	Normal	
(ksi)	C_1	11.1	4	8.25	8.30	
	C_2	0.137	4.10	0.394	0.647	
	No. Specimens	5	5	5	5	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	
	Mean	0.91	1.0	0.89	0.77	
	Minimum	0.84	0.89	0.82	0.70	
	Maximum	0.96	1.08	0.94	0.82	
G_{12}^{s}	C.V. (%)	5.1	7.1	5.3	7.4	
(Msi)	No. Specimens	5	5	5	5	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	
	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
γ_{12}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					
	Data File					

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.17(h)
C/Ep - UT
T300 15k/976
SBS, 31-plane
$[0]_{15}$
72/A, -67/A, 260/A, 350/A
Screening

MATERIAL: T300 15k/976 unidirectional tape

RESIN CONTENT: 25 wt%
FIBER VOLUME: 69 %
PLY THICKNESS: 0.0052 in.

COMP. DENSITY: 1.63 g/cm³
VOID CONTENT: approx. 0.1 %

TEST METHOD: ASTM D 2344-76

MODULUS CALCULATION: Linear portion of curve

NORMALIZED BY: Not normalized

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		72 ambient 48	-67 ambient 48	260 ambient 48	350 ambient 48	
	Mean	12.9	16.6	9.36	8.60	
	Minimum	9.42	14.2	8.59	7.71	
	Maximum	17.1	19.6	10.8	9.56	
	C.V. (%)	18.4	12.8	10.1	8.06	
F_{31}^{sbs} (ksi)	B-value	(1)	(1)	(1)	(1)	
	Distribution	Weibull	Normal	Normal	Normal	
	C_1	13.8	16.6	9.36	8.60	
	C_2	6.17	2.12	0.949	0.693	
	No. Specimens	10	5	5	5	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	

(1) B-values are presented only for fully approved data.

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4.2.18 IM7 12k/8551-7A unidirectional tape*

These data are presented in the MIL-HDBK-17-2D Annex.

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4.2.19 AS4 3k/3501-6 5-harness satin weave (bleed)*

C/Ep 280-5HS
AS4/3501-6
(Bleed)
Summary

MATERIAL:	AS4 3k/3501-6 5-harness satin weave (Bleed)		
PREPREG:	Hercules AW280-5H/3501-6		
FIBER:	Hercules AS4 3k, no twist	MATRIX:	Hercules 3501-6
T_g(dry):	T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure, 240 ± 10°F at 85 psig for 60 min., 350 ± 10°F for 120 ± 10 min. at 100 ± 5 psig		

* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	6/90
Date of prepreg manufacture	Date of analysis	2/95
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	75°F/A		200°F/A		75°F/W	200°F/W		
Tension, 1-axis	SS--							
Tension, 2-axis								
Tension, 3-axis								
Compression, 1-axis	SS--		SS--		SS--	II--		
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
SB Strength, 31-plane	S---		S---		S---			

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.80		
Resin Density	(g/cm^3)	1.26		
Composite Density	(g/cm^3)		1.59 - 1.60	
Fiber Areal Weight	(g/m^2)	280		
Ply Thickness	(in)		0.0099 - 0.0171	

LAMINATE PROPERTY SUMMARY

	75°F/A							
0/±45/90 Family								
Tension, x-axis	SS--							
OHT, x-axis	S---							

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.19(a)
C/Ep 280-5HS
AS4/3501-6 (Bleed)
Tension, 1-axis
$[0_f]_8$
75/A
Screening

MATERIAL: AS4 3k/3501-6 (Bleed) 5-harness satin weave

RESIN CONTENT: 29 wt% COMP. DENSITY: 1.61 g/cm^3
FIBER VOLUME: 61 vol% VOID CONTENT:
PLY THICKNESS: 0.0100 - 0.0106 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57% (0.019 in. CPT)

Temperature(°F)		75					
Moisture Content(%)		ambient					
Equilibrium at T, RH							
Source Code		43					
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	108	115				
	Minimum	93.3	98.8				
	Maximum	128	137				
	C.V. (%)	12.2	12.2				
F_1^{tu}	B-value	(1)	(1)				
	Distribution	ANOVA	ANOVA				
(ksi)	C_1	14.9	15.8				
	C_2	5.74	5.72				
	No. Specimens	9					
	No. Batches	3					
	Approval Class	Screening					
	Mean	9.83	10.4				
	Minimum	8.25	8.80				
	Maximum	12.0	13.1				
E_1^t	C.V. (%)	9.88	10.8				
(Msi)	No. Specimens	9					
	No. Batches	3					
	Approval Class	Screening					
	Mean						
ν_{12}^t	No Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.19(b)
C/Ep 280-5HS
AS4/3501-6 (Bleed)
Compression, 1-axis
$[0_f]_8$
75/A, 200/A, 75/W
Screening

MATERIAL: AS4 3k/3501-6 (Bleed) 5-harness satin weave

RESIN CONTENT: 29 wt% COMP. DENSITY: 1.61 g/cm^3
FIBER VOLUME: 61 vol% VOID CONTENT:
PLY THICKNESS: 0.0099 - 0.0104 in.

TEST METHOD: MODULUS CALCULATION:
SACMA SRM 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57% (0.019 in. CPT)

Temperature(°F)		75 ambient		200 ambient		75 wet (1)	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		43		43		43	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{cu} (ksi)	Mean	106	113	80.8	86.1	95.8	102
	Minimum	91.0	97.7	67.6	73.7	79.3	84.7
	Maximum	115	123	93.1	99.9	106	113
	C.V. (%)	6.52	6.65	8.84	8.69	9.43	9.42
	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	ANOVA	Weibull	Weibull	Weibull	Normal	Normal
	C_1	7.21	116	83.9	89.4	95.8	102
	C_2	3.73	18.4	13.6	13.4	9.03	9.64
	No. Specimens	13		13		9	
	No. Batches	3		3		2	
	Approval Class	Screening		Screening		Screening	
E_1^c (Msi)	Mean	8.7	9.3	8.48	9.04	9.23	9.87
	Minimum	7.6	8.2	6.42	7.00	9.07	9.70
	Maximum	9.4	9.9	9.43	10.0	9.44	10.2
	C.V. (%)	8.2	8.4	10.6	10.4	1.55	1.68
	No. Specimens	13		13		9	
	No. Batches	3		3		2	
	Approval Class	Screening		Screening		Screening	
ν_{12}^c	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_1^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned at 140°F, 95% relative humidity for 30 days.
(2) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.19(c)
C/Ep 280-5HS
AS4/3501-6 (Bleed)
Compression, 1-axis
$[0]_8$
200/W
Interim

MATERIAL: AS4 3k/3501-6 (Bleed) 5-harness satin weave

RESIN CONTENT: 29 wt% COMP. DENSITY: 1.59 g/cm^3
FIBER VOLUME: 61 vol% VOID CONTENT:
PLY THICKNESS: 0.0111 - 0.0171 in.

TEST METHOD: MODULUS CALCULATION:
SACMA SRM 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57% (0.019 in. CPT)

Temperature(°F)		200					
Moisture Content(%)		wet					
Equilibrium at T, RH		(1)					
Source Code		43					
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	57.0	60.8				
	Minimum	49.8	53.8				
	Maximum	67.8	72.2				
	C.V. (%)	8.85	8.82				
F_1^{cu}	B-value	(2)	(2)				
	Distribution	ANOVA	ANOVA				
(ksi)	C_1	5.46	5.761				
	C_2	4.57	4.38				
	No. Specimens	15					
	No. Batches	3					
	Approval Class	Interim					
	Mean	8.1	8.6				
	Minimum	6.5	7.0				
	Maximum	9.0	9.4				
E_1^c	C.V. (%)	10	10				
(Msi)	No. Specimens	15					
	No. Batches	3					
	Approval Class	Interim					
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) Conditioned at 140°F, 95% relative humidity for 30 days.
(2) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.19(d)
C/Ep 280-5HS
AS4/3501-6 (Bleed)
SBS, 31-plane
$[0]_8$
75/A, 200/A, 75/W
Screening

MATERIAL: AS4 3k/3501-6 (Bleed) 5-harness satin weave

RESIN CONTENT: 28 - 30 wt%
FIBER VOLUME: 60 - 62 vol%
PLY THICKNESS: 0.0099 - 0.0104 in.

COMP. DENSITY: 1.59-1.60 g/cm³
VOID CONTENT:

TEST METHOD: ASTM D 2344-84

MODULUS CALCULATION: N/A

NORMALIZED BY: Not normalized

Temperature(°F)		75	200	75		
Moisture Content(%)		ambient	ambient	wet		
Equilibrium at T, RH				(1)		
Source Code		43	43	43		
	Mean	9.93	7.94	9.35		
	Minimum	8.50	7.60	9.00		
	Maximum	10.7	8.40	9.60		
	C.V. (%)	7.38	3.89	2.22		
F_{31}^{sbs}	B-value	(2)	(2)	(2)		
	Distribution	Normal	ANOVA	Normal		
(ksi)	C_1	9.93	0.353	9.35		
	C_2	0.733	6.02	0.207		
	No. Specimens	9	9	6		
	No. Batches	3	3	2		
	Approval Class	Screening	Screening	Screening		

(1) Conditioned at 140°F, 95% relative humidity for 30 days.
(2) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.19(e)
C/Ep 280-5HS
AS4/3501-6 (Bleed)
Tension, x-axis
[(0/±45/90)f]$_s$
75/A
Screenin

MATERIAL: AS4 3k/3501-6 (Bleed) 5-harness satin weave

RESIN CONTENT: 29 wt% COMP. DENSITY: 1.59 g/cm^3
FIBER VOLUME: 61 vol% VOID CONTENT:
PLY THICKNESS: 0.0105 - 0.0106 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57% (0.019 in. CPT)

Temperature(°F)		75 ambient					
Moisture Content(%)							
Equilibrium at T, RH		43					
Source Code							
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	83.4	88.6				
	Minimum	75.7	81.3				
	Maximum	88.2	94.2				
	C.V. (%)	5.28	4.86				
F_x^{tu}	B-value	(1)	(1)				
	Distribution	Normal	Normal				
(ksi)	C_1	83.4	88.6				
	C_2	4.41	4.30				
	No. Specimens	6					
	No. Batches	2					
	Approval Class	Screening					
	Mean	6.9	7.3				
	Minimum	6.6	7.0				
	Maximum	7.0	7.5				
E_x^t	C.V. (%)	2.8	2.9				
(Msi)	No. Specimens	6					
	No. Batches	2					
	Approval Class	Screening					
	Mean						
ν_{xy}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_x^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.19(f)
C/Ep 280-5HS
AS4/3501-6 (Bleed)
OHT, x-axis
$[(0/\pm 45/90)f]_s$
75/A
Screening

MATERIAL: AS4 3k/3501-6 (Bleed) 5-harness satin weave

RESIN CONTENT: 29 - 30 wt%
FIBER VOLUME: 61 - 62 vol%
PLY THICKNESS: 0.0105 - 0.0109 in.

COMP. DENSITY: 1.59-1.60 g/cm³
VOID CONTENT:

TEST METHOD: SACMA 5-88

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57% (0.019 in. CPT)

Temperature(°F)		75 ambient					
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		43					
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_x^{oht} (ksi)	Mean	58.4	63.0				
	Minimum	57.0	60.9				
	Maximum	61.0	64.5				
	C.V. (%)	2.57	2.43				
	B-value	(1)	(1)				
	Distribution	Normal	Normal				
	C_1	58.4	63.0				
	C_2	1.50	1.53				
	No. Specimens	6					
	No. Batches	2					
	Approval Class	Screening					
E_x^{oht} (Msi)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_x^{oht} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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4.2.20 AS4 3k/3501-6 (No Bleed) 5-harness satin*

MATERIAL:	AS4 3k/3501-6 (No Bleed) 5-harness satin		C/Ep **AS4/3501-6 (No Bleed)** **Summary**
PREPREG:	Hercules AW280-5H/3501-6		
FIBER:	Hercules AS4 3k, no twist	MATRIX:	Hercules 3501-6
T_g(dry):	T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure, 240 ± 10°F at 85 psig for 60 min., 350 ± 10°F at 100 ± 5 psig for 120 ± 10 min.		

* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUBMITTED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	6/90
Date of prepreg manufacture	Date of analysis	2/95 - 3/95
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	75°F/A		-65°F/A	200°F/A				
Tension, 1-axis	SS--		SS--	SS--				
Tension, 2-axis								
Tension, 3-axis								
Compression, 1-axis	SS--							
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
SB Strength, 31-plane	S---							

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUBMITTED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.80		
Resin Density	(g/cm^3)	1.27		
Composite Density	(g/cm^3)	1.55	1.55 - 1.56	
Fiber Areal Weight	(g/m^2)	280		
Ply Thickness	(in)	0.011	0.011 - 0.017	

LAMINATE PROPERTY SUMMARY

	75°F/A							
0/±45/90 Family								
Tension, x-axis	SS--							
OHT, x-axis	S---							

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUBMITTED FOR THIS MATERIAL.

Table 4.2.20(a)
C/Ep 280-5HS
AS4/3501-6 (No Bleed)
Tension, 1-axis
$[0_f]_8$
75/A, -65/A, 200/A
Screening

MATERIAL: AS4 3k/3501-6 (No Bleed) 5-harness satin

RESIN CONTENT: 36 - 39 wt%

COMP. DENSITY: 1.55-1.56 g/cm^3

FIBER VOLUME: 52 - 55 vol%

VOID CONTENT:

PLY THICKNESS: 0.0111 - 0.0171 in.

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57% (0.011 in. CPT)

Temperature(°F)		75 ambient		-65 ambient		200 ambient	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		43		43		43	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	134	125	125	117	130	121
	Minimum	129	117	120	109	124	116
	Maximum	146	136	136	127	141	136
	C.V. (%)	3.79	4.85	3.85	4.89	4.49	5.11
F_1^{tu}	B-value	(1)	(1)	(1)	(1)	(1)	(1)
	Distribution	Normal	ANOVA	Normal	ANOVA	Lognormal	Nonpara.
(ksi)	C_1	134	6.56	125	6.07	4.86	6
	C_2	5.07	4.77	4.81	4.40	0.0440	2.25
	No. Specimens	9		9		9	
	No. Batches	3		3		3	
	Approval Class	Screening		Screening		Screening	
	Mean	9.67	9.06	10.2	9.57	10.8	10.1
	Minimum	9.39	8.60	9.63	8.80	9.88	9.00
	Maximum	9.88	9.50	11.0	10.3	11.8	11.3
E_1^t	C.V. (%)	1.65	3.63	4.26	5.68	6.74	8.23
(Msi)	No. Specimens	9		9		9	
	No. Batches	3		3		3	
	Approval Class	Screening		Screening		Screening	
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUBMITTED FOR THIS MATERIAL.

Table 4.2.20(b)
C/Ep 280-5HS
AS4/3501-6 (No Bleed)
Compression, 1-axis
$[0_1]_8$
75/A
Interim Data

MATERIAL: AS4 3k/3501-6 (No Bleed) 5-harness satin

RESIN CONTENT: 36 - 39 wt% COMP. DENSITY: 1.55-1.56 g/cm^3

FIBER VOLUME: 52 - 55 vol% VOID CONTENT:

PLY THICKNESS: 0.0114 - 0.0121 in.

TEST METHOD: MODULUS CALCULATION:

SACMA SRM 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57% (0.011 in. CPT)

Temperature(°F)		75					
Moisture Content(%)		ambient					
Equilibrium at T, RH							
Source Code		43					
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{cu} (ksi)	Mean	129	121				
	Minimum	121	111				
	Maximum	145	137				
	C.V. (%)	5.02	6.03				
	B-value	(1)	(1)				
	Distribution	Weibull	ANOVA				
	C_1	133	7.84				
	C_2	18.9	4.39				
	No. Specimens	15					
	No. Batches	3					
	Approval Class	Interim					
E_1^c (Msi)	Mean	9.42	8.81				
	Minimum	8.71	8.30				
	Maximum	10.0	9.50				
	C.V. (%)	4.25	5.35				
	No. Specimens	15					
	No. Batches	3					
	Approval Class	Interim					
ν_{12}^c	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_1^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUBMITTED FOR THIS MATERIAL.

Table 4.2.20(c)
C/Ep 280-5HS
AS4/3501-6 (No Bleed)
SBS, 31-plane
$[0_f]_8$
75/A
Screening

MATERIAL: AS4 3k/3501-6 (No Bleed) 5-harness satin

RESIN CONTENT: 36 - 39 wt%

COMP. DENSITY: 1.55-1.56 g/cm^3

FIBER VOLUME: 52 - 55 vol%

VOID CONTENT:

PLY THICKNESS: 0.0110 - 0.0114 in.

TEST METHOD: ASTM D 2344-84

MODULUS CALCULATION: N/A

NORMALIZED BY: Not normalized

Temperature(°F)		75				
Moisture Content(%)		ambient				
Equilibrium at T, RH						
Source Code		43				
	Mean	11.3				
	Minimum	10.1				
	Maximum	12.1				
	C.V. (%)	5.05				
F_{31}^{sbs}	B-value	(1)				
	Distribution	ANOVA				
(ksi)	C_1	0.611				
	C_2	4.35				
	No. Specimens	15				
	No. Batches	3				
	Approval Class	Screening				

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUBMITTED FOR THIS MATERIAL.

Table 4.2.20(d)
C/Ep 280-5HS
AS4/3501-6 (No Bleed)
Tension, x-axis
$[(0/45/90/-45)_t]_s$
75/A
Screening

MATERIAL: AS4 3k/3501-6 (No Bleed) 5-harness satin

RESIN CONTENT: 36 - 39 wt%

COMP. DENSITY: 1.55-1.56 g/cm^3

FIBER VOLUME: 52 - 55 vol%

VOID CONTENT:

PLY THICKNESS: 0.0113 - 0.0116 in.

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57% (0.011 in. CPT)

Temperature(°F)		75					
Moisture Content(%)		ambient					
Equilibrium at T, RH							
Source Code		43					
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	80.4	75.3				
	Minimum	77.1	68.8				
	Maximum	86.4	82.0				
	C.V. (%)	3.85	5.41				
F_x^{tu}	B-value	(1)	(1)				
	Distribution	Normal	ANOVA				
(ksi)	C_1	80.4	4.45				
	C_2	3.09	5.07				
	No. Specimens	9					
	No. Batches	3					
	Approval Class	Screening					
	Mean	6.94	6.50				
	Minimum	6.73	6.30				
	Maximum	7.13	6.60				
E_x^t	C.V. (%)	1.87	2.04				
(Msi)	No. Specimens	9					
	No. Batches	3					
	Approval Class	Screening					
	Mean						
ν_{xy}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_x^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUBMITTED FOR THIS MATERIAL.

Table 4.2.20(e)
C/Ep 280-5HS
AS4/3501-6 (No Bleed)
OHT, x-axis
$[(0/\pm 45/90)_4]_s$
75/A
Screening

MATERIAL: AS4 3k/3501-6 (No Bleed) 5-harness satin

RESIN CONTENT: 36 - 39 wt%

COMP. DENSITY: 1.55-1.56 g/cm^3

FIBER VOLUME: 52 - 55 vol%

VOID CONTENT:

PLY THICKNESS: 0.0113 - 0.0116 in.

TEST METHOD:

SACMA SRM 5-88

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57% (0.011 in. CPT)

Temperature(°F)		75					
Moisture Content(%)		ambient					
Equilibrium at T, RH							
Source Code		43					
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_x^{oht} (ksi)	Mean	54.4	55.5				
	Minimum	51.4	52.9				
	Maximum	57.7	58.7				
	C.V. (%)	4.58	3.72				
	B-value	(1)	(1)				
	Distribution	ANOVA	Normal				
	C_1	2.80	55.5				
	C_2	5.64	2.06				
	No. Specimens	9					
	No. Batches	3					
	Approval Class	Screening					
E_x^{oht} (Msi)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_x^{oht} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

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4.2.21 IM6 3501-6 Grades 145, 190 unidirectional tape*

These data are presented in the MIL-HDBK-17-2D Annex.

4.2.22 IM7 12k/8552 unidirectional tape*

These data are presented in the MIL-HDBK-17-2D Annex.

4.2.23 T300 3K/977-2 plain weave

These data are presented in the MIL-HDBK-17-2D Annex.

4.2.24 T300 3K/977-2 8-harness weave

These data are presented in the MIL-HDBK-17-2D Annex.

4.2.25 IM7 12K/977-2 unidirectional tape

These data are presented in the MIL-HDBK-17-2D Annex.

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4.3 CARBON - POLYESTER COMPOSITES

4.4 CARBON - BISMALEIMIDE COMPOSITES

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4.4.1 T-300 3k/F650 unidirectional tape*

MATERIAL:	T-300 3k/F650 unidirectional tape		**C/BMI T-300/F650 Summary**
PREPREG:	Hexcel T3T190/F652 unidirectional tape		
FIBER:	Toray T-300 3k	MATRIX: Hexcel F650	
T_g(ambient):	600°F T_g(wet): °F	T_g METHOD:	
PROCESSING:	Autoclave cure: 375°F, 4 hours, 85 psig; Postcure: 475°F, 4 hours, free-standing oven		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	4/89
Date of prepreg manufacture	Date of analysis	1/93
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	75°F/A		-67°F/A	400°F/A				
Tension, 1-axis	SS--		S---	SS--				
Tension, 2-axis								
Tension, 3-axis								
Compression, 1-axis								
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
SB Strength, 31-plane	S---			S---				

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.76		
Resin Density	(g/cm^3)	1.27		
Composite Density	(g/cm^3)	1.56	1.57	
Fiber Areal Weight	(g/m^2)	189		
Ply Thickness	(in)	0.0070		

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.1(a)
C/BMI 189-UT
T-300/F650
Tension, 1-axis
$[0]_6$
75/A, -67/A, 400/A
Screening

MATERIAL: T-300 3k/F650

RESIN CONTENT: 32 wt%
FIBER VOLUME: 61 %
PLY THICKNESS: 0.0070 in.

COMP. DENSITY: 1.57 g/cm^3
VOID CONTENT:

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION:

NORMALIZED BY: Batch fiber volume fraction to 60%

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		75 ambient 21		-67 ambient 21		400 ambient 21	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	248	252	194	197	229	233
	Minimum	216	220	167	170	216	220
	Maximum	293	298	212	216	243	247
	C.V.(%)	7.14	7.15	8.68	8.68	3.97	3.97
F_1^{tu}	B-value	(1)	(2)	(1)	(1)	(1)	(1)
	Distribution	Normal	Normal	Normal	Normal	Normal	Normal
(ksi)	C_1	248	252	194	197	229.	233
	C_2	17.7	18.0	16.8	17.1	11.1	9.24
	No. Specimens	15		5		7	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	18.9	19.2			19.1	19.4
	Minimum	16.5	16.8			16.8	17.1
	Maximum	20.3	20.6			21.0	21.4
E_1^t	C.V.(%)	5.58	5.49			7.26	7.23
(Msi)	No. Specimens	15				9	
	No. Batches	1				1	
	Approval Class	Screening				Screening	
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
ϵ_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.1(b)
C/BMI 189-UT
T-300/F650
SBS, 31-plane
$[0]_{34}$
75/A, 400/A
Screening

MATERIAL: T-300 3k/F650

RESIN CONTENT: 32 wt%
FIBER VOLUME: 61 %
PLY THICKNESS: 0.0070 in.

COMP. DENSITY: 1.57 g/cm^3
VOID CONTENT:

TEST METHOD:
ASTM D 2344

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75 ambient	400 ambient			
Moisture Content(%)						
Equilibrium at T, RH						
Source Code		21	21			
	Mean	14.1	9.39			
	Minimum	13.5	8.77			
	Maximum	15.0	10.1			
	C.V.(%)	3.04	4.25			
F_{31}^{sbs}	B-value	(1)	(1)			
	Distribution	Weibull	Weibull			
(ksi)	C_1	14.3	9.59			
	C_2	32.3	24.6			
	No. Specimens	15	15			
	No. Batches	1	1			
	Approval Class	Screening	Screening			

(1) B-values are presented only for fully approved data.

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4.4.2 T-300 3k/F650 8-harness satin weave*

MATERIAL:	T-300 3k/F650 8-harness satin weave		C/BMI T-300/F650 Summary
PREPREG:	Hexcel F3T584/F650 8-harness satin weave		
FIBER:	Toray T-300 3k	MATRIX: Hexcel F650	
T_g(ambient):	600°F T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure: 375°F, 4 hours, 85 psig; Postcure: 475°F, 4 hours, free-standing oven		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	4/89
Date of prepreg manufacture	Date of analysis	1/93
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	75°F/A		350°F/A	450°F/A				
Tension, 1-axis								
Tension, 2-axis								
Tension, 3-axis								
Compression, 1-axis								
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane	SS--							
Shear, 23-plane								
SB Strength, 31-plane	S---		S---	S---				

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.75		
Resin Density	(g/cm^3)	1.27		
Composite Density	(g/cm^3)	1.54		
Fiber Areal Weight	(g/m^2)	370		
Ply Thickness	(in)	0.015		

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.2(a)
C/BMI 370-8HS
T-300/F650
Shear, 12-plane
$[\pm 45_t]_{4s}$
75/A
Screening

MATERIAL: T-300 3k/F650

RESIN CONTENT: 40 wt%
FIBER VOLUME: 52 %
PLY THICKNESS: 0.015 in.

COMP. DENSITY: 1.51 g/cm^3
VOID CONTENT:

TEST METHOD:
ASTM D 3518-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75				
Moisture Content(%)		ambient				
Equilibrium at T, RH						
Source Code		21				
	Mean	9.77				
	Minimum	8.57				
	Maximum	11.1				
	C.V.(%)	8.78				
F_{12}^{su}	B-value	(1)				
	Distribution	Weibull				
(ksi)	C_1	10.2				
	C_2	12.9				
	No. Specimens	15				
	No. Batches	1				
	Approval Class	Screening				
	Mean	0.69				
	Minimum	0.59				
	Maximum	0.81				
G_{12}^{s}	C.V.(%)	10.				
(Msi)	No. Specimens	14				
	No. Batches	1				
	Approval Class	Screening				
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{12}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.2(b)
C/BMI 370-8HS
T-300/F650
SBS, 31-plane
$[0_t]_8$
75/A, 350/A, 450/A
Screening

MATERIAL: T-300 3k/F650

RESIN CONTENT: 40 wt%
FIBER VOLUME: 52 %
PLY THICKNESS: 0.015 in.

COMP. DENSITY: 1.51 g/cm^3
VOID CONTENT:

TEST METHOD:
ASTM D 2344

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75 ambient	350 ambient	450 ambient		
Moisture Content(%)						
Equilibrium at T, RH						
Source Code		21	21	21		
	Mean	5.83	5.59	5.80		
	Minimum	4.75	4.93	5.23		
	Maximum	8.06	6.44	6.57		
	C.V.(%)	15.0	10.9	6.81		
F_{31}^{sbs}	B-value	(1)	(1)	(1)		
	Distribution	Nonpara.	Weibull	Weibull		
(ksi)	C_1	8	5.86	5.98		
	C_2	1.54	11.0	15.5		
	No. Specimens	15	10	10		
	No. Batches	1	1	1		
	Approval Class	Screening	Screening	Screening		

(1) B-values are presented only for fully approved data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:29Z
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4.4.3 T-300 3k/F652 8-harness satin weave*

MATERIAL:	T-300 3k/F652 8-harness satin weave		C/BMI T-300/F652 Summary
PREPREG:	Hexcel F3G584/F652 8-harness satin weave		
FIBER:	Amoco Thornel T-300	MATRIX: Hexcel F652	
T_g(ambient):	600°F T_g(wet):	T_g METHOD:	
PROCESSING:	Press cured: 400°F, 2.5 hours, 125 psig; Postcure: 550°F, 4 hours		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	4/89
Date of prepreg manufacture	Date of analysis	1/93
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	70°F/A		600°F/A					
Tension, 1-axis	SS--							
Tension, 2-axis								
Tension, 3-axis								
Compression, 1-axis								
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
SB Strength, 31-plane	S---		S---					

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.76		
Resin Density	(g/cm^3)	1.26		
Composite Density	(g/cm^3)	1.55	1.57	
Fiber Areal Weight	(g/m^2)	367		
Ply Thickness	(in)	.00124		

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.3(a)
C/BMI 367-8HS
T-300/F652
Tension, 1-axis
$[0_f]_{10}$
70/A
Screening

MATERIAL: T-300 3k/F652 8-harness satin weave

RESIN CONTENT: 27.2 wt% COMP. DENSITY: 1.57 g/cm^3
FIBER VOLUME: 64.8 % VOID CONTENT:
PLY THICKNESS: 0.012 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Batch fiber volume to 57%

Temperature(°F)		70					
Moisture Content(%)		ambient					
Equilibrium at T, RH							
Source Code		21					
		Normalized	Measured				
	Mean	73.6	84.0				
	Minimum	58.8	67.1				
	Maximum	84.3	96.1				
	C.V.(%)	10.1	10.0				
F_1^{tu}	B-value	(1)	(1)				
	Distribution	Weibull	Weibull				
(ksi)	C_1	76.8	87.6				
	C_2	12.3	12.4				
	No. Specimens	15					
	No. Batches	1					
	Approval Class	Screening					
	Mean	9.71	11.1				
	Minimum	8.94	10.2				
	Maximum	10.2	11.6				
E_1^t	C.V.(%)	4.36	4.28				
(Msi)	No. Specimens	15					
	No. Batches	1					
	Approval Class	Screening					
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
ϵ_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.3(b)
C/BMI 367-8HS
T-300/F652
SBS, 31-plane
$[0_t]_{10}$
70/A, 600/A
Screening

MATERIAL: T-300 3k/F652 8-harness satin weave

RESIN CONTENT: 27.2 wt% COMP. DENSITY: 1.57 - g/cm^3
FIBER VOLUME: 64.8 % VOID CONTENT:
PLY THICKNESS: 0.0012 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 2344

NORMALIZED BY: Not normalized

Temperature(°F)		70 ambient	600 ambient			
Moisture Content(%)						
Equilibrium at T, RH						
Source Code		21	21			
	Mean	5.97	4.59			
	Minimum	5.13	4.29			
	Maximum	6.64	4.82			
	C.V.(%)	8.17	3.60			
F_{31}^{sbs}	B-value	(1)	(1)			
	Distribution	Weibull	Weibull			
(ksi)	C_1	6.18	4.66			
	C_2	14.8	36.8			
	No. Specimens	15	15			
	No. Batches	1	1			
	Approval Class	Screening	Screening			

(1) B-values are presented only for fully approved data.

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4.4.4 AS4/5250-3 unidirectional tape*

MATERIAL:	AS4/5250-3 unidirectional tape
PREPREG:	Narmco AS4/5250-3 unidirectional tape, grade 147
FIBER:	Hercules AS4 MATRIX: Narmco 5250-3
T_g(ambient):	642°F T_g(wet): 561°F T_g METHOD: DMA
PROCESSING:	Autoclave cure: 250°F, 60 min.; 350°F, 360 min., 85 psi; Postcure: 475°F, 6 hours

C/BMI AS4/5250-3 Summary

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	12/88
Date of prepreg manufacture	Date of analysis	1/93
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	72°F/A		-67°F/A	350°F/A	450°F/A		74°F/W	350°F/W
Tension, 1-axis	SSSS		SSSS	SSSS	SSSS		SSSS	SSSS
Tension, 2-axis	SS-S		SS-S	SS-S	SS-S			
Tension, 3-axis								
Compression, 1-axis	SS-S		SS-S	SS-S	SS-S		SS-S	SS-S
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane	SS--		SS--	SS--	SS--		SS--	SS--
Shear, 23-plane								
SB Strength, 31-plane								

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.80		
Resin Density	(g/cm^3)	1.25		
Composite Density	(g/cm^3)	1.58	1.52 - 1.63	
Fiber Areal Weight	(g/m^2)	147	132 - 165	ASTM D 3529
Ply Thickness	(in)	0.0051 - 0.0059	0.0050 - 0.0062	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.4(a)
C/BMI 147-UT
AS4/5250-3
Tension, 1-axis
$[0]_8$
72/A, -67/A, 350/A
Screening

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 26 - 28 wt% COMP. DENSITY: 1.58 - 1.61 g/cm³
FIBER VOLUME: 63 - 66 % VOID CONTENT: 0.1 - 0.9 %
PLY THICKNESS: 0.0050 - 0.0053 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		72 ambient (1)		-67 ambient (1)		350 ambient (1)	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{tu} (ksi)	Mean	252	291	270	311	266	308
	Minimum	223	255	249	285	241	276
	Maximum	275	322	288	332	283	325
	C.V.(%)	7.63	8.48	6.12	6.48	6.87	7.54
	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	Normal	Normal	Normal	Normal	Normal	Nonpara.
	C_1	252	291	270.	312	266	5
	C_2	19.2	24.7	16.5	20.2	18.3	3.06
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
E_1^t (Msi)	Mean	15.9	18.3	16.4	18.9	16.4	19.0
	Minimum	15.3	17.7	15.9	18.5	15.8	18.2
	Maximum	16.4	18.9	16.8	19.4	16.7	19.5
	C.V.(%)	3.04	2.51	2.23	1.91	2.07	2.85
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
ν_{12}^t	Mean	0.300		0.295		0.302	
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
ϵ_1^{tu} ($\mu\epsilon$)	Mean	17100		15800		15900	
	Minimum	14900		14100		14800	
	Maximum	20000		18000		17100	
	C.V.(%)	13.3		9.6		4.98	
	B-value	(2)		(2)		(2)	
	Distribution	Normal		Normal		Normal	
	C_1	17100		15800		15900	
	C_2	2270		1520		789	
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	

(1) Reference 4.4.4
(2) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.4(b)
C/BMI 147-UT
AS4/5250-3
Tension, 1-axis
$[0]_8$
450/A, 74/0.70%, 350/0.73%
Screening

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 26 - 28 wt% COMP. DENSITY: 1.61 - 1.63 g/cm³
FIBER VOLUME: 63 - 67 % VOID CONTENT: 0.0 - 0.9 %
PLY THICKNESS: 0.0050 - 0.0053 in.

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION:

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		450		74		350	
Moisture Content(%)		ambient		0.70		0.73	
Equilibrium at T, RH				160°F, 95%		(1)	
Source Code		25 (2)		25 (2)		25 (2)	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{tu} (ksi)	Mean	253	292	268	312	249	287
	Minimum	208	237	235	268	232	264
	Maximum	269	314	293	347	261	305
	C.V.(%)	8.87	9.64	7.74	8.99	4.50	5.42
	B-value	(3)	(3)	(3)	(3)	(3)	(3)
	Distribution	Nonpara.	Normal	Normal	Normal	Normal	Normal
	C_1	5	292	268	312	249	288
	C_2	3.06	28.1	20.7	28.1	11.2	15.6
	No. Specimens	6		6		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
E_1^t (Msi)	Mean	16.5	19.0	16.6	19.3	15.9	18.4
	Minimum	15.7	18.1	16.2	18.9	15.4	17.8
	Maximum	16.9	19.7	17.3	19.9	16.4	19.1
	C.V.(%)	3.43	3.56	2.36	1.82	2.41	2.71
	No. Specimens	6		6		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
ν_{12}^t	Mean	0.295		0.335		0.368	
	No. Specimens	6		6		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
ϵ_1^{tu} ($\mu\epsilon$)	Mean	13900		15200		14900	
	Minimum	11700		13500		13200	
	Maximum	15000		16600		15500	
	C.V.(%)	8.14		7.14		6.46	
	B-value	(3)		(3)		(3)	
	Distribution	Normal		Normal		Normal	
	C_1	13900		15200		14900	
	C_2	1130		1080		961	
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	

(1) Conditioned at 160°F, 95% relative humidity for 29 days (75% saturation)
(2) Reference 4.4.4
(3) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.4(c)
C/BMI 147-UT
AS4/5250-3
Tension, 1-axis
$[0]_8$
350/1.0%
Screening

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 26 - 28 wt% COMP. DENSITY: 1.61 g/cm^3
FIBER VOLUME: 63 - 66 % VOID CONTENT: 0.1 - 0.9 %
PLY THICKNESS: 0.0050 - 0.0053 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		350					
Moisture Content(%)		1.0					
Equilibrium at T, RH		160°F, 95%					
Source Code		(1)					
		Normalized	Measured				
	Mean	235	270				
	Minimum	176	202				
	Maximum	259	296				
	C.V.(%)	12.8	13.0				
F_1^{tu}	B-value	(2)	(2)				
	Distribution	Normal	Normal				
(ksi)	C_1	235	270				
	C_2	29.9	35.1				
	No. Specimens	6					
	No. Batches	1					
	Approval Class	Screening					
	Mean	16.7	19.2				
	Minimum	15.5	17.7				
	Maximum	18.4	21.2				
E_1^t	C.V.(%)	6.43	6.26				
(Msi)	No. Specimens	6					
	No. Batches	1					
	Approval Class	Screening					
	Mean		0.363				
ν_{12}^t	No. Specimens	4					
	No. Batches	1					
	Approval Class	Screening					
	Mean		14400				
	Minimum		9950				
	Maximum		16200				
	C.V.(%)		16.0				
ϵ_1^{tu}	B-value		(2)				
	Distribution		Normal				
($\mu\epsilon$)	C_1		14400				
	C_2		2300				
	No. Specimens	6					
	No. Batches	1					
	Approval Class	Screening					

(1) Reference 4.4.4
(2) B-values are presented only for fully approved data.

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Check the source to verify that this is the current version before use.

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.4(d)
C/BMI 147-UT
AS4/5250-3
Tension, 2-axis
$[90]_8$
72/A, -67/A, 350/A, 450/A
Screening

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 27 - 40 wt%
FIBER VOLUME: 51 - 65 %
PLY THICKNESS: 0.0051 - 0.0059 in.
COMP. DENSITY: 1.52 - 1.61 g/cm^3
VOID CONTENT: 0.1 - 0.8 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		72 ambient	-67 ambient	350 ambient	450 ambient		
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		25	25	25	25		
	Mean	4.61	4.98	4.63	4.54		
	Minimum	3.52	4.68	3.43	4.13		
	Maximum	5.65	5.94	5.33	5.19		
	C.V.(%)	18.4	9.69	13.7	9.20		
F_2^{tu}	B-value	(1)	(1)	(1)	(1)		
	Distribution	Normal	Nonpara.	Normal	Normal		
(ksi)	C_1	4.61	5	4.63	4.54		
	C_2	0.847	3.06	0.637	0.417		
	No. Specimens	6	6	6	6		
	No. Batches	1	1	1	1		
	Approval Class	Screening	Screening	Screening	Screening		
	Mean	1.24	1.40	1.04	1.08		
	Minimum	1.17	1.26	0.940	0.930		
	Maximum	1.35	1.47	1.16	1.26		
E_2^t	C.V.(%)	5.90	5.50	8.50	10.3		
(Msi)	No. Specimens	6	6	5	6		
	No. Batches	1	1	1	1		
	Approval Class	Screening	Screening	Screening	Screening		
	Mean						
ν_{21}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean	3540	3580	4680	4330		
	Minimum	2000	3180	3300	3600		
	Maximum	4900	4740	6000	5600		
	C.V.(%)	26.9	16.5	19.0	18.0		
ϵ_2^{tu}	B-value	(1)	(1)	(1)	(1)		
	Distribution	Normal	Lognormal	Normal	Normal		
($\mu\epsilon$)	C_1	3540	8.17	4680	4330		
	C_2	955	0.149	889	782		
	No. Specimens	6	6	6	6		
	No. Batches	1	1	1	1		
	Approval Class	Screening	Screening	Screening	Screening		

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.4(e)
C/BMI 147-UT
AS4/5250-3
Compression, 1-axis
$[0]_8$
72/A, -67/A, 350/A
Screening

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 36 - 38 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 53 - 56 % VOID CONTENT: 0.1 - 0.9 %
PLY THICKNESS: 0.0057 - 0.0062 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3410A-87

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		72 ambient (1)		-67 ambient (1)		350 ambient (1)	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	175	158	198	179	174	148
	Minimum	122	110	176	160	141	127
	Maximum	203	184	222	201	235	185
	C.V.(%)	15.9	15.9	8.0	8.0	23.6	15.9
F_1^{cu} (ksi)	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	Normal	Normal	Normal	Normal	Normal	Normal
	C_1	175	158	198	179	174	149
	C_2	27.7	25.1	15.8	14.3	41.1	23.6
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	17.0	15.4	15.5	14.0	17.4	14.9
	Minimum	14.1	12.8	13.9	12.6	15.2	13.8
	Maximum	22.7	20.5	18.5	16.7	21.9	17.2
E_1^c	C.V.(%)	20.1	20.0	10.7	10.6	14.7	8.55
(Msi)	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		12100		19800		15300
	Minimum		8000		8360		10200
	Maximum		22700		26700		18400
	C.V.(%)		46.2		43.9		18.1
ϵ_1^{cu}	B-value		(2)		(2)		(2)
	Distribution		Normal		Normal		Normal
($\mu\epsilon$)	C_1		12100		19800		15300
	C_2		5570		8710		2770
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	

(1) Reference 4.4.4
(2) B-values are presented only for fully approved data.

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Table 4.4.4(f)
C/BMI 147-UT
AS4/5250-3
Compression, 1-axis
$[0]_8$
450/A, 74/0.82%, 350/0.79%
Screening

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 36 - 38 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 53 - 56 % VOID CONTENT: 0.1 - 0.9 %
PLY THICKNESS: 0.0057 - 0.0062 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3410A-87

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		450 ambient		74 0.82 160°F, 95%		350 0.79 (1)	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		(2)		(2)		(2)	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	153	131	194	176	153	139
	Minimum	119	108	175	159	113	102
	Maximum	207	163	216	195	173	157
	C.V.(%)	21.2	15.1	8.6	8.63	15.5	15.5
F_1^{cu}	B-value	(3)	(3)	(3)	(3)	(3)	(3)
	Distribution	Normal	Normal	Normal	Normal	Normal	Normal
(ksi)	C_1	153	131	194	176	153	139
	C_2	32.4	19.7	16.7	15.2	23.8	21.5
	No. Specimens	6		6		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	18.2	15.6	18.5	16.8	16.1	14.6
	Minimum	14.0	12.6	16.4	14.9	14.3	12.9
	Maximum	21.7	17.1	21.5	19.5	18.2	16.5
E_1^c	C.V.(%)	16.0	10.4	9.42	9.39	9.78	9.75
(Msi)	No. Specimens	6		6		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		8480		15900		12600
	Minimum		2900		10600		6400
	Maximum		14600		22900		16000
	C.V.(%)		44.7		32.5		30.2
ϵ_1^{cu}	B-value		(3)		(3)		(3)
	Distribution		Normal		Normal		Normal
($\mu\epsilon$)	C_1		8480		15900		12600
	C_2		3790		5170		3810
	No. Specimens	6		6		5	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	

(1) Conditioned at 160°F, 95% relative humidity for 7 days (75% saturation)
(2) Reference 4.4.4
(3) B-values are presented only for fully approved data.

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Table 4.4.4(g)
C/BMI 147-UT
AS4/5250-3
Compression, 1-axis
$[0]_8$
350/1.0%
Screening

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 36 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 56 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0050 - 0.0053 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3410A-87

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		350					
Moisture Content(%)		1.0					
Equilibrium at T, RH		160°F, 95%					
Source Code		(1)					
		Normalized	Measured				
	Mean	127	115				
	Minimum	108	97.9				
	Maximum	152	138				
	C.V.(%)	11.4	11.4				
F_1^{cu} (ksi)	B-value	(2)	(2)				
	Distribution	Normal	Normal				
	C_1	127	115				
	C_2	14.4	13.0				
	No. Specimens	6					
	No. Batches	1					
	Approval Class	Screening					
E_1^c (Msi)	Mean	18.1	16.4				
	Minimum	16.6	15.0				
	Maximum	20.7	18.7				
	C.V.(%)	7.93	7.89				
	No. Specimens	6					
	No. Batches	1					
	Approval Cass	Screening					
ν_{12}^c	Mean						
	No. Specimens						
	No. Batches						
	Approval Class						
ϵ_1^{cu} ($\mu\epsilon$)	Mean		8120				
	Minimum		6600				
	Maximum		9180				
	C.V(%)		11.5				
	B-value		(2)				
	Distribution		Normal				
	C_1		8120				
	C_2		934				
	No. Specimens	6					
	No. Batches	1					
	Approval Class	Screening					

(1) Reference 4.4.4
(2) B-values are presented only for fully approved data.

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Table 4.4.4(h)
C/BMI 147-UT
AS4/5250-3
Shear, 12-plane
$[\pm 45]_{4s}$
72/A, -67/A, 350/A, 450/A
Screening

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 28 - 32 wt% COMP. DENSITY: 1.58 - 1.61 g/cm^3
FIBER VOLUME: 59 - 63 % VOID CONTENT: 0.0 - 1.2 %
PLY THICKNESS: 0.0055 - 0.0058 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3518-76

NORMALIZED BY: Not normalized

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		72 ambient (1)	-67 ambient (1)	350 ambient (1)	450 ambient (1)	
	Mean	9.61	10.1	10.4	9.01	
	Minimum	8.49	9.67	9.55	8.44	
	Maximum	10.4	10.5	11.0	9.47	
	C.V.(%)	6.95	3.50	5.31	4.87	
F_{12}^{su}	B-value	(2)	(2)	(2)	(2)	
	Distribution	Normal	Normal	Normal	Normal	
(ksi)	C_1	9.61	10.1	10.4	9.01	
	C_2	0.668	0.352	0.553	0.439	
	No. Specimens	6	6	6	6	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	
	Mean	0.77	0.84	0.66	0.62	
	Minimum	0.71	0.78	0.62	0.50	
	Maximum	0.83	0.86	0.72	0.69	
G_{12}^{s}	C.V.(%)	5.6	3.6	5.3	12.	
(Msi)	No. Specimens	6	6	6	6	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{12}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) Reference 4.4.4
(2) B-values are presented only for fully approved data.

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Table 4.4.4(i)
C/BMI 147-UT
AS4/5250-3
Shear, 12-plane
$[\pm 45]_{4s}$
74/0.55%, 350/0.55%, 350/1.1%
Screening

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 28 - 32 wt% COMP. DENSITY: 1.58 - 1.61 g/cm^3
FIBER VOLUME: 59 - 63 % VOID CONTENT: 0.0 - 1.2 %
PLY THICKNESS: 0.0055 - 0.0058 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3518-76

NORMALIZED BY: Not normalized

Temperature(°F)		74	350	350		
Moisture Content(%)		0.55	0.55	1.1		
Equilibrium at T, RH		160°F, 95%	(1)	160°F, 95%		
Source Code		(2)	(2)	(2)		
	Mean	12.5	8.70	9.81		
	Minimum	11.3	8.24	8.13		
	Maximum	13.2	8.95	10.6		
	C.V.(%)	5.26	3.42	9.27		
F_{12}^{su}	B-value	(3)	(3)	(3)		
	Distribution	Normal	Normal	Normal		
(ksi)	C_1	12.5	8.70	9.81		
	C_2	0.656	0.298	0.909		
	No. Specimens	6	5	6		
	No. Batches	1	1	1		
	Approval Class	Screening	Screening	Screening		
	Mean	0.79	0.46	0.49		
	Minimum	0.77	0.43	0.40		
	Maximum	0.81	0.48	0.56		
G_{12}^{s}	C.V.(%)	1.9	4.0	14		
(Msi)	No. Specimens	6	6	4		
	No. Batches	1	1	1		
	Approval Class	Screening	Screening	Screening		
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{12}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) Conditioned at 160°F, 95% relative humidity for 3 days (75% saturation)
(2) Reference 4.4.4
(3) B-values are presented only for fully approved data.

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4.5 CARBON - POLYIMIDE COMPOSITES

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4.5.1 Celion 3000/F670 8-harness satin weave*

MATERIAL:	Celion 3000/F670 8-harness satin weave		C/PI Celion 3000/F670 Summary
PREPREG:	Hexcel F3L584/F670 8-harness satin weave		
FIBER:	Celanese Celion 3000	MATRIX: Hexcel F670 (PMR-15)	
T_g(ambient):	635°F T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure: 440°F, 2 hours; 600°F, 3 Hours, 200 psig; Postcure		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	8/87
Date of resin manufacture		Date of data submittal	4/89
Date of prepreg manufacture	2/87 - 5/87	Date of analysis	1/93
Date of composite manufacture			

LAMINA PROPERTY SUMMARY

	75°F/A		550°F/A					
Tension, 1-axis	SS--		SS--					
Tension, 2-axis	SS--		SS--					
Tension, 3-axis								
Compression, 1-axis	SS--		SS--					
Compression, 2-axis	SS--		SS--					
Compression, 3-axis								
Shear, 12-plane								
SB Strength, 23-plane	S---							
SB Strength, 31-plane	S---							

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.8		
Resin Density	(g/cm^3)	1.32		
Composite Density	(g/cm^3)	1.59	1.59 - 1.63	
Fiber Areal Weight	(g/m^2)	384		
Ply Thickness	(in)		0.0132 - 0.0144	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.5.1(a)
C/PI 384-8HS
Celion 3000/F670
Tension, 1-axis
$[0_f]_8$
75/A, 550/A
Screening

MATERIAL: Celion 3000/F670 8-harness satin weave

RESIN CONTENT: 30 - 34 wt%
FIBER VOLUME: 57 - 64 %
PLY THICKNESS: 0.0132 - 0.0144 in.

COMP. DENSITY: 1.59 - 1.63 g/cm^3
VOID CONTENT: 0.0 - 0.62 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION:

NORMALIZED BY: Fiber volume to 57%

Temperature(°F)		75 ambient		550 ambient			
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		22		22			
		Normalized	Measured	Normalized	Measured		
	Mean	132	136	116	120		
	Minimum	127	131	95.4	98.7		
	Maximum	140	144	129	134		
	C.V.(%)	2.75	2.76	7.94	7.95		
F_1^{tu}	B-value	(1)	(1)	(1)	(1)		
	Distribution	Normal	Normal	Normal	Normal		
(ksi)	C_1	132	136	116	120		
	C_2	3.63	3.76	9.18	9.52		
	No. Specimens	9		9			
	No. Batches	3		3			
	Approval Class	Screening		Screening			
	Mean	9.03	9.35	8.67	8.98		
	Minimum	8.66	8.96	8.50	8.80		
	Maximum	9.35	9.68	9.07	9.39		
E_1^t	C.V.(%)	3.22	3.23	2.54	2.55		
(Msi)	No. Specimens	9		9			
	No. Batches	3		3			
	Approval Class	Screening		Screening			
	Mean						
ν_{12}^t	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
ϵ_1^{tu}	B-value						
	Distribution						
$(\mu\epsilon)$	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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Table 4.5.1(b)
C/PI 384-8HS
Celion 3000/F670
Tension, 2-axis
$[90_f]_8$
75/A, 550/A
Screening

MATERIAL: Celion 3000/F670 8-harness satin weave

RESIN CONTENT: 30 - 34 wt% COMP. DENSITY: 1.59 - 1.63 g/cm^3
FIBER VOLUME: 57 - 64 % VOID CONTENT: 0.0 - 0.62 %
PLY THICKNESS: 0.0132 - 0.0144 in.

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION:

NORMALIZED BY: Fiber volume fraction to 57%

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		75 ambient 22		550 ambient 22		
		Normalized	Measured	Normalized	Measured	
	Mean	107	111	90.4	93.5	
	Minimum	85.6	88.6	61.9	64.1	
	Maximum	129	133	123	127	
	C.V. (%)	15.7	15.7	23.8	23.8	
F_2^{tu}	B-value	(1)	(1)	(1)	(1)	
	Distribution	ANOVA	ANOVA	ANOVA	ANOVA	
(ksi)	C_1	19.3	20.0	24.7	25.5	
	C_2	6.09	6.09	6.02	6.02	
	No. Specimens	9		9		
	No. Batches	3		3		
	Approval Class	Screening		Screening		
	Mean	8.43	8.73	8.23	8.52	
	Minimum	7.43	7.69	7.58	7.85	
	Maximum	9.33	9.66	8.84	9.15	
E_2^t	C.V. (%)	7.45	7.46	5.49	5.48	
(Msi)	No. Specimens	9		9		
	No. Batches	3		3		
	Approval Class	Screening		Screening		
	Mean					
ν_{21}^t	No. Specimens					
	No. Batches					
	Approval Class					
	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
ϵ_2^{tu}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.5.1(c)
C/PI 384-8HS
Celion 3000/F670
Compression, 1-axis
$[0_f]_8$
75/A, 550/A
Screening

MATERIAL: Celion 3000/F670 8-harness satin weave

RESIN CONTENT: 30 - 34 wt% COMP. DENSITY: 1.59 - 1.63 g/cm^3
FIBER VOLUME: 57 - 64 % VOID CONTENT: 0.0 - 0.62 %
PLY THICKNESS: 0.0132 - 0.0144 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Fiber volume fraction to 57%

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		75 ambient 22		550 ambient 22			
		Normalized	Measured	Normalized	Measured		
	Mean	99.4	103	66.0	68.3		
	Minimum	87.9	91.3	59.0	61.1		
	Maximum	118	122	71.7	74.2		
	C.V.(%)	9.33	9.33	6.60	6.59		
F_1^{cu} (ksi)	B-value	(1)	(1)	(1)	(1)		
	Distribution	ANOVA	ANOVA	Normal	Normal		
	C_1	10.2	10.6	66.0	68.3		
	C_2	5.28	5.28	4.36	4.51		
	No. Specimens	9		9			
	No. Batches	3		3			
	Approval Class	Screening		Screening			
	Mean	8.61	8.92	8.09	8.38		
	Minimum	8.40	8.69	7.26	7.51		
	Maximum	9.09	9.41	8.78	9.09		
E_1^c	C.V.(%)	2.54	2.54	5.19	5.21		
(Msi)	No. Specimens	9		9			
	No. Batches	3		3			
	Approval Class	Screening		Screening			
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
ϵ_1^{cu} ($\mu\epsilon$)	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Approval Class						

(1) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.5.1(d)
C/PI 384-8HS
Celion 3000/F670
Compression, 2-axis
$[90]_{t/8}$
75/A, 550/A
Screening

MATERIAL: Celion 3000/F670 8-harness satin weave

RESIN CONTENT: 30 - 34 wt%
FIBER VOLUME: 57 - 64 %
PLY THICKNESS: 0.0132 - 0.0144 in.
COMP. DENSITY: 1.59 - 1.63 g/cm^3
VOID CONTENT: 0.0 - 0.62 %

TEST METHOD: SACMA 1-88

MODULUS CALCULATION:

NORMALIZED BY: Fiber volume fraction to 57%

Temperature(°F)		75 ambient		550 ambient	
Moisture Content(%)					
Equilibrium at T, RH					
Source Code		22		22	
		Normalized	Measured	Normalized	Measured
	Mean	78.9	81.7	54.2	56.1
	Minimum	76.1	78.8	52.4	54.2
	Maximum	80.7	83.5	56.6	58.6
	C.V. (%)	3.10	3.10	4.02	4.03
F_2^{cu} (ksi)	B-value	(1)		(1)	
	Distribution				
	C_1				
	C_2				
	No. Specimens	3		3	
	No. Batches	1		1	
	Approval Class	Screening		Screening	
E_2^c (Msi)	Mean	8.08	8.37	7.67	7.94
	Minimum	8.03	8.31	7.59	7.86
	Maximum	8.14	8.43	7.77	8.04
	C.V. (%)	0.681	0.720	1.19	1.15
	No. Specimens	3		3	
	No. Batches	1		1	
	Approval Class	Screening		Screening	
ν_{12}^c	Mean				
	No. Specimens				
	No. Batches				
	Approval Class				
ϵ_2^{cu} ($\mu\epsilon$)	Mean				
	Minimum				
	Maximum				
	C.V. (%)				
	B-value				
	Distribution				
	C_1				
	C_2				
	No. Specimens				
	No. Batches				
	Approval Class				

(1) Insufficient observations to complete the statistical calculations.

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Table 4.5.1(e)
C/PI 384-8HS
Celion 3000/F670
SBS, 23-plane
$[0_f]_8$
75/A
Screening

MATERIAL: Celion 3000/F670 8-harness satin weave

RESIN CONTENT:	30 - 34 wt%	COMP. DENSITY:	1.59 - 1.63 g/cm^3
FIBER VOLUME:	57 - 64 %	VOID CONTENT:	0.0 - 0.62 %
PLY THICKNESS:	0.0132 - 0.0144 in.		

TEST METHOD: ASTM D 2344-84

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75				
Moisture Content(%)		ambient				
Equilibrium at T, RH						
Source Code		22				
	Mean	11.1				
	Minimum	10.4				
	Maximum	11.7				
	C.V.(%)	5.88				
F_{23}^{sbs} (ksi)	B-value	(1)				
	Distribution					
	C_1					
	C_2					
	No. Specimens	3				
	No. Batches	1				
	Approval Class					

(1) Insufficient observations to complete the statistical calculations.

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Table 4.5.1(f)
C/PI 384-8HS
Celion 3000/F670
SBS, 31-plane
$[0_f]_8$
75/A
Screening

MATERIAL: Celion 3000/F670 8-harness satin weave

RESIN CONTENT: 30 - 34 wt%
FIBER VOLUME: 57 - 64 %
PLY THICKNESS: 0.0132 - 0.0144 in.

COMP. DENSITY: 1.59 - 1.63 g/cm^3
VOID CONTENT: 0.0 - 0.62 %

TEST METHOD: ASTM D 2344-84

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75				
Moisture Content(%)		ambient				
Equilibrium at T, RH						
Source Code		22				
	Mean	10.9				
	Minimum	9.70				
	Maximum	12.0				
	C.V.(%)	6.15				
F_{31}^{sbs} (ksi)	B-value	(1)				
	Distribution	ANOVA				
	C_1	0.722				
	C_2	4.78				
	No. Specimens	9				
	No. Batches	3				
	Approval Class	Screening				

(1) B-values are presented only for fully approved data.

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4.6 CARBON - PHENOLIC COMPOSITES

4.7 CARBON - SILICONE COMPOSITES

4.8 CARBON - POLYBENZIMIDAZOLE COMPOSITES

4.9 CARBON - PEEK COMPOSITES

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4.9.1 IM-6 12k/APC-2 unidirectional tape*

C/PEEK IM-6/APC-2 Summary

MATERIAL:	IM-6 12k/APC-2 unidirectional tape		
PREPREG:	Fiberite IM-6/APC-2 unidirectional tape		
FIBER:	Hercules IM-6 12k	MATRIX:	Fiberite APC-2
T_g(ambient):	291°F T_g(wet): 309°F	T_g METHOD:	DMA
PROCESSING:	Autoclave cure: 720°F, 30 - 45 min., 60 psig		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	12/88
Date of prepreg manufacture	Date of analysis	1/93
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	74°F/A		-67°F/A	180°F/A	250°F/A	180°F/O	74°F/W	180°F/W
Tension, 1-axis	SSSS		SSSS	SSSS	SSSS	SSSS	SSSS	SSSS
Tension, 2-axis	SS-S		SS-S	SS-S	SS-S			
Tension, 3-axis								
Compression, 1-axis	SS-S		SS-S	SS-S	SS-S	SS-S	SS-S	SS-S
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane	SS--		SS--	SS--	SS--	SS--	SS--	SS--
Shear, 23-plane								
Shear, 31-plane								

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	1.73		
Resin Density	(g/cm^3)	1.28		
Composite Density	(g/cm^3)	1.55	1.54 - 1.58	ASTM D792
Fiber Areal Weight	(g/m^2)			
Ply Thickness	(in)	0.0054	0.0052 - 0.0058	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.9.1(a)
C/PEEK -UT
IM-6/APC-2
Tension, 1-axis
$[0]_8$
74/A, -67/A, 180/A
Screening

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 32 wt% COMP. DENSITY: 1.55 g/cm³
FIBER VOLUME: 61 - 62 % VOID CONTENT: 0.0 - 0.2 %
PLY THICKNESS: 0.0053 - 0.0054 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3039-76

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		74 ambient		-67 ambient		180 ambient	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		(1)		(1)		(1)	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{tu} (ksi)	Mean	350	370	376	398	327	344
	Minimum	266	282	326	345	234	248
	Maximum	426	455	412	439	402	421
	C.V.(%)	15.9	16.0	8.69	8.93	17.3	16.8
	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	Normal	Normal	Normal	Normal	Normal	Normal
	C_1	350	370	376	398	327	344
	C_2	55.5	59.3	32.7	35.6	56.4	58.0
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
E_1^t (Msi)	Mean	21.6	22.9	22.0	23.3	23.2	24.4
	Minimum	21.3	22.4	20.9	22.2	22.3	23.6
	Maximum	22.0	23.3	23.2	24.5	23.7	25.0
	C.V.(%)	1.41	1.58	3.35	3.26	2.24	2.17
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
ν_{12}^t	Mean	0.342		0.357		0.355	
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
ϵ_1^{tu} ($\mu\epsilon$)	Mean	13600		15900		14100	
	Minimum	8100		13500		10400	
	Maximum	17500		17200		16800	
	C.V.(%)	24.6		9.23		14.9	
	B-value	(2)		(2)		(2)	
	Distribution	Normal		Normal		Normal	
	C_1	13600		15900		14100	
	C_2	3350		1470		2100	
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	

(1) Reference 4.9.1
(2) B-values are presented only for fully approved data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WAS NOT SUPPLIED FOR THIS MATERIAL.

Table 4.9.1(b)
C/PEEK -UT
IM-6/APC-2
Tension, 1-axis
$[0]_8$
250/A, 74/0.13%, 180/0.11%
Screening

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 32 wt% COMP. DENSITY: 1.55 g/cm³
FIBER VOLUME: 61 - 62 % VOID CONTENT: 0.0 - 0.2 %
PLY THICKNESS: 0.0053 - 0.0054 in.

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION:

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		250 ambient		180 0.11 (1)		74 0.13 160°F, 95%	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		(2)		(2)		(2)	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{tu} (ksi)	Mean	304	322	369	390	352	371
	Minimum	253	269	303	320	271	286
	Maximum	341	363	403	425	415	434
	C.V.(%)	11.4	11.4	12.3	12.2	14.6	14.2
	B-value	(3)	(3)	(3)	(3)	(3)	(3)
	Distribution	Normal	Normal	Normal	Normal	Normal	Normal
	C_1	304	322	369	390	352	371
	C_2	34.7	36.6	45.3	47.6	51.4	52.6
	No. Specimens	6		5		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
E_1^t (Msi)	Mean	21.4	22.7	21.8	23.0	21.2	22.3
	Minimum	20.5	21.9	20.9	22.1	20.4	21.6
	Maximum	22.1	23.4	22.2	23.5	22.0	23.0
	C.V.(%)	2.70	2.42	2.42	2.42	3.15	3.04
	No. Specimens	6		5		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
ν_{12}^t	Mean	0.338		0.366		0.372	
	No. Specimens	6		5		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
ϵ_1^{tu} ($\mu\epsilon$)	Mean	14800		16300		18100	
	Minimum	12500		14400		15700	
	Maximum	16400		17200		20800	
	C.V.(%)	11.8		6.70		10.8	
	B-value	(3)		(3)		(3)	
	Distribution	Normal		Normal		Normal	
	C_1	14800		16300		18100	
	C_2	1760		1090		1960	
	No. Specimens	6		5		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	

(1) Conditioned at 160°F, 96% relative humidity for 3 days (75% saturation)
(2) Reference 4.9.1
(3) B-values are presented only for fully approved data.

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Table 4.9.1(c)
C/PEEK -UT
IM-6/APC-2
Tension, 1-axis
$[0]_8$
180/0.14%
Screening

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 32 wt%
FIBER VOLUME: 61 - 62 %
PLY THICKNESS: 0.0053 - 0.0054 in.

COMP. DENSITY: 1.55 g/cm^3
VOID CONTENT: 0.0 - 0.2 %

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION:

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		180					
Moisture Content(%)		0.14					
Equilibrium at T, RH		160°F, 95%					
Source Code		(1)					
		Normalized	Measured				
	Mean	364	385				
	Minimum	325	344				
	Maximum	411	436				
	C.V.(%)	10.2	10.1				
F_1^{tu} (ksi)	B-value	(2)	(2)				
	Distribution	Normal	Normal				
	C_1	364	385				
	C_2	37.2	38.8				
	No. Specimens	6					
	No. Batches	1					
	Approval Class	Screening					
E_1^t (Msi)	Mean	21.2	22.4				
	Minimum	20.5	21.8				
	Maximum	22.2	23.2				
	C.V.(%)	3.14	2.77				
	No. Specimens	6					
	No. Batches	1					
	Approval Class	Screening					
ν_{12}^t	Mean		0.332				
	No. Specimens	6					
	No. Batches	1					
	Approval Class	Screening					
ϵ_1^{tu} ($\mu\epsilon$)	Mean		15400				
	Minimum		13600				
	Maximum		17200				
	C.V.(%)		9.24				
	B-value		(2)				
	Distribution		Normal				
	C_1		15400				
	C_2		1420				
	No. Specimens	6					
	No. Batches	1					
	Approval Class	Screening					

(1) Reference 4.9.1
(2) B-values are presented only for fully approved data.

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Table 4.9.1(d)
C/PEEK -UT
IM-6/APC-2
Tension, 2-axis
$[90]_{16}$
74/A, -67/A, 180/A, 250/A
Screening

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 31 - 34 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 60 - 62 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0054 - 0.0058 in.

TEST METHOD: ASTM D 3039-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		74 ambient (1)	-67 ambient (1)	180 ambient (1)	250 ambient (1)	
	Mean	9.41	9.67	11.1	9.07	
	Minimum	8.53	8.72	10.0	7.30	
	Maximum	10.6	10.7	12.2	9.72	
	C.V.(%)	9.35	6.52	8.87	10.1	
F_2^{tu}	B-value	(2)	(2)	(2)	(2)	
	Distribution	Normal	Normal	Normal	Normal	
(ksi)	C_1	9.41	9.67	11.1	9.07	
	C_2	0.880	0.631	0.985	0.916	
	No. Specimens	6	6	6	6	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	
	Mean	1.28	1.41	1.22	1.32	
	Minimum	1.24	1.35	1.17	1.27	
	Maximum	1.36	1.46	1.25	1.38	
E_2^t	C.V.(%)	3.33	3.32	2.13	3.44	
(Msi)	No. Specimens	6	6	6	6	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	
	Mean					
ν_{21}^t	No. Specimens					
	No. Batches					
	Approval Class					
	Mean	7610	7120	10900	12300	
	Minimum	6650	6450	8850	8510	
	Maximum	8830	8180	14900	23600	
	C.V.(%)	11.2	8.15	20.0	45.5	
ϵ_2^{tu}	B-value	(2)	(2)	(2)	(2)	
	Distribution	Normal	Normal	Normal	Nonpara.	
($\mu\epsilon$)	C_1	7610	7120	10900	5	
	C_2	850	581	2180	3.06	
	No. Specimens	6	6	6	6	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	

(1) Reference 4.9.1
(2) B-values are presented only for fully approved data.

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Table 4.9.1(e)
C/PEEK -UT
IM-6/APC-2
Compression, 1-axis
$[0]_{16}$
74/A, -67/A, 180/A
Screening

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 32 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 60 - 62 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0054 - 0.0058 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3410A-87

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F) Moisture Content(%) Equilibrium at T, RH Source Code		74 ambient (1)		-67 ambient (1)		180 ambient (1)	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	167	169	156	160	156	155
	Minimum	139	144	115	118	103	96.7
	Maximum	197	200.	179	181	195	190.
	C.V.(%)	13.3	13.3	16.0	15.6	20.2	20.4
F_1^{cu}	B-value	(2)	(2)	(2)	(2)	(2)	(2)
	Distribution	Normal	Normal	Normal	Normal	Normal	Normal
(ksi)	C_1	167	169	156	160	156	155
	C_2	22.1	22.4	25.0	24.9	31.5	31.6
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	19.4	19.7	20.4	20.9	21.4	21.2
	Minimum	17.6	18.1	16.9	17.3	17.0	16.0
	Maximum	20.9	21.2	24.0	24.8	27.5	26.7
E_1^c	C.V.(%)	6.54	7.17	12.2	12.6	16.1	16.1
(Msi)	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		8790		7910		8010
	Minimum		7780		4510		5950
	Maximum		10500		9630		9350
	C.V.(%)		11.8		24.7		14.9
ϵ_1^{cu}	B-value		(2)		(2)		(2)
	Distribution		Normal		Normal		Normal
($\mu\epsilon$)	C_1		8790		7910		8010
	C_2		1040		1950		1200
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	

(1) Reference 4.9.1
(2) B-values are presented only for fully approved data.

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Table 4.9.1(f)
C/PEEK -UT
IM-6/APC-2
Compression, 1-axis
$[0]_{16}$
250/A, 74/0.12%, 180/0.097%
Screening

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 32 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 60 - 62 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0054 - 0.0058 in.

TEST METHOD: ASTM D 3410A-87

MODULUS CALCULATION:

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		250 ambient		180 0.097 (1)		74 0.12 160°F, 95%	
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		(2)		(2)		(2)	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	129	126	162	160	174	176
	Minimum	70.0	71.5	156	146	141	144
	Maximum	154	145	168	169	186	192
	C.V.(%)	23.6	21.8	3.25	5.36	9.6	9.7
F_1^{cu}	B-value	(3)	(3)	(3)	(3)	(3)	(3)
	Distribution	Normal	Nonpara.	Normal	Normal	Normal	Normal
(ksi)	C_1	129	5	162	160	174	176
	C_2	30.5	3.06	5.26	8.59	16.7	17.1
	No. Specimens	6		5		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean	21.2	20.7	19.5	19.3	21.4	21.6
	Minimum	19.6	19.0	18.7	18.6	18.8	19.3
	Maximum	24.7	23.2	20.0	20.7	23.9	23.9
E_1^c	C.V.(%)	8.47	7.37	2.91	4.42	8.60	7.38
(Msi)	No. Specimens	6		5		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		6860		8310		8690
	Minimum		3380		7500		6950
	Maximum		8990		9390		12100
	C.V.(%)		28.7		8.94		23.5
ϵ_1^{cu}	B-value		(3)		(3)		(3)
	Distribution		Normal		Normal		Normal
($\mu\epsilon$)	C_1		6860		8310		8690
	C_2		1970		743		2050
	No. Specimens	6		5		6	
	No. Batches	1		1		1	
	Approval Class	Screening		Screening		Screening	

(1) Conditioned at 160°F, 95% relative humidity for 10 days (75% saturation)
(2) Reference 4.9.1
(3) B-values are presented only for fully approved data.

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Table 4.9.1(g)
C/PEEK -UT
IM-6/APC-2
Compression, 1-axis
$[0]_{16}$
180/0.11%
Screening

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 32 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 60 - 62 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0054 - 0.0058 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3410A-87

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		180					
Moisture Content(%)		0.11					
Equilibrium at T, RH		160°F, 95%					
Source Code		(1)					
		Normalized	Measured				
	Mean	154	151				
	Minimum	105	98.5				
	Maximum	189	183				
	C.V.(%)	18.2	19.3				
F_1^{cu}	B-value	(2)	(2)				
	Distribution	Normal	Normal				
(ksi)	C_1	154	151				
	C_2	28.0	29.3				
	No. Specimens	6					
	No. Batches	1					
	Approval Class	Screening					
	Mean	20.3	19.8				
	Minimum	15.6	15.7				
	Maximum	25.3	24.6				
E_1^c	C.V.(%)	18.4	17.6				
(Msi)	No. Specimens	6					
	No. Batches	1					
	Approval Class	Screening					
	Mean						
ν_{12}^c	No. Specimens						
	No. Batches						
	Approval Class						
	Mean		8180				
	Minimum		6580				
	Maximum		9500				
	C.V.(%)		13.0				
ϵ_1^{cu}	B-value		(2)				
	Distribution		Normal				
($\mu\epsilon$)	C_1		8180				
	C_2		1070				
	No. Specimens	6					
	No. Batches	1					
	Approval Class	Screening					

(1) Reference 4.9.1
(2) B-values are presented only for fully approved data.

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Table 4.9.1(h)
C/PEEK -UT
IM-6/APC-2
Shear, 12-plane
$[\pm 45]_{4s}$
74/A, -67/A, 180/A, 250/A
Screening

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 31 - 32 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 61 % VOID CONTENT: 0.0 - 0.2 %
PLY THICKNESS: 0.0052 - 0.0056 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3518-76

NORMALIZED BY: Not normalized

Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		74 ambient (1)	-67 ambient (1)	180 ambient (1)	250 ambient (1)	
	Mean	23.9	25.4	22.4	19.8	
	Minimum	18.9	18.1	17.2	14.2	
	Maximum	27.8	29.0	25.3	23.1	
	C.V.(%)	14.8	14.8	15.6	15.1	
F_{12}^{su} (ksi)	B-value	(2)	(2)	(2)	(2)	
	Distribution	Normal	Normal	Normal	Normal	
	C_1	23.9	25.4	22.4	19.8	
	C_2	3.53	3.77	3.49	2.98	
	No. Specimens	6	6	6	6	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	
	Mean	0.78	0.91	0.78	0.71	
	Minimum	0.73	0.83	0.72	0.63	
	Maximum	0.83	0.96	0.86	0.79	
G_{12}^{s}	C.V.(%)	5.5	5.5	6.2	9.3	
(Msi)	No. Specimens	6	6	6	6	
	No. Batches	1	1	1	1	
	Approval Class	Screening	Screening	Screening	Screening	
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{12}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) Reference 4.9.1
(2) B-values are presented only for fully approved data.

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Table 4.9.1(i)
C/PEEK -UT
IM-6/APC-2
Shear, 12-plane
$[\pm 45]_{4s}$
74/0.21%, 180/0.17%, 180/0.20%
Screening

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 31 - 32 wt% COMP. DENSITY: 1.55 g/cm^3
FIBER VOLUME: 61 % VOID CONTENT: 0.0 - 0.2 %
PLY THICKNESS: 0.0052 - 0.0056 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 3518-76

NORMALIZED BY: Not normalized

Temperature(°F)		180	74	180		
Moisture Content(%)		0.17	0.21	0.20		
Equilibrium at T, RH		(1)	160°F, 95%	160°F, 95%		
Source Code		(2)	(2)	(2)		
	Mean	23.3	23.0	20.0		
	Minimum	21.8	16.2	14.5		
	Maximum	24.0	26.7	26.1		
	C.V.(%)	3.85	15.4	22.4		
F_{12}^{su}	B-value	(3)	(3)	(3)		
	Distribution	Normal	Normal	Normal		
(ksi)	C_1	23.3	23.0	20.0		
	C_2	0.897	3.55	4.48		
	No. Specimens	5	6	6		
	No. Batches	1	1	1		
	Approval Class	Screening	Screening	Screening		
	Mean	0.76	0.79	0.71		
	Minimum	0.74	0.65	0.64		
	Maximum	0.78	0.89	0.78		
G_{12}^{s}	C.V.(%)	2.7	10.	9.0		
(Msi)	No. Specimens	4	6	6		
	No. Batches	1	1	1		
	Approval Class	Screening	Screening	Screening		
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{12}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Approval Class					

(1) Conditioned at 160°F, 95% relative humidity for 27 days (75% saturation)
(2) Reference 4.9.1
(3) B-values are presented only for fully approved data.

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REFERENCES

4.4.4 Rondeau, R.A., Askins, D. R., and Sjoblom, P., "Development of Engineering Data on New Aerospace Materials," University of Dayton Research Institute, UDR-TR-88-88, AFWAL-TR-88-4217, December 1988, Distribution authorized to DoD and DoD contractors only; critical technology; September 1988. Other requests for this document shall be referred to AFWAL/MLSE, OH 45433-6533.

4.9.1 Rondeau, R.A., Askins, D. R., and Sjoblom, P., "Development of Engineering Data on New Aerospace Materials," University of Dayton Research Institute, UDR-TR-88-88, AFWAL-TR-88-4217, December 1988, Distribution authorized to DoD and DoD contractors only; critical technology; September 1988. Other requests for this document shall be referred to AFWAL/MLSE, OH 45433-6533.

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9.9 SILICON CARBIDE - PEEK COMPOSITES

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10.1 INTRODUCTION

10.2 QUARTZ - EPOXY COMPOSITES

10.3 QUARTZ - POLYESTER COMPOSITES

10.4 QUARTZ - BISMALEIMIDE COMPOSITES

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10.4.1 Astroquartz II/F650 8-harness satin weave *

MATERIAL:	Astroquartz II/F650 8-harness satin weave		**Q/BMI Astroquartz II/F650 Summary**
PREPREG:	Hexcel AQII581/F650 8-harness satin weave		
FIBER:	J.P. Stevens Astroquartz II	MATRIX: Hexcel F650	
T_g(dry):	600°F T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure: 375°F, 4 hours, 85 psig; Postcure: 475°F, 4 hours		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	Date of testing	
Date of resin manufacture	Date of data submittal	4/89
Date of prepreg manufacture	Date of analysis	1/93
Date of composite manufacture		

LAMINA PROPERTY SUMMARY

	75°F/A		450°F/A					
Tension, 1-axis								
Tension, 2-axis								
Tension, 3-axis								
Compression, 1-axis								
Compression, 2-axis								
Compression, 3-axis								
Shear, 12-plane								
Shear, 23-plane								
SB strength, 31-plane	S---		S---					

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cm^3)	2.17		
Resin Density	(g/cm^3)	1.27		
Composite Density	(g/cm^3)	1.78	1.73	
Fiber Areal Weight	(g/m^2)	285		
Ply Thickness	(in)	0.0100	0.010	

LAMINATE PROPERTY SUMMARY

Classes of data: F - Fully approved, I - Interim, S - Screening in Strength/Modulus/Poisson's ratio/Strain-to-failure order

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Table 10.4.1(a)
Q/BMI 285-8HS
Astroquartz II/F650
SB strength, 31-plane
$[0_t]_{12}$
75/A, 450/A
Screening

MATERIAL: Astroquartz II/F650 8-harness satin weave

RESIN CONTENT: 37 wt% COMP. DENSITY: 1.73 g/cm^3
FIBER VOLUME: 51 % VOID CONTENT:
PLY THICKNESS: 0.010 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D 2344

NORMALIZED BY: Not normalized

Temperature(°F)		75	450				
Moisture Content(%)		ambient	ambient				
Equilibrium at T, RH							
Source Code		21	21				
	Mean	6.41	6.56				
	Minimum	6.31	6.43				
	Maximum	6.50	6.72				
	C.V.(%)	1.06	1.69				
F_{31}^{sbs}	B-value	(1)	(1)				
	Distribution	Normal	Normal				
(ksi)	C_1	6.41	6.56				
	C_2	0.068	0.111				
	No. Specimens	5	5				
	No. Batches	1	1				
	Approval Class	Screening	Screening				

(1) B-values are presented only for fully approved data.

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10.5 QUARTZ - POLYIMIDE COMPOSITES

10.6 QUARTZ - PHENOLIC COMPOSITES

10.7 QUARTZ - SILICONE COMPOSITES

10.8 QUARTZ - POLYBENZIMIDAZOLE COMPOSITES

10.9 QUARTZ - PEEK COMPOSITES

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APPENDIX A1 MIL-HDBK-17A DATA

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A1.1 GENERAL INFORMATION

The data on polymer-matrix composite materials which were presented in MIL-HDBK-17A, dated January 1971, are presented in this appendix. MIL-HDBK-17 A has been superseded so these data are presented here so they can be referenced in a current publication. However, these data do not meet the data requirements in Volume 1.

The materials which were included in MIL-HDBK-17A are listed in Table A1. Of the sixteen materials, six are still available, five are no longer available, and the availability of the other five materials could not be determined. The data from the six available materials are provided in this appendix. The data from the remaining materials may be added as availability of the material or usefulness of the data is determined. Note that Narmco 5505 has been licensed to AVCO and those data are presented herein as AVCO 5505.

TABLE A1 *Materials from MIL-HDBK-17A*

Available:
U.S. Polymeric E-720E/7781 (ECDE-1/0-550) Fiberglass Epoxy
Hexcel F-161/7743(550) Fiberglass Epoxy
Hexcel F-161/7781(ECDE-1/0-550) Fiberglass Epoxy
Narmco N588/7781 (ECDE-1/0-550) Fiberglass Epoxy
Narmco 506/7781 (ECDE-1/0-A1100) Fiberglass Phenolic
AVCO 5505 Boron Epoxy
Not available:
U.S. Polymeric E-779/7743 (Volan) Fiberglass Epoxy
3M XP251S Fiberglass Epoxy
U.S. Polymeric S-860/1581 (ECG-1/2-112) Neutral pH Fiberglass Silicone
U.S. Polymeric P670A/7781 (ECDE-1/0) Fiberglass Modified DAP Polyester
SP272 Boron Epoxy
Availability unknown:
Bloomingdale BP915/7781 (ECDE-1/0-550) Fiberglass Epoxy
Bloomingdale BP911/7781 (ECDE-1/0 Volan) Fiberglass Epoxy
Cordo E293/7781 (ECDE-1/0-550) Fiberglass Epoxy
Styrene-Alkyd Polyester/7781 Fiberglass
Cordo IFRR/7781 (ECDE-1/0) Fiberglass Modified DAP Polyester

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The table and figure numbers used in this appendix are similar to those in MIL-HDBK-17A. The chapter identification has been changed from 4 to A1 but the rest of all figure and table numbers has not been changed. For example, Table A1.40 is the same as Table 4.40 in MIL-HDBK-17A. The MIL-HDBK-17A text describing the test program and methods is reproduced in Sections A1.2 through A1.4.

A1.2 INTRODUCTION

The laminate properties presented in this chapter have been generated in test programs conducted at the U.S. Forest Products Laboratory and elsewhere (Reference A1.2).[1] Properties are given for fiberglass with epoxy, phenolic, silicone and polyester resins and for boron with epoxy. Additional information on these and other material combinations will be issued as supplements or revisions of the present handbook edition.

A1.3 HANDBOOK TEST PROGRAM

A1.3.1 Objectives

The objectives of the handbook test program are to obtain statistically significant data for materials currently in use and to determine the degree of reproducibility attained in their fabrication. A minimum requirement is that test results include data from three sets of panels which are representative of the manufacturing procedures employed by three different fabricators. The properties listed in the charts and tables of this chapter represent test results from only one set of panels for each material system. Properties are therefore not given minimum values and are considered to be "typical" for each material. When the minimum number of tests has been completed for a material, its properties will be assigned values on a B-basis; that is, the value above which 90 percent of the population of values is expected to fall with a confidence of 95 percent.

A1.3.2 Preimpregnated materials

All test panels are fabricated from prepregs. Emphasis is placed on materials for use as facings in sandwich type structures. The prepregs for facings are normally processed to conform with two methods of sandwich fabrication. These are the laminate grades for two-step sandwich constructions and the controlled flow adhesive grades for one-step sandwich constructions. Only laminates simulating precured facings, that is, for use in two-step sandwiches, have been subjected to the narrow coupon tests listed in this chapter. The controlled flow adhesive prepregs are best tested as sandwich panels, and such testing is not at present included in the handbook program.

The prepreg materials comply with the specifications established by the individual fabricators. In general, the materials are autoclave molding grades with flows controlled to attain minimum bleedout and optimum bonding of the plies. When possible handling characteristics are specified consistent with the objectives of collimated plies in the laminate and the retention of fiber orientation during lay-up and cure.

Imposed tolerances on the gravimetric resin content of the prepregs are dependent on the type of reinforcement. For bidirectional woven broadgoods such as style 7781 fabric, the resin fraction is specified as not varying by more than two percent from the assigned devolatilized resin content. For directionally woven broadgoods such as style 7743 fabric, and nonwoven parallel fiber tapes such as XP251S, variation from the assigned devolatilized resin content is not to exceed three percent.

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[1]Exceptions are the data for fiberglass-polyester laminates, taken from earlier sources, and the data for boron-epoxy panels which were compiled under special contract and published separately (Reference A1.2).

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A1.3.3 Test panels

A minimum size of the test panels has been established as two feet parallel to the warp direction by three feet parallel to the width for woven fabrics. For the non-woven laminates, including unidirectional, crossplied and quasi-isotropic configurations, the three foot dimension is parallel to the fiber direction in the outer plies.

It is desirable that the test laminates be fabricated so that fiber alignment and orthotropy are maintained and that they are symmetrically balanced. Such conditions are generally attained in the test panels and they are designated in the following data summary tables as balanced and parallel. One set of panels (Table A1.1) is not balanced. In this case the laminates are parallel plied.

A1.3.4 Test procedures

Conventional uniaxial tests are conducted at constant crosshead rates. The direction parallel to the warp of woven fabrics is designated as the 0° or 1-direction. The direction perpendicular to the 0° direction is designated as the 90° or 2-direction. For non-woven unidirectional laminates, the 0° direction corresponds to the fiber direction. For crossplied and quasi-isotropic laminates, the 0° direction corresponds to the fiber direction in the outer plies.

A1.3.4.1 Tensile tests

Tensile tests for woven fabric laminates have been conducted initially using the method of ASTM D 638 and Type I specimens (Reference 1.3.4.1(a)). Later tests are conducted with a modified specimen (Reference A1.2) and the method is designated as MIL-HDBK-17 tensile test. Tab ended specimens are used to test the 0° tensile properties of the non-woven unidirectional laminates (Reference A1.3.4.1(b)).

A1.3.4.2 Compression tests

Compression tests have been conducted with the end clamped and jig stabilized ASTM D 695 specimen (Reference A1.3.4.2) and with the MIL-HDBK-17 compression specimen (Reference A1.2) in which the specimen and fixture have been modified.

A1.3.4.3 Shear tests

The picture frame method (Reference A1.2) has been used to determine the 0° - 90° shear properties of one material system at three resin fractions (Figure A1.6.3). In these tests it is assumed that 88 percent of the load is reacted by the specimen, while the pins in the fixture react the remainder. The other materials are tested by a modified rail shear method (Reference A1.3.4.3).

A1.3.4.4 Interlaminar shear

Interlaminar shear properties are determined by the short beam test method (Reference A1.3.4.1(b)), or by the method of ASTM D 2733-68T when indicated (Reference 1.3.4.4).

A1.3.4.5 Flexural tests

Flexural properties are determined by the method of ASTM D 790 (Reference 1.3.4.5).

A1.3.4.6 Bearing strength

Bearing strengths are determined by the method of ASTM D 953 (Reference 1.3.4.6).

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A1.3.5 Dry conditioning

Specimens are dry conditioned by allowing them to attain equilibrium at 70°F to 75°F and 45 percent to 55 percent relative humidity for a minimum of ten days. When tested at other than room temperature, the dry conditioned specimens are soaked at the test temperature for one-half hour prior to applying load.

A1.3.6 Wet conditioning

Specimens are wet conditioned at 125°F and 95 percent to 100 percent relative humidity for 1000 hours (42 days). When tested at temperatures below freezing, the wet conditioned specimens are cycled four times from the wet condition at 125°F to the sub-freezing test temperature; the dwell time at each temperature being one-half hour. Wet specimens tested at 160°F are soaked for one-half hour at this temperature immediately prior to testing. Some materials are shown as being tested at 220°F after wet conditioning. Such testing has been discontinued since these results appear inconclusive.

A1.3.7 Test schedule

The 0° and 90° tension and compression properties are determined at three reference temperatures, 65°F, 70°F - 75°F and 160°F, for both dry and wet conditioned specimens. Dry conditioned specimens are tested at maximum temperature for those materials which are potentially serviceable at elevated temperatures. Ten test results are obtained for the stress-strain relations at each of these conditions. Tests at intermediate temperatures are conducted to verify property changes, in which cases five specimens are tested. Ten test results are also required for the 0° - 90° shear at -65°F, 70°F - 75°F, and 160°F in the dry condition. Five tests are conducted at 70°F - 75°F to determine the stress-strain relations for Poisson's ratio. Flexure, bearing and interlaminar shear are determined in the 0° direction and dry condition at -65°F, 70°F - 75°F and 160°F. Five specimens are tested for each temperature.

A1.4 DATA PRESENTATION

Uniaxial tension, compression and shear are shown as stress-strain relations at each temperature and the properties are summarized in tabular form. Flexural, bearing and interlaminar shear properties are listed in summary tables. Poisson's ratio is shown as the response of the 0° elongation and 90° contraction to the applied tensile stress.

When ten or more results are available at a test condition, average values and the associated standard deviations are given in the tables. Stress-strain relations are plotted as an average curve and a plot of the average minus three times the standard deviation is also shown. When five to nine results are obtained from a test condition, average, maximum, and minimum values and curves are shown.

A1.4.1 Epoxy-fiberglass laminates

All data on fiberglass-epoxy systems are results obtained from the handbook test program. Properties are summarized in Tables A1.1 through A1.8. Detailed data are shown in Figures A1.1.1(a) through A1.8.5. [Four of the nine materials are known to be available.]

A1.4.2 Phenolic-fiberglass laminates

Handbook tested properties are summarized in Table A1.40 and Figures A1.40.1(a) through A1.40.5 for one fiberglass-phenolic system. [This material is available.]

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A1.4.3 Silicone-fiberglass laminates

Partial handbook test results were listed in MIL-HDBK-17A for one fiberglass-silicone system. [This material is not available]

A1.4.4 Polyester-fiberglass laminates

Previous data for fiberglass-polyester laminates were listed in MIL-HDBK-17A. [None of these materials are known to be available.]

A1.4.5 Boron-epoxy laminates

Data on two boron-epoxy systems have been abstracted from the literature (Reference A1.4.5) and are presented in Tables A1.110 and A1.111 and in Figures A1.110.1(a) through A1.111.3. [One of these materials is available.]

The laminate thickness is controlled by the number of plies in the construction and the desired resin content. In general, the thickness of woven fabric laminates is maintained at eight plies, except for low resin content laminates which may require as many as ten plies. Nonwoven laminate monolayers are constructed with six plies to reduce the shear lag apparent in testing, and eight plies for the crossplied and quasi-isotropic panels.

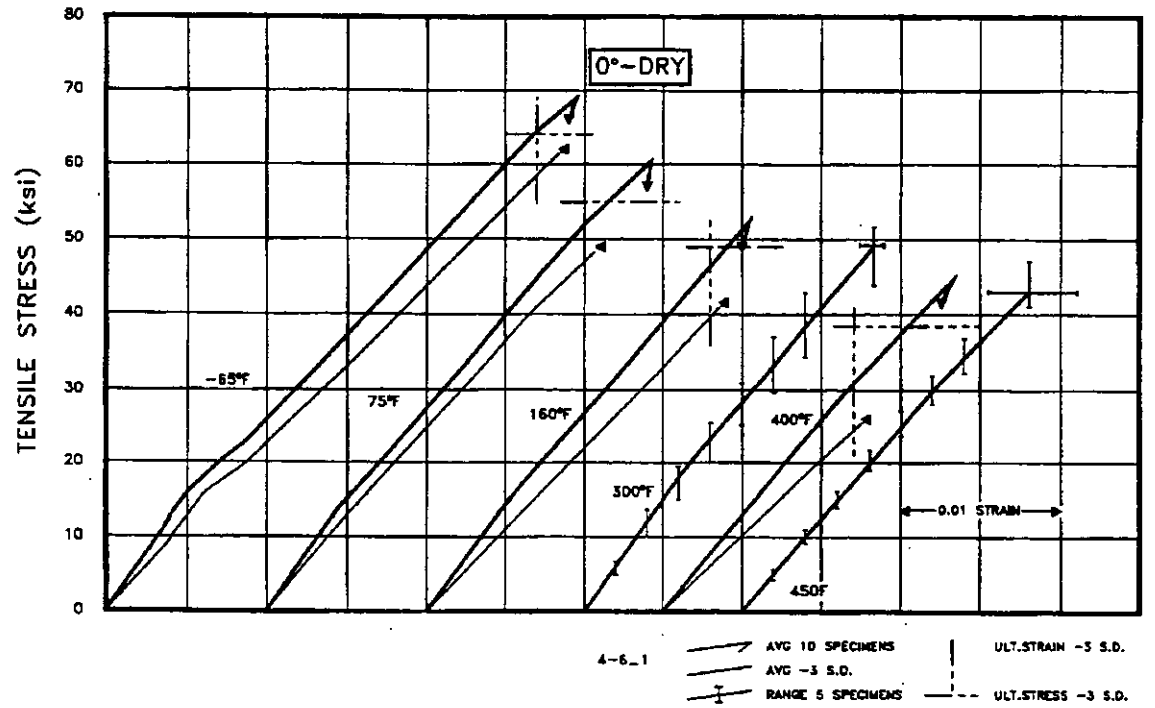
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TABLE A1.1 *Summary of Mechanical Properties of U.S. Polymeric E-720E/7781 (ECDE-1/0-550) Fiberglass Epoxy*

Fabrication	Lay-up: Parallel		Vacuum: None		Pressure: 55-65 PSI		Bleedout: Edge & Vertical		Cure: 2 hr/350°F		Postcure: 4 hrs/400°F		Plies: 8	
Physical Properties	Weight Percent Resin: 34.9				Avg. Specific Gravity: 1.78				Avg. Percent Voids: 2.0				Avg. Thickness: 0.082 inches	
Test Methods	Tension: ASTM D 638 TYPE-1		Compression: MIL-HDBK-17		Shear: Rail		Flexure: ASTM D 790		Bearing: ASTM D 953				Interlaminar Shear: Short Beam	
Temperature	-65°F				75°F				160°F				400°F	
Condition	Dry		Wet		Dry		Wet		Dry		Wet		Dry	
	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
Tension														
ultimate stress, ksi 0°	69.2	1.6	69.1	1.7	60.4	1.7	55.7	1.5	52.5	1.0	42.9	0.8	44.8	2.0
90°	56.0	2.0	56.5	2.0	49.0	1.8	45.9	1.4	42.3	1.2	36.9	1.1	34.9	1.6
ultimate strain, % 0°	2.93	0.08	2.70	0.11	2.43	0.14	2.12	0.08	2.05	0.08	1.61	0.06	1.80	0.20
90°	2.92	0.22	2.54	0.19	2.33	0.09	2.04	0.09	1.98	0.08	1.70	0.13	1.72	0.22
proportional limit, ksi 0°														
90°														
initial modulus, 10^6 psi 0°	3.30		3.38		3.12		3.12		2.95		2.76		2.60	
90°	2.90		3.02		2.82		2.78		2.50		2.65		2.30	
secondary modulus, 10^6 psi 0°	2.30		2.85		2.45		2.50		2.46		2.37			
90°	1.90		1.74		2.05		2.19		2.01		1.97			
Compression														
ultimate stress, ksi 0°	77.1	4.0	75.0	3.7	64.8	2.9	57.3	3.8	54.0	1.4	46.2	1.4	23.8	2.2
90°	57.2	2.7	53.9	2.7	50.2	2.9	45.2	2.4	40.8	2.9	36.2	3.1	14.7	1.6
ultimate strain, % 0°	2.48	0.16	2.44	0.15	2.14	0.11	1.99	0.09	1.86	0.08	1.62	0.06	1.12	0.22
90°	1.93	0.16	1.81	0.19	1.70	0.14	1.58	0.14	1.46	0.17	1.37	0.15	0.91	0.08
proportional limit, ksi 0°														
90°														
initial modulus, 10^6 psi 0°	3.50		3.45		3.25		3.10		3.15		3.03		2.45	
90°	3.20		3.26		3.21		3.03		2.99		2.85		1.85	
Shear														
ultimate stress, ksi 0°-90° ±45°	17.5				14.3	0.6			11.2					

	-65°F Dry			75°F Dry			160° Dry		
	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure									
ultimate stress, ksi 0°	115.6	119.4	111.5	91.7	93.4	90.3	69.4	71.1	67.2
proportional limit, ksi 0°	88.1	100.7	77.5	32.5	36.2	30.8	56.2	62.8	49.4
initial modulus, 10^6 psi 0°	2.87	2.91	2.74	3.21	3.36	3.03	2.81	2.87	2.76
Bearing									
ultimate stress, ksi 0°	74.1	78.4	70.7	60.8	64.4	58.2	50.0	53.0	47.9
stress at 4% elong., ksi 0°	32.1	34.8	29.1	23.9	34.2	20.1	18.1	21.5	15.9
Interlaminar Shear									
ultimate stress, ksi 0°	7.09	7.36	6.80	5.90	6.07	5.72	6.05	6.16	5.91

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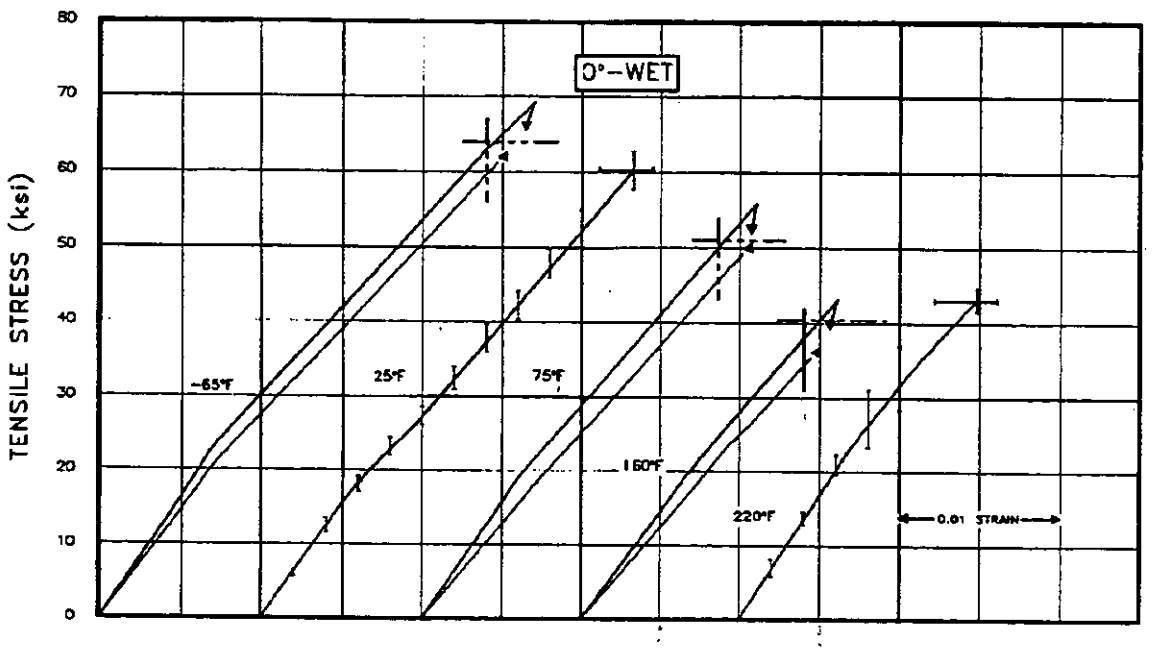
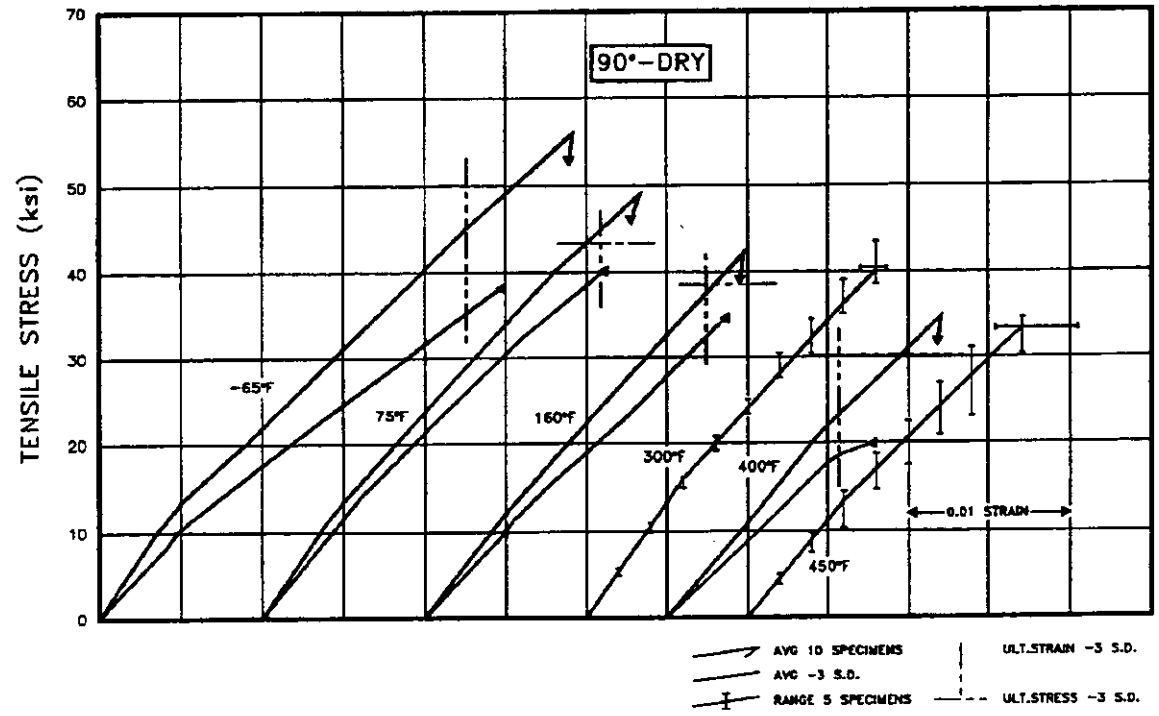


FIGURE A1.1.1(a) *Tensile stress-strain for E-720E/7781 fiberglass epoxy loaded in the 0° direction.*

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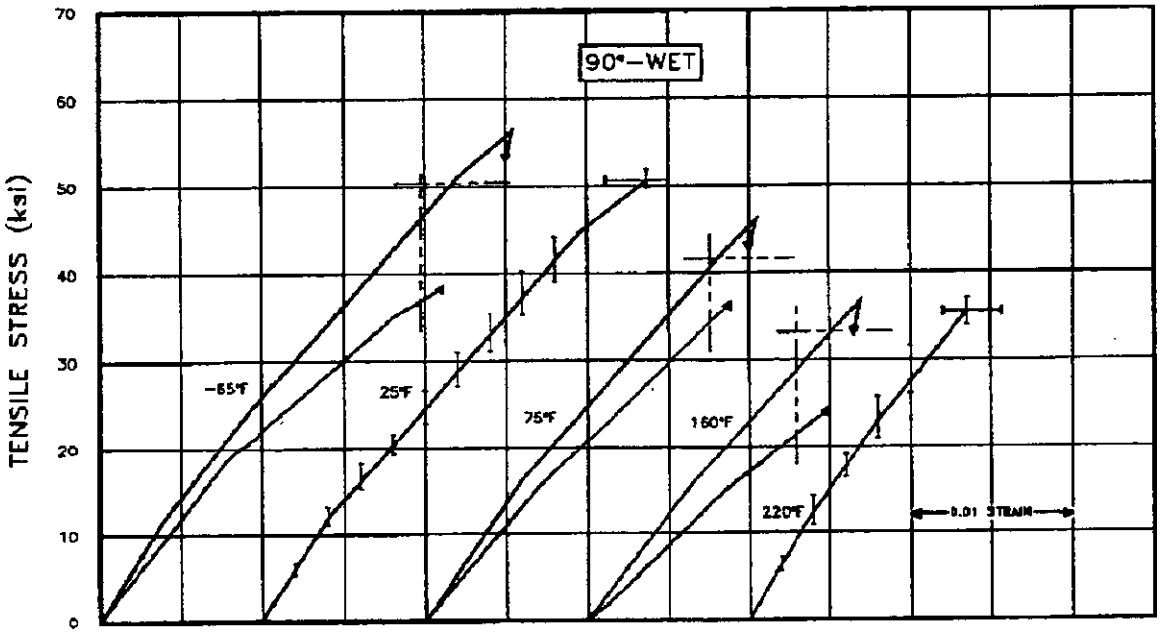


FIGURE A1.1.1(b) *Tensile stress-strain for E-720E/7781 fiberglass epoxy in the 90°direction.*

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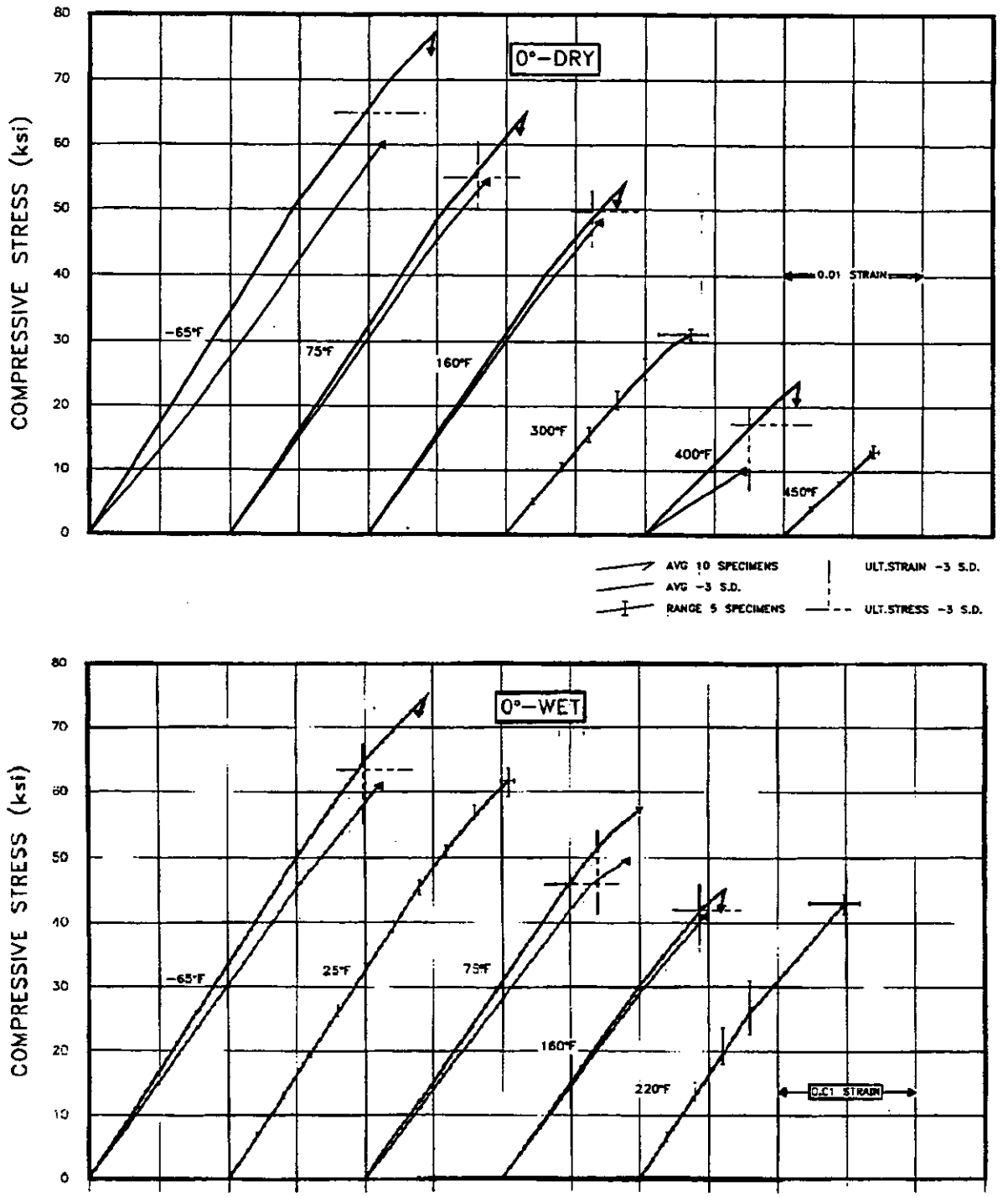
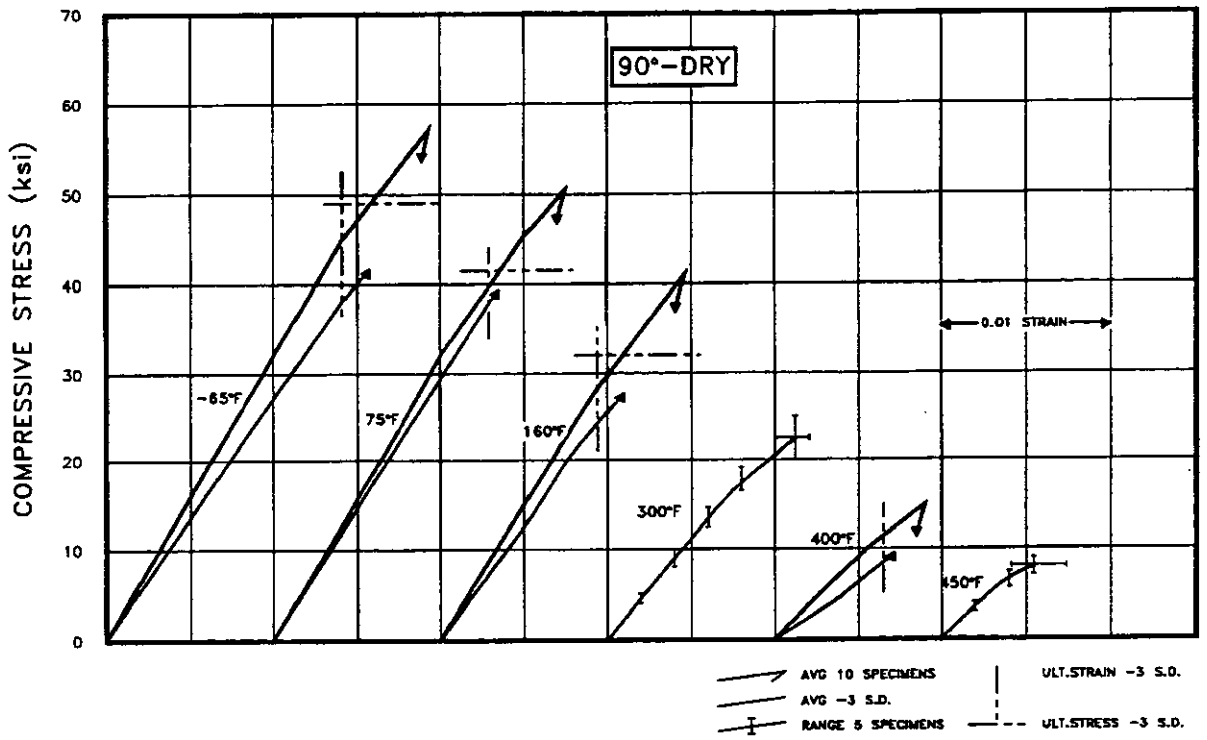


FIGURE A1.1.2(a) *Compressive stress-strain for E-720E/7781 fiberglass epoxy loaded in the 0° direction.*

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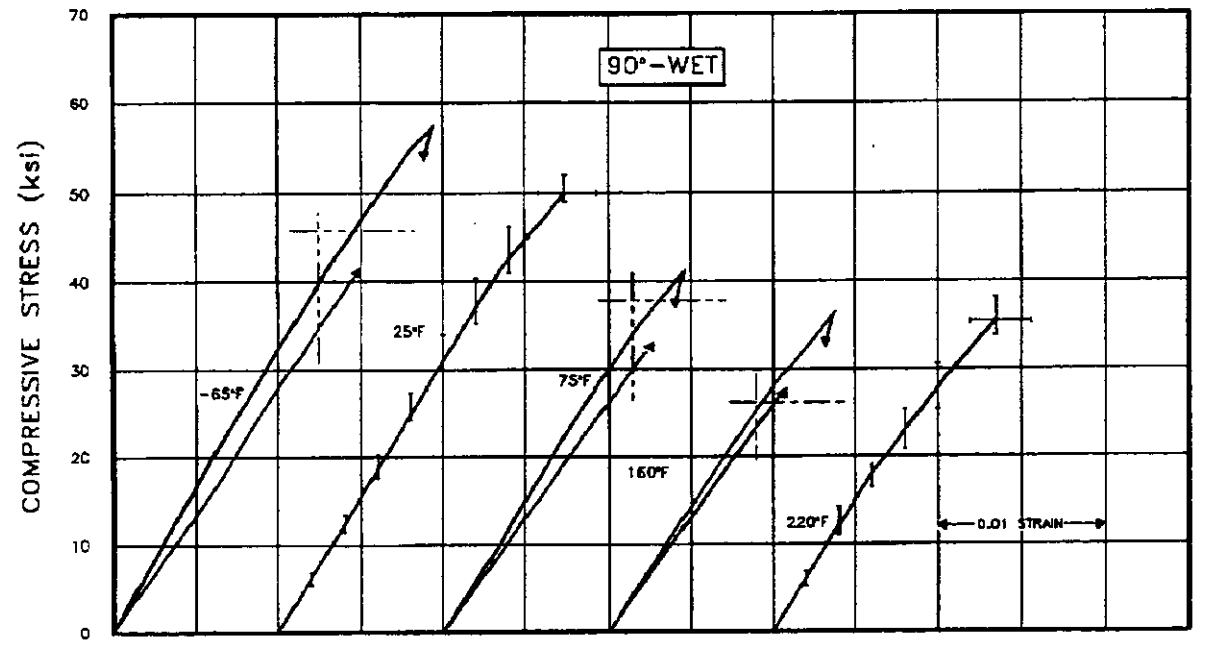


FIGURE A1.1.2(b) *Compressive stress-strain for E-720E/7781 fiberglass epoxy loaded in the 90° direction.*

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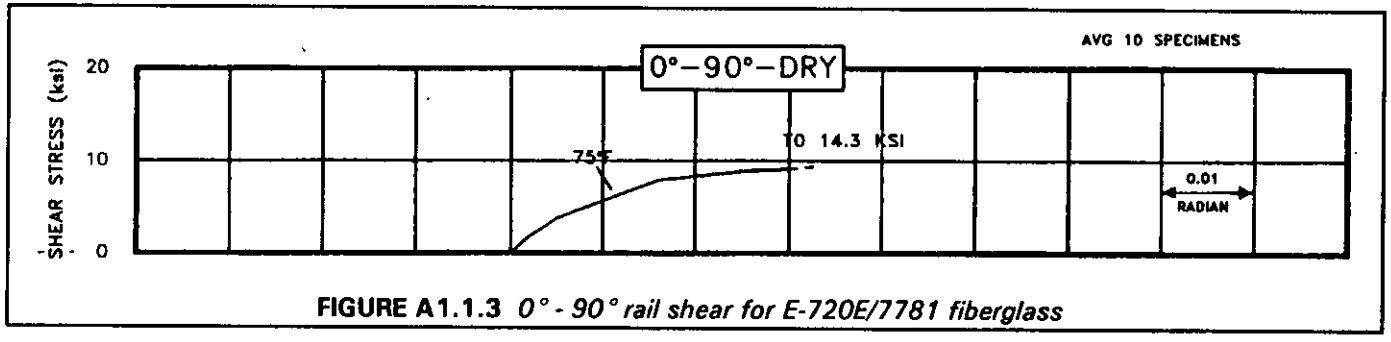
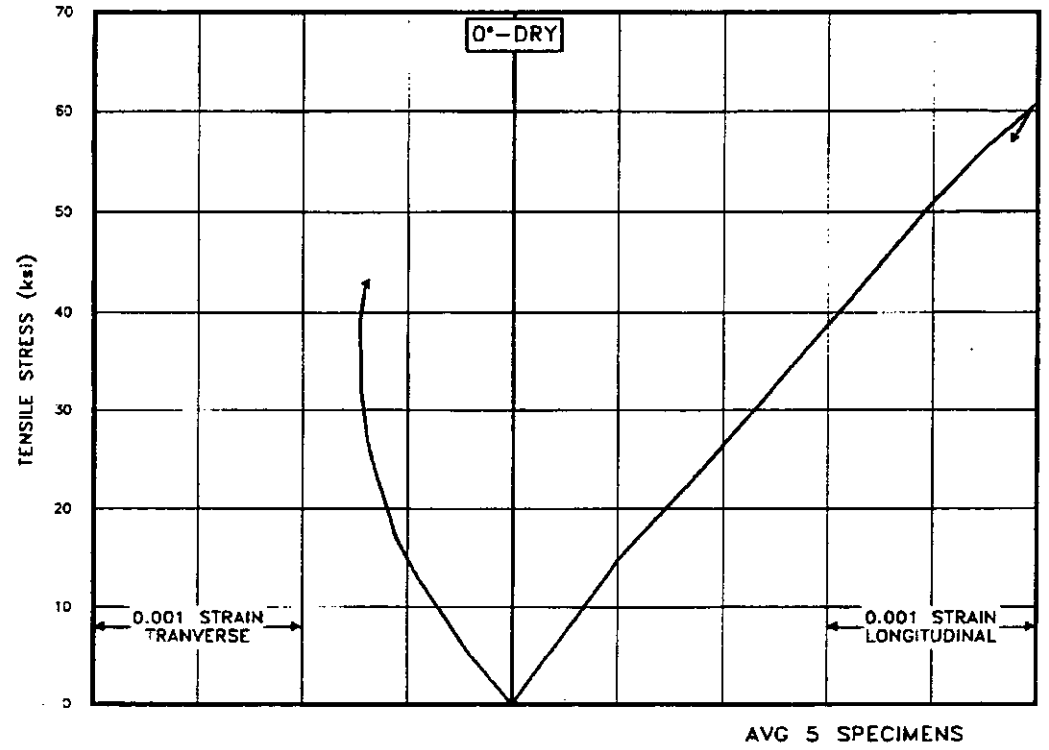


FIGURE A1.1.3 *0° - 90° rail shear for E-720E/7781 fiberglass*

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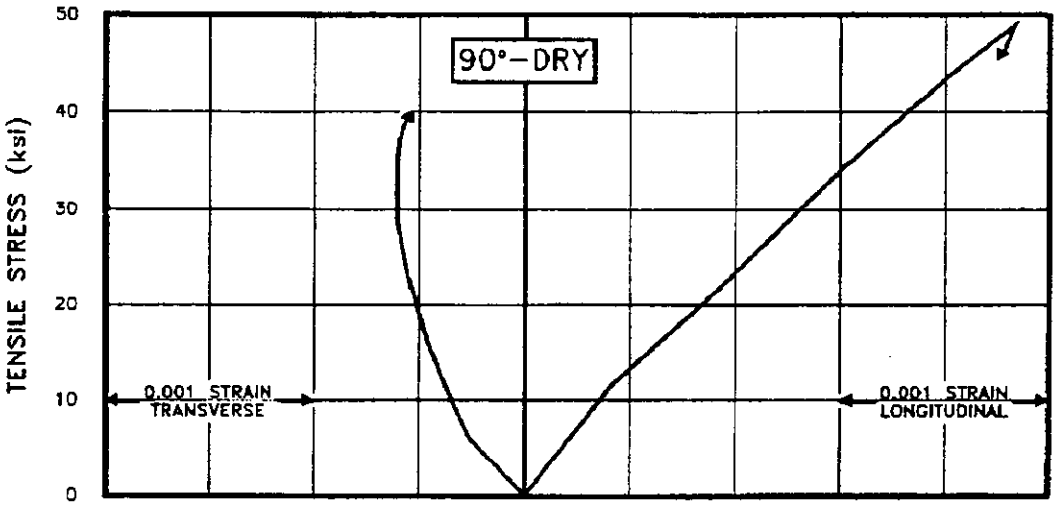


FIGURE A1.1.4 *Poisson effects for E-720E/7781 fiberglass epoxy.*

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TABLE A1.3 *Summary of Mechanical Properties of Hexcel F-161/7743(550) Fiberglass Epoxy*

Fabrication		Lay-up: Balanced		Vacuum: 14 psi		Pressure: 35 psi		Bleedout: Pinched Edge		Cure: 2 hr/350°F		Postcure: 2 hr/350°F		Plies: 8	
Physical Properties		Weight Percent Resin: 32.4 v = 0.496				Avg. Specific Gravity: 1.85				Avg. Percent Voids: 3.0				Avg. Thickness: 0.086 inches	
Test Methods		Tension: ASTM-D 638 TYPE 1		Compression: MIL-HDBK-17		Shear: Rail		Flexure: ASTM-D 790		Bearing: ASTM-D 953				Interlaminar Shear: Short Beam	
Temperature		-65°F				75°F				160°F				400°F	
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry	
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
Tension															
ultimate stress, ksi	0°	111.3	1.12	107.3	3.60	95.5	7.57	87.3	5.2	80.9	4.05	71.7	2.73	74.5	5.90
	90°	9.84	0.78	9.42	0.59	8.15	0.40	7.27	0.28	6.78	0.18	6.16	0.21	6.59	0.41
ultimate strain, %	0°	2.10	0.31	2.11	0.10	1.88	0.10	1.72	0.17	1.56	0.15	1.35	0.12	1.64	0.09
	90°	2.43	0.25	2.03	0.21	1.82	0.23	1.20	0.28	1.26	0.19	0.61	0.13	1.44	0.19
proportional limit, ksi	0°	86.2		87.8		74.7		81.5		64.0		65.4		61.0	
	90°	5.6		5.0		5.2		4.8		5.0		5.0		3.0	
initial modulus, 10^6 psi	0°	5.42		5.35		5.30		5.55		5.36		5.47		4.52	
	90°	1.61		1.73		1.73		1.41		1.11		1.30		0.74	
secondary modulus, 10^6 psi	0°					5.15									
	90°					0.09									
Compression															
ultimate stress, ksi	0°	95.0	7.42	89.7	7.0	75.9	5.43	67.4	4.43	66.3	5.53	55.0	2.80	26.7	1.93
	90°	40.3	1.93	37.6	2.93	32.1	2.87	30.4	1.27	27.4	1.93	23.0	1.30	8.3	0.90
ultimate strain, %	0°	1.90	0.11	1.83	0.14	1.58	0.11	1.36	0.11	1.47	0.08	1.22	0.06	0.68	0.08
	90°	2.57	0.16	2.46	0.25	2.51	0.19	2.38	1.90	2.58	0.22	2.53	0.30	1.62	0.12
proportional limit, ksi	0°	83.0		70.0		52.2		49.8		55.6		40.8		20.0	
	90°	18.1		15.0		11.9		10.6		9.2		8.2			
initial modulus, 10^6 psi	0°	5.02		4.98		4.96		5.09		4.59		4.66		4.12	
	90°	1.91		1.88		1.65		1.77		1.46		1.37			
Shear															
ultimate stress, ksi	0°-90° ±45°	12.5				9.2	0.2			7.7					

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°	203.0	210.0	196.0	160.0	163.0	155.0	138.0	142.0	135.0
proportional limit, ksi	0°	153.0	158.0	147.0	127.0	139.0	116.0	116.0	118.0	112.0
initial modulus, 10^6 psi	0°	5.71	5.80	5.63	5.18	5.27	5.10	5.43	5.46	5.32
Bearing										
ultimate stress, ksi	0°	79.4	90.2	64.8	58.8	63.2	52.7	53.7	57.5	50.6
stress at 4% elong., ksi	0°	37.9	45.6	31.5	23.0	27.1	19.5	21.9	23.6	20.5
Interlaminar Shear										
ultimate stress, ksi	0°	9.55	10.15	8.72	9.35	9.55	9.17	8.31	8.65	8.02

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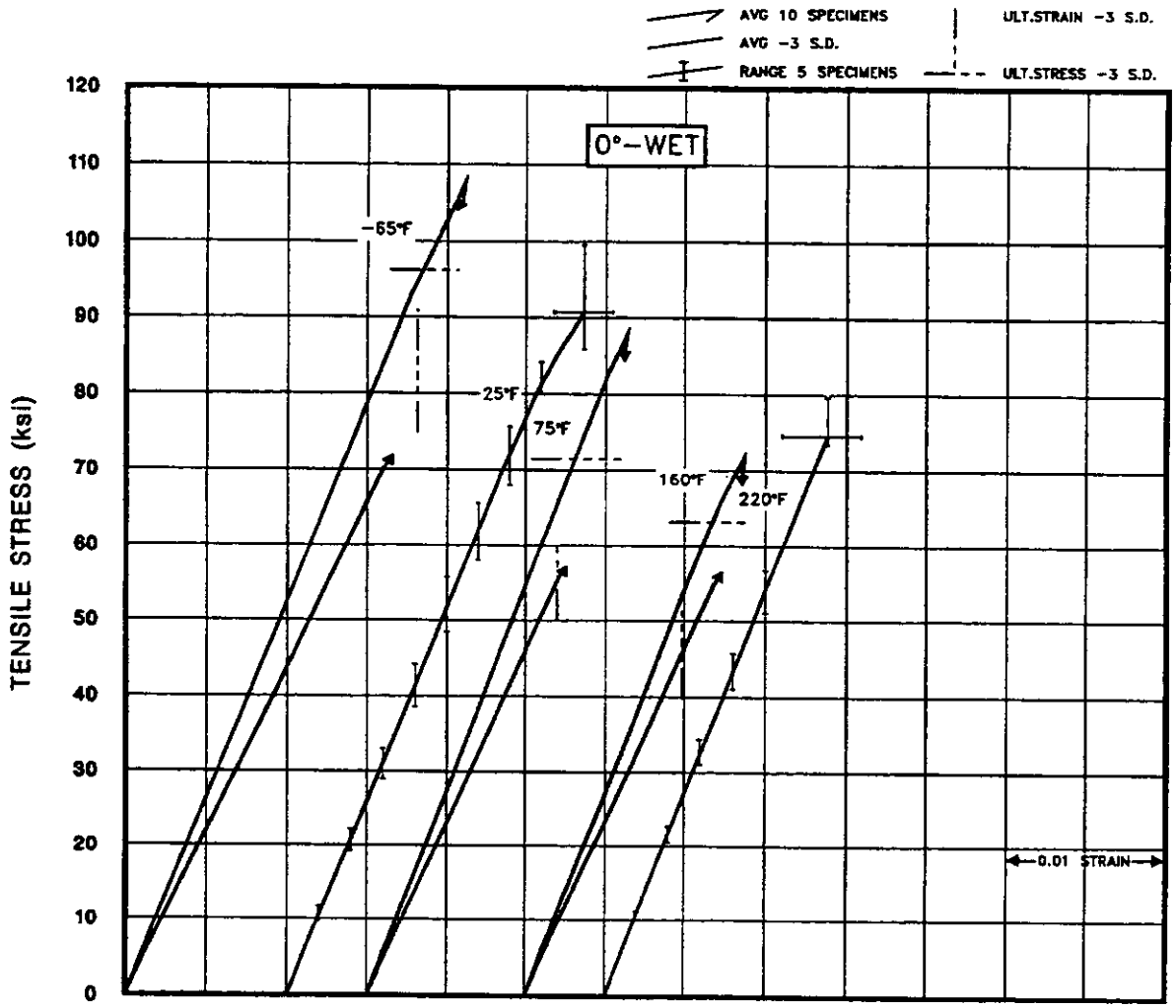


FIGURE A1.3.1(a) *Tensile stress-strain for F-161/7743 fiberglass epoxy loaded in the 0° direction*, continued on next page.

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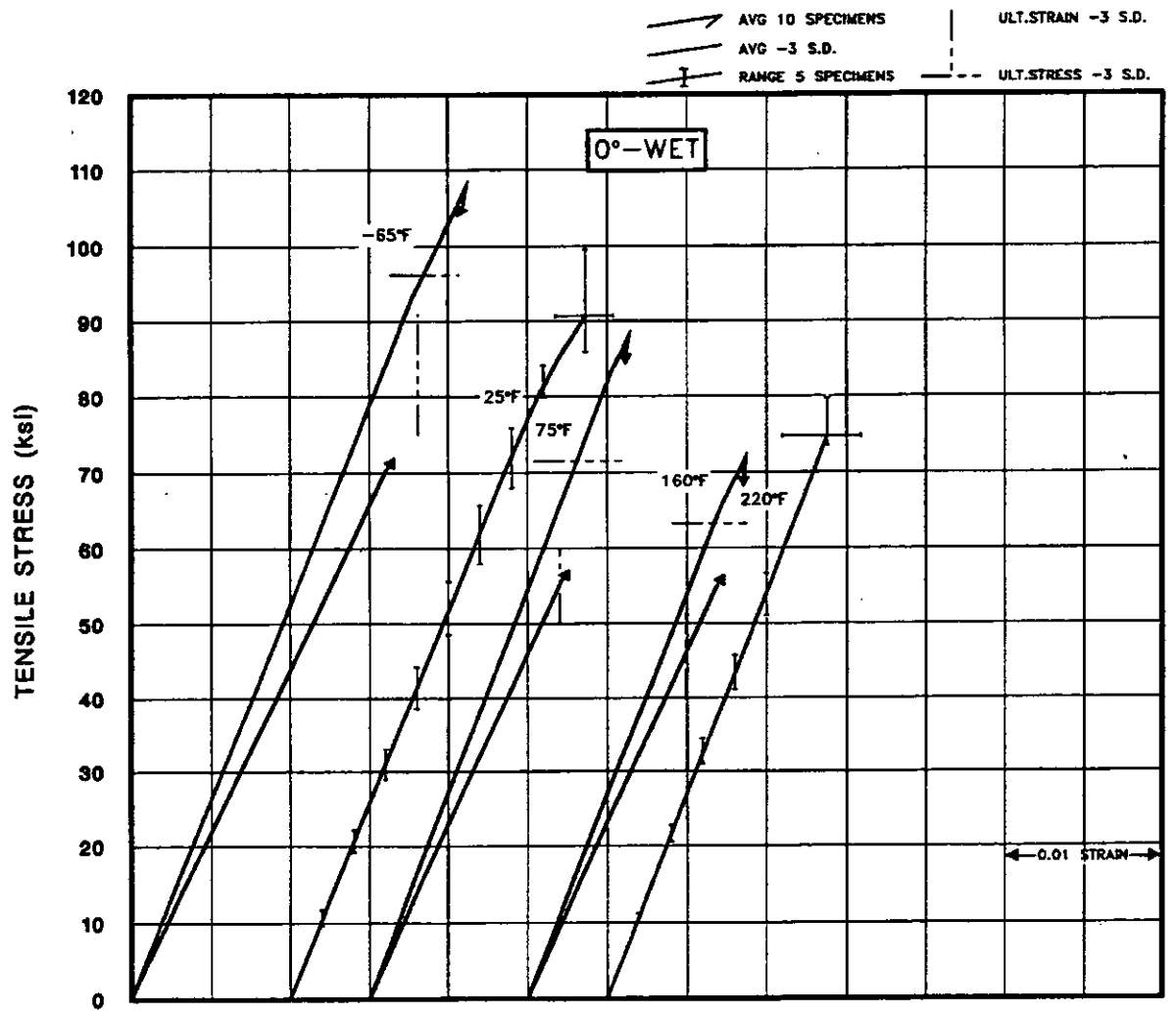


FIGURE A1.3.1(a) *Tensile stress-strain for F-161/7743 fiberglass epoxy loaded in the 0° direction,* concluded.

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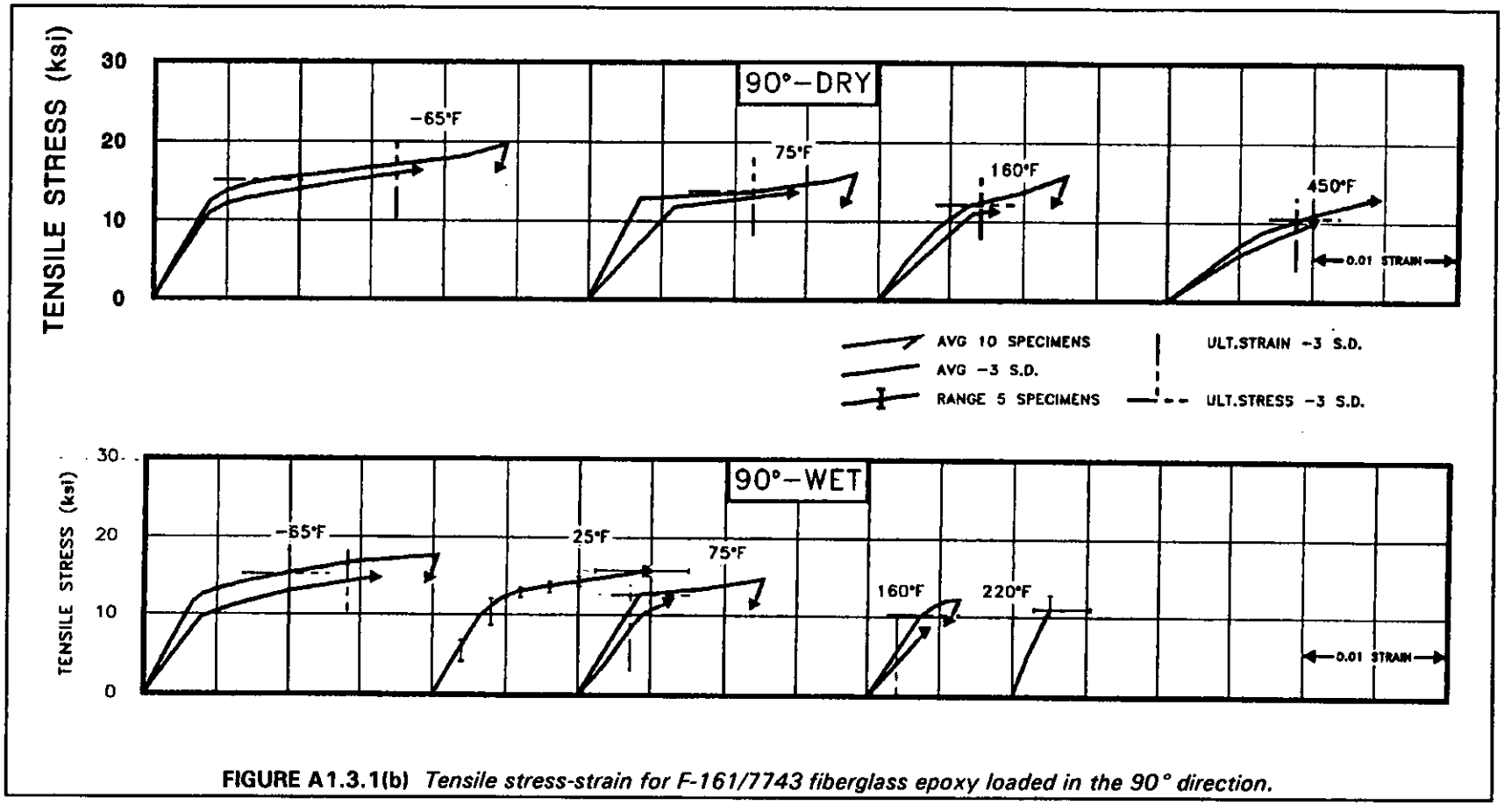


FIGURE A1.3.1(b) *Tensile stress-strain for F-161/7743 fiberglass epoxy loaded in the 90° direction.*

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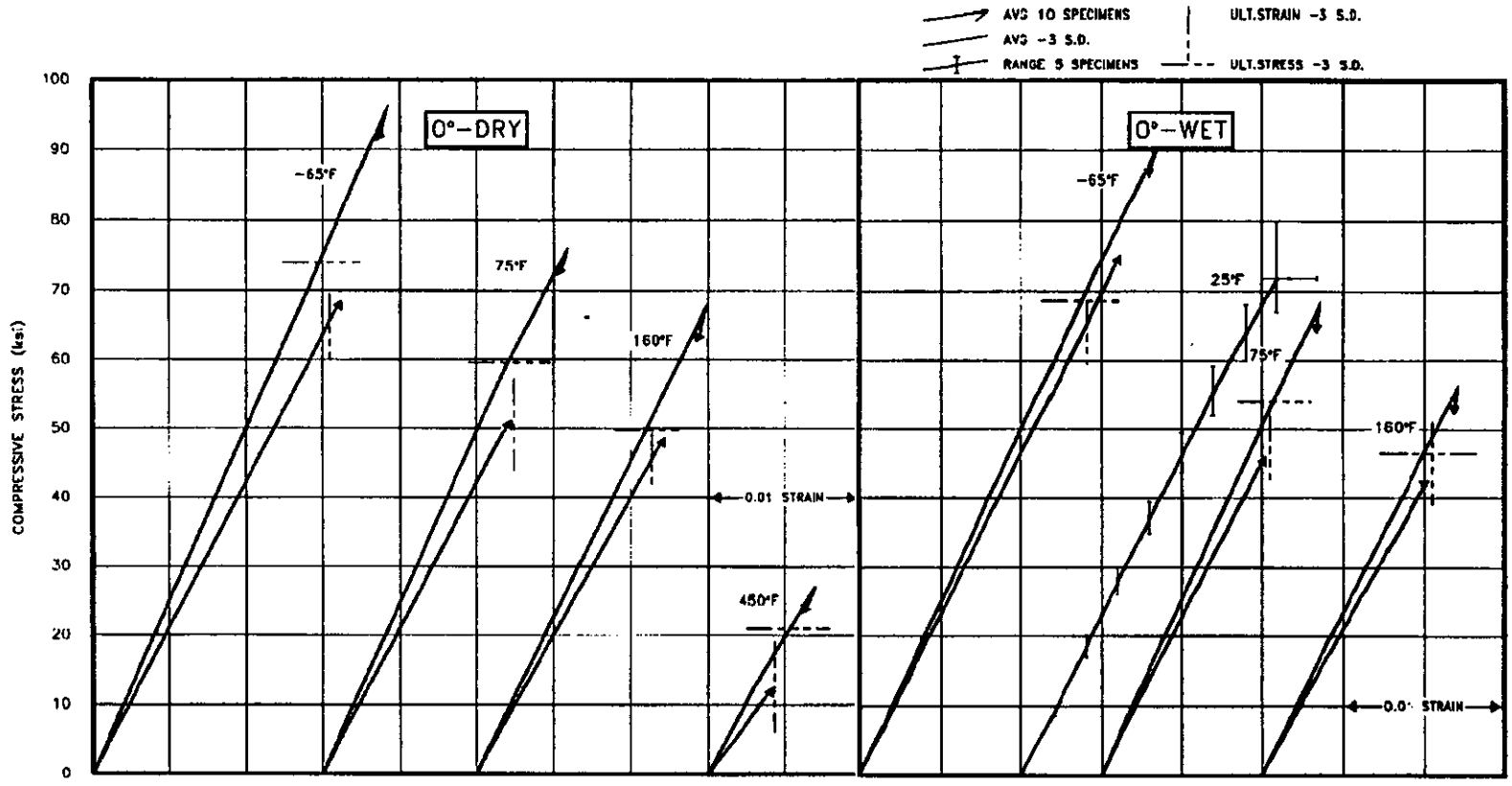
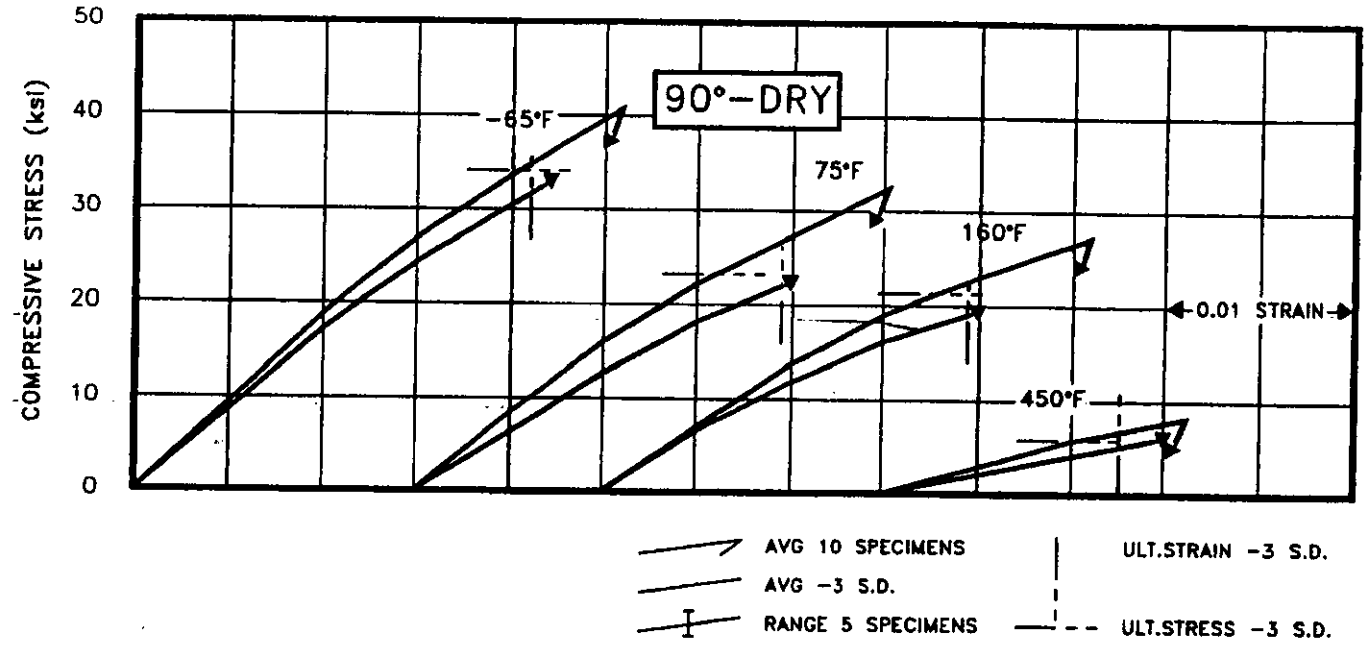


FIGURE A1.3.2(a) *Compressive stress-strain for F-161/7743 fiberglass epoxy loaded in the 0° direction.*

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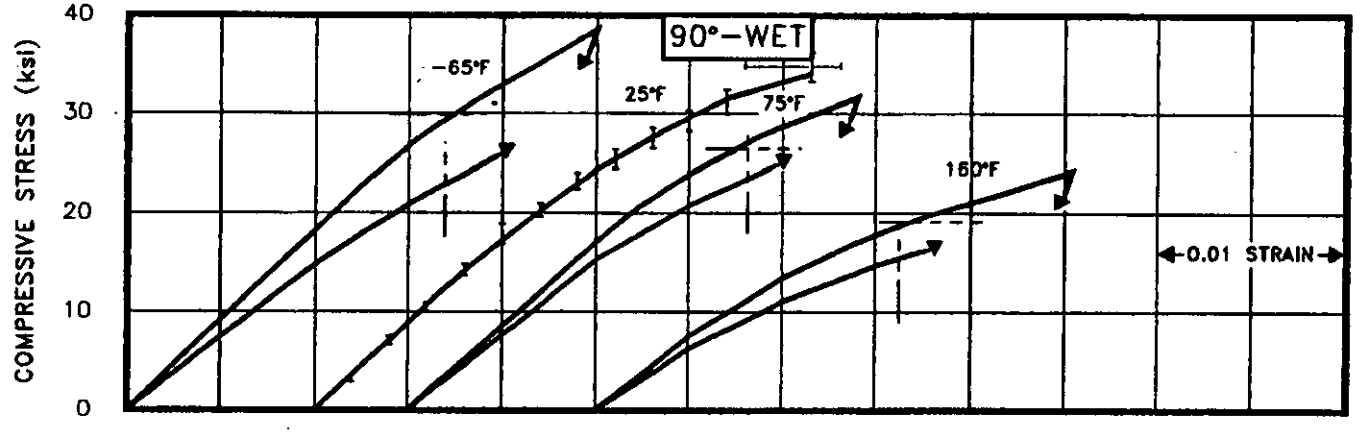


FIGURE A1.3.2(b) *Compressive stress-strain for F-161/7743 fiberglass epoxy loaded in the 90° direction.*

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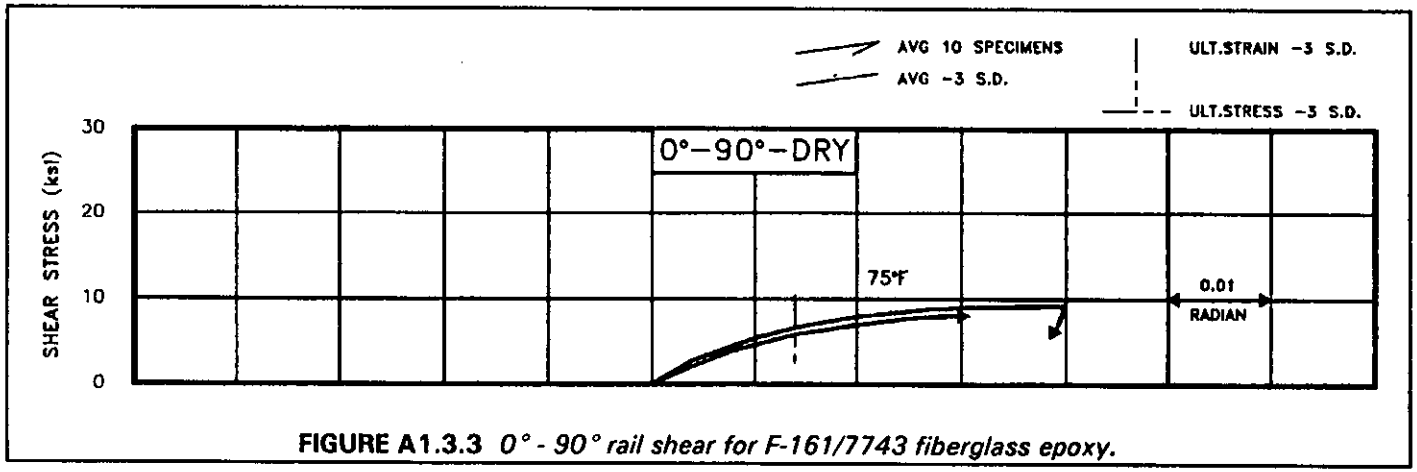


FIGURE A1.3.3 *0° - 90° rail shear for F-161/7743 fiberglass epoxy.*

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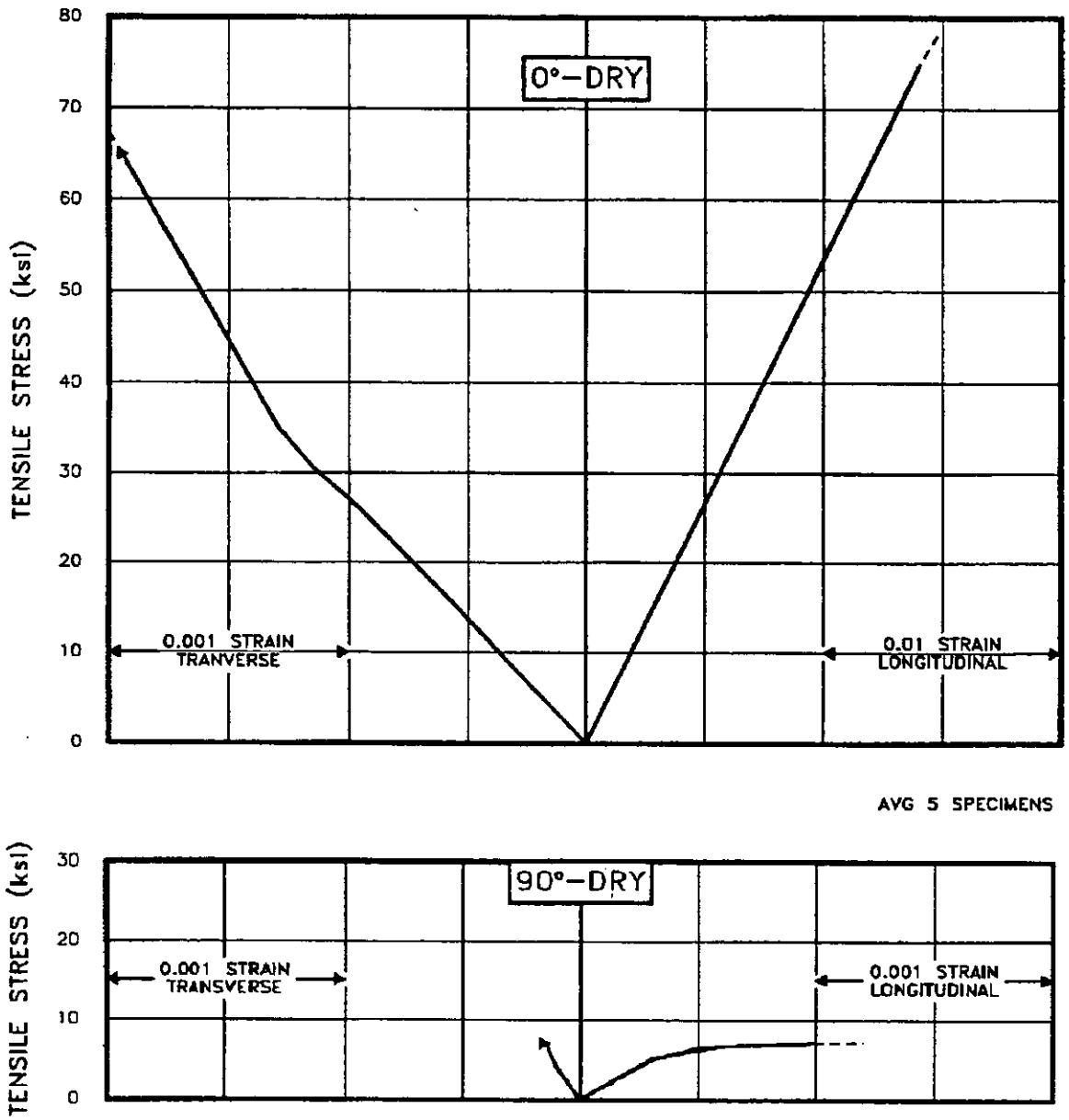


FIGURE A1.3.4 *Poisson effects for F-161/7743 fiberglass epoxy.*

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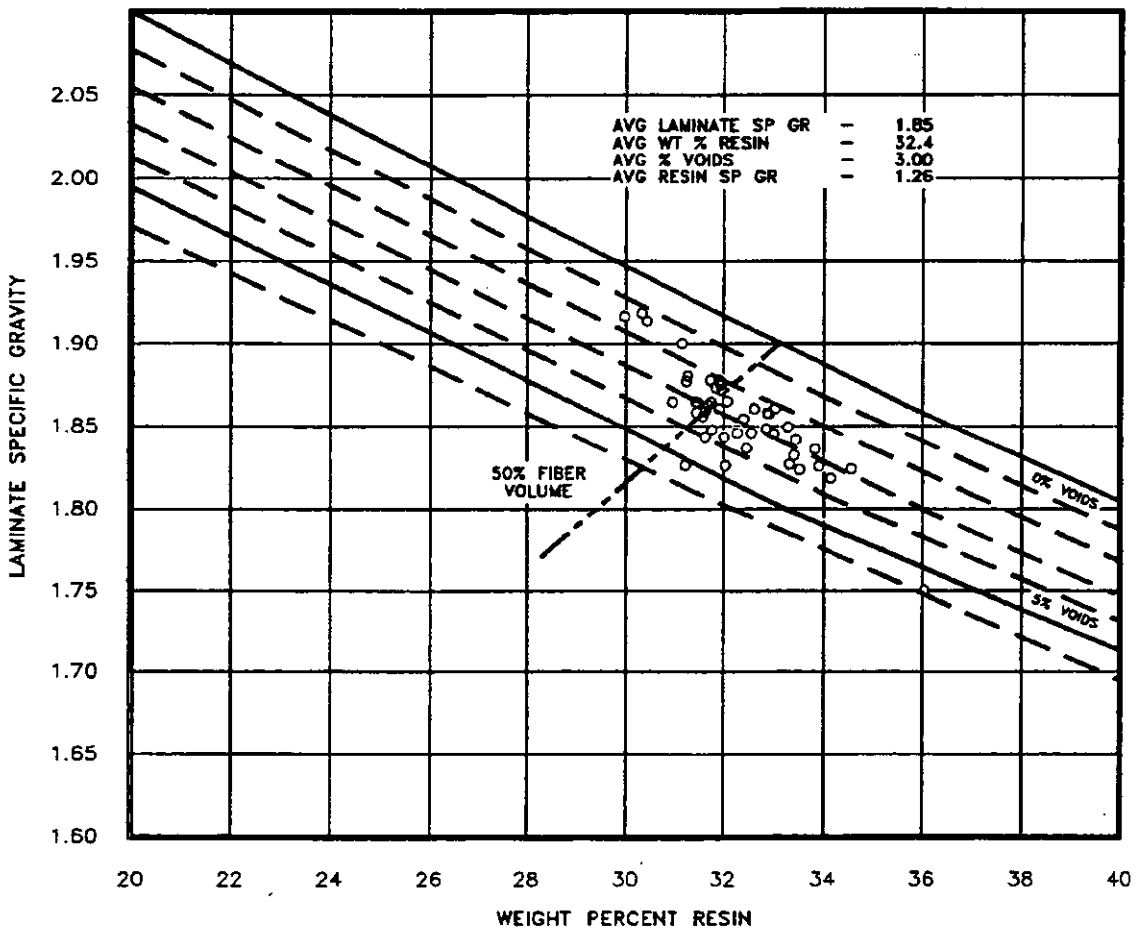


FIGURE A1.3.5 *Voids vs. resin content and specific gravity for F-161/7743 fiberglass epoxy.*

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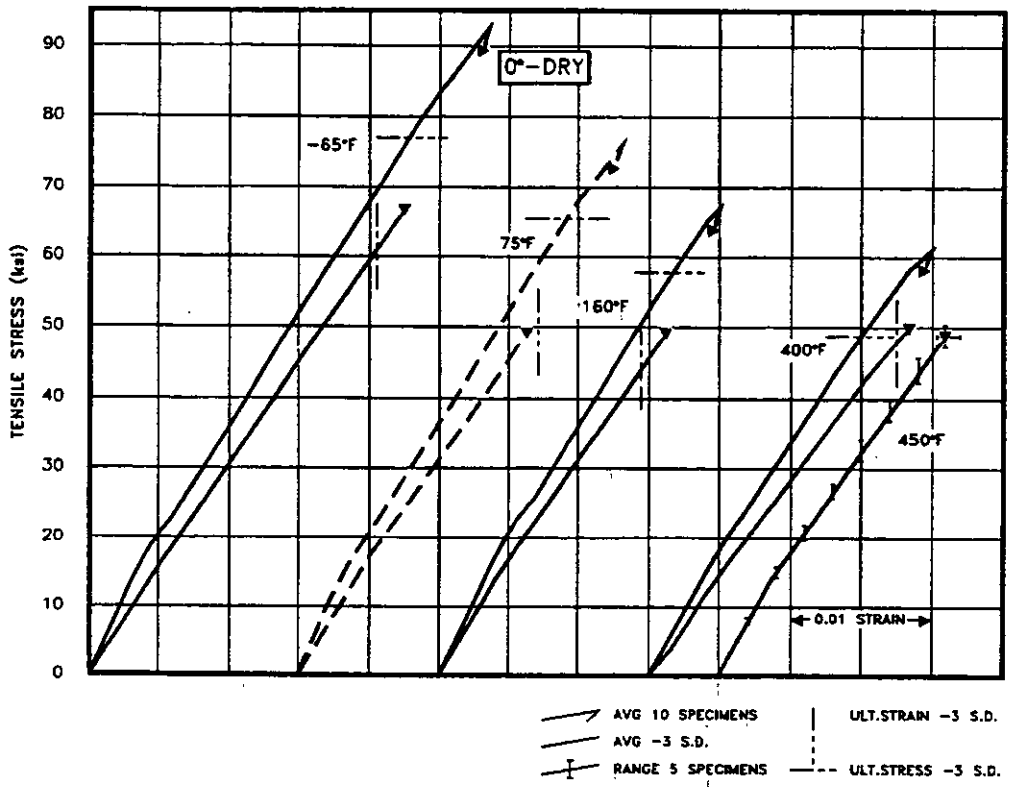
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TABLE A1.4 *Summary of Mechanical Properties of Hexcel F-161/7781 (ECDE-1/0-550) Fiberglass Epoxy (26% Resin)*

Fabrication		Lay-up: Balanced		Vacuum: None		Pressure: 55-65 psi		Bleedout: Vertical and Stepped Edge		Cure: 1 hr/350°F		Postcure: 2 hr/300°F 2.5 hr/400°F		Plies: 8 and 10	
Physical Properties		Weight Percent Resin: 26.0 v = 0.59				Avg. Specific Gravity: 2.01				Avg. Percent Voids: 0.5				Avg. Thickness: 0.008 inch/ply	
Test Methods		Tension: MIL-HDBK-17		Compression: MIL-HDBK-17		Shear: Picture Frame		Flexure: ASTM-D 790		Bearing:				Interlaminar Shear: ASTM-D 2345	
Temperature		-65°F				75°F				160°F				400°F	
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry	
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
Tension															
ultimate stress, ksi	0°	92.4	5.16	80.5	10.87			61.4	3.20	65.7	3.03	50.7	5.72	59.8	3.81
	90°	67.8	10.65	62.3	5.01			50.3	2.61	53.6	5.19	46.2	2.69	35.2	5.16
ultimate strain, %	0°	2.86	2.11	2.37	0.31			1.78	0.13	1.97	0.14	1.58	0.19	1.96	0.08
	90°	2.42	3.14	1.97	0.24			1.65	0.08	1.88	0.12	1.55	0.10	1.38	0.13
proportional limit, ksi	0°														
	90°														
initial modulus, 10^6 psi	0°	4.42		4.49				4.10		3.92		3.72		3.27	
	90°	4.22		4.21				3.76		3.17		3.38		2.86	
secondary modulus, 10^6 psi	0°	3.32		3.14				3.06		3.24		3.07		2.94	
	90°	2.70		2.74				2.62		2.72		2.55		2.46	
Compression															
ultimate stress, ksi	0°	73.2	6.83	74.0	5.02			57.3	4.0	48.9	3.50	44.7	3.25	28.8	3.03
	90°	64.2	3.19	55.8	4.40			37.5	2.28	42.0	2.64	40.1	1.90	18.9	0.69
ultimate strain, %	0°	1.70	0.42	1.65	0.28			1.09	0.17	1.12	0.15	0.84	0.14	0.79	0.03
	90°	1.40	0.14	1.42	0.27			1.26	0.41	1.14	0.23	1.22	0.18	0.71	0.27
proportional limit, ksi	0°	39.0		46.0				42.0		41.0		24.0		15.0	
	90°	28.0		41.0				24.0		36.0		21.0		11.0	
initial modulus, 10^6 psi	0°	4.42		4.47				4.27		4.05		3.94		3.73	
	90°	4.02		4.19				4.12		3.68		3.40		3.07	
Shear															
ultimate stress, ksi	0°-90° ±45°	20.1	2.3					16.0	1.64	13.4	1.28				

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°				94.10	96.86	89.64			
proportional limit, ksi	0°									
initial modulus, 10^6 psi	0°									
Bearing										
ultimate stress, ksi	0°									
stress at 4% elong., ksi	0°									
Interlaminar Shear										
ultimate stress, ksi	0°				5.56	5.65	5.50			

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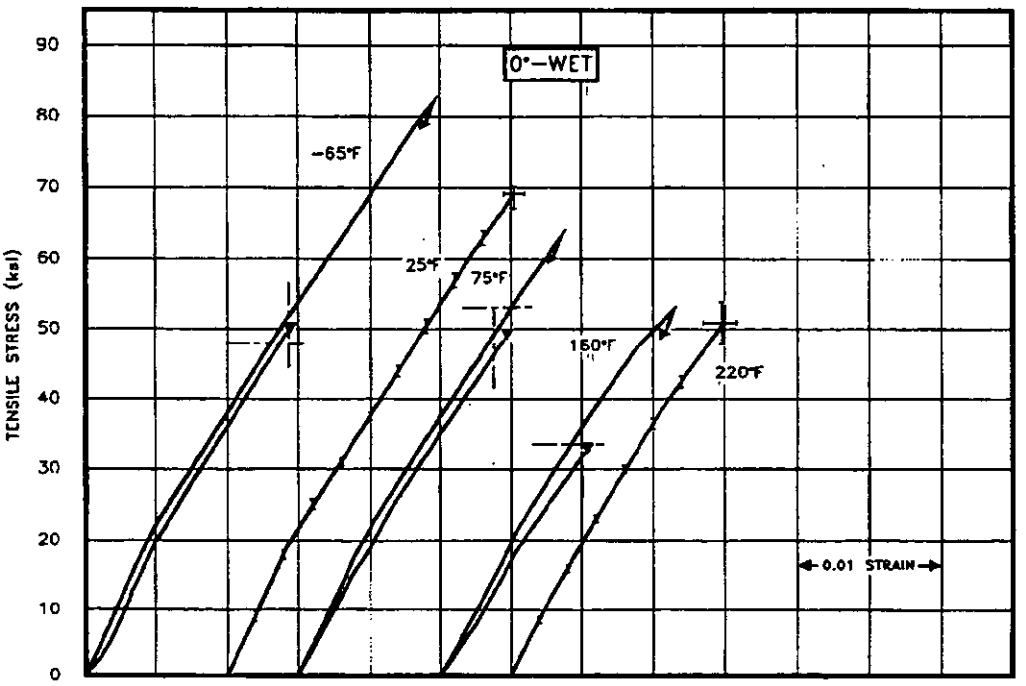


FIGURE A1.4.1(a) *Tensile stress-strain for F-161/7781 fiberglass epoxy loaded in the 0° direction (26% resin).*

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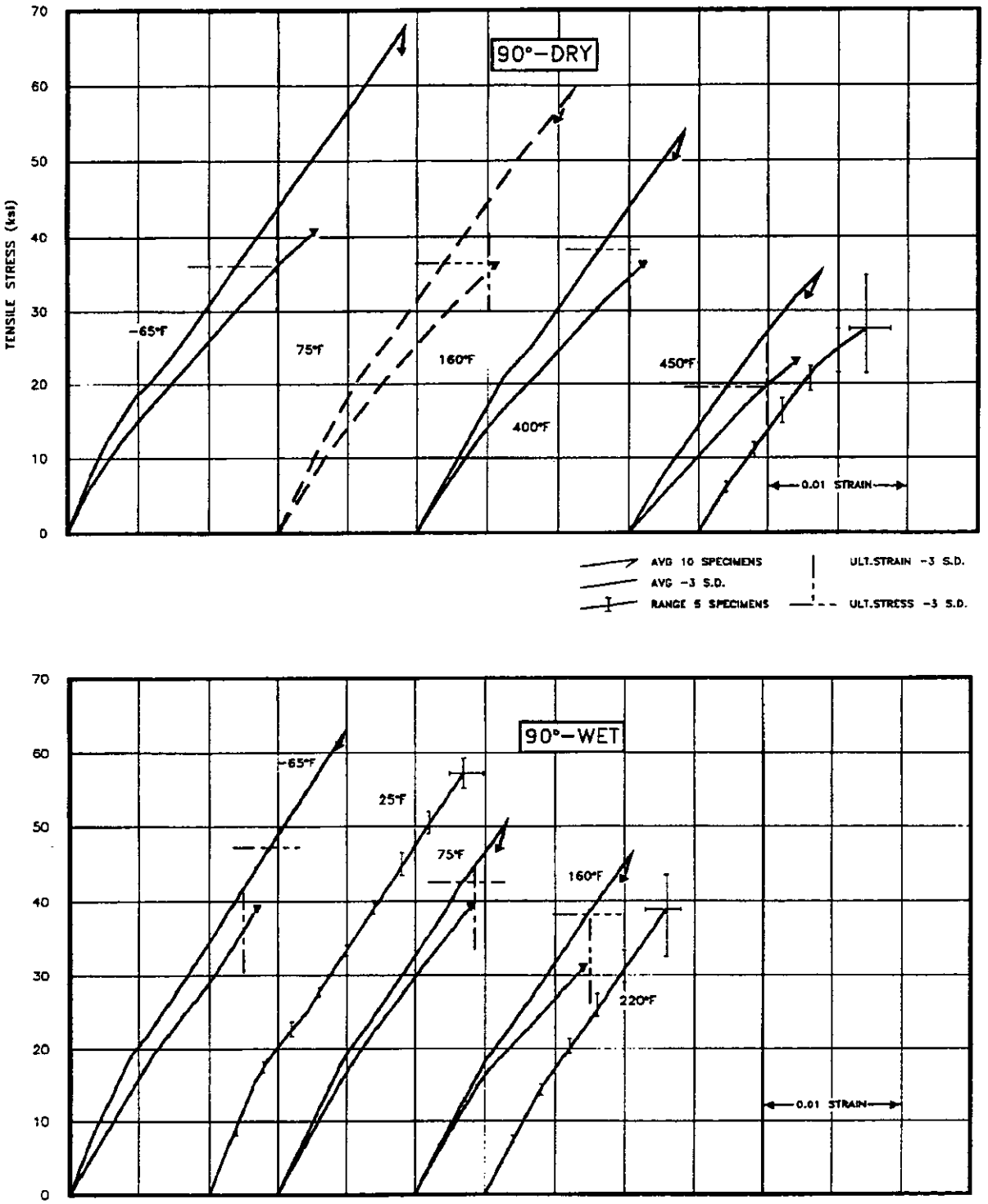
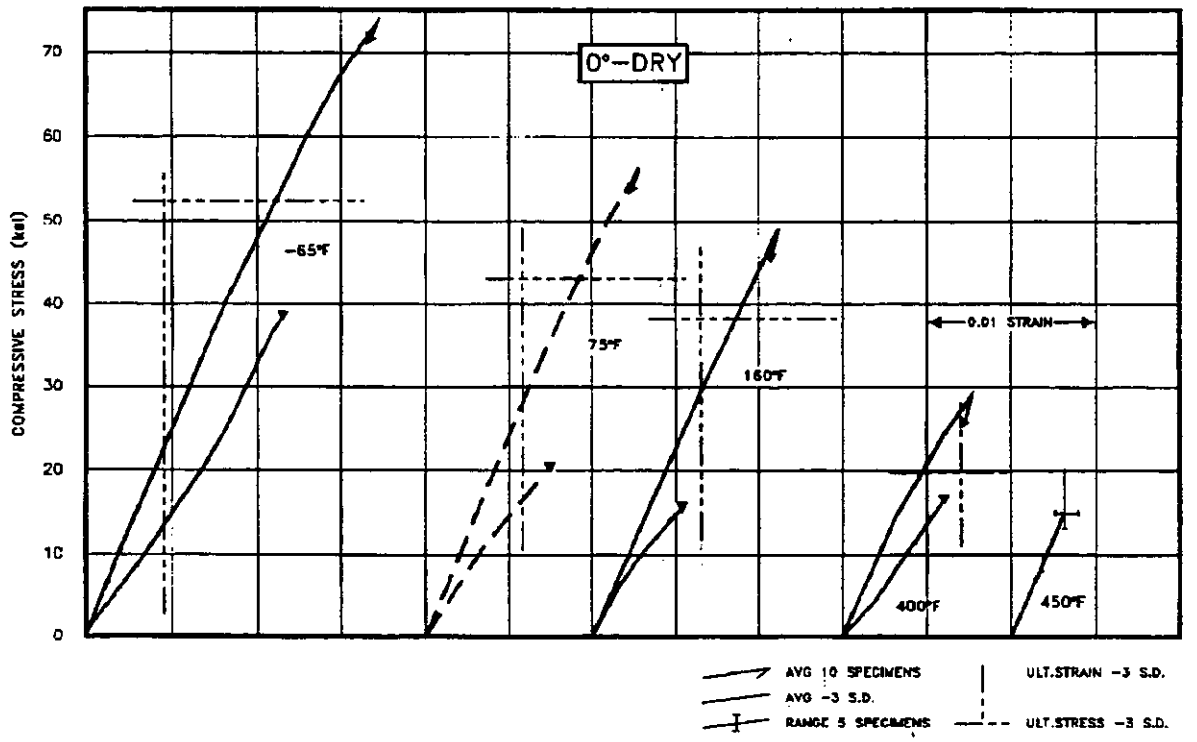


FIGURE A1.4.1(b) *Tensile stress-strain for F-161/7781 fiberglass epoxy loaded in the 90° direction (26% resin).*

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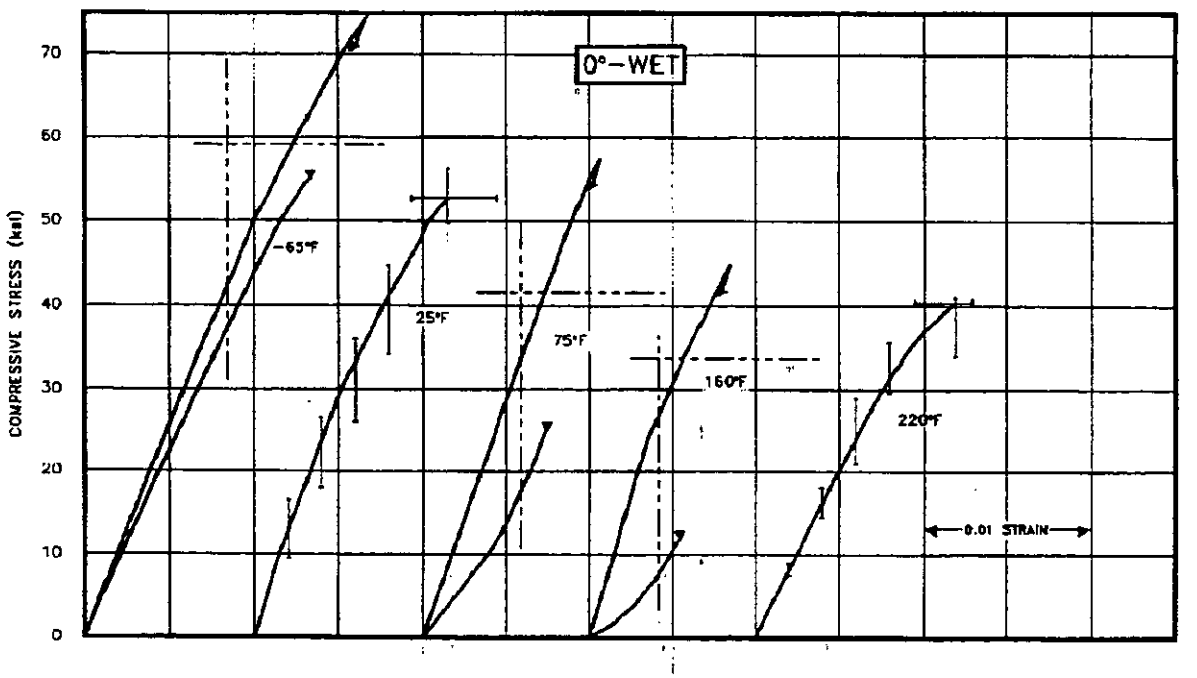


FIGURE A1.4.2(a) *Compressive stress-strain for F-161/7781 fiberglass epoxy loaded in the 0° direction (26% resin).*

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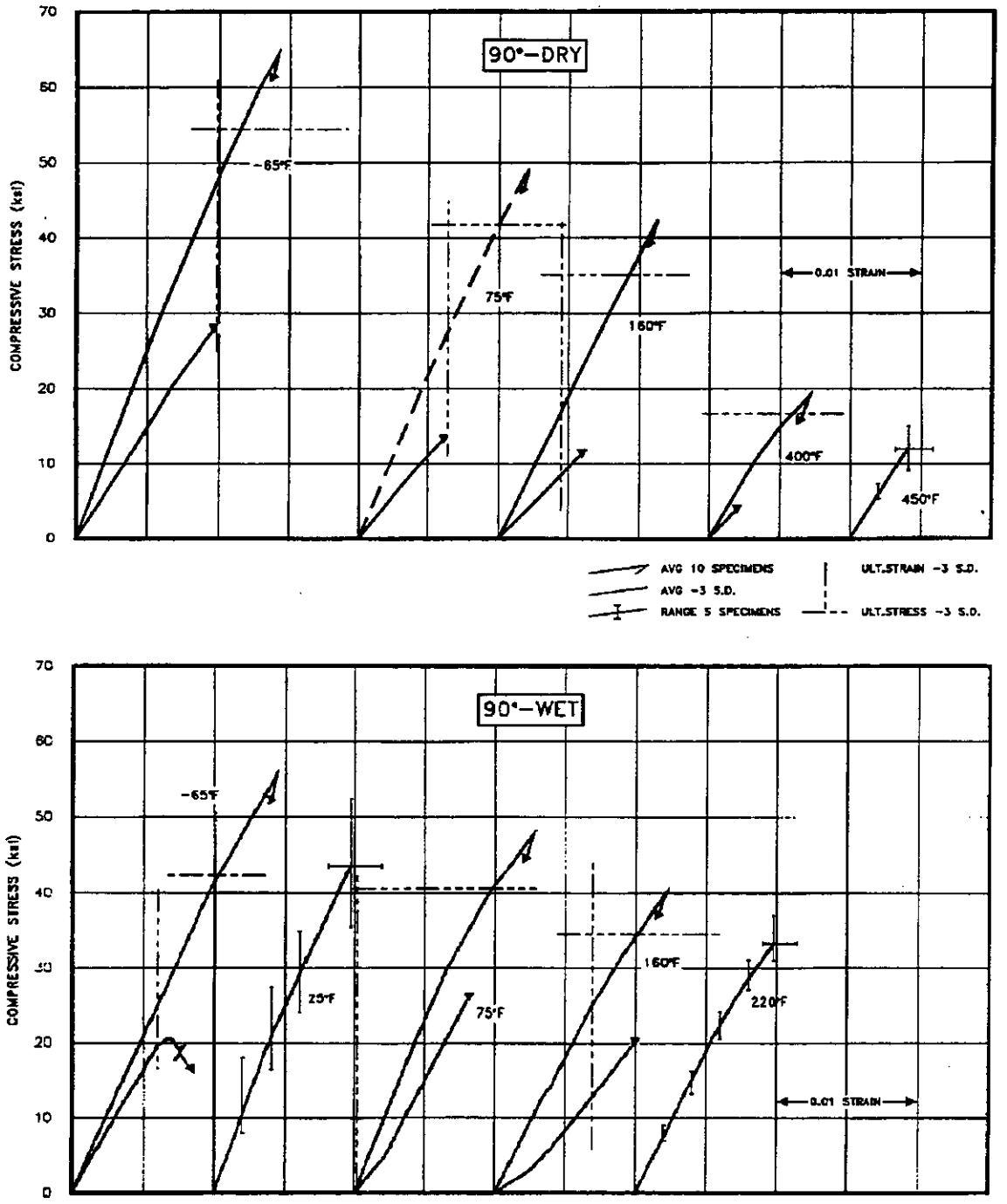
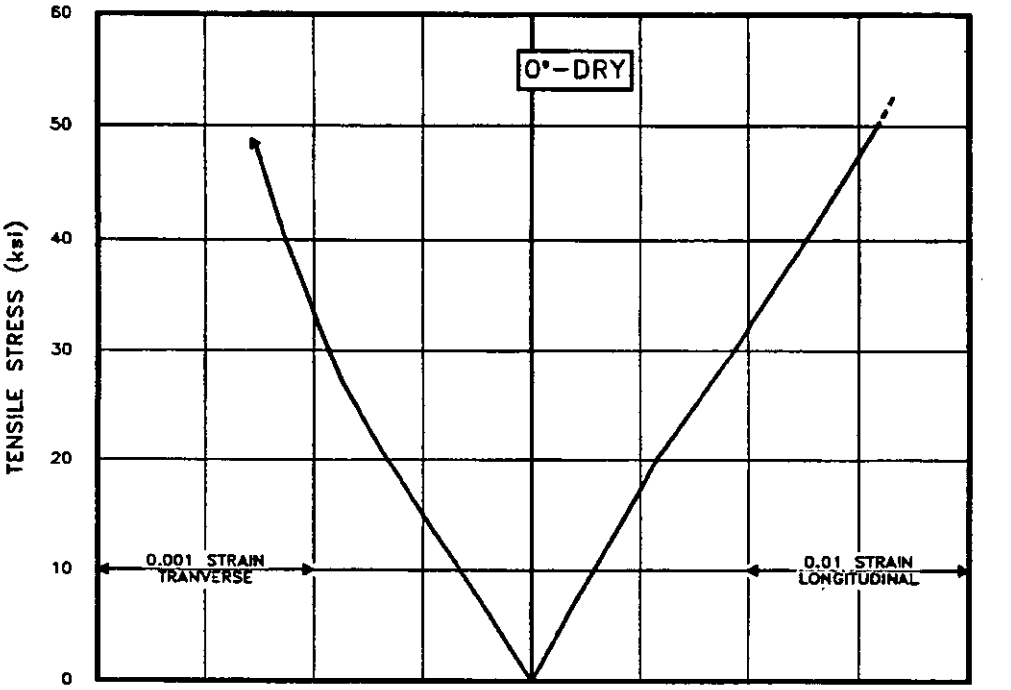


FIGURE A1.4.2(b) *Compressive stress-strain for F-161/7781 fiberglass epoxy loaded in the 90° direction (26% resin).*

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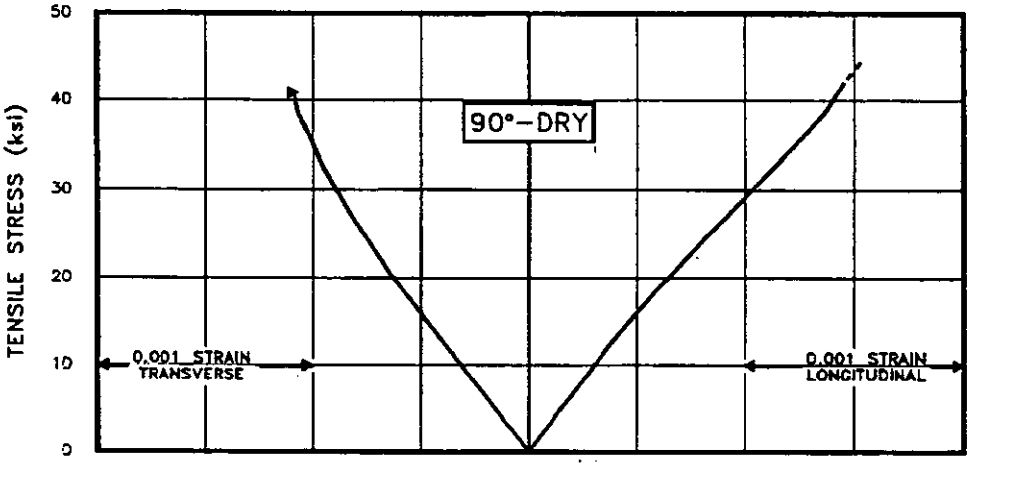


FIGURE A1.4.4 *Poisson effects for F-161/7781 fiberglass epoxy (26% resin).*

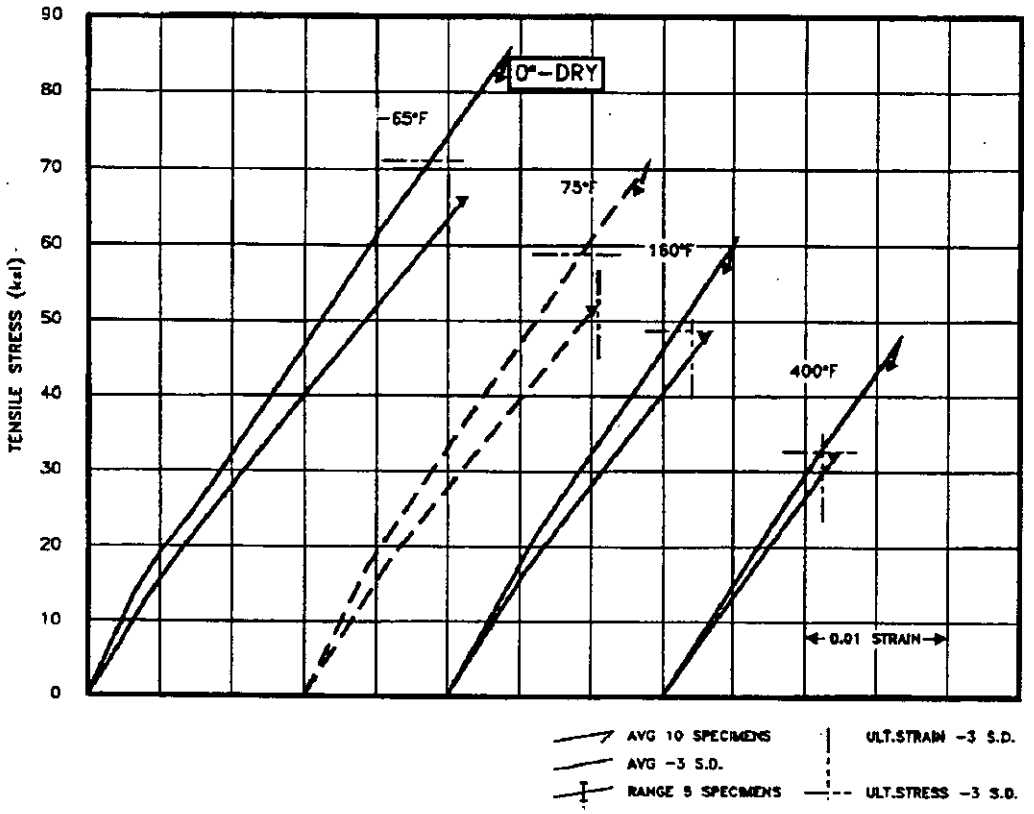
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TABLE A1.5 *Summary of Mechanical Properties of Hexcel F-161/7781 (ECDE-1/0-550) Fiberglass Epoxy (31% Resin)*

Fabrication		Lay-up: Balanced		Vacuum: None		Pressure: 55-65 psi		Bleedout: Vertical and Stepped Edge		Cure: 1 hr/350°F		Postcure: 2 hr/300°F 2.5 hr/400°F		Plies: 8 and 10	
Physical Properties		Weight Percent Resin: 31.0		Avg. Specific Gravity: 1.92		Avg. Percent Voids: 0.6		Avg. Thickness: 0.009 inch/ply							
Test Methods		Tension: MIL-HDBK-17		Compression: MIL-HDBK-17		Shear: Picture Frame		Flexure: ASTM-D 790		Bearing:		Interlaminar Shear:			
Temperature		-65°F				75°F				160°F				400°F	
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry	
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
Tension															
ultimate stress, ksi	0°	85.2	4.68	82.3	4.97			64.0	2.04	60.1	3.75	51.4	4.23	47.3	4.87
	90°	70.0	5.24	67.9	2.98			53.5	2.91	49.3	0.95	39.8	3.50	31.0	1.95
ultimate strain, %	0°	2.93	0.14	2.53	0.18			2.10	0.06	2.02	0.10	1.66	0.17	1.66	0.18
	90°	2.50	0.21	2.41	0.22			1.90	0.11	1.86	0.06	1.47	0.09	1.25	0.09
proportional limit, ksi	0°														
	90°														
initial modulus, 10^6 psi	0°	4.22		4.30				3.84		3.69	3.72	3.65		3.09	
	90°	3.97		4.15				3.68		3.37	3.34	3.30		2.75	
secondary modulus, 10^6 psi	0°	3.13		3.01				3.03		2.97	0.04	2.88		2.94	
	90°	2.62		2.96				2.62		2.55	0.25	2.46		2.47	
Compression															
ultimate stress, ksi	0°	73.1	5.18	66.0	10.75			54.4	7.04	50.6		45.9	5.39	32.8	6.04
	90°	58.4	3.17	57.5	11.56			47.3	4.73	42.2		38.7	4.19	25.8	8.27
ultimate strain, %	0°	1.86	0.21	1.72	0.32			1.33	0.28	1.52		1.04	0.23	0.95	0.24
	90°	1.61	0.29	1.44	0.36			1.10	0.21	1.30		0.99	0.22	0.87	0.28
proportional limit, ksi	0°	44.0		38.0				33.0		32.0		25.0		16.0	
	90°	33.0		33.0				30.0		--		21.0		15.0	
initial modulus, 10^6 psi	0°	3.90		4.04				4.03		3.42		4.06		3.50	
	90°	3.56		3.84				3.96		3.23		4.01		3.07	
Shear															
ultimate stress, ksi	0°-90° ±45°	20.5	2.23					15.9	0.72	13.7	0.82				

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°				90.23	93.74	87.29			
proportional limit, ksi	0°									
initial modulus, 10^6 psi	0°									
Bearing										
ultimate stress, ksi	0°									
stress at 4% elong., ksi	0°									
Interlaminar Shear										
ultimate stress, ksi	0°				5.56	5.65	5.50			

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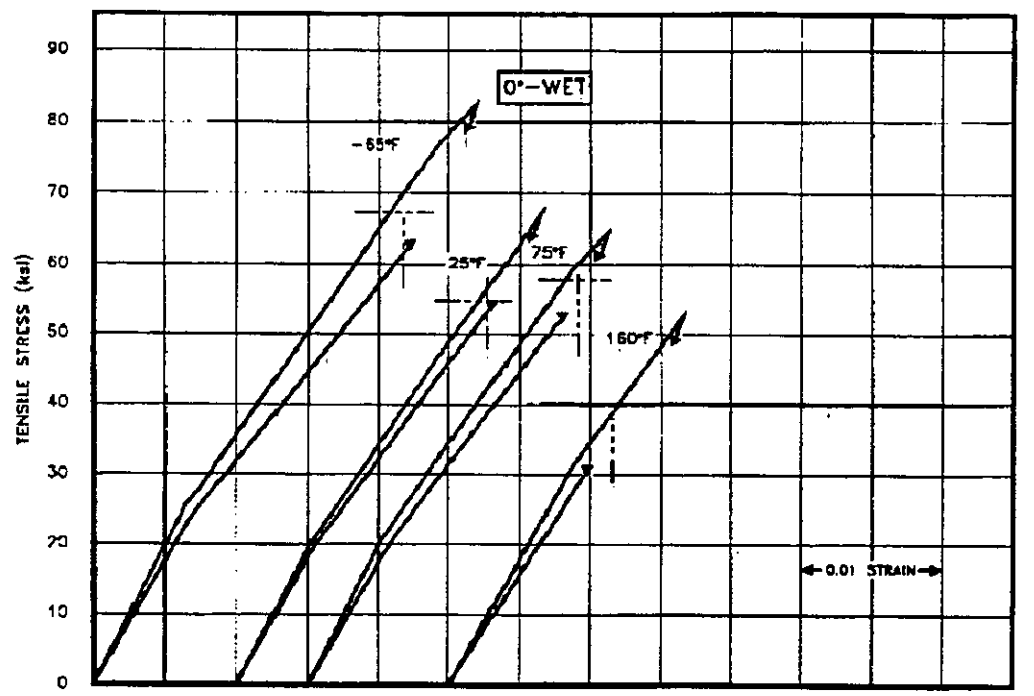
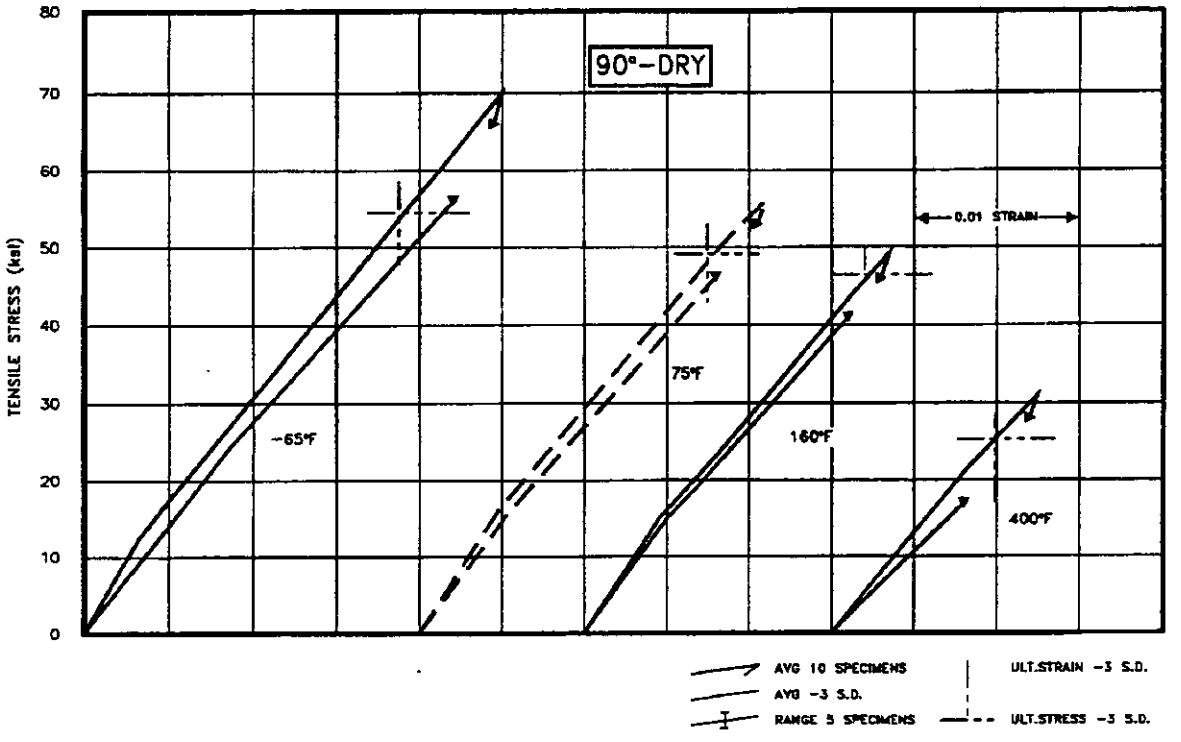


FIGURE A1.5.1(a) *Tensile stress-strain for F-161/7781 fiberglass epoxy loaded in the 0° direction (31% resin).*

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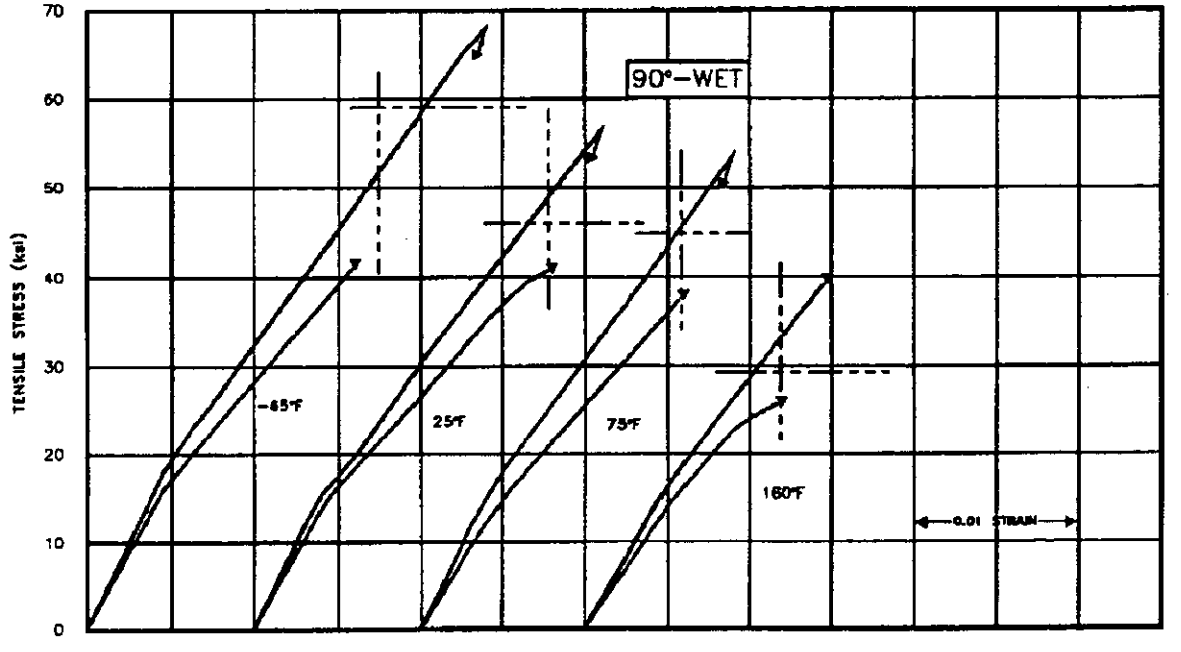
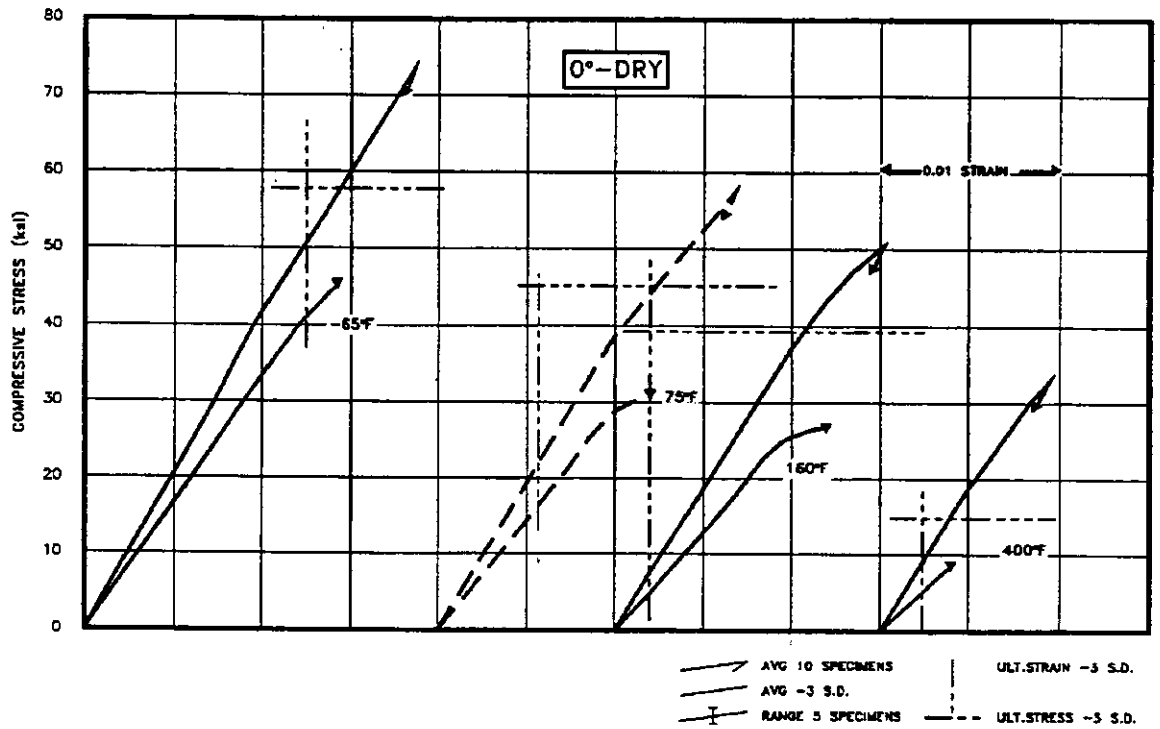


FIGURE A1.5.1(b) *Tensile stress-strain for F-161/7781 fiberglass epoxy loaded in the 90° direction (31% resin).*

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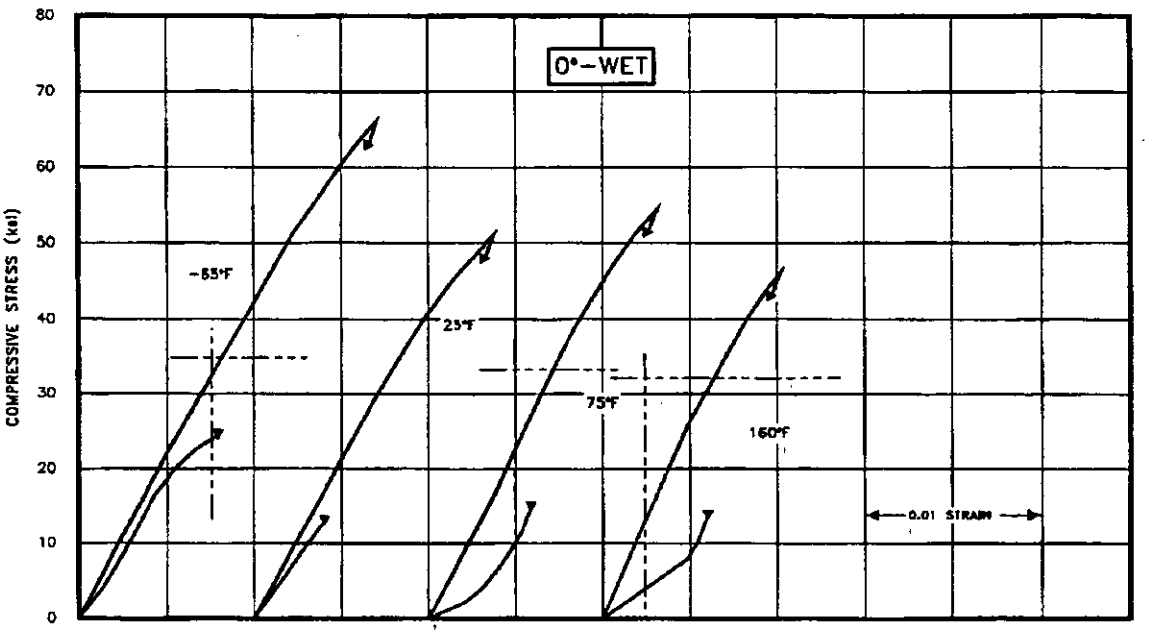


FIGURE A1.5.2(a) *Compressive stress-strain for F-161/7781 fiberglass epoxy loaded in the 0° direction (31% resin).*

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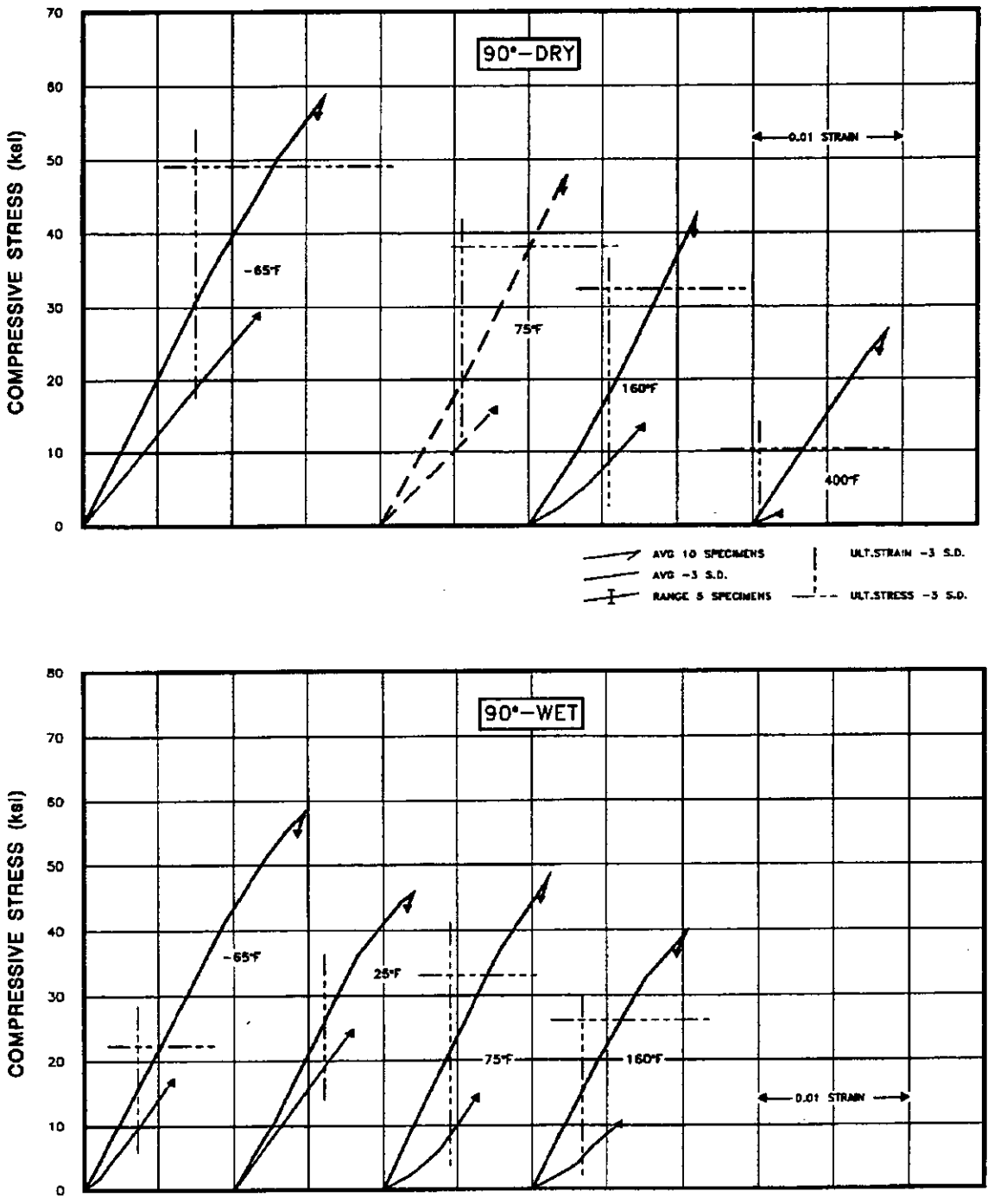
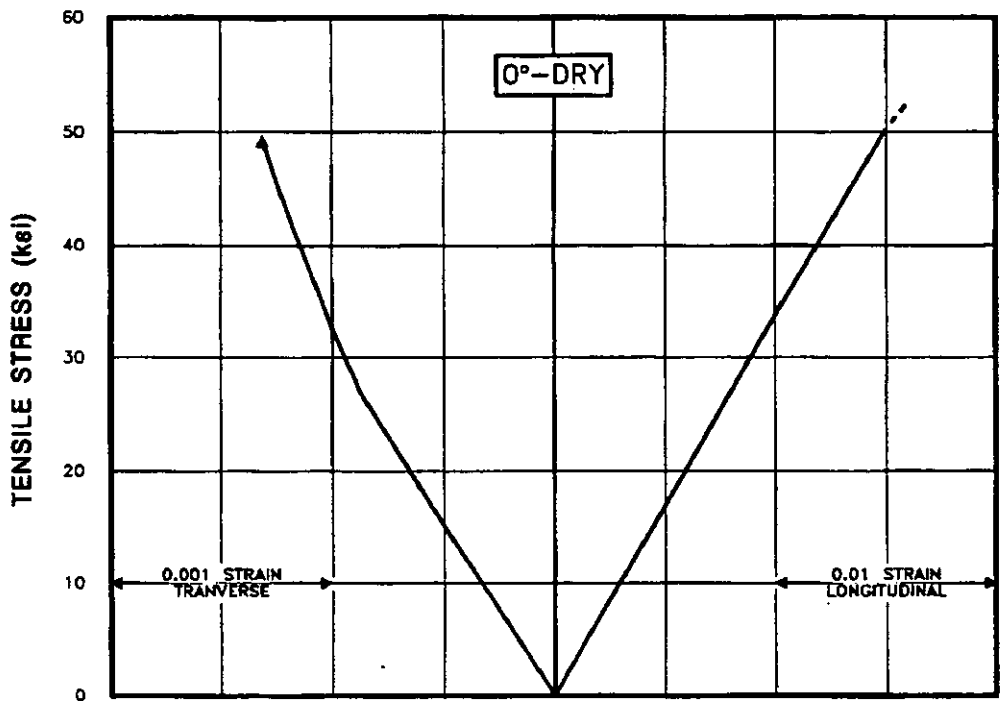


FIGURE A1.5.2(b) *Compressive stress-strain for F-161/7781 fiberglass epoxy loaded in the 90° direction (31% resin).*

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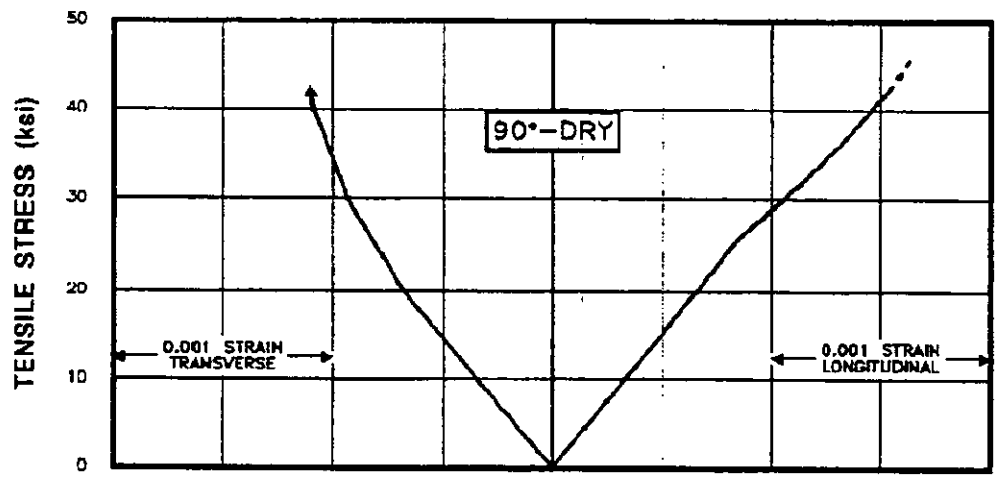


FIGURE A1.5.4 *Poisson effects for F-161/7781 fiberglass epoxy (31% resin).*

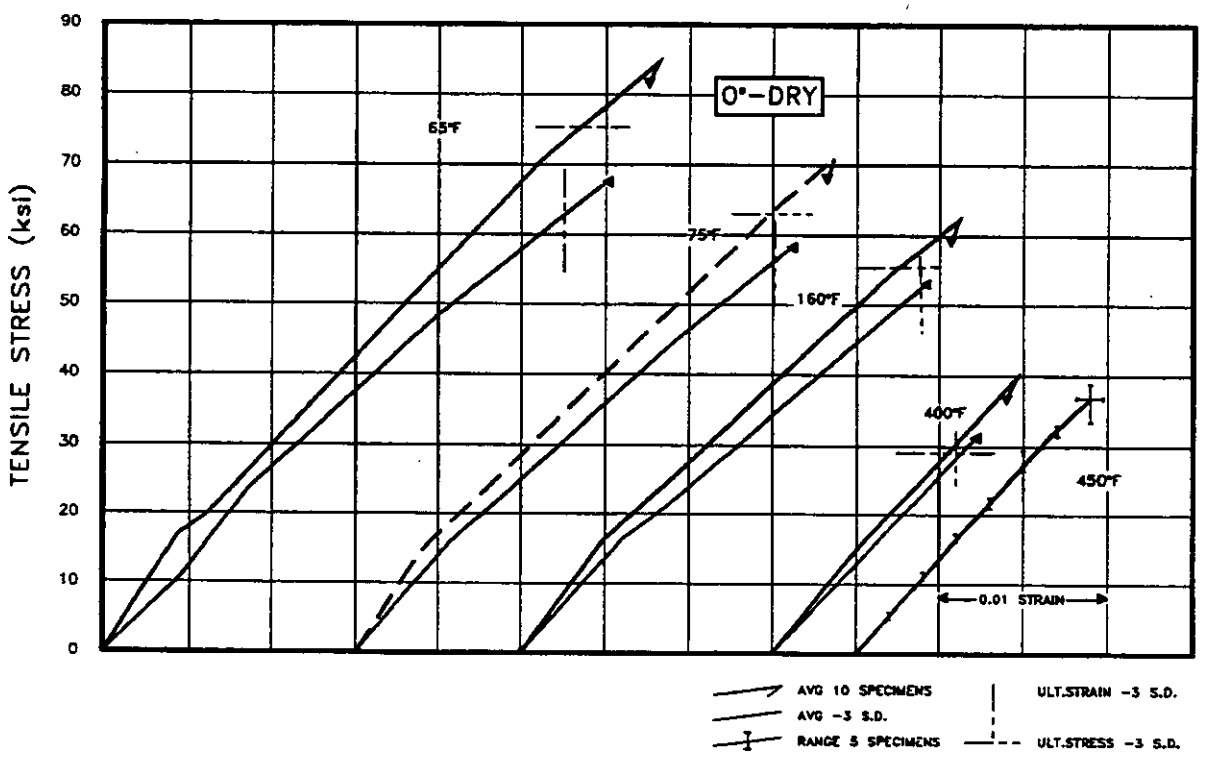
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TABLE A1.6 *Summary of Mechanical Properties of Hexcel F-161/7781 (ECDE-1/0-550) Fiberglass Epoxy (36% Resin)*

Fabrication		Lay-up: Balanced		Vacuum: None		Pressure: 55-65 psi		Bleedout: Vertical and Stepped Edge		Cure: 1 hr/350°F		Postcure: 2 hr/300°F 2.5 hr/400°F		Plies: 8	
Physical Properties		Weight Percent Resin: 35.6		Avg. Specific Gravity: 1.86				Avg. Percent Voids: 0.9				Avg. Thickness: 0.010 inch/ply			
Test Methods		Tension: MIL-HDBK-17		Compression: MIL-HDBK-17		Shear: Picture Frame				Flexure: ASTM-D 790		Bearing:		Interlaminar Shear:	
Temperature		-65°F				75°F				160°F				400°F	
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry	
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
Tension															
ultimate stress, ksi	0°	83.9	2.85	73.0	2.89			55.5	2.57	61.9	2.24	45.0	1.85	39.2	3.40
	90°	68.7	4.19	63.9	1.61			48.9	2.67	51.9	3.25	37.6	0.99	32.0	1.44
ultimate strain, %	0°	3.30	0.18	2.79	0.02			2.12	0.14	2.61	0.08	1.59	0.07	1.45	0.13
	90°	2.80	0.18	2.41	0.05			1.95	0.09	2.18	0.19	1.50	0.05	1.35	0.08
proportional limit, ksi	0°														
	90°														
initial modulus, 10^6 psi	0°	3.84		3.81				3.58		3.25		3.35		2.96	
	90°	3.67		3.81				3.30		3.13		3.18		2.51	
secondary modulus, 10^6 psi	0°	2.81		2.75				3.04		2.49		3.04		2.74	
	90°	2.65		2.67				2.72		2.39		2.70		2.22	
Compression															
ultimate stress, ksi	0°	76.2	5.88	68.8	4.36			55.1	2.63	54.7	5.49	46.0	5.66	31.0	8.08
	90°	56.0	4.56	52.9	6.32			47.0	6.78	36.9	1.47	35.3	3.30	23.2	3.26
ultimate strain, %	0°	2.13	0.28	1.64	0.23			1.36	0.32	1.90	0.56	1.32	2.41	1.02	0.23
	90°	1.75	0.48	1.58	0.57			2.00	0.89	1.29	0.09	1.27	2.40	0.91	0.14
proportional limit, ksi	0°	28.0		24.0				24.0		32.0		22.0		17.0	
	90°	18.0		17.0				16.0		28.0		17.0		12.0	
initial modulus, 10^6 psi	0°	4.10		4.50				3.87		3.45		3.36		2.87	
	90°	4.00		4.10				3.64		2.87		2.88		2.63	
Shear															
ultimate stress, ksi	0°-90° ±45°	19.6	1.04					15.0	0.70	12.7	0.62				

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°				86.31	92.16	79.07			
proportional limit, ksi	0°									
initial modulus, 10^6 psi	0°									
Bearing										
ultimate stress, ksi	0°									
stress at 4% elong., ksi	0°									
Interlaminar Shear										
ultimate stress, ksi	0°				5.56	5.65	5.50			

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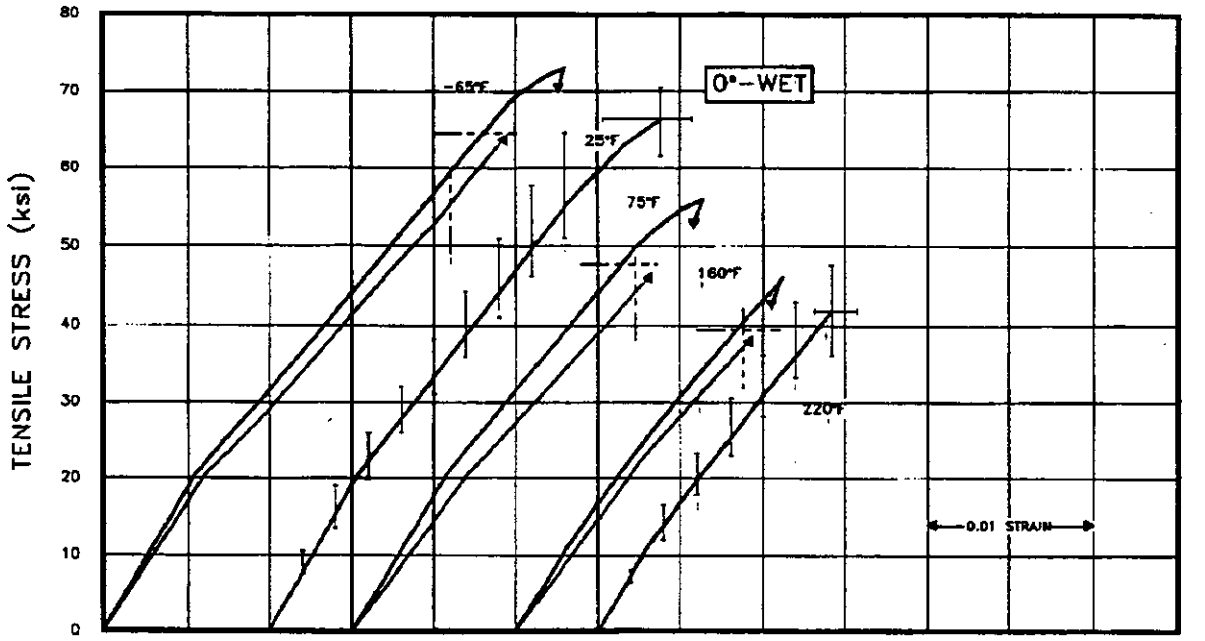


FIGURE A1.6.1(a) *Tensile stress-strain for F-161/7781 fiberglass epoxy loaded in the 0° direction (36% resin).*

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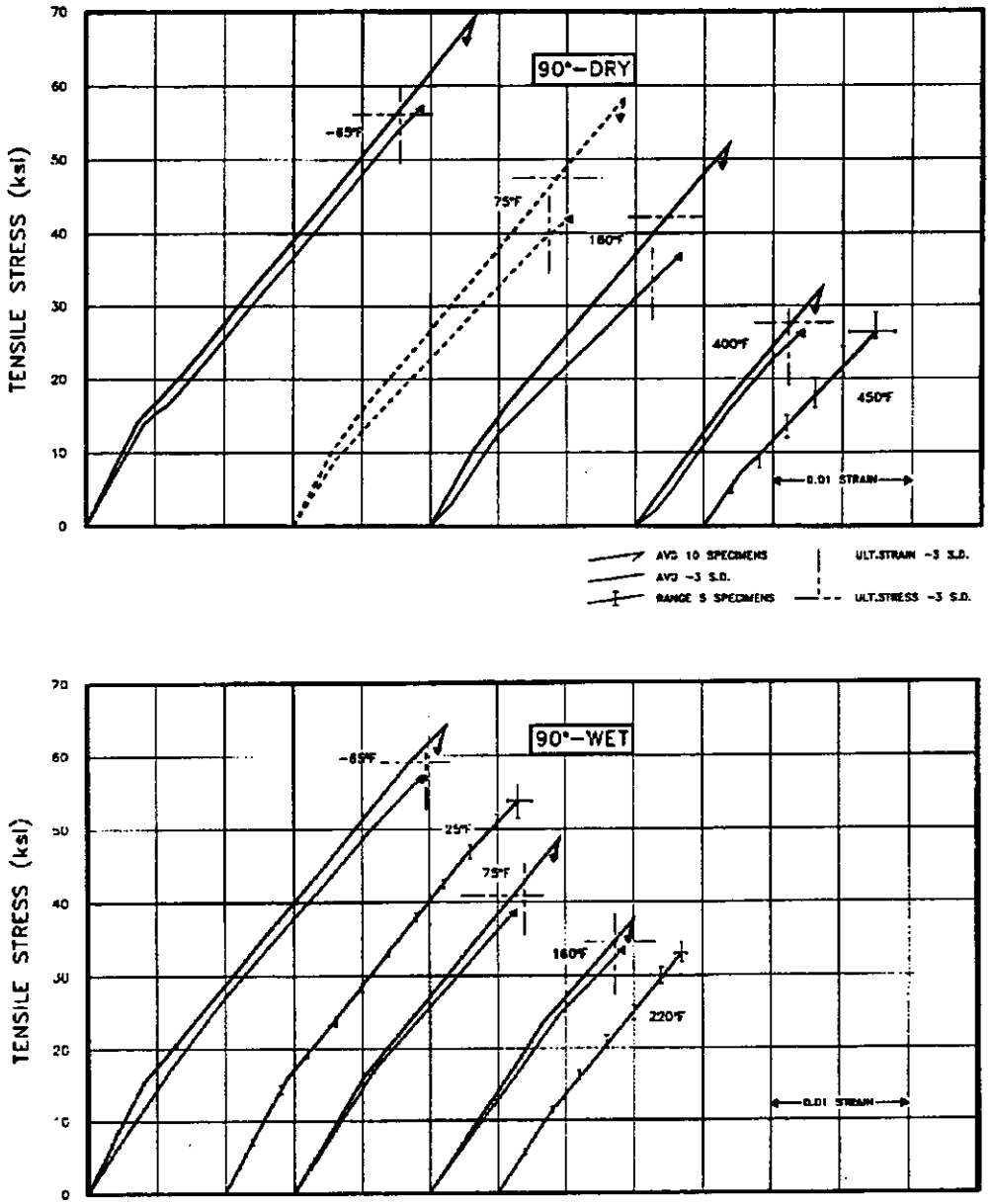
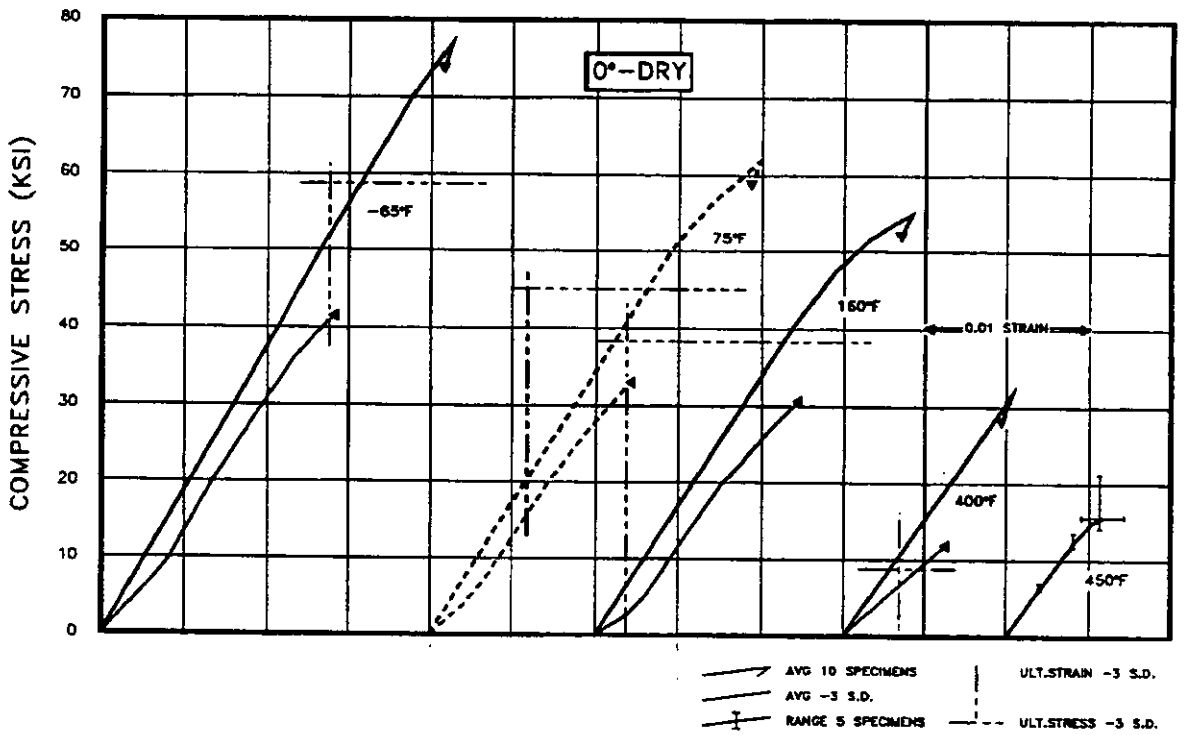


FIGURE A1.6.1(b) *Tensile stress-strain for F-161/7781 fiberglass epoxy loaded in the 90° direction (36% resin).*

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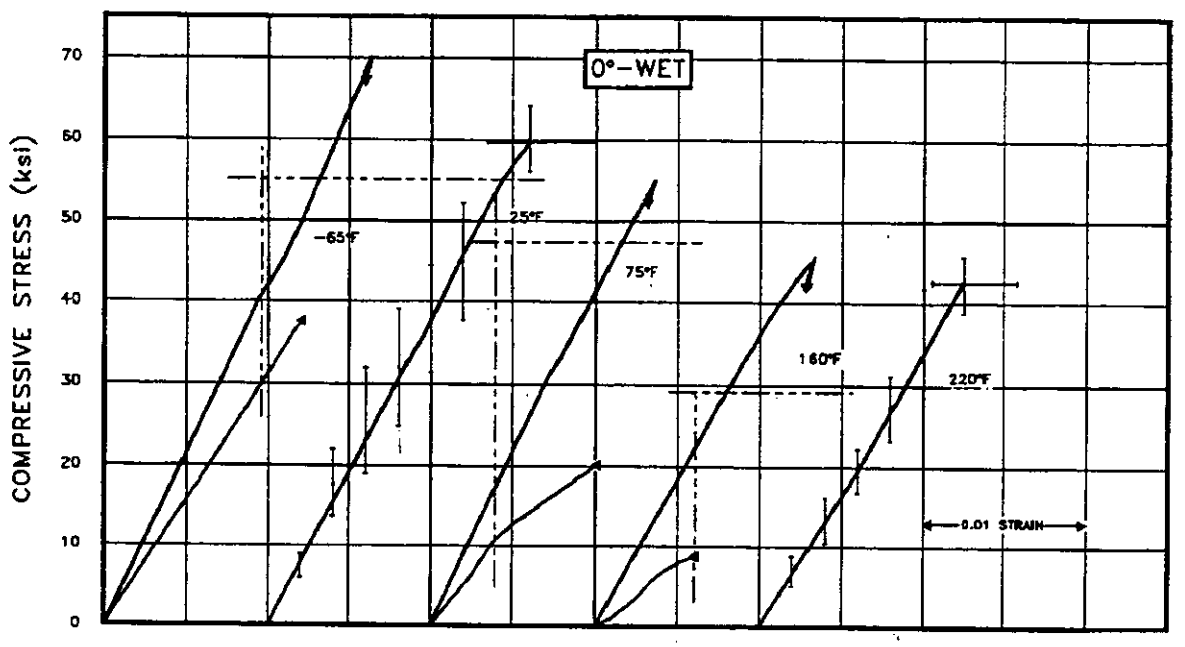


FIGURE A1.6.2(a) *Compressive stress-strain for F-161/7781 fiberglass epoxy loaded in the 0° direction (36% resin).*

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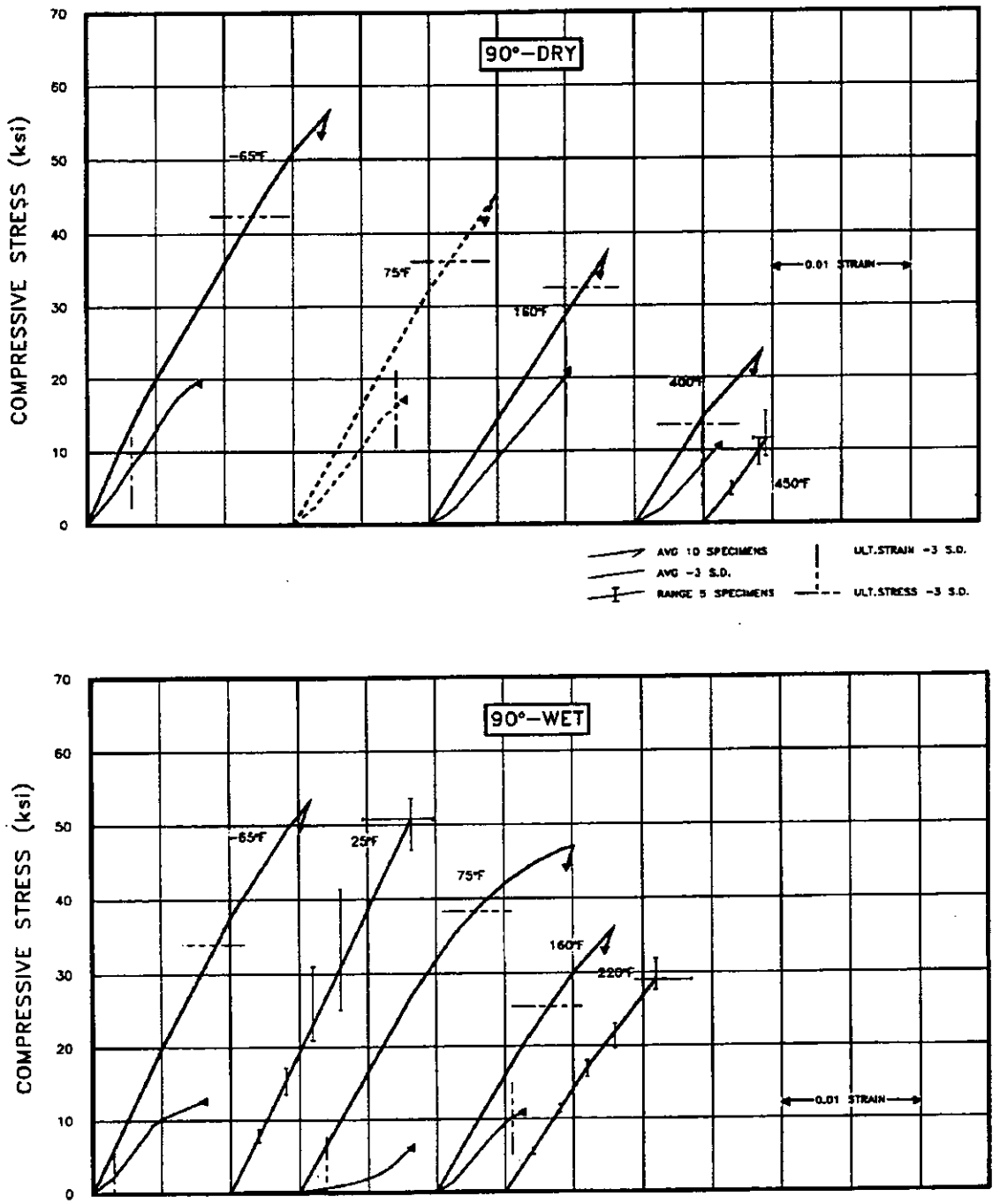


FIGURE A1.6.2(b) *Compressive stress-strain for F-161/7781 fiberglass epoxy loaded in the 90° direction (36% resin).*

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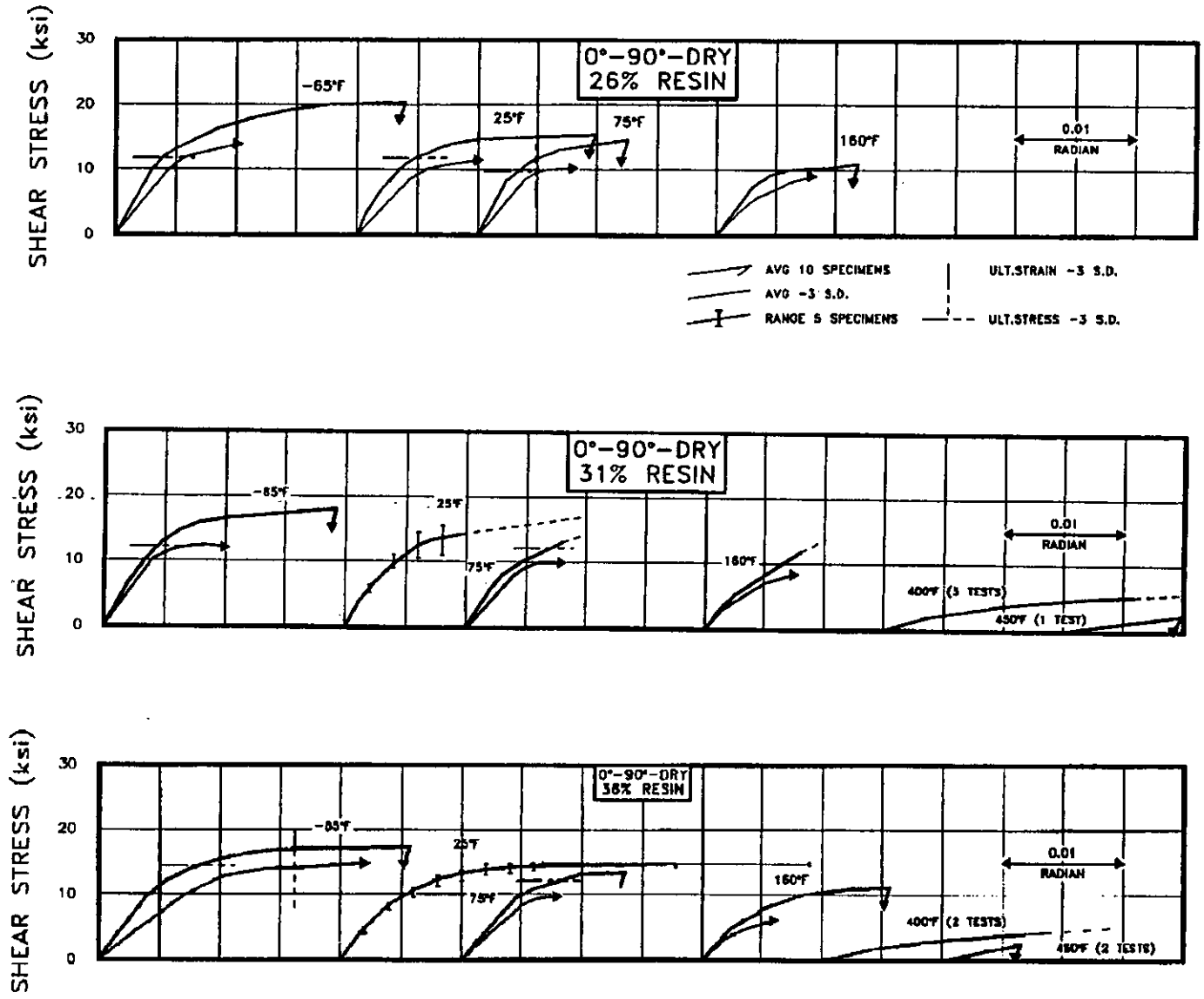


FIGURE A1.6.3 *Picture frame shear for F-161/7781 fiberglass epoxy (26%, 31%, 36% resin).*

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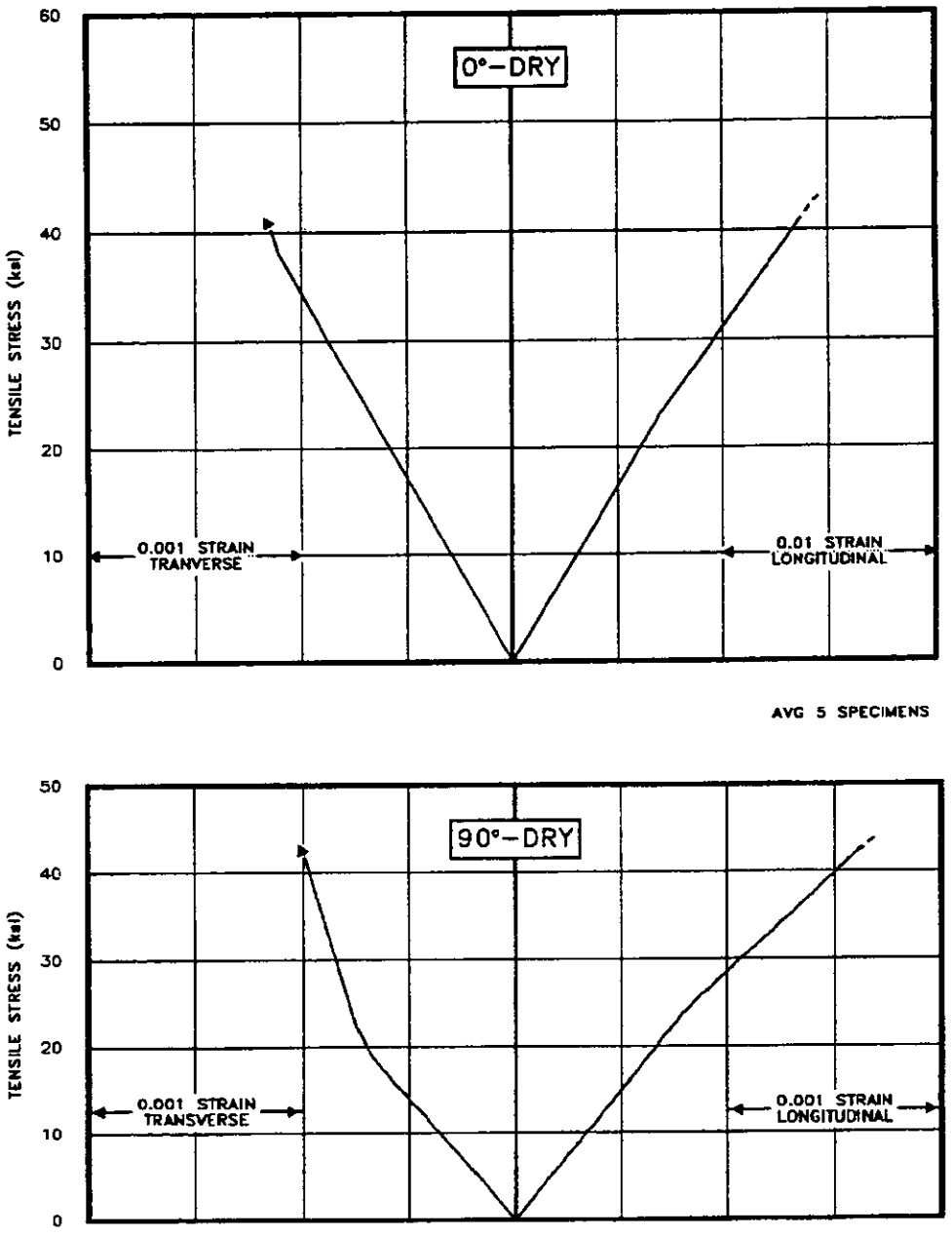


FIGURE A1.6.4 *Poisson effects for F-161/7781 fiberglass epoxy (36% resin).*

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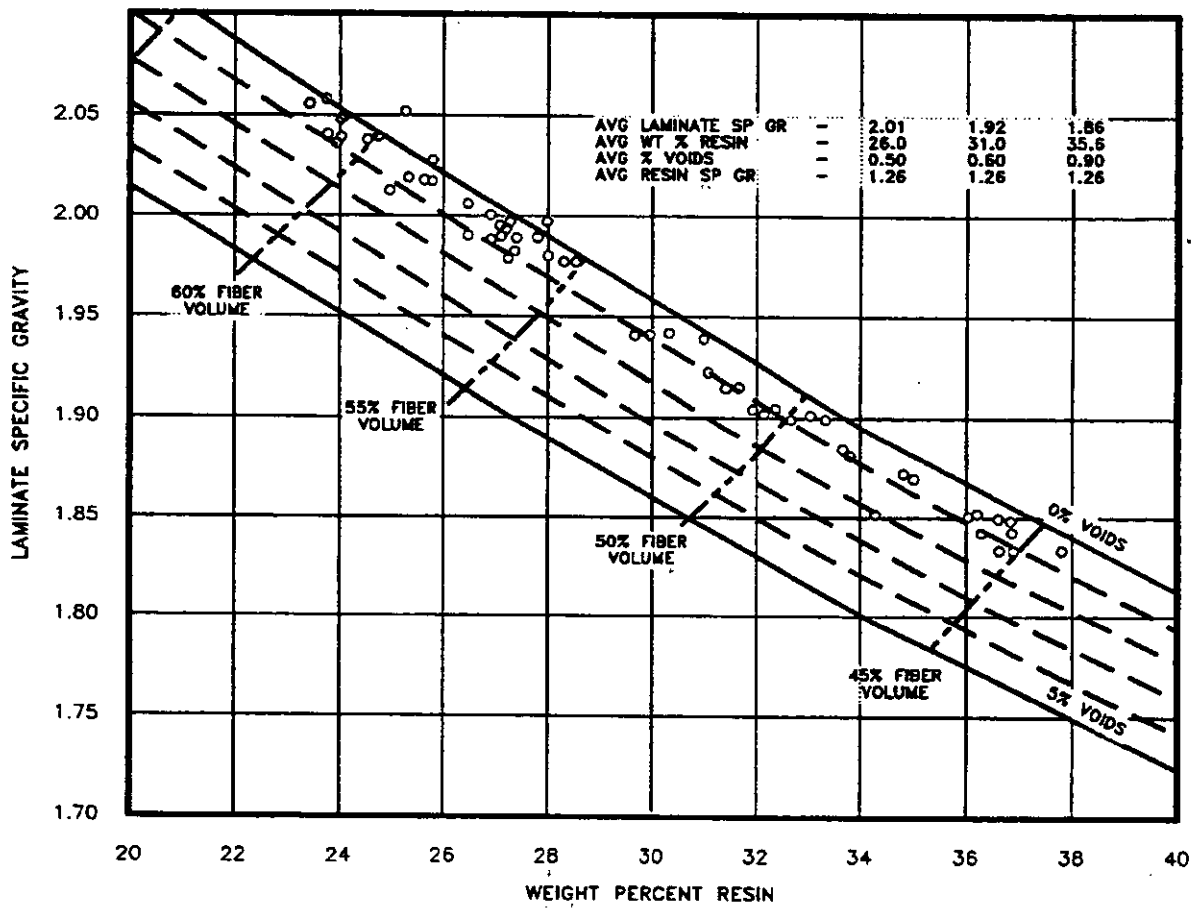


FIGURE A1.6.5 *Voids vs resin content and specific gravity for F-161/7781 fiberglass epoxy (26%, 31%, 36% resin).*

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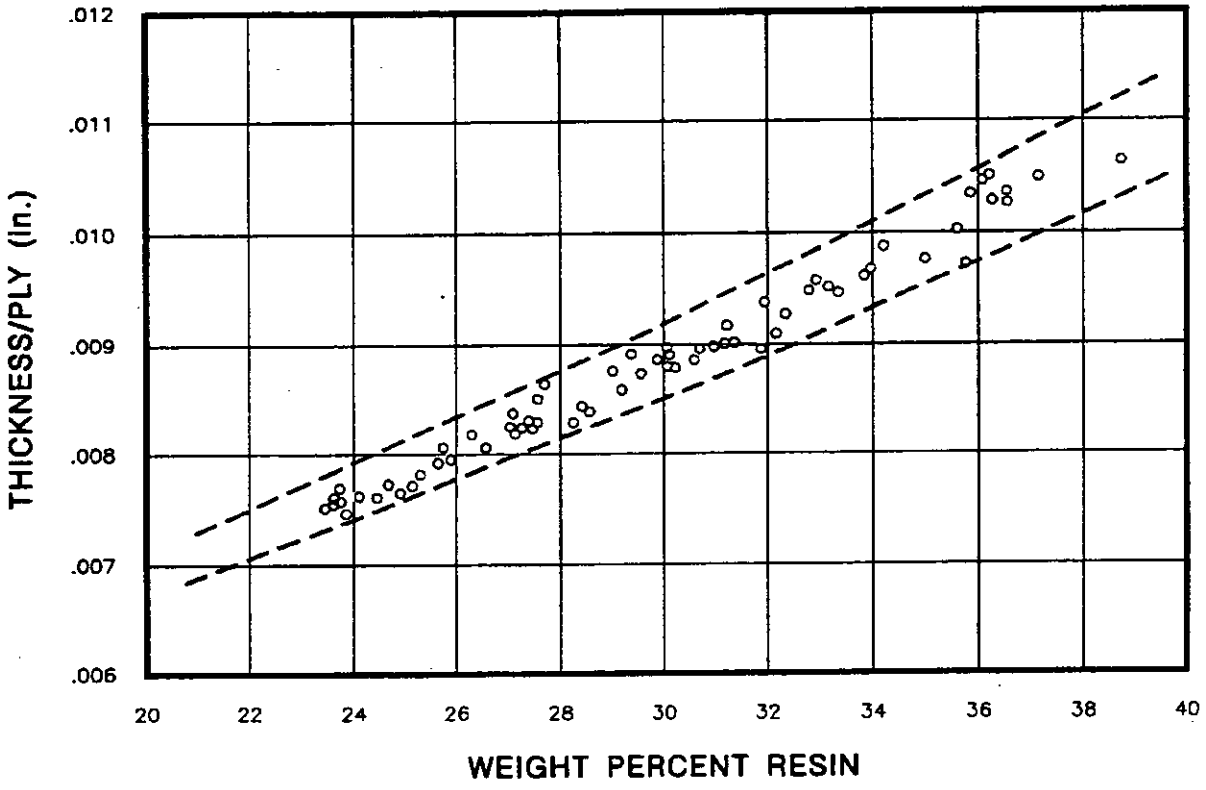


FIGURE A1.6.6 *Thickness vs resin content for F-161/7781 fiberglass epoxy.*

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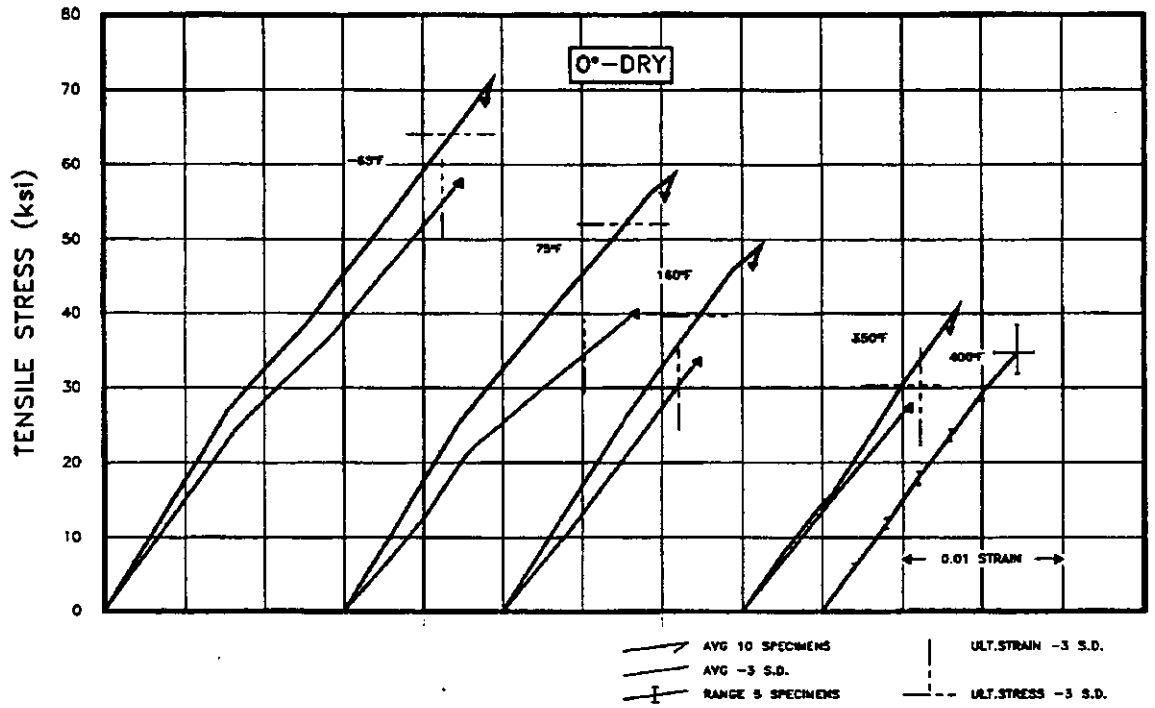
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TABLE A1.8 *Summary of Mechanical Properties of Narmco N588/7781 (ECDE-1/0-550) Fiberglass Epoxy*

Fabrication		Lay-up: Balanced		Vacuum: None		Pressure: 45-55 psi		Bleedout: Vertical		Cure: Stepwise to 350°F; 1hr/350°F		Postcure: None		Plies: 8		
Physical Properties		Weight Percent Resin: 32.8 v = 0.51				Avg. Specific Gravity: 1.91				Avg. Percent Voids: 1.0				Avg. Thickness: 0.075 inches		
Test Methods		Tension: ASTM-D 638 TYPE 1		Compression: MIL-HDBK-17		Shear: Rail		Flexure: ASTM-D 790		Bearing: ASTM-D 953				Interlaminar Shear: Short Beam		
Temperature		-65°F				75°F				160°F				400°F		
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry		
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	
Tension																
ultimate stress, ksi	0°	71.4	2.4	63.8	3.3	58.4	2.1	50.0	2.3	48.8	3.0	35.0	2.0	40.4	3.4	
	90°	59.3	3.3	50.6	2.4	47.2	3.8	41.1	2.7	41.4	2.0	28.9	2.8	33.3	3.8	
ultimate strain, %	0°	2.41	0.09	2.06	0.15	2.05	0.18	1.61	0.12	1.59	0.15	1.13	0.07	1.26	0.07	
	90°	2.35	0.17	1.96	0.12	1.81	0.16	1.55	0.16	1.67	0.10	1.17	0.14	1.25	0.12	
proportional limit, ksi	0°	26.6	1.7	28.7	2.5	23.3	1.1	25.4	2.8	21.0	1.7	29.9	2.0	24.3		
	90°	19.3	0.8	19.2	1.6	17.6	0.8	18.1	1.4	17.3	2.5	20.9	1.3	14.3		
initial modulus, 10^6 psi	0°	3.64		3.85		3.71		3.57		3.58		3.10		3.13	0.17	
	90°	3.41		3.37		3.56		3.23		2.92		2.63		2.80	0.23	
secondary modulus, 10^6 psi	0°															
	90°															
Compression																
ultimate stress, ksi	0°	99.2	5.9	87.4	5.8	74.0	3.6	63.5	3.2	59.0	2.4	49.5	1.9			
	90°	83.4	3.5	71.8	4.1	62.9	2.9	53.7	1.7	50.9	1.5	40.7	1.8			
ultimate strain, %	0°	2.52	0.26	2.30	0.25	1.89	0.15	1.65	0.19	1.60	0.12	1.38	0.06			
	90°	2.30	0.27	2.06	0.20	1.87	0.14	1.58	0.15	1.63	0.16	1.29	0.08			
proportional limit, ksi	0°	42.7	2.6	46.2	2.5	44.5	3.2	39.8	3.6	37.6	2.7	30.7	2.7			
	90°	40.8	3.8	42.4	2.7	35.3	3.7	34.4	2.3	31.2	2.4	24.4	1.6			
initial modulus, 10^6 psi	0°	4.32		4.15		4.18		4.11		3.88		3.70				
	90°	4.08		3.83		3.68		3.72		3.41		3.41				
Shear																
ultimate stress, ksi	0°-90° ±45°	22.6				16.0	1.05			13.8						

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°	105.0	115.6	95.6	90.4	102.6	84.5	79.3	87.8	74.0
proportional limit, ksi	0°	69.6	75.9	59.0	68.9	72.4	64.6	64.8	72.2	57.2
initial modulus, 10^6 psi	0°	3.48	3.62	3.42	3.36	3.60	3.20	3.19	3.27	3.09
Bearing										
ultimate stress, ksi	0°	84.6	92.5	77.9	68.4	71.3	66.0	48.4	53.6	44.2
stress at 4% elong., ksi	0°	29.3	30.9	26.5	26.2	27.4	25.3	21.8	22.8	20.6
Interlaminar Shear										
ultimate stress, ksi	0°	8.84	9.16	8.56	8.35	8.56	8.05	7.39	7.72	6.47

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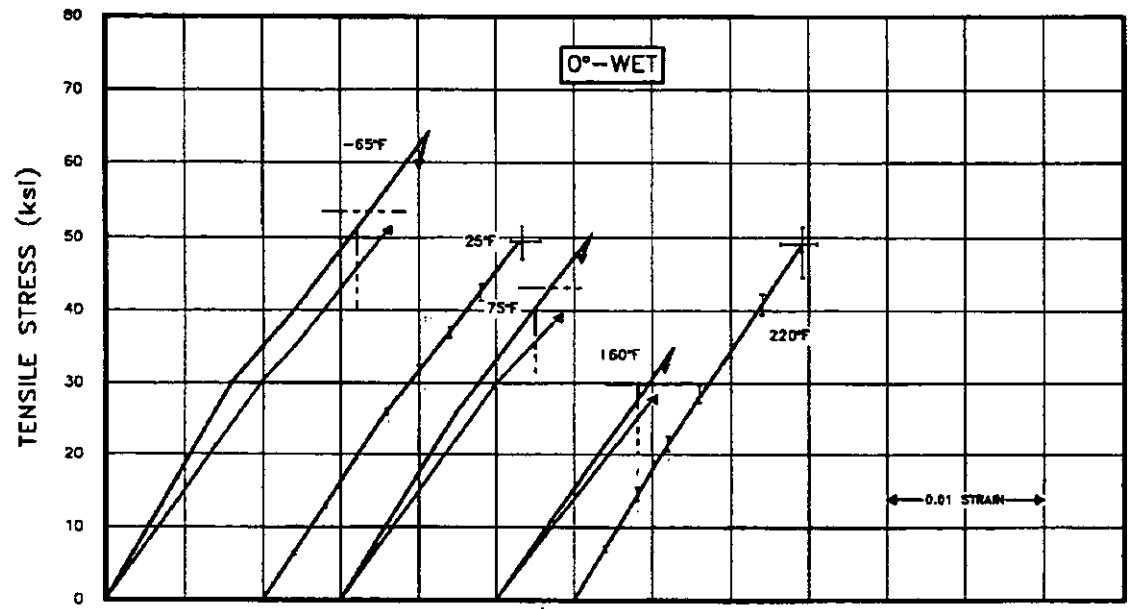


FIGURE A1.8.1(a) *Tensile stress-strain for N588/7781 fiberglass epoxy loaded in the 0° direction.*

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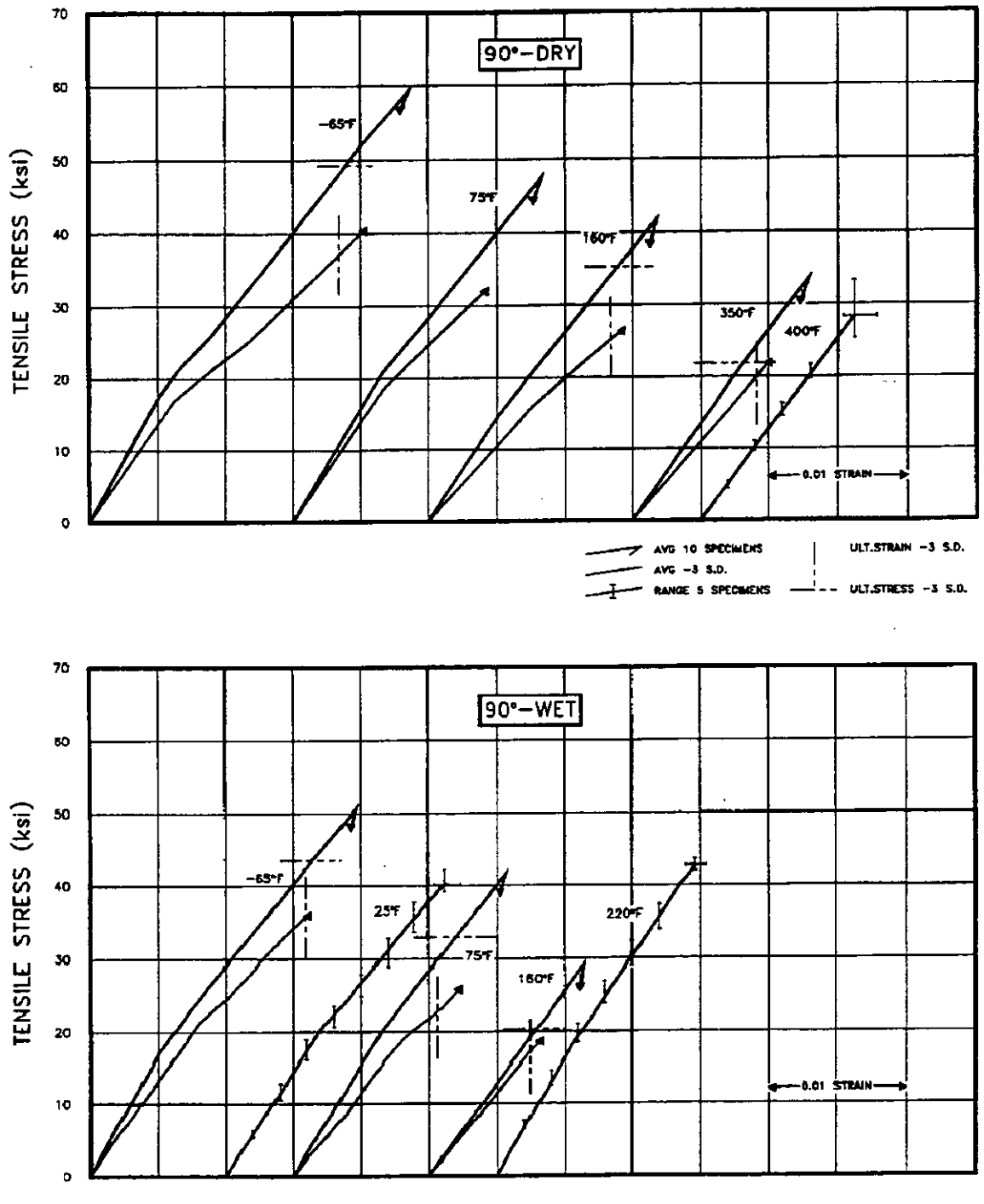


FIGURE A1.8.1(b) *Tensile stress-strain for N588/7781 fiberglass epoxy loaded in the 90° direction.*

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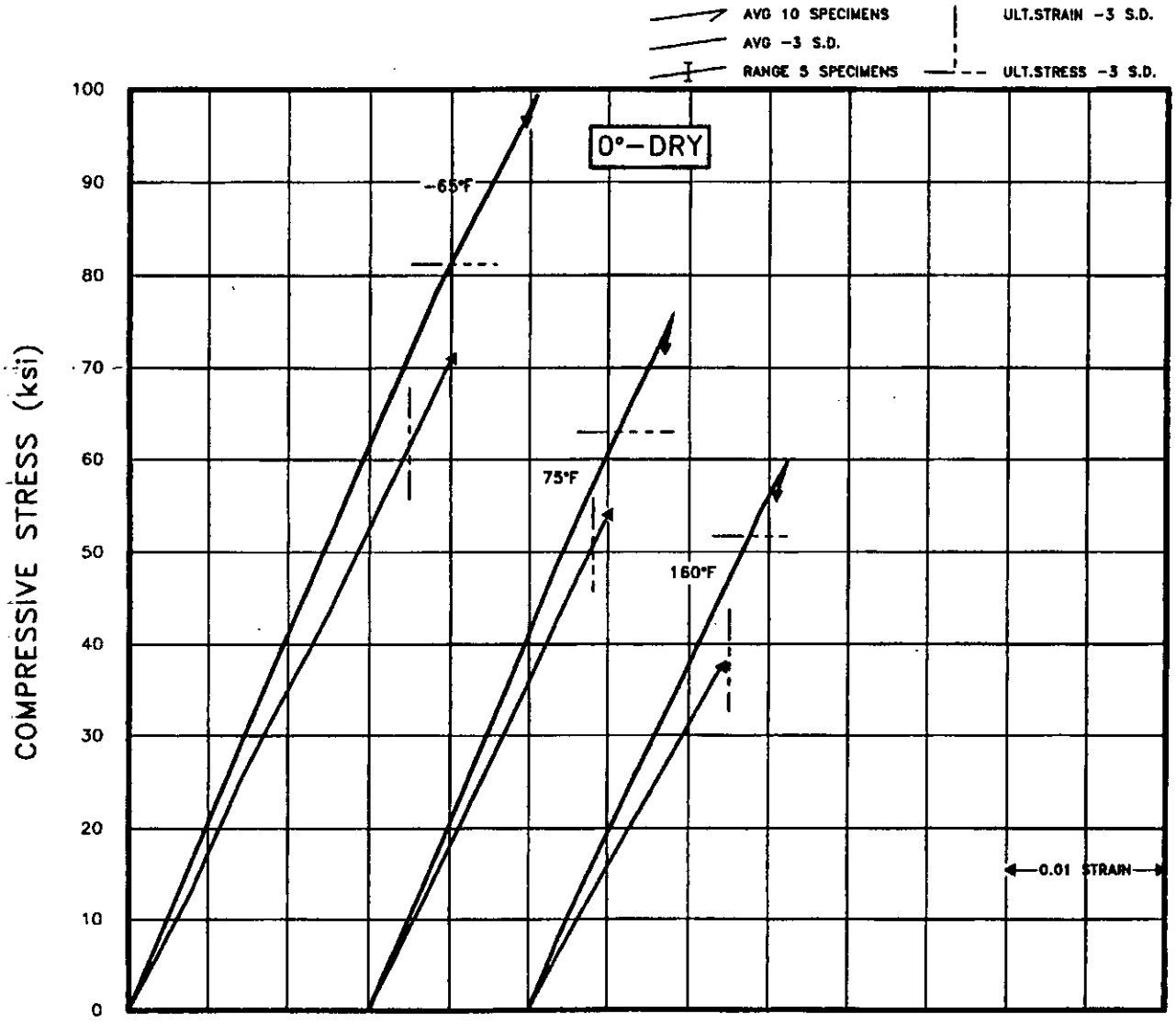


FIGURE A1.8.2(a) *Compressive stress-strain for N588/7781 fiberglass epoxy loaded in the 0° direction*, continued on next page.

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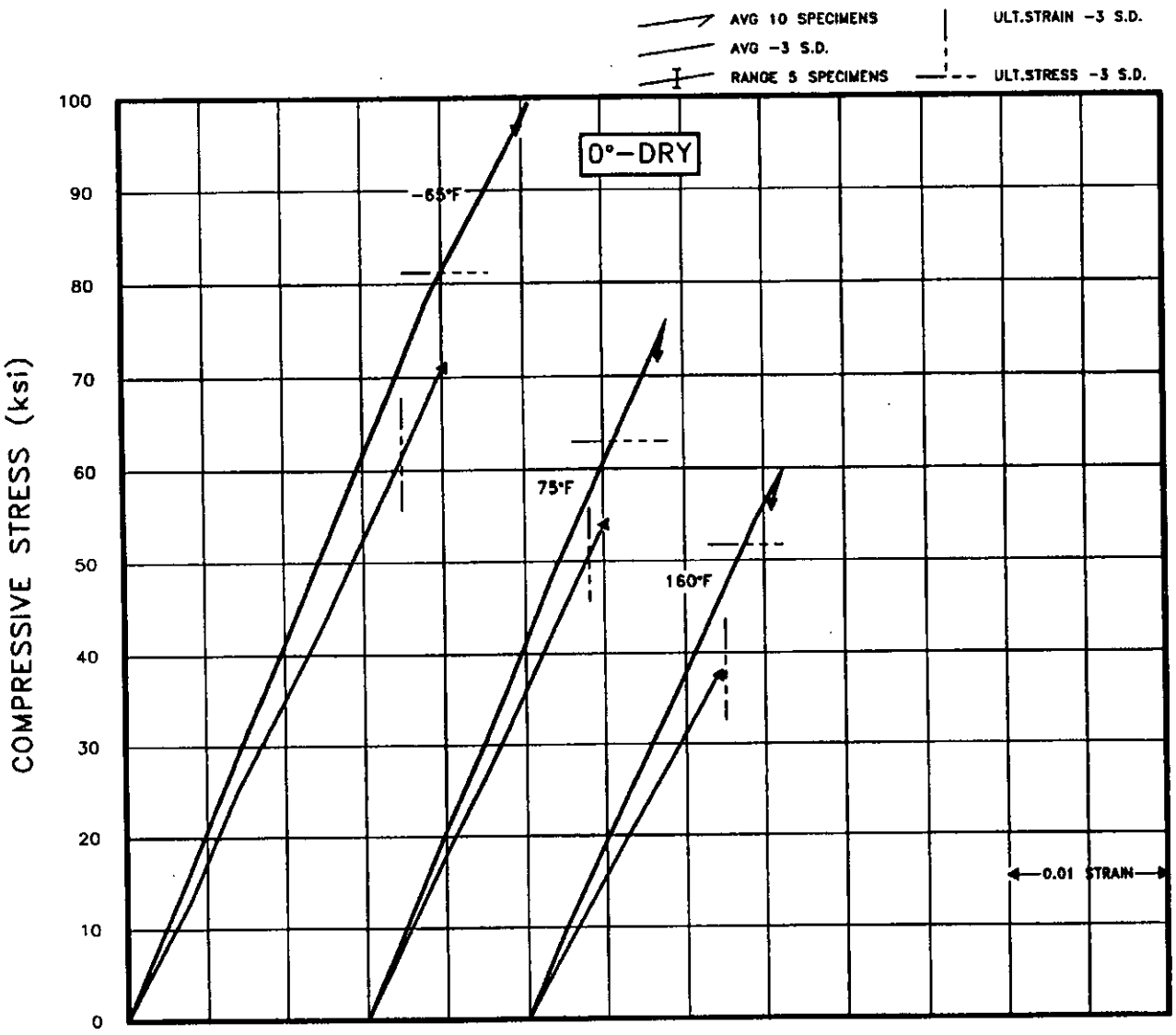
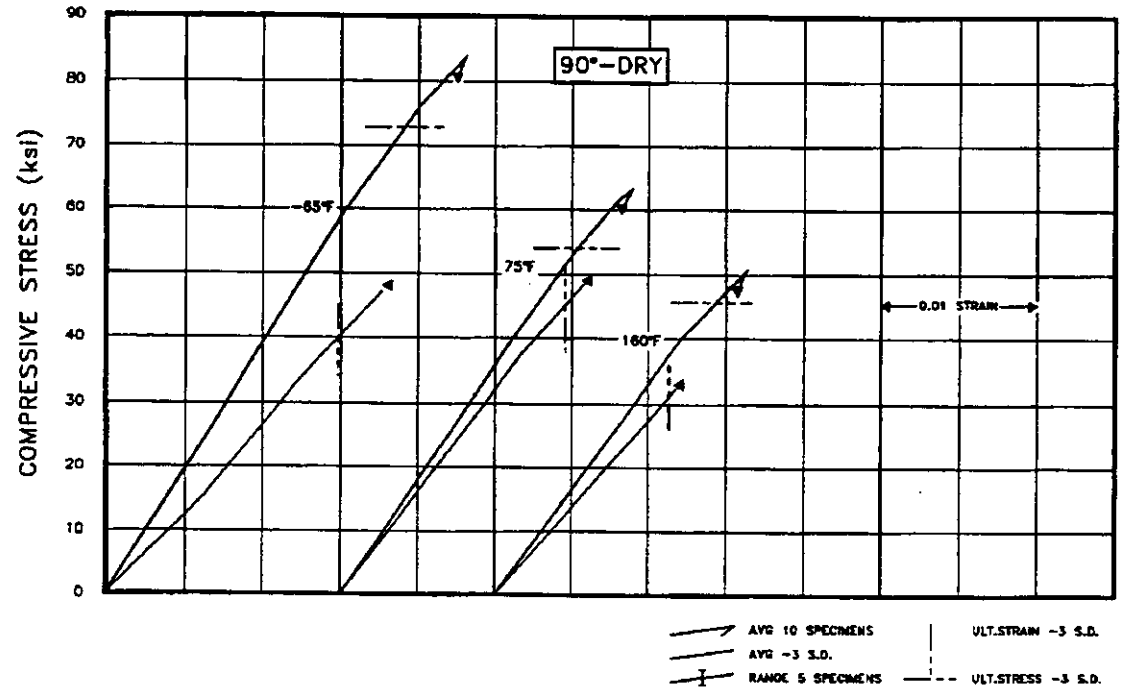


FIGURE A1.8.2(a) *Compressive stress-strain for N588/7781 fiberglass epoxy loaded in the 0° direction,* concluded.

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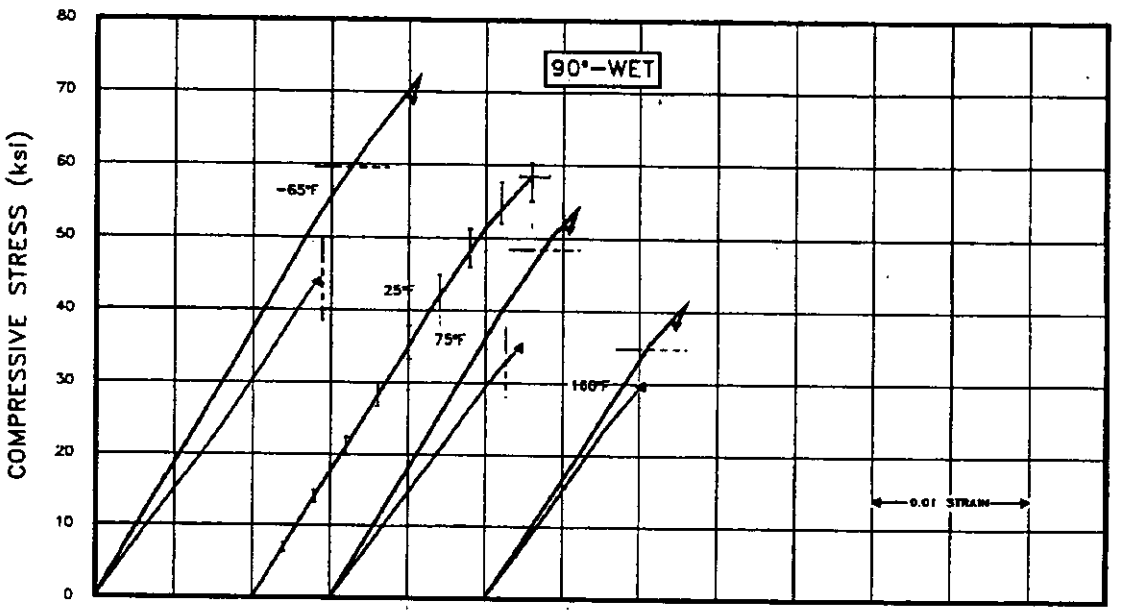


FIGURE A1.8.2(b) *Compressive stress-strain for N588/7781 fiberglass epoxy loaded in the 90° direction.*

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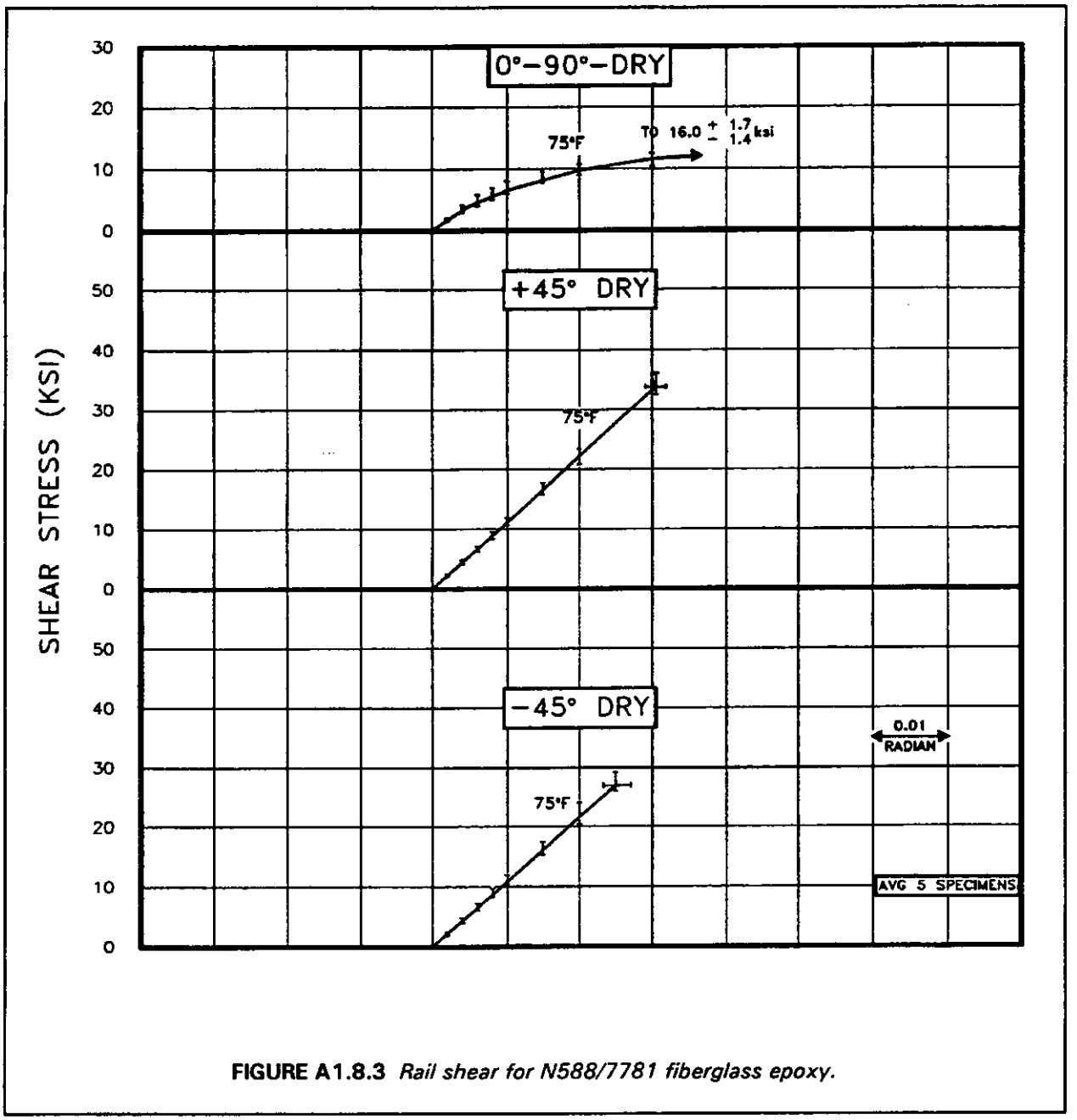
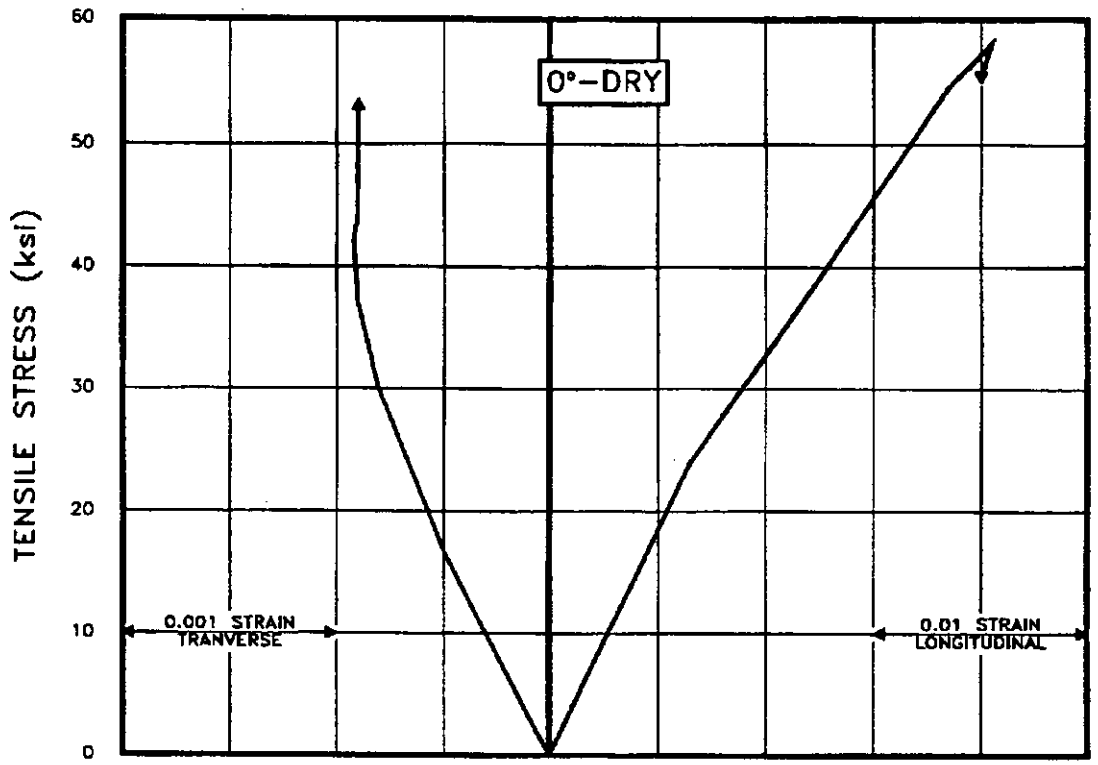


FIGURE A1.8.3 *Rail shear for N588/7781 fiberglass epoxy.*

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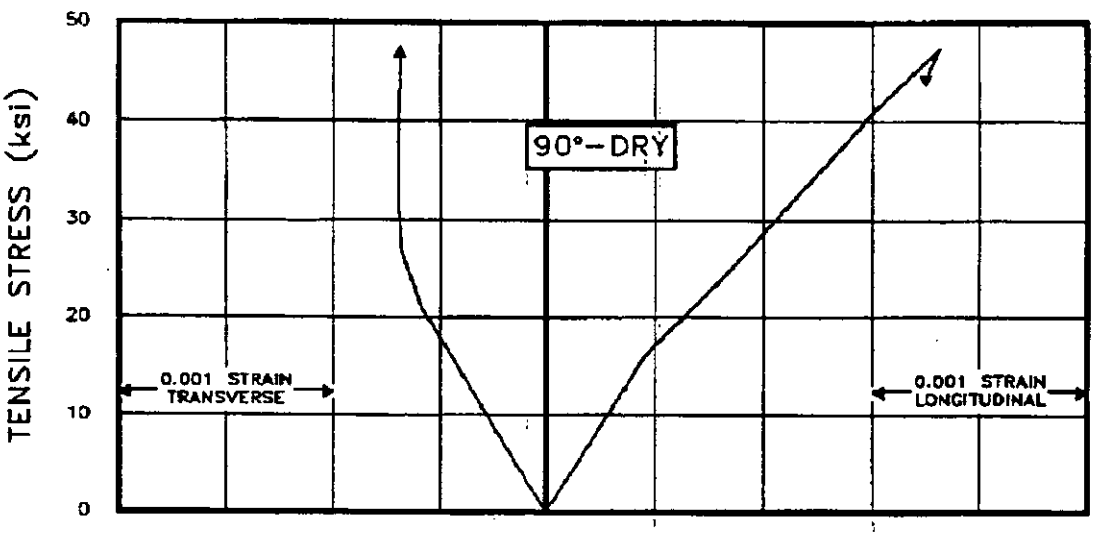


FIGURE A1.8.4 *Poisson effects for N588/7781 fiberglass epoxy.*

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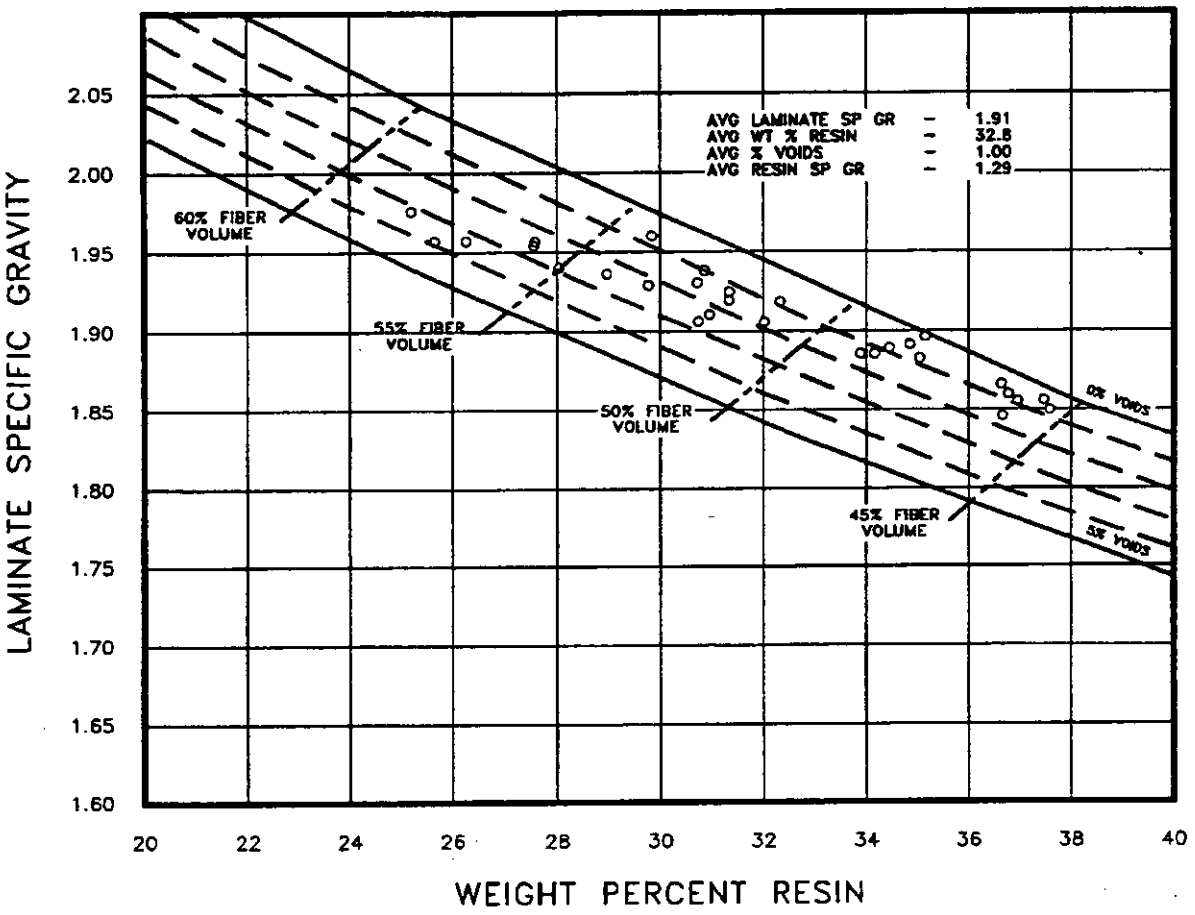


FIGURE A1.8.5 *Voids vs resin content and specific gravity for N588/7781 fiberglass epoxy.*

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TABLE A1.40 *Summary of Mechanical Properties of Narmco N506/7781 (ECDE-1/0-A1100) Fiberglass Phenolic*

Fabrication		Lay-up: Balanced		Vacuum:		Pressure:		Bleedout: Vertical		Cure:		Postcure:		Plies: 8		
Physical Properties		Weight Percent Resin: 25.3 - 32.3				Avg. Specific Gravity: 1.72 - 1.85				Avg. Percent Voids: Figure 4.40.5				Avg. Thickness: 0.071 - 0.095 inches		
Test Methods		Tension: ASTM-D 638 TYPE 1		Compression: MIL-HDBK-17		Shear: Rail				Flexure: ASTM-D 790		Bearing: ASTM-D 953		Interlaminar Shear: Short Beam		
Temperature		-65°F				75°F				160°F				400°F		
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry		
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	
Tension																
ultimate stress, ksi	0°	48.1	2.4	49.8	3.3	38.9	1.5	37.2	1.8	35.3	1.4	30.6	3.0	21.6	1.6	
	90°	37.9	1.8	40.0	2.7	31.5	1.5	32.1	1.4	27.9	1.7	26.2	2.2	21.6	1.7	
ultimate strain, %	0°	1.76	0.07	1.76	0.13	1.33	0.14	1.34	0.13	1.19	0.10	1.15	0.14	0.69	0.05	
	90°	1.63	0.08	1.65	0.13	1.26	0.15	1.32	0.07	1.11	0.07	1.11	0.14	0.78	0.06	
proportional limit, ksi	0°	13.6	0.9	18.1	1.2	13.5	0.6	17.0	1.0	13.9	1.0	14.9	0.70	9.7	1.1	
	90°	9.9	0.4	12.5	0.9	9.2	0.8	12.8	0.7	10.3	0.8	11.6	0.70	8.6	0.5	
initial modulus, 10^6 psi	0°	3.40	0.21	3.35	0.20	3.94	0.69	3.14	0.26	3.74	0.41	3.01	0.19	3.57	0.24	
	90°	3.08	0.29	3.04	0.22	3.54	0.41	2.81	0.24	3.33	0.37	2.78	0.21	3.18	0.30	
secondary modulus, 10^6 psi	0°															
	90°															
Compression																
ultimate stress, ksi	0°	66.7	6.2	65.9	5.0	59.7	4.7	54.5	7.1	50.6	2.3	49.2	4.2			
	90°	57.7	5.8	56.2	5.8	49.0	4.6	48.7	4.0	43.0	4.3	42.9	3.7			
ultimate strain, %	0°	1.85	0.09	1.69	0.18	1.58	0.14	1.49	0.12	1.45	0.06	1.40	0.12			
	90°	1.70	0.21	1.63	0.13	1.40	0.09	1.43	0.07	1.37	0.12	1.31	0.15			
proportional limit, ksi	0°	45.8	3.8	38.5	7.9	39.0	2.4	41.2	4.6	39.9	2.4	35.0	1.7			
	90°	35.2	3.8	34.4	5.0	32.6	4.4	35.5	3.0	32.4	3.1	31.1	3.3			
initial modulus, 10^6 psi	0°	3.90	0.19	4.17	0.29	3.95	0.28	3.89	0.26	3.68	0.21	3.67	0.12			
	90°	3.69	0.25	3.68	0.17	3.70	0.20	3.57	0.20	3.30	0.23	3.45	0.21			
Shear																
ultimate stress, ksi	0°-90° ±45°	13.8				12.3	0.97			11.4						

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°	68.2	72.8	65.2	58.4	64.0	52.1	52.7	56.3	47.4
proportional limit, ksi	0°	59.3	66.1	54.6	48.9	56.8	42.5	42.4	46.2	38.8
initial modulus, 10^6 psi	0°	2.97	3.04	2.88	2.89	2.99	2.78	2.97	3.06	2.82
Bearing										
ultimate stress, ksi	0°	65.7	73.2	57.0	58.9	64.0	46.8	49.5	55.8	44.5
stress at 4% elong., ksi	0°	25.1	26.0	23.7	24.5	24.9	23.8	21.6	22.6	20.7
Interlaminar Shear										
ultimate stress, ksi	0°	4.83	5.10	4.29	4.64	4.92	3.94	4.62	4.88	4.08

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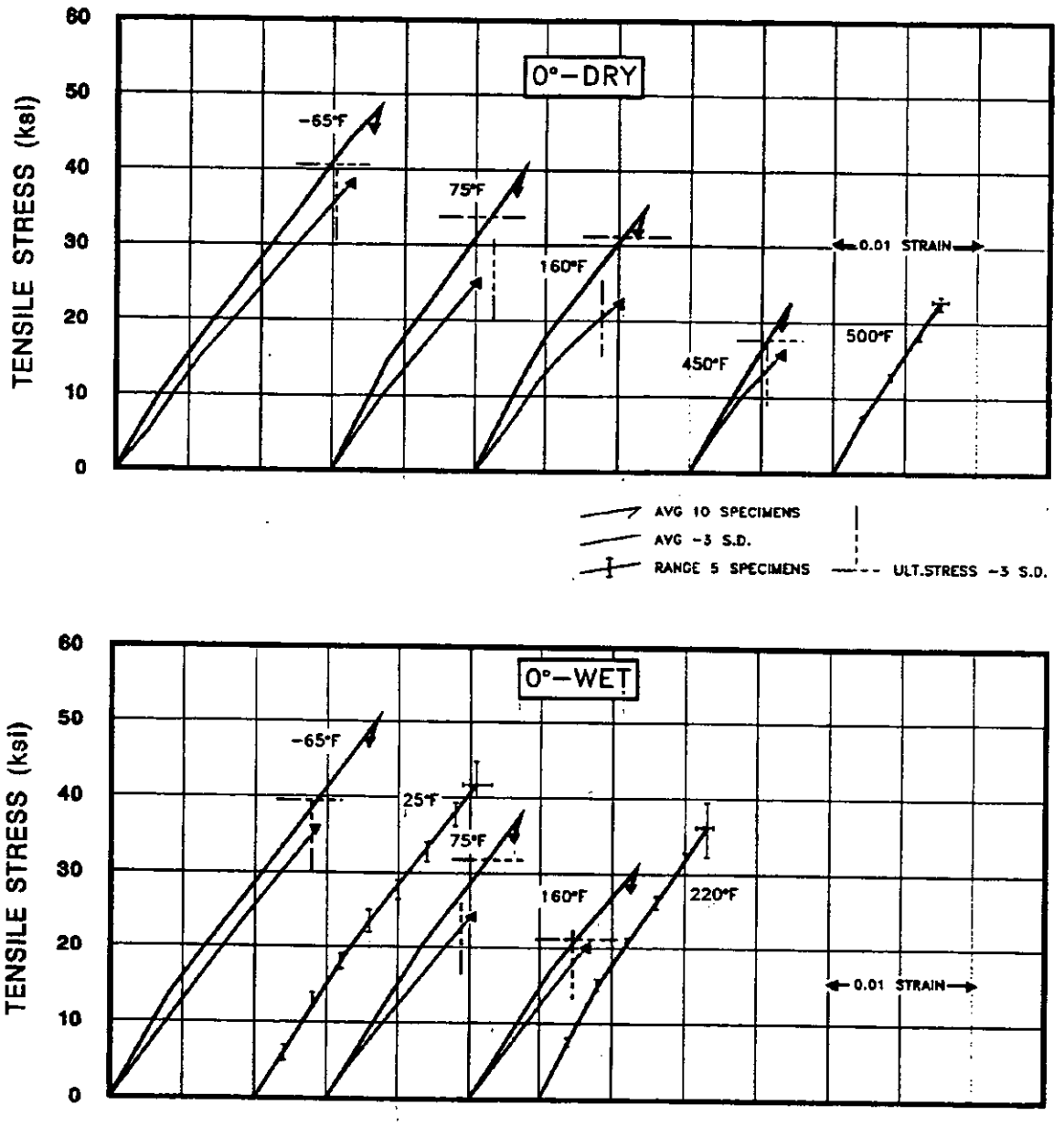


FIGURE A1.40.1(a) *Tensile stress-strain for N506/7781 fiberglass phenolic loaded in the 0° direction.*

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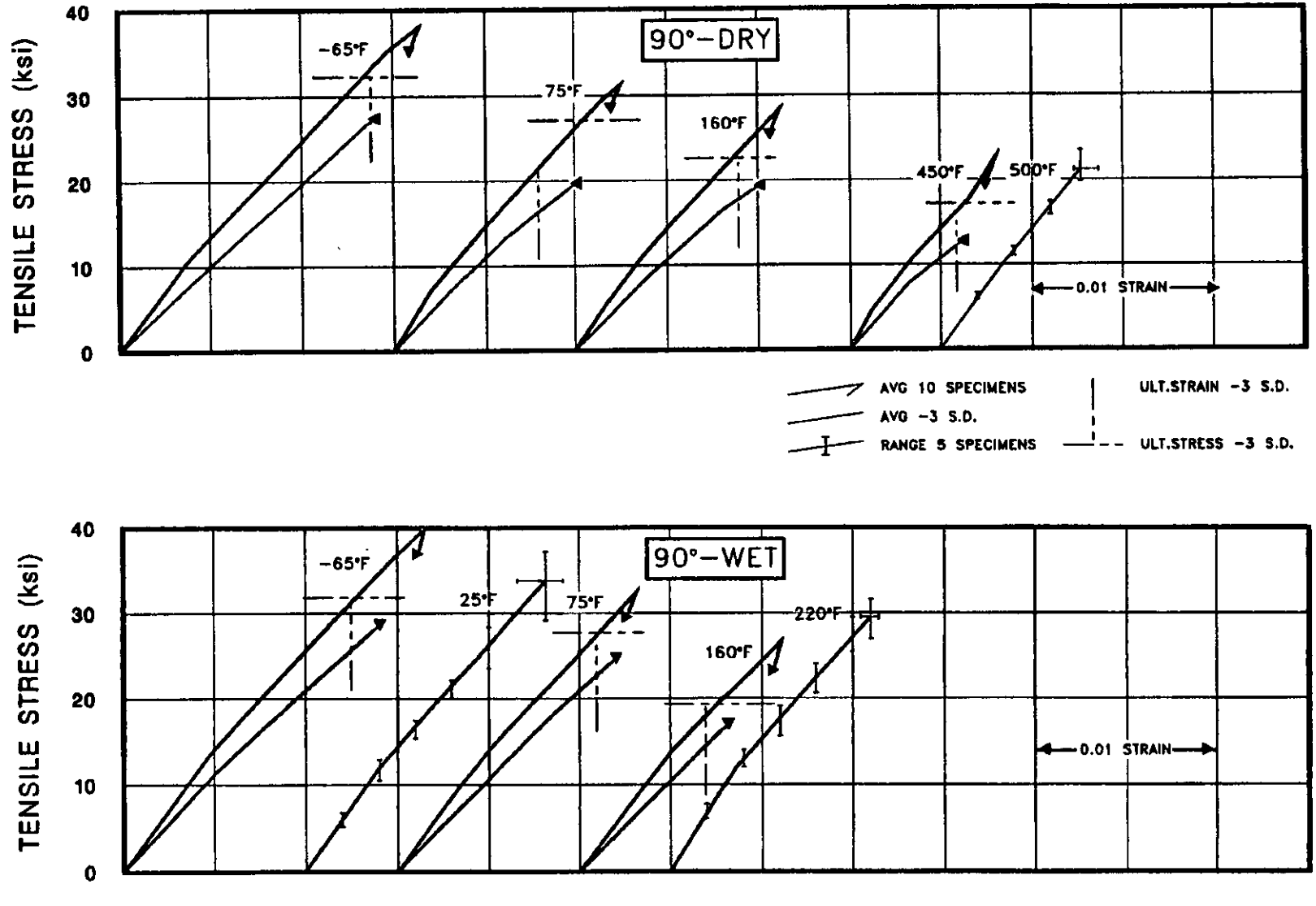
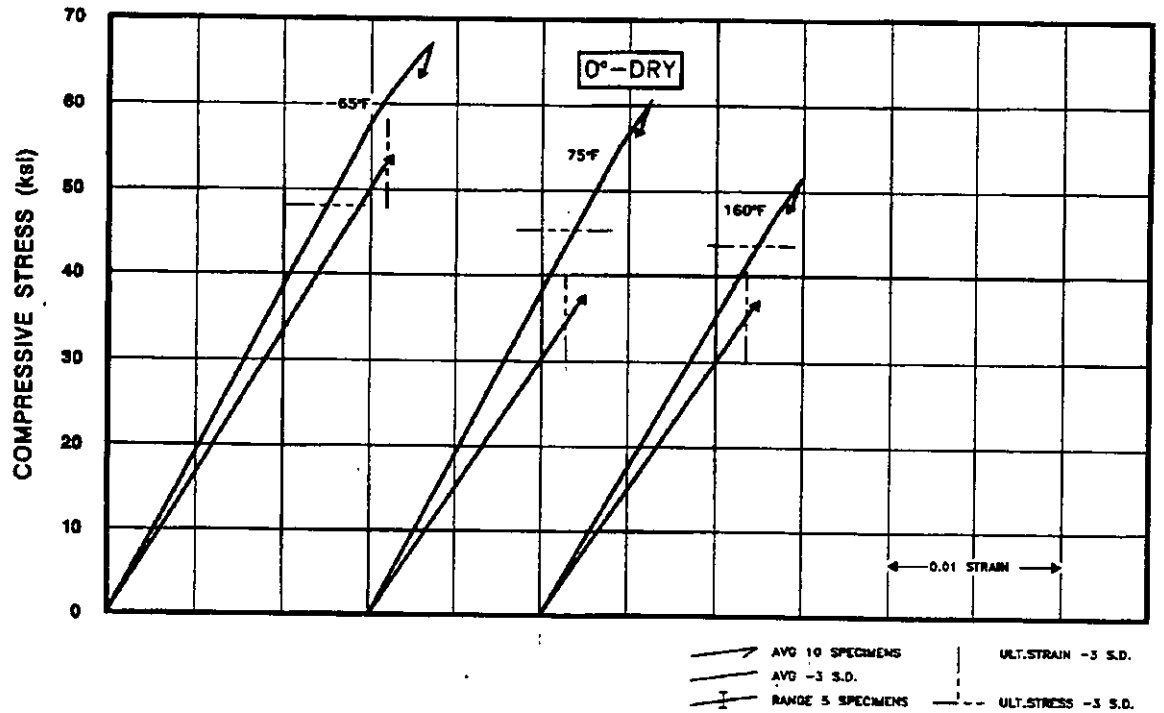


FIGURE A1.40.1(b) *Tensile stress-strain for N506/7781 fiberglass phenolic loaded in the 90° direction.*

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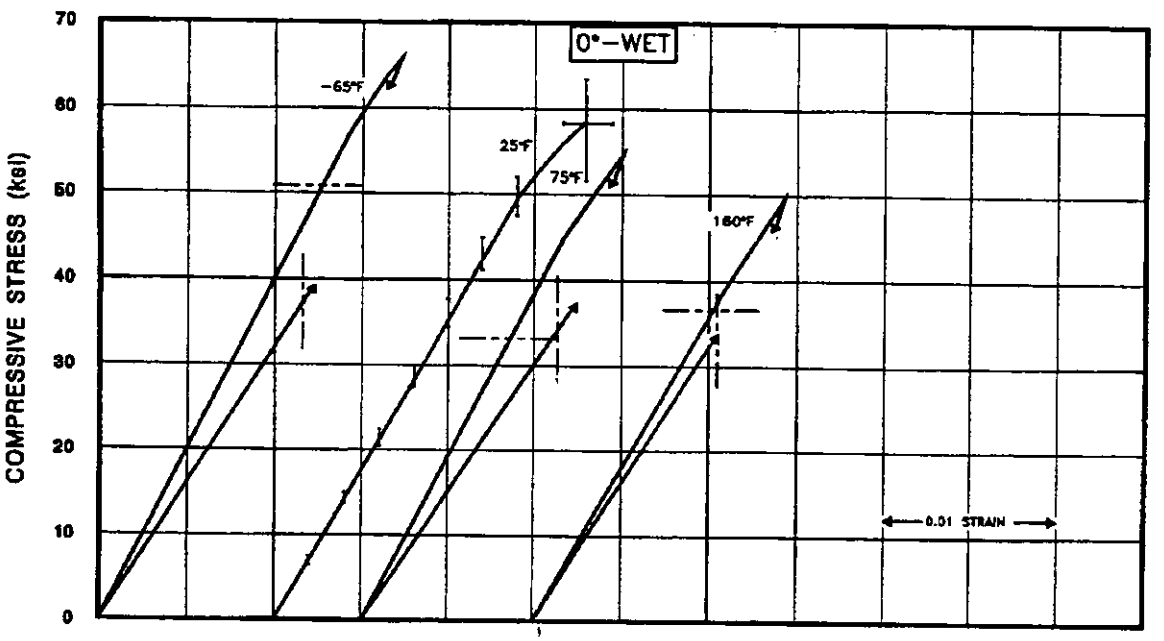


FIGURE A1.40.2(a) *Compressive stress-strain for N506/7781 fiberglass phenolic loaded in the 0° direction.*

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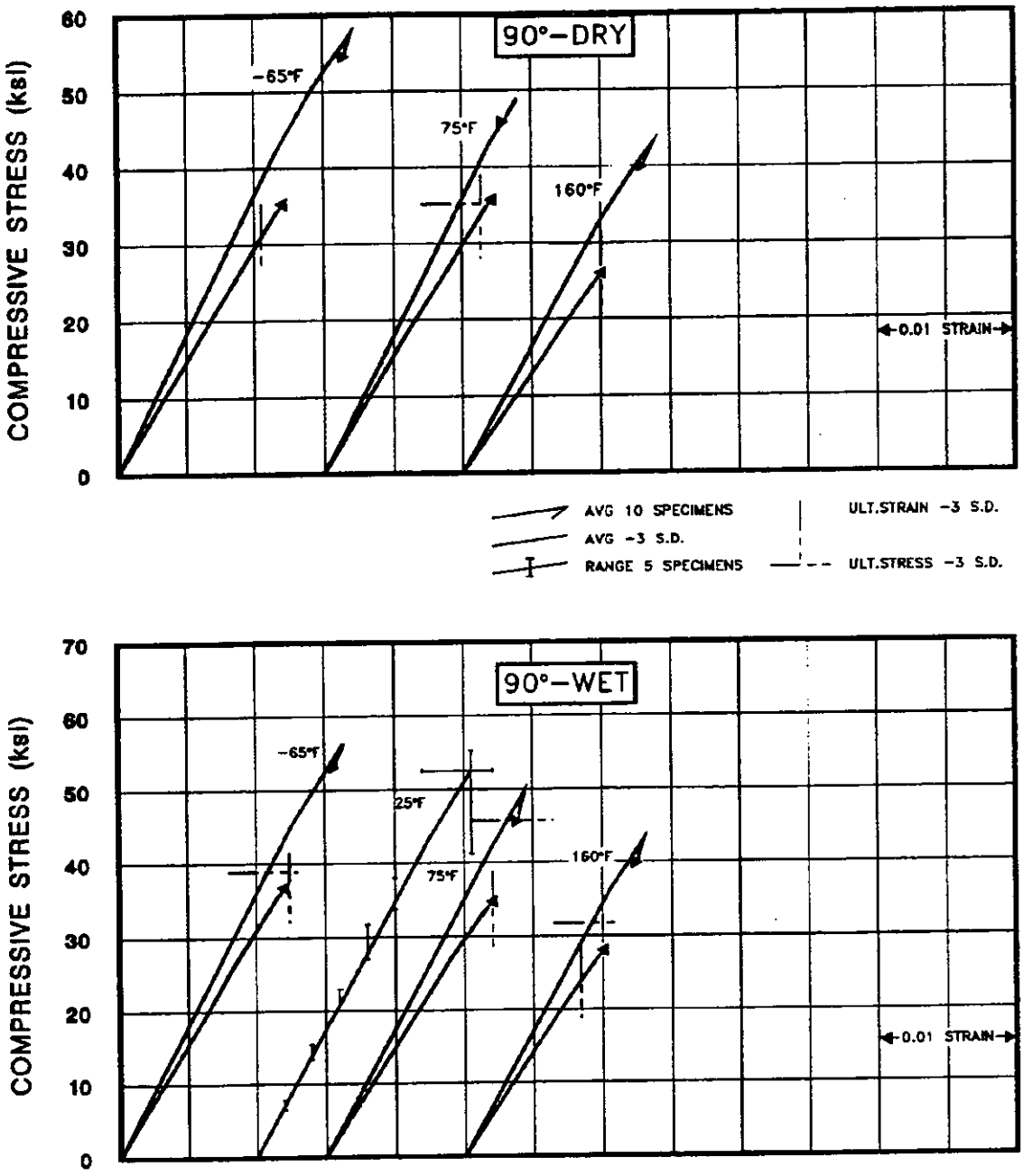


FIGURE A1.40.2(b) *Compressive stress-strain for N506/7781 fiberglass phenolic loaded in the 90° direction.*

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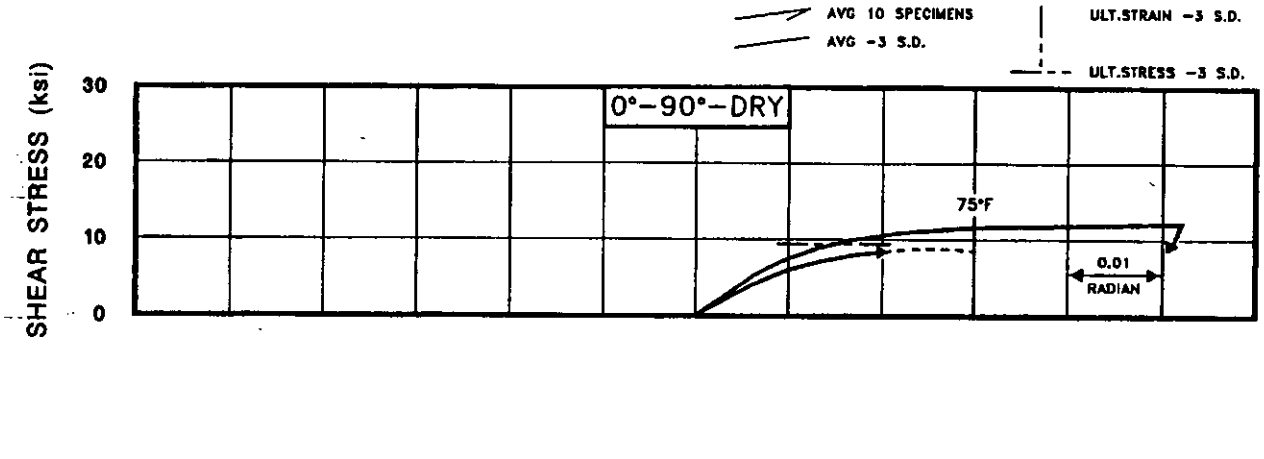
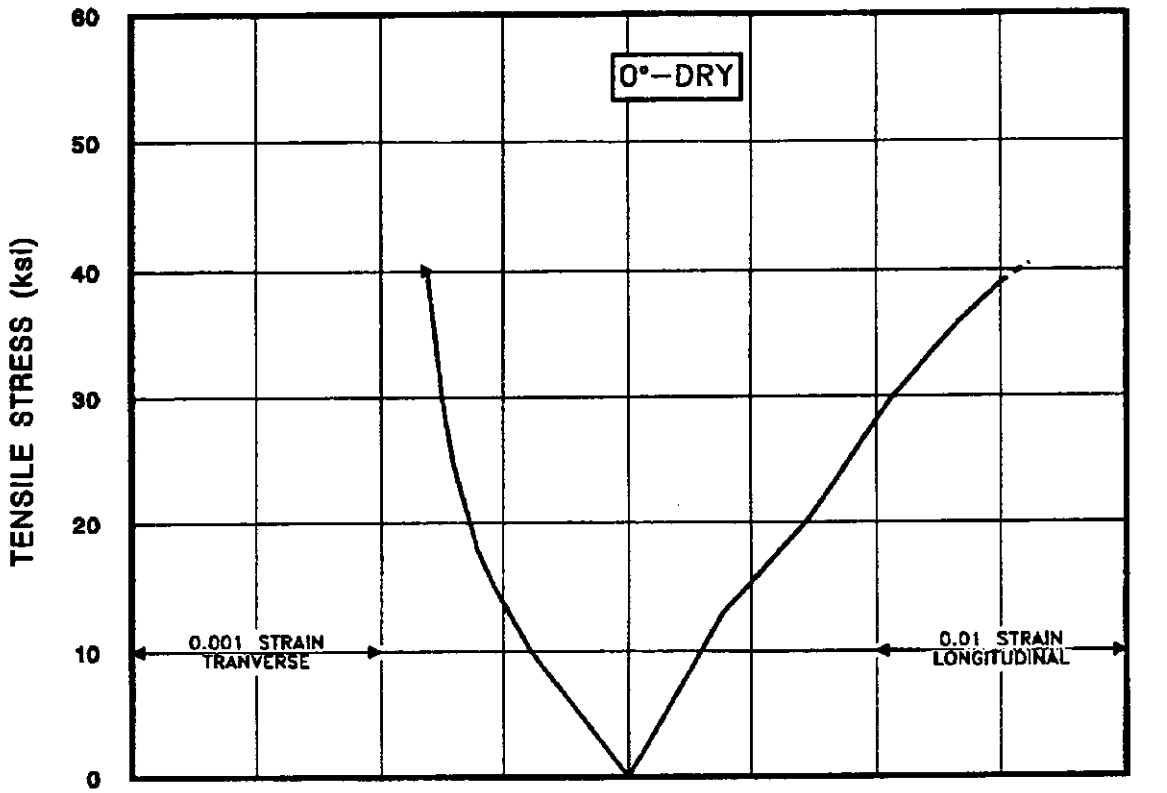


FIGURE A1.40.3 *0° - 90° rail shear for N506/7781 fiberglass phenolic.*

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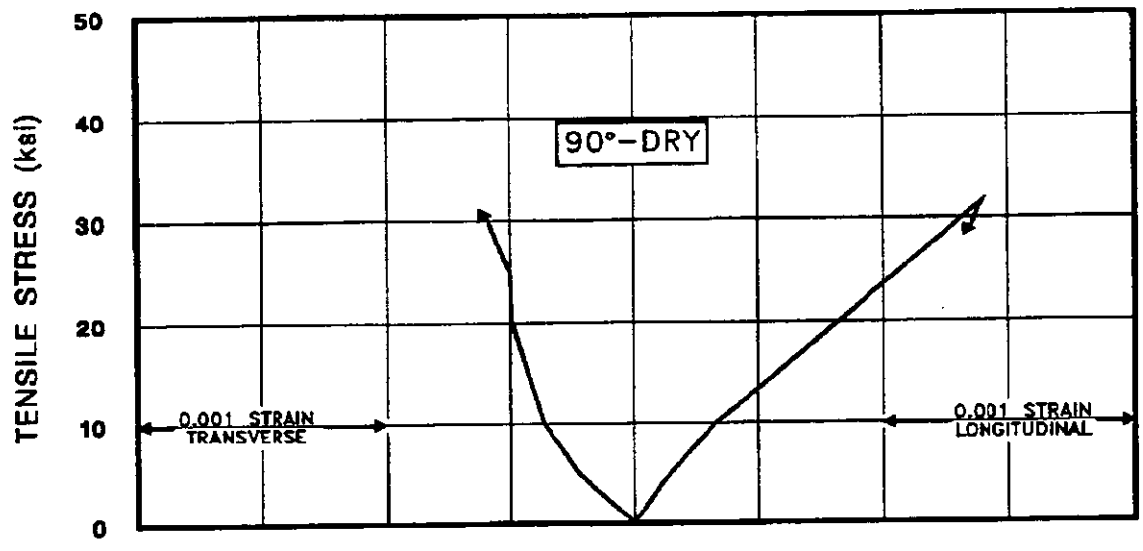


FIGURE A1.40.4 *Poisson effects for N506/7781 fiberglass phenolic.*

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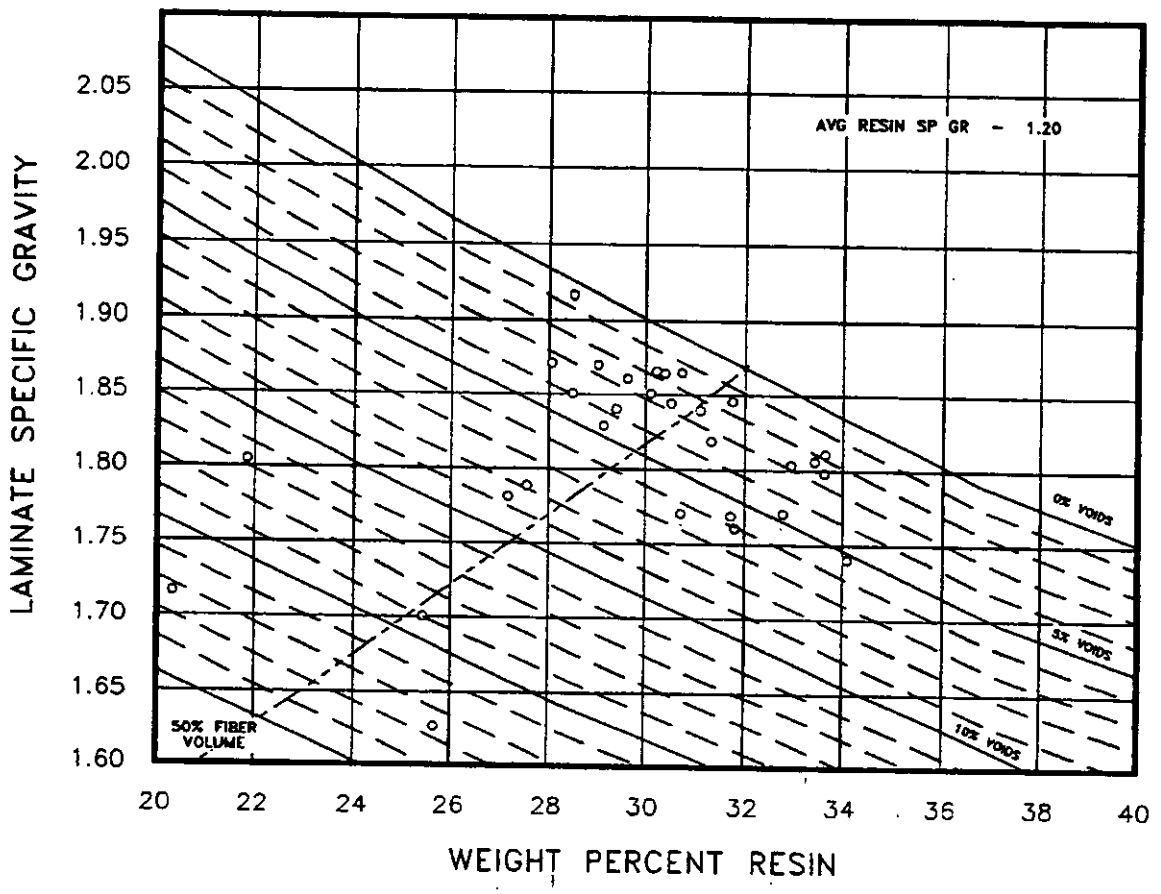


FIGURE A1.40.5 *Voids vs. resin content and specific gravity for N506/7781 fiberglass phenolic.*

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TABLE A1.110 *Summary of Mechanical Properties of Narmco 5505 Boron-Epoxy (100%-0° Direction) (Tentative)*

Fabrication		Lay-up: Parallel		Vacuum: 2 ins		Pressure: 50 ± 5 psi		Bleedout:		Cure: 1.5hr/ 350°F ± 10°F		Postcure: 2hr/350°F		Plies: 6			
Physical Properties		Weight Percent Resin:		Avg. Specific Gravity:		Avg. Percent Voids:				Avg. Thickness: 0.005 in/ply							
Test Methods		Tension: Tab-ended		Compression: Sandwich Beam		Shear:		Flexure: 4 Point Loading		Bearing:		Interlaminar Shear: Short Beam					
Temperature		-67°F				75°F				260°F				375°F			
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry			
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD		
Tension																	
ultimate stress, ksi	0°	201.1				208.3				191.6				167.3			
	90°	10.5				8.7				6.5				3.3			
ultimate strain, %	0°	6390				6930				6660				6150			
	90°	3250				3710				4970				6920			
proportional limit, ksi	0°	141.8				175.5				140.0				79.5			
	90°																
initial modulus, 10^6 psi	0°	32.0				30.9				29.6				28.6			
	90°																
secondary modulus, 10^6 psi	0°																
	90°																
Compression																	
ultimate stress, ksi	0°	482.3				378.0				303.3				143.9			
	90°																
ultimate strain, %	0°	13670				10830				8920				4466			
	90°																
proportional limit, ksi	0°	333.5															
	90°																
initial modulus, 10^6 psi	0°	35.7				34.8				34.6				35.8			
	90°																
Shear																	
ultimate stress, ksi	0°-90°																
	±45°																

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°									
proportional limit, ksi	0°									
initial modulus, 10^6 psi	0°									
Bearing										
ultimate stress, ksi	0°									
stress at 4% elong., ksi	0°									
Interlaminar Shear										
ultimate stress, ksi	0°									

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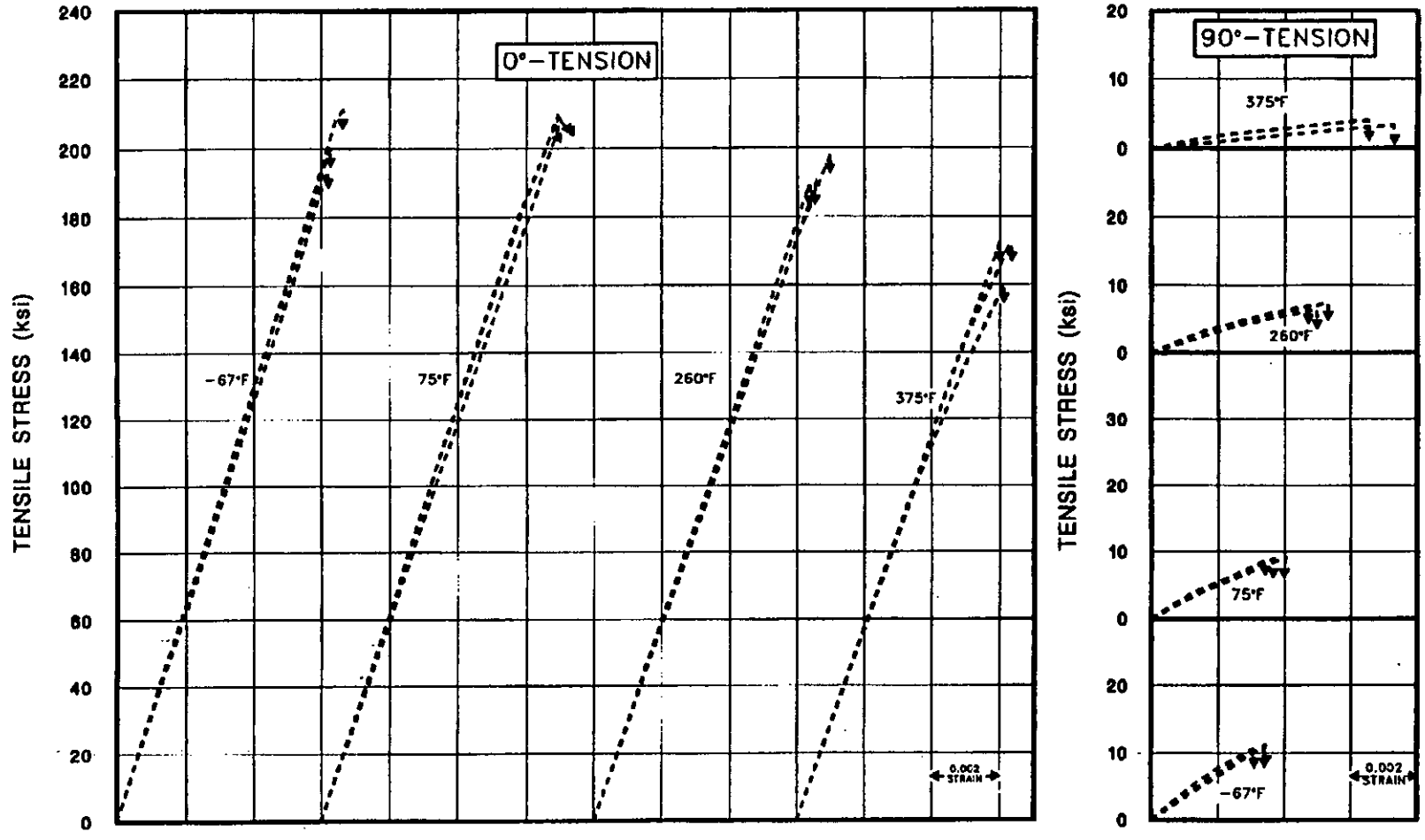


FIGURE A1.110.1 *Tensile stress-strain for AVCO 5505 boron/epoxy (100% - 0° orientation/ 50.3% to 35% fiber volume) loaded in the 0° and the 90° directions. Individual tests shown.*

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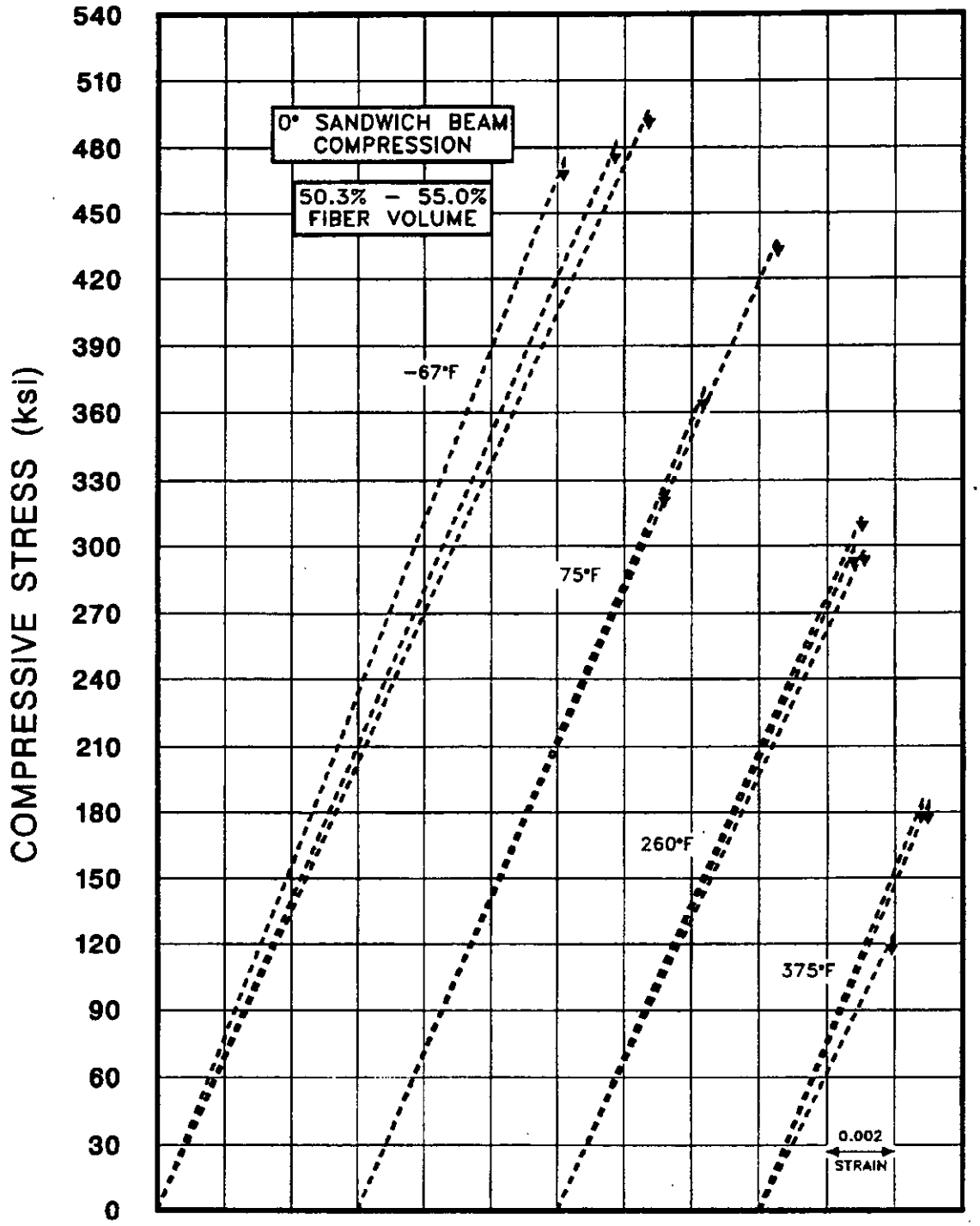


FIGURE A1.110.2 *Compressive stress-strain for AVCO 5505 boron/epoxy (100% - 0° orientation) loaded in the 0° direction. Individual tests shown.*

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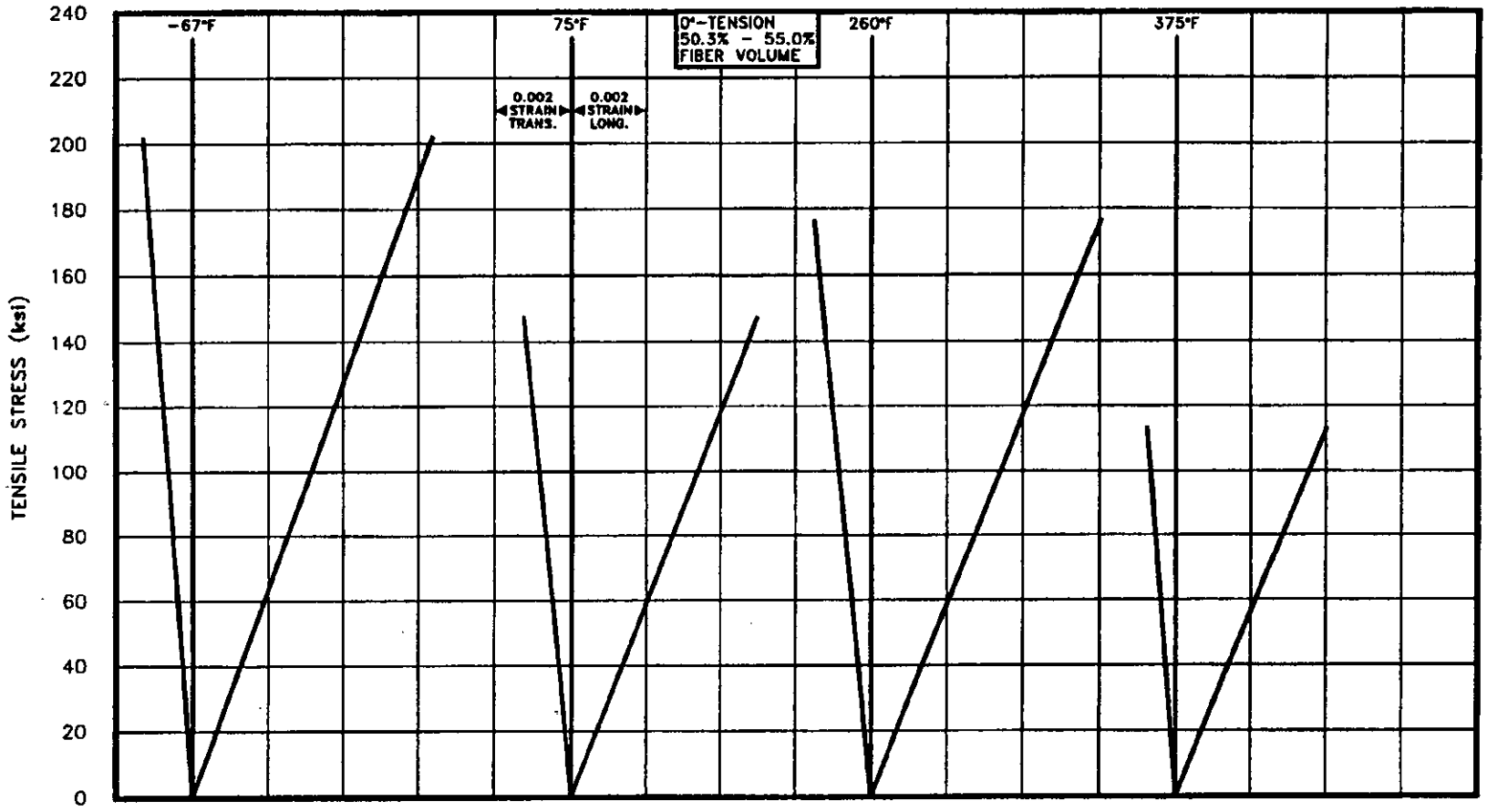


FIGURE A1.110.3 *Poisson effects for AVCO 5505 boron/epoxy (100% - 0° direction).*

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TABLE A1.111 *Summary of Mechanical Properties of Narmco 5505 Boron-Epoxy (0°-90° Crossply) (Tentative)*

Fabrication		Lay-up: [2(0/90)]S		Vacuum: 2 ins		Pressure: 50 ± 5 psi		Bleedout:		Cure: 1.5hr/ 350°F ± 10°F		Postcure: 2hr/380°F		Plies: 6			
Physical Properties		Weight Percent Resin:		Avg. Specific Gravity:		Avg. Percent Voids:		Avg. Thickness: 0.005 in/ply									
Test Methods		Tension: Tab-ended		Compression:		Shear: Picture Frame		Flexure:		Bearing:		Interlaminar Shear:					
Temperature		-67°F				75°F				260°F				375°F			
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry			
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD		
Tension																	
ultimate stress, ksi	0°	99.9				103.9				98.5				91.9			
	90°	23.6				17.8				11.4				8.1			
ultimate strain, %	0°	5400				5710				5830				5780			
	90°	15850				24470											
proportional limit, ksi	0°	53.0				77.7				48.6				48.6			
	90°																
initial modulus, 10^6 psi	0°	18.9				18.0				17.5				16.5			
	90°																
secondary modulus, 10^6 psi	0°																
	90°																
Compression																	
ultimate stress, ksi	0°																
	90°																
ultimate strain, %	0°																
	90°																
proportional limit, ksi	0°																
	90°																
initial modulus, 10^6 psi	0°																
	90°																
Shear																	
ultimate stress, ksi	0°-90°	19.5				17.3									5.4		
	±45°	65.7				63.7									33.3		

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°									
proportional limit, ksi	0°									
initial modulus, 10^6 psi	0°									
Bearing										
ultimate stress, ksi	0°									
stress at 4% elong., ksi	0°									
Interlaminar Shear										
ultimate stress, ksi	0°									

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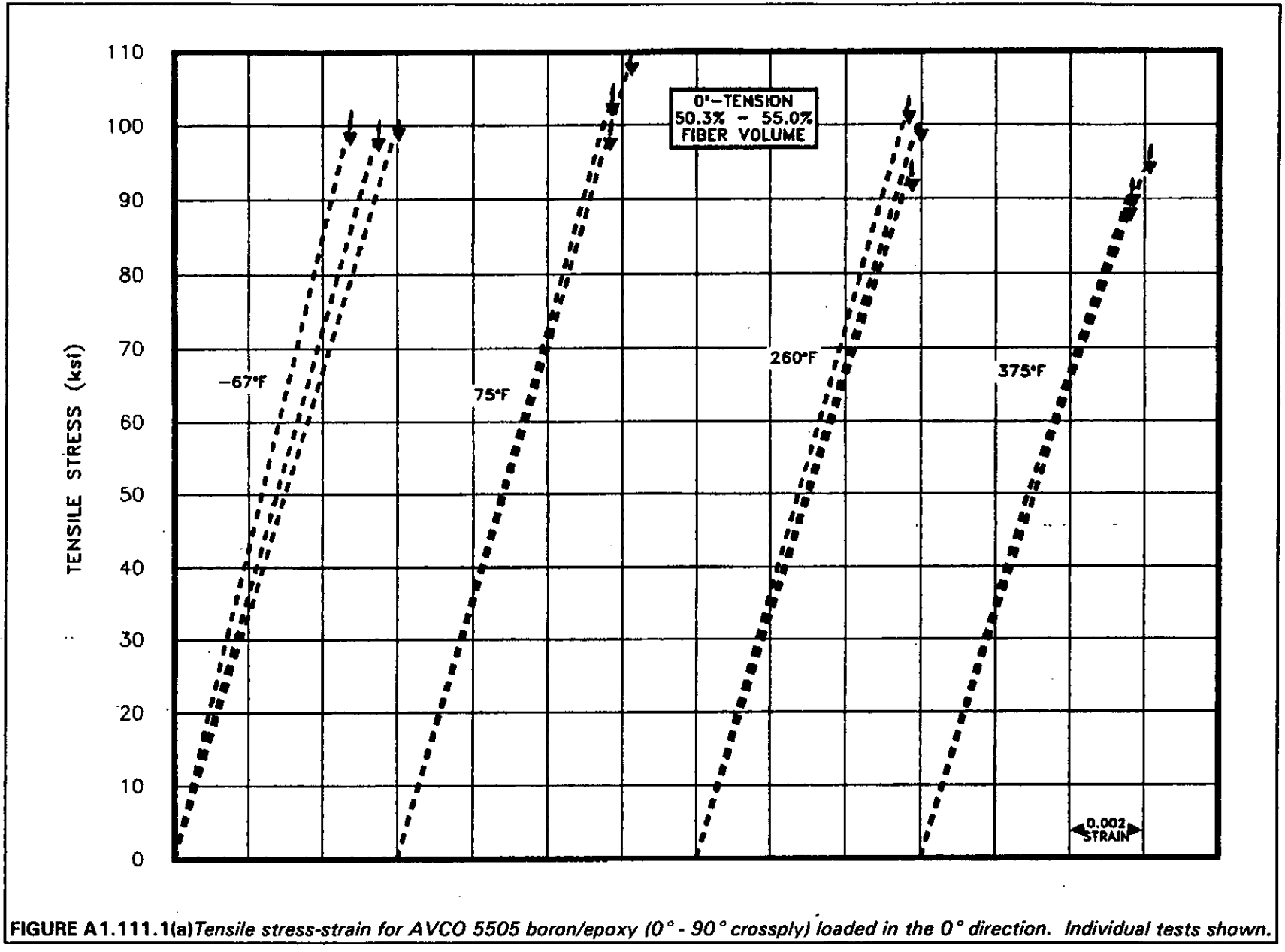


FIGURE A1.111.1(a) *Tensile stress-strain for AVCO 5505 boron/epoxy (0° - 90° crossply) loaded in the 0° direction. Individual tests shown.*

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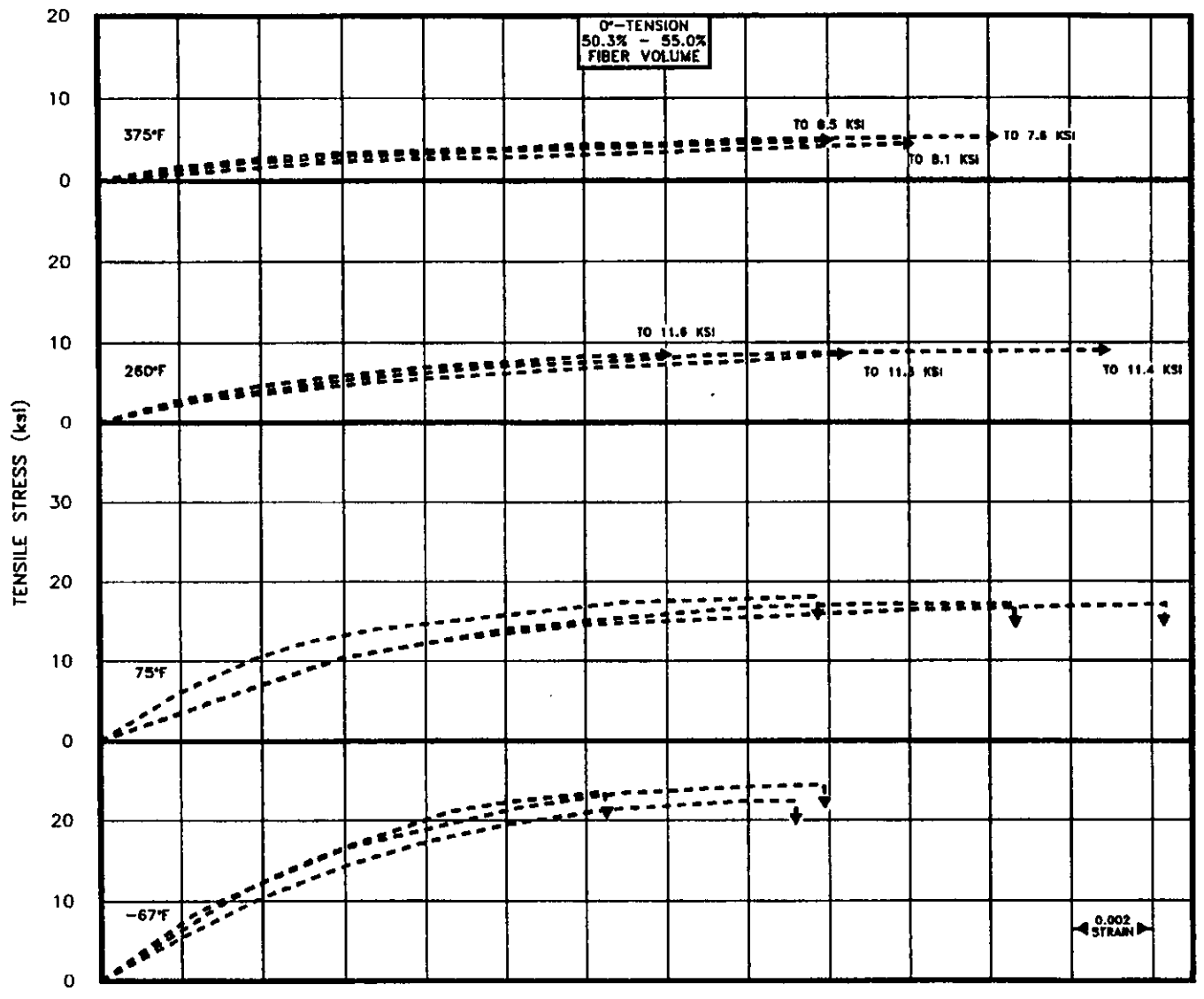


FIGURE A1.111.1(b) *Tensile stress-strain for AVCO 5505 boron/epoxy (0° - 90° crossply) loaded in the 45° direction. Individual tests shown.*

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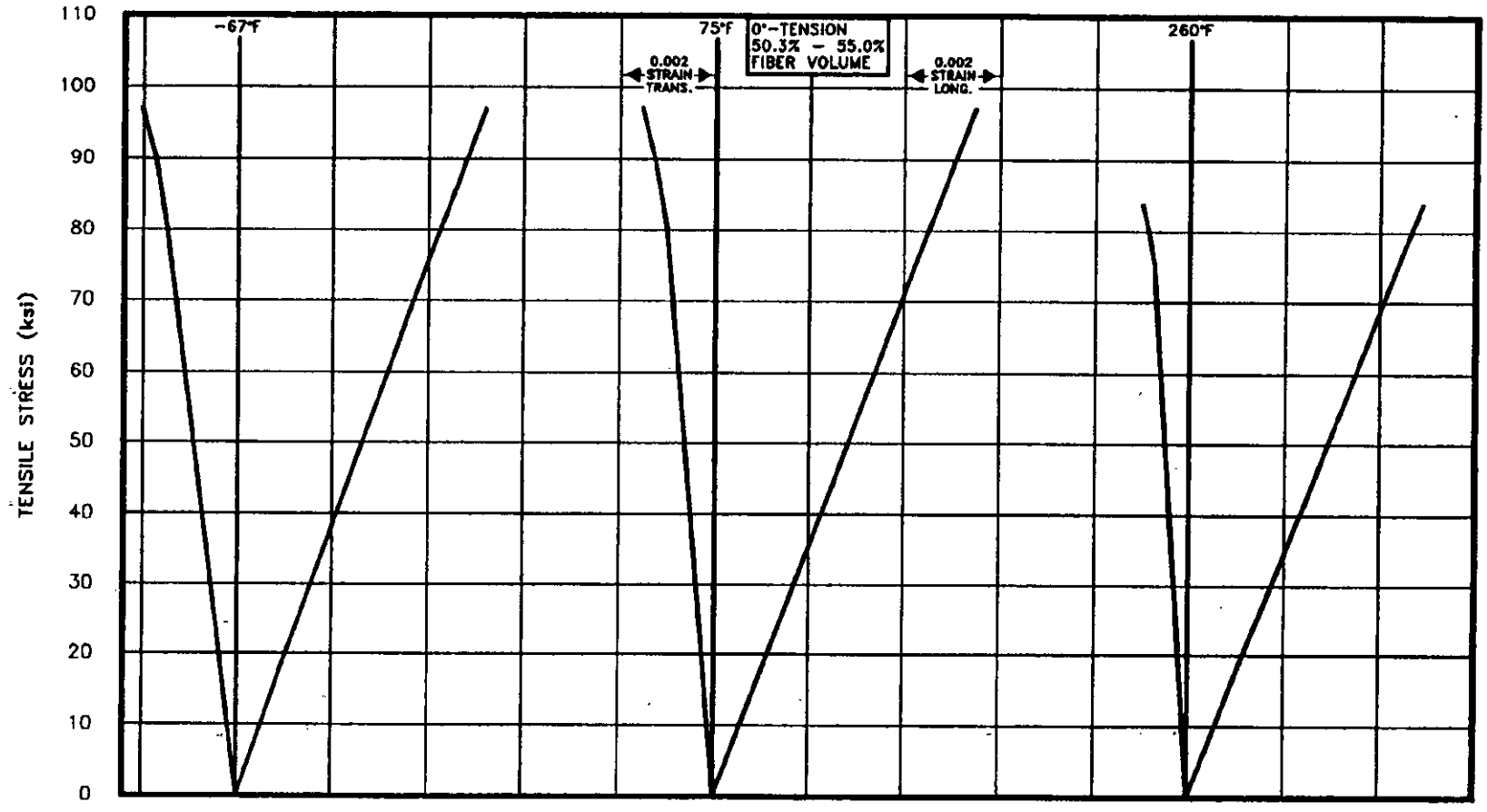


FIGURE A1.111.3 *Poisson effects for AVCO 5505 boron/epoxy (0° - 90° crossply).*

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